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1 E:\Anaconda\envs\airsim\python.exe E:\Github\AirSim_RL\Src\
batch_test_script.py
2 Gym has been unmaintained since 2022 and does not support NumPy 2
.0 amongst other critical functionality.
3 Please upgrade to Gymnasium, the maintained drop-in replacement of
Gym, or contact the authors of your software and request that they
upgrade.
4 Users of this version of Gym should be able to simply replace 'import
gym' with 'import gymnasium as gym' in the vast majority of cases.
5 See the migration guide at https://gymnasium.farama.org/introduction/
migration\_guide/ for additional information.
6 Detected UE-world task coordinates in E:\Github\AirSim_RL\dataset\
relative_coordinates_export.csv; converted to AirSim local NED using
origin (1910.0, -458.0, 100.0) and scale /100.
7 Initializing environment...
8 ??? CSV ?? 96 ??????? 4 ????????????
9 Connected!
10 Client Ver:1 (Min Req: 1), Server Ver:1 (Min Req: 1)
11
12 检测到无人机 'Drone1'
13 [Astar_planner] Loading TD3 lower policy: E:\Github\AirSim_RL\
Network\TD3_base\checkpoints\td3_resume_latest.zip
14 Starting batch test for 38 tasks...
15
16 --- Task 1/38 ---
17 From [101.4 29.68 -6.6 ] to [-75.1 -8.22 -16. ]
18 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
19 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells (
53.4%)
20 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
21 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
22 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
23 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.0
24 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
25 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=0.6
26 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
27 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=0.3
28 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
clearance penalties.
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29 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
30 [Astar_planner] WARNING: using legacy occupancy fallback; local
targets may be closer to obstacles.
31 Path planned via occupancy_legacy (Points: 18, Dist: 209.2m). Local
target z/alt: 0:-6.63(6.63m), 1:-21.47(21.47m), 2:-20.96(20.96m), 3:-22.
38(22.38m), 4:-22.36(22.36m), 5:-22.67(22.67m), 6:-21.43(21.43m), 7:-
24.26(24.26m), 8:-23.97(23.97m), 9:-21.37(21.37m), 10:-24.92(24.92m
), 11:-22.41(22.41m), 12:-20.32(20.32m), 13:-22.44(22.44m), 14:-23.37(
23.37m), 15:-24.39(24.39m), 16:-20.38(20.38m), 17:-15.99(15.99m)
32 [Astar_planner] Segment 0: [101.    30.    -6.626673] -> [ 92.
21.    -21.470577]
33 [Astar_planner] Placing first segment start at [101.    30.    -6.
626673]
34 [Astar_planner] Initial climb to 20.0m.
35 [Step 20 | Total    0] Target: ( 92.0, 21.0, -21.5) | Pos: ( 93.7, 21.6, -
19.9) | Vel: ( -0.6, -0.1,  0.3) | Acc: (  0.2, -1.6, -0.3) | Rwd:  4.44
36 [Astar_planner] Segment 0 reached in 20 steps, dist=2.33, axis_error=[
1.6587982 0.5909748 1.5315723], reason=running
37 [Astar_planner] Segment 1 starts from actual pos [ 93.651054 21.
589304 -19.934866] (planned waypoint was [ 92.    21.    -21.
470577])
38 [Astar_planner] Segment 1: [ 93.651054 21.589304 -19.934866] -> [ 92
.    10.    -20.963358]
39 [Step 18 | Total    0] Target: ( 92.0, 10.0, -21.0) | Pos: ( 91.4, 11.4, -
20.4) | Vel: ( -0.1, -1.6, -0.1) | Acc: (  0.1, -1.6,  0.5) | Rwd:  4.71
40 [Astar_planner] Segment 1 reached in 18 steps, dist=1.65, axis_error=[
0.630043 1.4033995 0.5874367], reason=running
41 [Astar_planner] Segment 2 starts from actual pos [ 91.36834 11.
384319 -20.376833] (planned waypoint was [ 92.    10.    -20.
963358])
42 [Astar_planner] Segment 2: [ 91.36834 11.384319 -20.376833] -> [ 90
.    7.    -22.376116]
43 [Step  5 | Total    0] Target: ( 90.0,  7.0, -22.4) | Pos: ( 89.4,  7.5, -20
.7) | Vel: ( -0.1, -0.1,  0.9) | Acc: ( -0.5,  2.0, -0.5) | Rwd: 601.96 ->
Episode End: ARRIVED!
44
45 [Astar_planner] Segment 2 reached in 5 steps, dist=1.80, axis_error=[0.
57832336 0.4591465 1.6429424 ], reason=arrived
46 [Astar_planner] Segment 3 starts from actual pos [ 89.42054  7.
458676 -20.724995] (planned waypoint was [ 90.    7.    -22.376116
])
47 [Astar_planner] Segment 3: [ 89.42054  7.458676 -20.724995] -> [ 77
.    7.    -22.358665]
48 [Step 49 | Total    0] Target: ( 77.0,  7.0, -22.4) | Pos: ( 75.1,  6.5, -
20.5) | Vel: ( -0.3,  1.3, -1.1) | Acc: ( -2.0,  2.0,  2.0) | Rwd:  5.97
49 [Astar_planner] Segment 3 reached in 49 steps, dist=2.74, axis_error=[
1.9300003 0.49151564 1.8834572 ], reason=running

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50 [Astar_planner] Segment 4 starts from actual pos [ 75.06794  6.
    5162473 -20.481834 ] (planned waypoint was [ 77.      7.    -22.
    358665])
51 [Astar_planner] Segment 4: [ 75.06794  6.5162473 -20.481834 ] -> [
    62.      7.    -22.671907]
52 [Step  25 | Total    0] Target: ( 62.0,  7.0, -22.7) | Pos: ( 63.7,  5.6, -
    20.9) | Vel: ( -0.2,  0.1,  0.2) | Acc: ( -0.8,  0.9, -0.6) | Rwd:  0.17
53 [Astar_planner] Segment 4 reached in 25 steps, dist=2.88, axis_error=[
    1.746769  1.392314  1.8207169], reason=running
54 [Astar_planner] Segment 5 starts from actual pos [ 63.744083  5.
    6090446 -20.849009 ] (planned waypoint was [ 62.      7.    -22.
    671907])
55 [Astar_planner] Segment 5: [ 63.744083  5.6090446 -20.849009 ] -> [
    55.      7.    -21.432335]
56 [Step  30 | Total    0] Target: ( 55.0,  7.0, -21.4) | Pos: ( 55.4,  5.5, -
    19.9) | Vel: (  0.3,  0.4, -0.2) | Acc: (  0.7,  0.0, -0.1) | Rwd:  2.02
57 [Astar_planner] Segment 5 reached in 30 steps, dist=2.24, axis_error=[
    0.4067917  1.5407867  1.5718937], reason=running
58 [Astar_planner] Segment 6 starts from actual pos [ 55.409256  5.
    4623833 -19.86174 ] (planned waypoint was [ 55.      7.    -21.
    432335])
59 [Astar_planner] Segment 6: [ 55.409256  5.4623833 -19.86174 ] -> [
    40.      6.    -24.257252]
60 [Step  64 | Total    0] Target: ( 40.0,  6.0, -24.3) | Pos: ( 42.0,  4.8, -
    23.2) | Vel: ( -0.0, -0.2, -0.0) | Acc: (  0.9, -1.6,  0.5) | Rwd: -1.61
61 [Astar_planner] Segment 6 reached in 64 steps, dist=2.53, axis_error=[
    1.981884  1.1741991  1.0434704], reason=running
62 [Astar_planner] Segment 7 starts from actual pos [ 41.98162  4.
    8237395 -23.213917 ] (planned waypoint was [ 40.      6.    -24.
    257252])
63 [Astar_planner] Segment 7: [ 41.98162  4.8237395 -23.213917 ] -> [
    25.      6.    -23.973452]
64 [Step  50 | Total    0] Target: ( 25.0,  6.0, -24.0) | Pos: ( 26.5,  4.3, -
    22.1) | Vel: (  0.1, -0.0,  0.8) | Acc: (  0.6, -1.3, -0.7) | Rwd:  1.01
65 [Astar_planner] Segment 7 reached in 50 steps, dist=2.94, axis_error=[
    1.4626083  1.7041845  1.892664 ], reason=running
66 [Astar_planner] Segment 8 starts from actual pos [ 26.46315  4.
    295738 -22.073366] (planned waypoint was [ 25.      6.    -23.973452
    ])
67 [Astar_planner] Segment 8: [ 26.46315  4.295738 -22.073366] -> [ 10
    .      6.    -21.370422]
68 [Step  37 | Total    0] Target: ( 10.0,  6.0, -21.4) | Pos: ( 10.1,  4.7, -
    19.5) | Vel: ( -0.1,  0.2,  0.5) | Acc: (  0.3, -1.6, -1.2) | Rwd: 600.67 ->
    Episode End: ARRIVED!
69
70 [Astar_planner] Segment 8 reached in 37 steps, dist=2.25, axis_error=[
    0.12232113  1.3079939  1.8212681 ], reason=arrived
71 [Astar_planner] Segment 9 starts from actual pos [ 10.121243  4.

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71 6939173 -19.544418 ] (planned waypoint was [ 10.      6.    -21.
    370422])
72 [Astar_planner] Segment 9: [ 10.121243  4.6939173 -19.544418
    ]-> [ 0.      6.    -24.91641]
73 [Step 20 | Total    0] Target: ( 0.0,  6.0, -24.9) | Pos: ( -0.4,  7.9, -
    23.9) | Vel: ( -0.6,  0.8,  0.3) | Acc: ( -1.1,  2.0,  1.9) | Rwd:  2.14
74 [Astar_planner] Segment 9 reached in 20 steps, dist=2.22, axis_error
    =[0.40021318 1.9216218 1.0293636 ], reason=running
75 [Astar_planner] Segment 10 starts from actual pos [ -0.4054249  7.
    9285226 -23.884363 ] (planned waypoint was [ 0.      6.    -24.91641
    ])
76 [Astar_planner] Segment 10: [ -0.4054249  7.9285226 -23.884363
    ]-> [ -8.      -2.    -22.412859]
77 [Step 23 | Total    0] Target: ( -8.0, -2.0, -22.4) | Pos: ( -6.0, -3.2, -
    21.4) | Vel: ( -0.0, -0.2, -0.0) | Acc: (  0.9, -0.5,  0.8) | Rwd: -1.68
78 [Astar_planner] Segment 10 reached in 23 steps, dist=2.52, axis_error
    =[1.9952631 1.1943431 0.97395134], reason=running
79 [Astar_planner] Segment 11 starts from actual pos [ -6.00515  -3.
    1957545 -21.439096 ] (planned waypoint was [ -8.      -2.    -22.
    412859])
80 [Astar_planner] Segment 11: [ -6.00515  -3.1957545 -21.439096
    ]-> [-23.      -2.    -20.322847]
81 [Step 35 | Total    0] Target: (-23.0, -2.0, -20.3) | Pos: (-21.5, -2.6, -
    19.4) | Vel: ( -0.9, -0.0,  0.7) | Acc: ( -1.9, -0.2,  0.4) | Rwd:  4.27
82 [Astar_planner] Segment 11 reached in 35 steps, dist=1.84, axis_error
    =[1.4784336 0.5542128 0.9417267], reason=running
83 [Astar_planner] Segment 12 starts from actual pos [-21.532604 -2.
    554733 -19.37287 ] (planned waypoint was [-23.      -2.    -20.
    322847])
84 [Astar_planner] Segment 12: [-21.532604 -2.554733 -19.37287 ]-> [-
    38.      -2.    -22.438341]
85 [Step 85 | Total    0] Target: (-38.0, -2.0, -22.4) | Pos: (-36.2, -1.7, -
    21.7) | Vel: ( -1.1, -0.3,  0.3) | Acc: ( -1.2, -0.7,  0.2) | Rwd:  5.09
86 [Astar_planner] Segment 12 reached in 85 steps, dist=1.95, axis_error
    =[1.7703362 0.31397176 0.74487305], reason=running
87 [Astar_planner] Segment 13 starts from actual pos [-36.23922  -1.
    6887758 -21.690147 ] (planned waypoint was [-38.      -2.    -22.
    438341])
88 [Astar_planner] Segment 13: [-36.23922  -1.6887758 -21.690147
    ]-> [-53.      -2.    -23.374292]
89 [Step 59 | Total    0] Target: (-53.0, -2.0, -23.4) | Pos: (-51.1, -1.0, -
    22.2) | Vel: ( -0.8,  0.3, -0.3) | Acc: ( -0.3,  1.1, -0.2) | Rwd:  4.91
90 [Astar_planner] Segment 13 reached in 59 steps, dist=2.43, axis_error
    =[1.900013 0.9854585 1.1439304], reason=running
91 [Astar_planner] Segment 14 starts from actual pos [-51.107006  -1.
    0117031 -22.233217 ] (planned waypoint was [-53.      -2.    -23.
    374292])
92 [Astar_planner] Segment 14: [-51.107006  -1.0117031 -22.233217

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92 ]-> [-64.    -2.    -24.3898]
93 [Step 38 | Total    0] Target: (-64.0, -2.0, -24.4) | Pos: (-62.6, -1.3, -
23.2) | Vel: ( -0.7, -0.2,  0.1) | Acc: ( 1.2, -1.3, -0.5) | Rwd: 3.84
94 [Astar_planner] Segment 14 reached in 38 steps, dist=1.96, axis_error
=[1.3766747 0.68280005 1.2236271 ], reason=running
95 [Astar_planner] Segment 15 starts from actual pos [-62.629345 -1.
318902 -23.165356] (planned waypoint was [-64.    -2.    -24.3898])
96 [Astar_planner] Segment 15: [-62.629345 -1.318902 -23.165356] -> [-
70.    -3.    -20.382774]
97 [Step 19 | Total    0] Target: (-70.0, -3.0, -20.4) | Pos: (-68.8, -1.8, -
20.9) | Vel: ( -0.9, -0.0, -0.4) | Acc: ( -0.7,  1.5,  0.2) | Rwd: 3.62
98 [Astar_planner] Segment 15 reached in 19 steps, dist=1.81, axis_error
=[1.2167358 1.2285465 0.5507984], reason=running
99 [Astar_planner] Segment 16 starts from actual pos [-68.79092  -1.
7717646 -20.937069 ] (planned waypoint was [-70.    -3.    -20.
382774])
100 [Astar_planner] Segment 16: [-68.79092  -1.7717646 -20.937069
] -> [-75.    -8.    -15.992509]
101 [Step 9 | Total    0] Target: (-75.0, -8.0, -16.0) | Pos: (-73.9, -8.5, -
15.2) | Vel: ( -0.3, -0.8,  0.4) | Acc: ( 1.7, -2.0,  0.3) | Rwd: 602.45
-> Episode End: ARRIVED!
102
103 [Astar_planner] Segment 16 reached in 9 steps, dist=1.44, axis_error
=[1.1287613 0.46500492 0.75750065], reason=arrived
104 Task 1 result: success (Success: Reached destination (17/17
segments). Shield interventions=138, emergencies=0, recoveries=23,
recovery_steps=138.)
105
106 --- Task 2/38 ---
107 From [ -61.1 -101.02  -3.9 ] to [-10.97211182 -13.19131592 -14.
84776611]
108 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
109 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
110 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
111 Path planned via occupancy (Points: 11, Dist: 142.5m). Local target z/
alt: 0:-3.91(3.91m), 1:-22.01(22.01m), 2:-20.58(20.58m), 3:-24.50(24.
50m), 4:-23.80(23.80m), 5:-23.75(23.75m), 6:-22.98(22.98m), 7:-21.55
(21.55m), 8:-22.56(22.56m), 9:-22.21(22.21m), 10:-14.85(14.85m)
112 [Astar_planner] Segment 0: [ -61.    -101.    -3.91104] -> [-59.    -
91.    -22.007843]
113 [Astar_planner] Placing first segment start at [ -61.    -101.    -3.
91104]
114 [Astar_planner] Initial climb to 20.0m.
115 [Step 28 | Total    0] Target: (-59.0, -91.0, -22.0) | Pos: (-60.8, -91.8
, -20.2) | Vel: ( 0.4,  0.2,  0.1) | Acc: ( 1.5,  1.8,  1.3) | Rwd: 0.52

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116 [Astar_planner] Segment 0 reached in 28 steps, dist=2.63, axis_error
    =[1.7705727 0.7507477 1.7992344], reason=running
117 [Astar_planner] Segment 1 starts from actual pos [-60.76836 -91.
    74974 -20.207722] (planned waypoint was [-59.    -91.    -22.
    007843])
118 [Astar_planner] Segment 1: [-60.76836 -91.74974 -20.207722] -> [-46
    .    -88.    -20.576109]
119 [Step 47 | Total    0] Target: (-46.0, -88.0, -20.6) | Pos: (-47.5, -89.4
    , -19.0) | Vel: ( 0.2, -0.6, 0.4) | Acc: ( -1.6, -2.0, -0.3) | Rwd: 0.85
120 [Astar_planner] Segment 1 reached in 47 steps, dist=2.60, axis_error
    =[1.5227051 1.4219818 1.5618038], reason=running
121 [Astar_planner] Segment 2 starts from actual pos [-47.520546 -89.
    42906 -19.009518] (planned waypoint was [-46.    -88.    -20.
    576109])
122 [Astar_planner] Segment 2: [-47.520546 -89.42906 -19.009518] -> [-
    32.    -88.    -24.501173]
123 [Step 38 | Total    0] Target: (-32.0, -88.0, -24.5) | Pos: (-30.8, -89.4
    , -23.5) | Vel: ( 0.4, 0.3, -0.0) | Acc: ( -0.2, 0.9, 0.1) | Rwd: -0.22
124 [Astar_planner] Segment 2 reached in 38 steps, dist=2.07, axis_error
    =[1.1529217 1.4092636 0.9936886], reason=running
125 [Astar_planner] Segment 3 starts from actual pos [-30.843903 -89.
    406334 -23.507648] (planned waypoint was [-32.    -88.    -24.
    501173])
126 [Astar_planner] Segment 3: [-30.843903 -89.406334 -23.507648] -> [-
    19.    -88.    -23.80164]
127 [Step 49 | Total    0] Target: (-19.0, -88.0, -23.8) | Pos: (-20.2, -87.3
    , -22.3) | Vel: ( 0.5, -0.0, 0.4) | Acc: ( 0.4, -0.8, -0.5) | Rwd: 602.86
    -> Episode End: ARRIVED!
128
129 [Astar_planner] Segment 3 reached in 49 steps, dist=2.03, axis_error
    =[1.208498 0.66189575 1.4842682 ], reason=arrived
130 [Astar_planner] Segment 4 starts from actual pos [-20.203781 -87.
    33831 -22.313595] (planned waypoint was [-19.    -88.    -23.80164
    ])
131 [Astar_planner] Segment 4: [-20.203781 -87.33831 -22.313595] -> [-
    11.    -80.    -23.754398]
132 [Step 24 | Total    0] Target: (-11.0, -80.0, -23.8) | Pos: (-12.3, -79.8
    , -22.0) | Vel: ( 0.7, -0.5, 0.2) | Acc: ( 0.3, -1.3, -1.3) | Rwd: 603.86
    -> Episode End: ARRIVED!
133
134 [Astar_planner] Segment 4 reached in 24 steps, dist=2.24, axis_error
    =[1.3380814 0.23743439 1.7847214 ], reason=arrived
135 [Astar_planner] Segment 5 starts from actual pos [-12.331967 -79.
    76676 -21.967913] (planned waypoint was [-11.    -80.    -23.
    754398])
136 [Astar_planner] Segment 5: [-12.331967 -79.76676 -21.967913] -> [-
    10.    -66.    -22.975521]
137 [Step 36 | Total    0] Target: (-10.0, -66.0, -23.0) | Pos: (-11.9, -66.0

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137 , -21.7) | Vel: ( 0.1, 0.1, 0.6) | Acc: ( -0.7, 0.5, -0.1) | Rwd: 2.20
138 [Astar_planner] Segment 5 reached in 36 steps, dist=2.31, axis_error
    =[1.899929 0.02854919 1.3215466 ], reason=running
139 [Astar_planner] Segment 6 starts from actual pos [-11.898653 -65.
    97044 -21.648468] (planned waypoint was [-10.    -66.    -22.
    975521])
140 [Astar_planner] Segment 6: [-11.898653 -65.97044 -21.648468] -> [-
    10.    -52.    -21.5466]
141 [Step 43 | Total    0] Target: (-10.0, -52.0, -21.5) | Pos: ( -8.5, -52.6
    , -20.9) | Vel: ( -0.4, 0.2, 0.2) | Acc: ( -1.0, 1.7, 1.1) | Rwd: 3.17
142 [Astar_planner] Segment 6 reached in 43 steps, dist=1.73, axis_error
    =[1.5118856 0.568882 0.6244087], reason=running
143 [Astar_planner] Segment 7 starts from actual pos [ -8.492088 -52.
    566742 -20.920694] (planned waypoint was [-10.    -52.    -21.5466])
144 [Astar_planner] Segment 7: [ -8.492088 -52.566742 -20.920694] -> [ -9
    .    -39.    -22.555838]
145 [Step 31 | Total    0] Target: ( -9.0, -39.0, -22.6) | Pos: ( -7.9, -40.1, -
    20.9) | Vel: ( -0.7, 0.4, 0.6) | Acc: ( -1.2, -0.3, -0.3) | Rwd: 4.36
146 [Astar_planner] Segment 7 reached in 31 steps, dist=2.24, axis_error
    =[1.0689774 1.1130753 1.6198292], reason=running
147 [Astar_planner] Segment 8 starts from actual pos [ -7.9371505 -40.
    109386 -20.930609 ] (planned waypoint was [ -9.    -39.    -22.
    555838])
148 [Astar_planner] Segment 8: [ -7.9371505 -40.109386 -20.930609
    ] -> [ -9.    -25.    -22.207874]
149 [Step 38 | Total    0] Target: ( -9.0, -25.0, -22.2) | Pos: (-10.7, -25.3
    , -21.0) | Vel: ( -0.1, 0.1, 0.0) | Acc: ( -0.7, -1.0, -0.0) | Rwd: 3.92
150 [Astar_planner] Segment 8 reached in 38 steps, dist=2.14, axis_error
    =[1.7013731 0.33917046 1.2477093 ], reason=running
151 [Astar_planner] Segment 9 starts from actual pos [-10.702833 -25.
    337843 -20.959406] (planned waypoint was [ -9.    -25.    -22.
    207874])
152 [Astar_planner] Segment 9: [-10.702833 -25.337843 -20.959406] -> [-
    11.    -13.    -14.847765]
153 [Step 17 | Total    0] Target: (-11.0, -13.0, -14.8) | Pos: (-10.0, -14.5
    , -14.2) | Vel: ( -0.3, 0.2, 0.1) | Acc: ( -0.2, -0.7, -1.1) | Rwd: 2.12
154 [Astar_planner] Segment 9 reached in 17 steps, dist=1.90, axis_error
    =[0.9520426 1.4882832 0.6902666], reason=running
155 Task 2 result: success (Success: Reached destination (10/10
    segments). Shield interventions=40, emergencies=0, recoveries=7,
    recovery_steps=40.)
156
157 --- Task 3/38 ---
158 From [119.7 78.28 -7.4 ] to [-106.2 125.18 -4.9 ]
159 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
160 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)

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161 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
162 Path planned via occupancy (Points: 25, Dist: 320.1m). Local target z/
    alt: 0:-7.40(7.40m), 1:-24.05(24.05m), 2:-22.81(22.81m), 3:-22.91(22.
    91m), 4:-22.73(22.73m), 5:-22.17(22.17m), 6:-20.45(20.45m), 7:-22.28
    (22.28m), 8:-22.27(22.27m), 9:-20.37(20.37m), 10:-23.21(23.21m), 11
    :-24.05(24.05m), 12:-20.39(20.39m), 13:-24.46(24.46m), 14:-22.72(22.
    72m), 15:-20.20(20.20m), 16:-23.13(23.13m), 17:-22.35(22.35m), 18:-
    23.00(23.00m), 19:-22.35(22.35m), 20:-22.97(22.97m), 21:-20.54(20.
    54m), 22:-20.77(20.77m), 23:-23.88(23.88m), 24:-4.90(4.90m)
163 [Astar_planner] Segment 0: [120.      78.      -7.3955536] -> [110
    .      84.      -24.051624]
164 [Astar_planner] Placing first segment start at [120.      78.      -7.
    3955536]
165 [Astar_planner] Initial climb to 20.0m.
166 [Step 14 | Total 0] Target: (110.0, 84.0, -24.1) | Pos: (111.2, 84.8
    , -22.8) | Vel: ( -0.6, -0.0, 0.2) | Acc: ( -0.4, -0.1, -0.6) | Rwd: 607.10
    -> Episode End: ARRIVED!
167
168 [Astar_planner] Segment 0 reached in 14 steps, dist=1.90, axis_error
    =[1.1892014 0.8421173 1.2119808], reason=arrived
169 [Astar_planner] Segment 1 starts from actual pos [111.1835 84.
    842026 -22.837807] (planned waypoint was [110.      84.      -24.
    051624])
170 [Astar_planner] Segment 1: [111.1835 84.842026 -22.837807] -> [
    105.      95.      -22.814888]
171 [Step 20 | Total 0] Target: (105.0, 95.0, -22.8) | Pos: (106.5, 93.1
    , -24.2) | Vel: ( -0.6, 0.4, 0.7) | Acc: ( -1.6, -0.4, -0.7) | Rwd: -0.93
172 [Astar_planner] Segment 1 reached in 20 steps, dist=2.74, axis_error
    =[1.5014496 1.8637772 1.3434944], reason=running
173 [Astar_planner] Segment 2 starts from actual pos [106.49781 93.
    138756 -24.154367] (planned waypoint was [105.      95.      -22.
    814888])
174 [Astar_planner] Segment 2: [106.49781 93.138756 -24.154367] -> [
    105.      108.      -22.905796]
175 [Step 43 | Total 0] Target: (105.0, 108.0, -22.9) | Pos: (108.0, 111.
    0, -21.3) | Vel: ( -0.3, -0.0, -0.9) | Acc: ( -1.0, 1.6, -0.7) | Rwd: -1006.
    09 -> Episode End: CRASHED!
176
177 [Astar_planner] Segment 2 ended: collision
178 [Astar_planner] Stop after failed segment 2.
179 Task 3 result: failed (Failed at segment 2: collision (2/24 reached).
    Shield interventions=31, emergencies=2, recoveries=6,
    recovery_steps=31.)
180
181 --- Task 4/38 ---
182 From [16. 3.18 -7.2 ] to [-106.2 125.18 -4.9 ]
183 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-

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183 200.0, 189.0], occupied=72869 cells (53.4%)
184 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
185 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
186 Path planned via occupancy (Points: 19, Dist: 253.2m). Local target z/
    alt: 0:-7.20(7.20m), 1:-20.38(20.38m), 2:-23.76(23.76m), 3:-23.04(23.
    04m), 4:-20.96(20.96m), 5:-23.12(23.12m), 6:-23.63(23.63m), 7:-21.86
    (21.86m), 8:-22.82(22.82m), 9:-24.16(24.16m), 10:-21.13(21.13m), 11
    :-23.97(23.97m), 12:-20.35(20.35m), 13:-24.76(24.76m), 14:-21.49(21.
    49m), 15:-20.32(20.32m), 16:-21.35(21.35m), 17:-24.91(24.91m), 18:-
    4.90(4.90m)
187 [Astar_planner] Segment 0: [16.      3.      -7.1976023] -> [ 3.      3
    .      -20.38302]
188 [Astar_planner] Placing first segment start at [16.      3.      -7.
    1976023]
189 [Astar_planner] Initial climb to 20.0m.
190 [Step 16 | Total 0] Target: ( 3.0, 3.0, -20.4) | Pos: ( 4.8, 3.3, -
    19.3) | Vel: ( -0.3, -0.3, 0.3) | Acc: ( 0.2, -1.3, -0.5) | Rwd: 0.69
191 [Astar_planner] Segment 0 reached in 16 steps, dist=2.09, axis_error
    =[1.7510862 0.28378892 1.0987701 ], reason=running
192 [Astar_planner] Segment 1 starts from actual pos [ 4.7481422 3.
    2809572 -19.281767 ] (planned waypoint was [ 3.      3.      -20.38302
    ])
193 [Astar_planner] Segment 1: [ 4.7481422 3.2809572 -19.281767
    ] -> [ -9.      -2.      -23.76483]
194 [Step 31 | Total 0] Target: ( -9.0, -2.0, -23.8) | Pos: ( -7.1, -2.4, -
    22.1) | Vel: ( -0.3, -0.1, 0.5) | Acc: ( 0.7, -1.5, 0.8) | Rwd: 2.31
195 [Astar_planner] Segment 1 reached in 31 steps, dist=2.58, axis_error
    =[1.9449606 0.43454456 1.6397419 ], reason=running
196 [Astar_planner] Segment 2 starts from actual pos [ -7.05742 -2.
    435047 -22.120264] (planned waypoint was [ -9.      -2.      -23.76483])
197 [Astar_planner] Segment 2: [ -7.05742 -2.435047 -22.120264] -> [-21
    .      0.      -23.037853]
198 [Step 41 | Total 0] Target: (-21.0, 0.0, -23.0) | Pos: (-19.0, -0.1, -
    21.6) | Vel: ( -0.3, 0.3, 0.4) | Acc: ( -0.1, 2.0, 0.8) | Rwd: 0.83
199 [Astar_planner] Segment 2 reached in 41 steps, dist=2.45, axis_error
    =[1.9723473 0.07527722 1.4546719 ], reason=running
200 [Astar_planner] Segment 3 starts from actual pos [-19.030571 -0.
    07292936 -21.579685 ] (planned waypoint was [-21.      0.      -23.
    037853])
201 [Astar_planner] Segment 3: [-19.030571 -0.07292936 -21.579685
    ] -> [-31.      8.      -20.958006]
202 [Step 18 | Total 0] Target: (-31.0, 8.0, -21.0) | Pos: (-29.4, 8.4, -
    20.2) | Vel: ( -0.3, -0.2, 0.3) | Acc: ( 0.4, -1.3, 0.2) | Rwd: 2.31
203 [Astar_planner] Segment 3 reached in 18 steps, dist=1.81, axis_error
    =[1.5979023 0.44529057 0.71982 ], reason=running
204 [Astar_planner] Segment 4 starts from actual pos [-29.404902 8.

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204 443367 -20.235617] (planned waypoint was [-31.      8.    -20.
    958006])
205 [Astar_planner] Segment 4: [-29.404902  8.443367 -20.235617] -> [-
    41.      18.    -23.12432]
206 [Step  24 | Total      0] Target: (-41.0, 18.0, -23.1) | Pos: (-39.5, 17.5
    , -21.4) | Vel: ( -0.6,  0.3, -0.8) | Acc: ( -1.4,  0.8,  0.3) | Rwd: 10.24
207 [Astar_planner] Segment 4 reached in 24 steps, dist=2.38, axis_error
    =[1.5251007 0.49708557 1.7537041 ], reason=running
208 [Astar_planner] Segment 5 starts from actual pos [-39.48179  17.
    506779 -21.379433] (planned waypoint was [-41.      18.    -23.12432
    ])
209 [Astar_planner] Segment 5: [-39.48179  17.506779 -21.379433] -> [-
    50.      27.    -23.627468]
210 [Step  20 | Total      0] Target: (-50.0, 27.0, -23.6) | Pos: (-50.2, 25.2
    , -22.3) | Vel: ( -0.4,  0.1, -0.3) | Acc: ( -0.0, -1.8,  0.7) | Rwd:  4.88
211 [Astar_planner] Segment 5 reached in 20 steps, dist=2.22, axis_error
    =[0.16321564 1.7605667 1.344347 ], reason=running
212 [Astar_planner] Segment 6 starts from actual pos [-50.166412 25.
    240538 -22.286018] (planned waypoint was [-50.      27.    -23.
    627468])
213 [Astar_planner] Segment 6: [-50.166412 25.240538 -22.286018] -> [-
    59.      37.    -21.856756]
214 [Step  27 | Total      0] Target: (-59.0, 37.0, -21.9) | Pos: (-57.7, 36.6
    , -21.1) | Vel: ( -0.7, -0.2,  0.3) | Acc: ( -0.2, -1.3,  1.1) | Rwd: 603.22
    -> Episode End: ARRIVED!
215
216 [Astar_planner] Segment 6 reached in 27 steps, dist=1.55, axis_error
    =[1.3127251 0.42313385 0.7157097 ], reason=arrived
217 [Astar_planner] Segment 7 starts from actual pos [-57.693153 36.
    574642 -21.138653] (planned waypoint was [-59.      37.    -21.
    856756])
218 [Astar_planner] Segment 7: [-57.693153 36.574642 -21.138653] -> [-
    60.      50.    -22.815634]
219 [Step  31 | Total      0] Target: (-60.0, 50.0, -22.8) | Pos: (-58.2, 51.9
    , -21.5) | Vel: ( -0.7, -0.1, -0.2) | Acc: ( -0.9, -1.9,  1.6) | Rwd: -1.43
220 [Astar_planner] Segment 7 reached in 31 steps, dist=2.92, axis_error
    =[1.8134956 1.882698 1.3033638], reason=running
221 [Astar_planner] Segment 8 starts from actual pos [-58.19271  51.
    88204 -21.513826] (planned waypoint was [-60.      50.    -22.
    815634])
222 [Astar_planner] Segment 8: [-58.19271  51.88204 -21.513826] -> [-53
    .      61.    -24.163425]
223 [Step  10 | Total      0] Target: (-53.0, 61.0, -24.2) | Pos: (-54.5, 60.4
    , -22.3) | Vel: (  0.8,  0.7, -0.4) | Acc: (  0.9, -1.4, -0.2) | Rwd:  4.22
224 [Astar_planner] Segment 8 reached in 10 steps, dist=2.41, axis_error
    =[1.4624252 0.5954399 1.8267994], reason=running
225 [Astar_planner] Segment 9 starts from actual pos [-54.455444 60.
    410778 -22.33971 ] (planned waypoint was [-53.      61.    -24.

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225 163425])
226 [Astar_planner] Segment 9: [-54.455444 60.410778 -22.33971 ] -> [-
    43.    68.   -21.129719]
227 [Step 17 | Total    0] Target: (-43.0, 68.0, -21.1) | Pos: (-44.6, 69.6
    , -20.7) | Vel: ( 0.6, 0.4, 0.2) | Acc: ( -0.5, 0.7, 0.1) | Rwd: 1.54
228 [Astar_planner] Segment 9 reached in 17 steps, dist=2.35, axis_error
    =[1.6389046 1.6219788 0.45256615], reason=running
229 [Astar_planner] Segment 10 starts from actual pos [-44.63327 69.
    62586 -20.675653] (planned waypoint was [-43.    68.   -21.
    129719])
230 [Astar_planner] Segment 10: [-44.63327 69.62586 -20.675653] -> [-
    44.    80.   -23.968426]
231 [Step 11 | Total    0] Target: (-44.0, 80.0, -24.0) | Pos: (-45.0, 79.1
    , -22.3) | Vel: ( -0.2, 0.9, -1.4) | Acc: ( -0.3, -0.5, 1.3) | Rwd: 605.71
    -> Episode End: ARRIVED!
232
233 [Astar_planner] Segment 10 reached in 11 steps, dist=2.12, axis_error
    =[0.96204376 0.89514923 1.6619091 ], reason=arrived
234 [Astar_planner] Segment 11 starts from actual pos [-44.964317 79.
    11618 -22.322607] (planned waypoint was [-44.    80.   -23.
    968426])
235 [Astar_planner] Segment 11: [-44.964317 79.11618 -22.322607] -> [-
    53.    89.   -20.352772]
236 [Step 12 | Total    0] Target: (-53.0, 89.0, -20.4) | Pos: (-51.5, 89.6
    , -18.9) | Vel: ( -0.8, 1.0, -2.0) | Acc: ( 1.3, -2.0, 0.3) | Rwd: 4.64
237 [Astar_planner] Segment 11 reached in 12 steps, dist=2.24, axis_error
    =[1.5477638 0.6387329 1.4930477], reason=running
238 [Astar_planner] Segment 12 starts from actual pos [-51.459877 89.
    648 -18.877502] (planned waypoint was [-53.    89.   -20.
    352772])
239 [Astar_planner] Segment 12: [-51.459877 89.648 -18.877502] -> [-
    67.    89.   -24.756481]
240 [Step 20 | Total    0] Target: (-67.0, 89.0, -24.8) | Pos: (-65.6, 88.9
    , -22.9) | Vel: ( -0.3, -0.1, -1.2) | Acc: ( 0.6, 0.2, 2.0) | Rwd: 604.92
    -> Episode End: ARRIVED!
241
242 [Astar_planner] Segment 12 reached in 20 steps, dist=2.31, axis_error
    =[1.3614426 0.13082886 1.8607349 ], reason=arrived
243 [Astar_planner] Segment 13 starts from actual pos [-65.640144 88.
    86841 -22.902754] (planned waypoint was [-67.    89.   -24.
    756481])
244 [Astar_planner] Segment 13: [-65.640144 88.86841 -22.902754] -> [-
    80.    89.   -21.487236]
245 [Step 20 | Total    0] Target: (-80.0, 89.0, -21.5) | Pos: (-81.7, 89.3
    , -20.8) | Vel: ( 0.3, 0.2, -0.0) | Acc: ( 1.7, 1.5, 0.9) | Rwd: 1.42
246 [Astar_planner] Segment 13 reached in 20 steps, dist=1.86, axis_error
    =[1.7188339 0.25582886 0.65341187], reason=running
247 [Astar_planner] Segment 14 starts from actual pos [-81.71581 89.

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247 25821 -20.834024] (planned waypoint was [-80.    89.    -21.
    487236])
248 [Astar_planner] Segment 14: [-81.71581  89.25821 -20.834024] -> [-
    93.    90.    -20.32381]
249 [Step  20 | Total    0] Target: (-93.0, 90.0, -20.3) | Pos: (-91.4, 89.4
    , -19.5) | Vel: ( -1.8, -0.2, -0.8) | Acc: ( -0.4, 0.2, -0.2) | Rwd: 4.05
250 [Astar_planner] Segment 14 reached in 20 steps, dist=1.85, axis_error
    =[1.563591  0.58509064 0.8054428 ], reason=running
251 [Astar_planner] Segment 15 starts from actual pos [-91.45221  89.
    41274 -19.525734] (planned waypoint was [-93.    90.    -20.32381
    ])
252 [Astar_planner] Segment 15: [-91.45221  89.41274 -19.525734] -> [-
    100.    101.    -21.35138]
253 [Step  30 | Total    0] Target: (-100.0, 101.0, -21.4) | Pos: (-99.8, 102
    .5, -20.9) | Vel: ( 0.4, -0.4, -0.0) | Acc: ( 1.2, -1.7, 0.1) | Rwd: 2.86

254 [Astar_planner] Segment 15 reached in 30 steps, dist=1.61, axis_error
    =[0.16880035 1.5247192 0.48469925], reason=running
255 [Astar_planner] Segment 16 starts from actual pos [-99.827965 102.
    521126 -20.866703] (planned waypoint was [-100.    101.    -21.
    35138])
256 [Astar_planner] Segment 16: [-99.827965 102.521126 -20.866703
    ] -> [-101.    114.    -24.910894]
257 [Step  16 | Total    0] Target: (-101.0, 114.0, -24.9) | Pos: (-101.1,
    112.6, -23.7) | Vel: ( -0.7, 1.2, -1.6) | Acc: ( -0.9, -1.4, 0.1) | Rwd:
    603.76 -> Episode End: ARRIVED!
258
259 [Astar_planner] Segment 16 reached in 16 steps, dist=1.82, axis_error
    =[0.07813263 1.3824158 1.1802177 ], reason=arrived
260 [Astar_planner] Segment 17 starts from actual pos [-101.0821  112.
    624695 -23.740248] (planned waypoint was [-101.    114.    -24.
    910894])
261 [Astar_planner] Segment 17: [-101.0821  112.624695 -23.740248
    ] -> [-106.    125.    -4.903579]
262 [Step  22 | Total    0] Target: (-106.0, 125.0, -4.9) | Pos: (-107.4,
    124.9, -5.9) | Vel: ( 1.3, 0.6, 0.1) | Acc: ( 0.7, -0.5, 0.5) | Rwd:
    601.37 -> Episode End: ARRIVED!
263
264 [Astar_planner] Segment 17 reached in 22 steps, dist=1.74, axis_error
    =[1.4469986 0.12696075 0.9509263 ], reason=arrived
265 Task 4 result: success (Success: Reached destination (18/18
    segments). Shield interventions=6, emergencies=0, recoveries=1,
    recovery_steps=6.)
266
267 --- Task 5/38 ---
268 From [-106.2  125.18  -4.9 ] to [-10.1 -76.12 -9.8 ]
269 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)

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270 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
271 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
272 Path planned via occupancy (Points: 24, Dist: 295.9m). Local target z/
    alt: 0:-4.91(4.91m), 1:-22.05(22.05m), 2:-21.84(21.84m), 3:-24.26(24.
    26m), 4:-20.20(20.20m), 5:-22.53(22.53m), 6:-23.93(23.93m), 7:-22.21
    (22.21m), 8:-23.08(23.08m), 9:-24.98(24.98m), 10:-23.67(23.67m), 11
    :-23.16(23.16m), 12:-20.19(20.19m), 13:-20.53(20.53m), 14:-23.81(23.
    81m), 15:-20.96(20.96m), 16:-23.56(23.56m), 17:-20.32(20.32m), 18:-
    24.97(24.97m), 19:-21.60(21.60m), 20:-24.80(24.80m), 21:-22.39(22.
    39m), 22:-21.15(21.15m), 23:-9.80(9.80m)
273 [Astar_planner] Segment 0: [-106.    125.    -4.905911] -> [-100
    .    115.    -22.047169]
274 [Astar_planner] Placing first segment start at [-106.    125.    -4.
    905911]
275 [Astar_planner] Initial climb to 20.0m.
276 [Step  17 | Total    0] Target: (-100.0, 115.0, -22.0) | Pos: (-101.7,
    114.3, -21.5) | Vel: ( 0.5, -0.2, -0.0) | Acc: ( -0.0, -2.0, 0.6) | Rwd:
    0.89
277 [Astar_planner] Segment 0 reached in 17 steps, dist=1.89, axis_error
    =[1.6756897 0.69488525 0.53499985], reason=running
278 [Astar_planner] Segment 1 starts from actual pos [-101.67281  114.
    30384 -21.512325] (planned waypoint was [-100.    115.    -22.
    047169])
279 [Astar_planner] Segment 1: [-101.67281  114.30384 -21.512325
    ] -> [-100.    103.    -21.842505]
280 [Step  16 | Total    0] Target: (-100.0, 103.0, -21.8) | Pos: (-100.9,
    102.6, -20.3) | Vel: ( 0.4, 0.1, 0.7) | Acc: ( 1.2, 0.8, -0.8) | Rwd:
    602.75 -> Episode End: ARRIVED!
281
282 [Astar_planner] Segment 1 reached in 16 steps, dist=1.84, axis_error
    =[0.8690033 0.37145996 1.5743542 ], reason=arrived
283 [Astar_planner] Segment 2 starts from actual pos [-100.86545  102.
    629524 -20.262102] (planned waypoint was [-100.    103.    -21.
    842505])
284 [Astar_planner] Segment 2: [-100.86545  102.629524 -20.262102
    ] -> [-95.    92.    -24.259474]
285 [Step  180 | Total    0] Target: (-95.0, 92.0, -24.3) | Pos: (-98.3, 96.4
    , -20.1) | Vel: ( 0.1, 0.0, 0.0) | Acc: ( 1.7, 0.1, -0.5) | Rwd: 0.20
286 [Astar_planner] Segment 2 timeout, dist=6.92
287 [Astar_planner] Stop after failed segment 2.
288 Task 5 result: failed (Failed at segment 2: segment_timeout (2/23
    reached). Shield interventions=72, emergencies=1, recoveries=12,
    recovery_steps=72.)
289
290 --- Task 6/38 ---
291 From [64.2  3.18 -7.4 ] to [-60.9 -95.62 -8.8 ]

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292 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
293 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
294 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
295 Path planned via occupancy (Points: 17, Dist: 221.5m). Local target z/
    alt: 0:-7.40(7.40m), 1:-23.82(23.82m), 2:-23.84(23.84m), 3:-21.90(21.
    90m), 4:-22.45(22.45m), 5:-22.33(22.33m), 6:-22.59(22.59m), 7:-24.84
    (24.84m), 8:-24.72(24.72m), 9:-22.96(22.96m), 10:-21.01(21.01m), 11
    :-22.03(22.03m), 12:-20.20(20.20m), 13:-21.95(21.95m), 14:-23.32(23.
    32m), 15:-20.74(20.74m), 16:-8.80(8.80m)
296 [Astar_planner] Segment 0: [64.      3.      -7.4023633] -> [ 51.
    2.      -23.822212]
297 [Astar_planner] Placing first segment start at [64.      3.      -7.
    4023633]
298 [Astar_planner] Initial climb to 20.0m.
299 [Step 17 | Total 0] Target: ( 51.0, 2.0, -23.8) | Pos: ( 53.0, 1.4, -
    22.3) | Vel: ( -0.6, -0.1, 0.9) | Acc: ( -1.5, -1.1, -0.9) | Rwd: 6.03
300 [Astar_planner] Segment 0 reached in 17 steps, dist=2.58, axis_error
    =[1.9506493 0.63784194 1.5705395 ], reason=running
301 [Astar_planner] Segment 1 starts from actual pos [ 52.94539 1.
    3608689 -22.243275 ] (planned waypoint was [ 51.      2.      -23.
    822212])
302 [Astar_planner] Segment 1: [ 52.94539 1.3608689 -22.243275 ] -> [
    38.      2.      -23.844625]
303 [Step 28 | Total 0] Target: ( 38.0, 2.0, -23.8) | Pos: ( 39.5, 0.6, -
    22.3) | Vel: ( -0.1, 0.0, 0.3) | Acc: ( 0.9, -1.0, -0.2) | Rwd: 3.84
304 [Astar_planner] Segment 1 reached in 28 steps, dist=2.54, axis_error
    =[1.503643 1.3900311 1.5104122], reason=running
305 [Astar_planner] Segment 2 starts from actual pos [ 39.503235 0.
    61011463 -22.332418 ] (planned waypoint was [ 38.      2.      -23.
    844625])
306 [Astar_planner] Segment 2: [ 39.503235 0.61011463 -22.332418
    ] -> [ 25.      1.      -21.901377]
307 [Step 25 | Total 0] Target: ( 25.0, 1.0, -21.9) | Pos: ( 26.8, 0.2, -
    20.3) | Vel: ( -0.1, -0.1, 0.7) | Acc: ( 1.3, -1.0, -1.6) | Rwd: 0.98
308 [Astar_planner] Segment 2 reached in 25 steps, dist=2.59, axis_error
    =[1.832798 0.8449925 1.6285133], reason=running
309 [Astar_planner] Segment 3 starts from actual pos [ 26.831665 0.
    15401556 -20.266743 ] (planned waypoint was [ 25.      1.      -21.
    901377])
310 [Astar_planner] Segment 3: [ 26.831665 0.15401556 -20.266743
    ] -> [ 12.      1.      -22.450315]
311 [Step 25 | Total 0] Target: ( 12.0, 1.0, -22.5) | Pos: ( 13.6, 0.2, -
    21.3) | Vel: ( -0.4, 0.6, 0.4) | Acc: ( -0.3, 0.2, -0.1) | Rwd: 4.74
312 [Astar_planner] Segment 3 reached in 25 steps, dist=2.14, axis_error
    =[1.6188459 0.7927408 1.157753 ], reason=running

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313 [Astar_planner] Segment 4 starts from actual pos [ 13.615085  0.
    21229033 -21.288496 ] (planned waypoint was [ 12.      1.    -22.
    450315])
314 [Astar_planner] Segment 4: [ 13.615085  0.21229033 -21.288496
    ] -> [ -2.      1.    -22.333937]
315 [Step  25 | Total    0] Target: ( -2.0,  1.0, -22.3) | Pos: ( -1.3,  1.4, -
    20.5) | Vel: ( -0.4, -0.0,  0.5) | Acc: (  1.1, -1.7,  1.0) | Rwd: 601.10
    -> Episode End: ARRIVED!
316
317 [Astar_planner] Segment 4 reached in 25 steps, dist=1.98, axis_error
    =[0.65967774 0.39673018 1.8292923 ], reason=arrived
318 [Astar_planner] Segment 5 starts from actual pos [ -1.3437567  1.
    3963908 -20.499878 ] (planned waypoint was [ -2.      1.    -22.
    333937])
319 [Astar_planner] Segment 5: [ -1.3437567  1.3963908 -20.499878
    ] -> [ -9.      -9.    -22.588696]
320 [Step  18 | Total    0] Target: ( -9.0, -9.0, -22.6) | Pos: ( -7.9, -8.5, -
    21.4) | Vel: ( -0.2, -0.4,  0.2) | Acc: (  0.8, -0.9, -0.4) | Rwd: 605.64
    -> Episode End: ARRIVED!
321
322 [Astar_planner] Segment 5 reached in 18 steps, dist=1.70, axis_error
    =[1.111927  0.4664049 1.2042007], reason=arrived
323 [Astar_planner] Segment 6 starts from actual pos [ -7.889871 -8.
    537352 -21.382349] (planned waypoint was [ -9.      -9.    -22.
    588696])
324 [Astar_planner] Segment 6: [ -7.889871 -8.537352 -21.382349] -> [ -9
    .      -22.    -24.835407]
325 [Step  21 | Total    0] Target: ( -9.0, -22.0, -24.8) | Pos: ( -9.4, -22.6, -
    22.9) | Vel: (  0.5, -0.9,  0.5) | Acc: (  0.0, -1.9, -0.7) | Rwd: 600.70
    -> Episode End: ARRIVED!
326
327 [Astar_planner] Segment 6 reached in 21 steps, dist=2.10, axis_error
    =[0.40500832 0.6013355 1.9672661 ], reason=arrived
328 [Astar_planner] Segment 7 starts from actual pos [ -9.40072 -22.
    609203 -22.863342] (planned waypoint was [ -9.      -22.    -24.
    835407])
329 [Astar_planner] Segment 7: [ -9.40072 -22.609203 -22.863342] -> [ -9
    .      -35.    -24.719042]
330 [Step  13 | Total    0] Target: ( -9.0, -35.0, -24.7) | Pos: ( -9.0, -34.3, -
    23.1) | Vel: (  0.2, -1.3, -0.1) | Acc: (  0.2, -1.8, -0.9) | Rwd: 605.31
    -> Episode End: ARRIVED!
331
332 [Astar_planner] Segment 7 reached in 13 steps, dist=1.80, axis_error
    =[0.02801323 0.7293625 1.6449795 ], reason=arrived
333 [Astar_planner] Segment 8 starts from actual pos [ -8.970498 -34.
    278572 -23.07445 ] (planned waypoint was [ -9.      -35.    -24.
    719042])
334 [Astar_planner] Segment 8: [ -8.970498 -34.278572 -23.07445 ] -> [ -9

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334 .      -48.      -22.955704]
335 [Step 13 | Total    0] Target: ( -9.0, -48.0, -23.0) | Pos: ( -8.0, -46.6, -
    22.4) | Vel: ( 0.6, -1.2, 0.3) | Acc: ( 1.0, -1.5, 0.7) | Rwd: 3.59
336 [Astar_planner] Segment 8 reached in 13 steps, dist=1.84, axis_error
    =[0.98962307 1.4308434 0.58930016], reason=running
337 [Astar_planner] Segment 9 starts from actual pos [ -8.004967 -46.
    57996 -22.364086] (planned waypoint was [ -9.      -48.      -22.
    955704])
338 [Astar_planner] Segment 9: [ -8.004967 -46.57996 -22.364086] -> [ -9
    .      -62.      -21.008547]
339 [Step 15 | Total    0] Target: ( -9.0, -62.0, -21.0) | Pos: (-10.4, -61.1
    , -20.8) | Vel: ( 0.4, -1.2, -0.2) | Acc: ( 0.5, -1.9, 1.2) | Rwd: 2.68
340 [Astar_planner] Segment 9 reached in 15 steps, dist=1.71, axis_error
    =[1.4314318 0.90872574 0.20012856], reason=running
341 [Astar_planner] Segment 10 starts from actual pos [-10.427997 -61.
    10185 -20.809864] (planned waypoint was [ -9.      -62.      -21.
    008547])
342 [Astar_planner] Segment 10: [-10.427997 -61.10185 -20.809864] -> [-
    10.      -75.      -22.032099]
343 [Step 11 | Total    0] Target: (-10.0, -75.0, -22.0) | Pos: ( -9.4, -73.8
    , -21.0) | Vel: ( 0.1, -1.8, -0.5) | Acc: ( 0.1, -2.0, 1.3) | Rwd: 607.92
    -> Episode End: ARRIVED!
344
345 [Astar_planner] Segment 10 reached in 11 steps, dist=1.70, axis_error
    =[0.61337185 1.1739731 1.0608921 ], reason=arrived
346 [Astar_planner] Segment 11 starts from actual pos [ -9.386054 -73.
    84184 -20.97509 ] (planned waypoint was [-10.      -75.      -22.
    032099])
347 [Astar_planner] Segment 11: [ -9.386054 -73.84184 -20.97509 ] -> [-
    13.      -86.      -20.196247]
348 [Step 10 | Total    0] Target: (-13.0, -86.0, -20.2) | Pos: (-12.7, -84.1
    , -18.8) | Vel: ( -0.5, -1.0, -0.1) | Acc: ( -0.4, -0.8, 0.8) | Rwd: 2.08
349 [Astar_planner] Segment 11 reached in 10 steps, dist=2.39, axis_error
    =[0.2881527 1.942955 1.3675632], reason=running
350 [Astar_planner] Segment 12 starts from actual pos [-12.718069 -84.
    06902 -18.830023] (planned waypoint was [-13.      -86.      -20.
    196247])
351 [Astar_planner] Segment 12: [-12.718069 -84.06902 -18.830023] -> [-
    26.      -88.      -21.954151]
352 [Step 17 | Total    0] Target: (-26.0, -88.0, -22.0) | Pos: (-27.1, -88.0
    , -20.0) | Vel: ( -0.8, 0.3, 0.3) | Acc: ( -1.4, -0.3, -1.9) | Rwd: 599.22
    -> Episode End: ARRIVED!
353
354 [Astar_planner] Segment 12 reached in 17 steps, dist=2.26, axis_error
    =[1.1343994 0.03492737 1.9562759 ], reason=arrived
355 [Astar_planner] Segment 13 starts from actual pos [-27.144333 -88.
    031204 -19.993788] (planned waypoint was [-26.      -88.      -21.
    954151])

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356 [Astar_planner] Segment 13: [-27.144333 -88.031204 -19.993788] -> [-
39.    -88.    -23.321596]
357 [Step 15 | Total    0] Target: (-39.0, -88.0, -23.3) | Pos: (-37.7, -88.8
, -22.2) | Vel: ( -0.6,  0.4,  0.6) | Acc: ( -0.8,  1.2,  0.1) | Rwd: 5.03
358 [Astar_planner] Segment 13 reached in 15 steps, dist=1.90, axis_error
=[1.3142548 0.8086777 1.1127968], reason=running
359 [Astar_planner] Segment 14 starts from actual pos [-37.691063 -88.
804855 -22.203016] (planned waypoint was [-39.    -88.    -23.
321596])
360 [Astar_planner] Segment 14: [-37.691063 -88.804855 -22.203016] -> [-
53.    -88.    -20.739195]
361 [Step 16 | Total    0] Target: (-53.0, -88.0, -20.7) | Pos: (-51.5, -88.3
, -20.2) | Vel: ( -1.3,  0.3,  0.1) | Acc: ( -1.3,  0.3,  0.1) | Rwd: 604.40
-> Episode End: ARRIVED!
362
363 [Astar_planner] Segment 14 reached in 16 steps, dist=1.58, axis_error
=[1.4668007 0.30675507 0.50767326], reason=arrived
364 [Astar_planner] Segment 15 starts from actual pos [-51.54854 -88.
30325 -20.229832] (planned waypoint was [-53.    -88.    -20.
739195])
365 [Astar_planner] Segment 15: [-51.54854 -88.30325 -20.229832] -> [-
61. -96. -8.8]
366 [Step 27 | Total    0] Target: (-61.0, -96.0, -8.8) | Pos: (-60.8, -94.6
, -8.6) | Vel: ( 0.0, -0.2, -0.0) | Acc: ( 0.7,  0.4,  0.8) | Rwd: 600.41
-> Episode End: ARRIVED!
367
368 [Astar_planner] Segment 15 reached in 27 steps, dist=1.38, axis_error
=[0.15657806 1.3596954 0.15203857], reason=arrived
369 Task 6 result: success (Success: Reached destination (16/16
segments). Shield interventions=0, emergencies=0, recoveries=0,
recovery_steps=0.)
370
371 --- Task 7/38 ---
372 From [108. -87.82 -5.4 ] to [ -61. -139.82 -11.6 ]
373 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
374 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
375 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
376 Path planned via occupancy (Points: 18, Dist: 248.7m). Local target z/
alt: 0:-5.41(5.41m), 1:-24.47(24.47m), 2:-24.03(24.03m), 3:-24.75(24.
75m), 4:-20.02(20.02m), 5:-23.19(23.19m), 6:-23.51(23.51m), 7:-24.89
(24.89m), 8:-20.33(20.33m), 9:-23.75(23.75m), 10:-20.39(20.39m), 11
:-23.87(23.87m), 12:-22.67(22.67m), 13:-21.13(21.13m), 14:-24.07(24.
07m), 15:-20.42(20.42m), 16:-24.63(24.63m), 17:-11.60(11.60m)
377 [Astar_planner] Segment 0: [108.    -88.    -5.4063115] -> [ 98
.    -98.    -24.468105]

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378 [Astar_planner] Placing first segment start at [108.    -88.    -5.
    4063115]
379 [Astar_planner] Initial climb to 20.0m.
380 [Step 16 | Total    0] Target: ( 98.0, -98.0, -24.5) | Pos: ( 99.9, -99.6
    , -23.3) | Vel: ( -0.8, -0.8, -1.0) | Acc: ( 1.1, -0.8, -0.7) | Rwd: 2.10
381 [Astar_planner] Segment 0 reached in 16 steps, dist=2.77, axis_error
    =[1.9235382 1.6328964 1.1329346], reason=running
382 [Astar_planner] Segment 1 starts from actual pos [ 99.918846 -99.
    63774 -23.341326] (planned waypoint was [ 98.    -98.    -24.
    468105])
383 [Astar_planner] Segment 1: [ 99.918846 -99.63774 -23.341326] -> [
    97.    -111.    -24.032797]
384 [Step 19 | Total    0] Target: ( 97.0, -111.0, -24.0) | Pos: ( 97.7, -109
    .2, -23.0) | Vel: ( -0.3, -1.1, -1.2) | Acc: ( 0.9, 1.1, -0.2) | Rwd: 3.88

385 [Astar_planner] Segment 1 reached in 19 steps, dist=2.15, axis_error
    =[0.72615814 1.7614441 1.0010605 ], reason=running
386 [Astar_planner] Segment 2 starts from actual pos [ 97.72341 -109.
    248405 -23.042557] (planned waypoint was [ 97.    -111.    -24.
    032797])
387 [Astar_planner] Segment 2: [ 97.72341 -109.248405 -23.042557
    ] -> [ 90.    -122.    -24.745218]
388 [Step 15 | Total    0] Target: ( 90.0, -122.0, -24.7) | Pos: ( 90.7, -120
    .5, -24.0) | Vel: ( -0.5, -1.2, -0.8) | Acc: ( 0.5, 0.7, 0.5) | Rwd: 3.68

389 [Astar_planner] Segment 2 reached in 15 steps, dist=1.82, axis_error
    =[0.6563568 1.5266495 0.7369442], reason=running
390 [Astar_planner] Segment 3 starts from actual pos [ 90.65323 -120.
    48034 -24.013094] (planned waypoint was [ 90.    -122.    -24.
    745218])
391 [Astar_planner] Segment 3: [ 90.65323 -120.48034 -24.013094
    ] -> [ 82.    -133.    -20.019382]
392 [Step 14 | Total    0] Target: ( 82.0, -133.0, -20.0) | Pos: ( 83.5, -131
    .2, -19.7) | Vel: ( 0.3, -1.1, 0.5) | Acc: ( 2.0, 1.1, 0.6) | Rwd: 0.32
393 [Astar_planner] Segment 3 reached in 14 steps, dist=2.37, axis_error
    =[1.4781952 1.8097229 0.36795998], reason=running
394 [Astar_planner] Segment 4 starts from actual pos [ 83.479996 -131.
    19666 -19.648706] (planned waypoint was [ 82.    -133.    -20.
    019382])
395 [Astar_planner] Segment 4: [ 83.479996 -131.19666 -19.648706
    ] -> [ 81.    -147.    -23.189243]
396 [Step 29 | Total    0] Target: ( 81.0, -147.0, -23.2) | Pos: ( 82.9, -145
    .8, -23.1) | Vel: ( -0.3, -0.7, 0.1) | Acc: ( 0.4, 0.5, 0.2) | Rwd: 0.78
397 [Astar_planner] Segment 4 reached in 29 steps, dist=2.26, axis_error
    =[1.8858185 1.2465057 0.10832405], reason=running
398 [Astar_planner] Segment 5 starts from actual pos [ 82.88504 -145.
    75548 -23.080576] (planned waypoint was [ 81.    -147.    -23.
    189243])

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399 [Astar_planner] Segment 5: [ 82.88504 -145.75548 -23.080576
    ] -> [ 73. -156. -23.5107]
400 [Step 14 | Total 0] Target: ( 73.0, -156.0, -23.5) | Pos: ( 74.9, -154
    .7, -23.3) | Vel: ( -0.8, -1.0, 0.3) | Acc: ( 0.5, -0.5, 0.5) | Rwd: 1.44

401 [Astar_planner] Segment 5 reached in 14 steps, dist=2.32, axis_error
    =[1.884346 1.3295898 0.21461296], reason=running
402 [Astar_planner] Segment 6 starts from actual pos [ 74.8773 -154.
    6798 -23.293608] (planned waypoint was [ 73. -156. -23.5107
    ])
403 [Astar_planner] Segment 6: [ 74.8773 -154.6798 -23.293608] -> [
    59. -156. -24.894697]
404 [Step 33 | Total 0] Target: ( 59.0, -156.0, -24.9) | Pos: ( 57.2, -155
    .9, -24.0) | Vel: ( 0.5, 0.4, 0.5) | Acc: ( 0.6, 2.0, 1.4) | Rwd: 2.97
405 [Astar_planner] Segment 6 reached in 33 steps, dist=1.97, axis_error
    =[1.7536125 0.0766449 0.8926544], reason=running
406 [Astar_planner] Segment 7 starts from actual pos [ 57.25205 -155.
    91896 -23.995989] (planned waypoint was [ 59. -156. -24.
    894697])
407 [Astar_planner] Segment 7: [ 57.25205 -155.91896 -23.995989
    ] -> [ 45. -156. -20.32988]
408 [Step 18 | Total 0] Target: ( 45.0, -156.0, -20.3) | Pos: ( 43.1, -157
    .7, -18.8) | Vel: ( -0.1, 0.4, -0.0) | Acc: ( -0.9, 1.1, -0.0) | Rwd: 0.72

409 [Astar_planner] Segment 7 reached in 18 steps, dist=3.00, axis_error
    =[1.9403801 1.6998444 1.5260544], reason=running
410 [Astar_planner] Segment 8 starts from actual pos [ 43.058487 -157.
    69632 -18.803825] (planned waypoint was [ 45. -156. -20.
    32988])
411 [Astar_planner] Segment 8: [ 43.058487 -157.69632 -18.803825
    ] -> [ 32. -159. -23.753708]
412 [Step 34 | Total 0] Target: ( 32.0, -159.0, -23.8) | Pos: ( 30.4, -160
    .3, -22.8) | Vel: ( 0.8, -0.7, 0.1) | Acc: ( -0.7, -1.6, 0.5) | Rwd: -0.00

413 [Astar_planner] Segment 8 reached in 34 steps, dist=2.26, axis_error
    =[1.5787086 1.2695618 0.99367714], reason=running
414 [Astar_planner] Segment 9 starts from actual pos [ 30.428312 -160.
    27608 -22.759443] (planned waypoint was [ 32. -159. -23.
    753708])
415 [Astar_planner] Segment 9: [ 30.428312 -160.27608 -22.759443
    ] -> [ 19. -160. -20.385489]
416 [Step 14 | Total 0] Target: ( 19.0, -160.0, -20.4) | Pos: ( 20.2, -160
    .9, -19.8) | Vel: ( -1.3, 0.2, 0.1) | Acc: ( -0.8, 0.0, -0.7) | Rwd: 604.
    77 -> Episode End: ARRIVED!
417
418 [Astar_planner] Segment 9 reached in 14 steps, dist=1.59, axis_error
    =[1.1566486 0.9122772 0.5969238], reason=arrived
419 [Astar_planner] Segment 10 starts from actual pos [ 20.144957 -160.

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419 91072 -19.787922] (planned waypoint was [ 19.    -160.    -20.
    385489])
420 [Astar_planner] Segment 10: [ 20.144957 -160.91072 -19.787922
    ] -> [ 5.    -161.    -23.866543]
421 [Step 159 | Total    0] Target: ( 5.0, -161.0, -23.9) | Pos: ( 9.9, -169.
    7, -25.7) | Vel: ( 0.0,  0.5,  0.1) | Acc: ( 0.9,  1.4, -0.7) | Rwd: -899.
    75 -> Episode End: CRASHED!
422
423 [Astar_planner] Segment 10 ended: collision
424 [Astar_planner] Stop after failed segment 10.
425 Task 7 result: failed (Failed at segment 10: collision (10/17 reached).
    Shield interventions=136, emergencies=3, recoveries=23,
    recovery_steps=136.)
426
427 --- Task 8/38 ---
428 From [-73.2 -85.22 -5.6 ] to [ 57.9  19.78 -11.3 ]
429 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
430 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
431 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
432 Path planned via occupancy (Points: 20, Dist: 229.9m). Local target z/
    alt: 0:-5.61(5.61m), 1:-20.30(20.30m), 2:-23.10(23.10m), 3:-22.32(22.
    32m), 4:-20.06(20.06m), 5:-20.45(20.45m), 6:-23.92(23.92m), 7:-24.84
    (24.84m), 8:-23.04(23.04m), 9:-21.79(21.79m), 10:-24.83(24.83m), 11
    :-24.09(24.09m), 12:-21.27(21.27m), 13:-22.72(22.72m), 14:-21.22(21.
    22m), 15:-23.68(23.68m), 16:-21.99(21.99m), 17:-23.05(23.05m), 18:-
    24.13(24.13m), 19:-11.30(11.30m)
433 [Astar_planner] Segment 0: [-73.    -85.    -5.61009] -> [-62.    -87
    .    -20.296148]
434 [Astar_planner] Placing first segment start at [-73.    -85.    -5.61009
    ]
435 [Astar_planner] Initial climb to 20.0m.
436 [Step  13 | Total    0] Target: (-62.0, -87.0, -20.3) | Pos: (-63.5, -87.9
    , -19.7) | Vel: ( 0.5,  0.0,  0.2) | Acc: ( 1.2, -0.0,  0.2) | Rwd:  3.03
437 [Astar_planner] Segment 0 reached in 13 steps, dist=1.87, axis_error
    =[1.5083427  0.92842865  0.5979328 ], reason=running
438 [Astar_planner] Segment 1 starts from actual pos [-63.503834 -87.
    928215 -19.696815] (planned waypoint was [-62.    -87.    -20.
    296148])
439 [Astar_planner] Segment 1: [-63.503834 -87.928215 -19.696815] -> [-
    50.    -87.    -23.100693]
440 [Step  65 | Total    0] Target: (-50.0, -87.0, -23.1) | Pos: (-49.7, -85.7
    , -22.0) | Vel: ( 0.2, -0.2, -0.3) | Acc: ( 1.9,  1.2,  0.9) | Rwd: 603.95
    -> Episode End: ARRIVED!
441
442 [Astar_planner] Segment 1 reached in 65 steps, dist=1.76, axis_error

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442 =[0.3241272 1.3418198 1.096775 ], reason=arrived
443 [Astar_planner] Segment 2 starts from actual pos [-49.67426 -85.
    66023 -22.0062 ] (planned waypoint was [-50.    -87.    -23.100693
    ])
444 [Astar_planner] Segment 2: [-49.67426 -85.66023 -22.0062 ] -> [-39
    .    -88.    -22.321712]
445 [Step 18 | Total    0] Target: (-39.0, -88.0, -22.3) | Pos: (-40.2, -87.6
    , -20.5) | Vel: ( 0.4, -0.4, 0.2) | Acc: ( -1.9, -1.8, -0.9) | Rwd: 602.70
    -> Episode End: ARRIVED!
446
447 [Astar_planner] Segment 2 reached in 18 steps, dist=2.24, axis_error
    =[1.1809616 0.44509125 1.8459854 ], reason=arrived
448 [Astar_planner] Segment 3 starts from actual pos [-40.177376 -87.
    558205 -20.473513] (planned waypoint was [-39.    -88.    -22.
    321712])
449 [Astar_planner] Segment 3: [-40.177376 -87.558205 -20.473513] -> [-
    27.    -88.    -20.060978]
450 [Step 17 | Total    0] Target: (-27.0, -88.0, -20.1) | Pos: (-28.6, -89.3
    , -19.1) | Vel: ( 0.6, -0.1, -0.0) | Acc: ( 0.5, -1.9, -0.4) | Rwd: 1.11
451 [Astar_planner] Segment 3 reached in 17 steps, dist=2.29, axis_error
    =[1.6120224 1.278511 0.9974594], reason=running
452 [Astar_planner] Segment 4 starts from actual pos [-28.606415 -89.
    27935 -19.063595] (planned waypoint was [-27.    -88.    -20.
    060978])
453 [Astar_planner] Segment 4: [-28.606415 -89.27935 -19.063595] -> [-
    16.    -88.    -20.453674]
454 [Step 14 | Total    0] Target: (-16.0, -88.0, -20.5) | Pos: (-17.5, -89.3
    , -19.7) | Vel: ( 0.9, 0.2, 0.5) | Acc: ( 1.2, 0.2, -1.3) | Rwd: 4.39
455 [Astar_planner] Segment 4 reached in 14 steps, dist=2.12, axis_error
    =[1.5496979 1.2638626 0.7146015], reason=running
456 [Astar_planner] Segment 5 starts from actual pos [-17.541656 -89.
    26198 -19.734707] (planned waypoint was [-16.    -88.    -20.
    453674])
457 [Astar_planner] Segment 5: [-17.541656 -89.26198 -19.734707] -> [-
    11.    -79.    -23.922302]
458 [Step 68 | Total    0] Target: (-11.0, -79.0, -23.9) | Pos: (-12.7, -80.1
    , -22.7) | Vel: ( 0.1, 0.4, -0.3) | Acc: ( -1.6, 0.1, 0.9) | Rwd: 4.61
459 [Astar_planner] Segment 5 reached in 68 steps, dist=2.34, axis_error
    =[1.6601 1.0667267 1.2502747], reason=running
460 [Astar_planner] Segment 6 starts from actual pos [-12.658798 -80.
    06337 -22.674948] (planned waypoint was [-11.    -79.    -23.
    922302])
461 [Astar_planner] Segment 6: [-12.658798 -80.06337 -22.674948] -> [-
    10.    -67.    -24.83542]
462 [Step 33 | Total    0] Target: (-10.0, -67.0, -24.8) | Pos: (-11.9, -66.1
    , -23.2) | Vel: ( -0.1, 0.3, 0.5) | Acc: ( -0.4, -2.0, -0.5) | Rwd: 0.74
463 [Astar_planner] Segment 6 reached in 33 steps, dist=2.68, axis_error
    =[1.8864202 0.9459305 1.646471 ], reason=running

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464 [Astar_planner] Segment 7 starts from actual pos [-11.88705 -66.05146 -23.184452] (planned waypoint was [-10. -67. -24.83542
])
465 [Astar_planner] Segment 7: [-11.88705 -66.05146 -23.184452] -> [-10. -56. -23.038649]
466 [Step 47 | Total 0] Target: (-10.0, -56.0, -23.0) | Pos: ( -8.5, -57.3, -21.4) | Vel: ( -0.5, 0.2, 0.9) | Acc: ( -0.9, 0.3, 0.2) | Rwd: 3.68
467 [Astar_planner] Segment 7 reached in 47 steps, dist=2.54, axis_error=[1.5047865 1.2829819 1.5963707], reason=running
468 [Astar_planner] Segment 8 starts from actual pos [ -8.501655 -57.28067 -21.431494] (planned waypoint was [-10. -56. -23.038649])
469 [Astar_planner] Segment 8: [ -8.501655 -57.28067 -21.431494] -> [ -9. -45. -21.792128]
470 [Step 50 | Total 0] Target: ( -9.0, -45.0, -21.8) | Pos: ( -8.1, -46.9, -20.1) | Vel: ( -0.1, 0.3, 0.6) | Acc: ( 0.4, -0.8, -1.6) | Rwd: 4.56
471 [Astar_planner] Segment 8 reached in 50 steps, dist=2.69, axis_error=[0.8738785 1.9188385 1.665638 ], reason=running
472 [Astar_planner] Segment 9 starts from actual pos [ -8.127431 -46.916477 -20.121305] (planned waypoint was [ -9. -45. -21.792128])
473 [Astar_planner] Segment 9: [ -8.127431 -46.916477 -20.121305] -> [ -9. -33. -24.831333]
474 [Step 79 | Total 0] Target: ( -9.0, -33.0, -24.8) | Pos: (-10.3, -31.1, -23.8) | Vel: ( -0.0, -0.8, 0.2) | Acc: ( -0.6, -1.3, -1.8) | Rwd: 4.12
475 [Astar_planner] Segment 9 reached in 79 steps, dist=2.51, axis_error=[1.3239403 1.869606 1.0213203], reason=running
476 [Astar_planner] Segment 10 starts from actual pos [-10.324086 -31.13777 -23.808136] (planned waypoint was [ -9. -33. -24.831333])
477 [Astar_planner] Segment 10: [-10.324086 -31.13777 -23.808136] -> [ -9. -21. -24.089125]
478 [Step 33 | Total 0] Target: ( -9.0, -21.0, -24.1) | Pos: (-10.9, -21.3, -22.9) | Vel: ( 0.7, -0.7, 0.0) | Acc: ( 0.3, -1.8, 0.5) | Rwd: 6.65
479 [Astar_planner] Segment 10 reached in 33 steps, dist=2.29, axis_error=[1.9409208 0.3093109 1.176796 ], reason=running
480 [Astar_planner] Segment 11 starts from actual pos [-10.934429 -21.31545 -22.912018] (planned waypoint was [ -9. -21. -24.089125])
481 [Astar_planner] Segment 11: [-10.934429 -21.31545 -22.912018] -> [ -9. -9. -21.26582]
482 [Step 38 | Total 0] Target: ( -9.0, -9.0, -21.3) | Pos: ( -9.8, -8.1, -19.4) | Vel: ( 0.2, -0.6, 0.7) | Acc: ( -0.2, -1.9, -0.7) | Rwd: 601.56
-> Episode End: ARRIVED!
483
484 [Astar_planner] Segment 11 reached in 38 steps, dist=2.22, axis_error=[0.7689457 0.9022398 1.8733139], reason=arrived
485 [Astar_planner] Segment 12 starts from actual pos [ -9.767721 -8.

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485 101245 -19.388294] (planned waypoint was [ -9.    -9.    -21.26582])
486 [Astar_planner] Segment 12: [ -9.767721 -8.101245 -19.388294] -> [ -
4.    0.    -22.723108]
487 [Step 80 | Total    0] Target: ( -4.0, 0.0, -22.7) | Pos: ( -5.0, 0.5, -
21.2) | Vel: ( 0.8, -0.7, -0.1) | Acc: ( 1.7, -0.1, 0.1) | Rwd: 606.51
-> Episode End: ARRIVED!
488
489 [Astar_planner] Segment 12 reached in 80 steps, dist=1.91, axis_error
=[1.045855 0.4624869 1.524107 ], reason=arrived
490 [Astar_planner] Segment 13 starts from actual pos [ -5.041003 0.
45800713 -21.199709 ] (planned waypoint was [ -4.    0.    -22.
723108])
491 [Astar_planner] Segment 13: [ -5.041003 0.45800713 -21.199709
] -> [ 7.    1.    -21.21561]
492 [Step 20 | Total    0] Target: ( 7.0, 1.0, -21.2) | Pos: ( 5.2, 0.8, -
20.1) | Vel: ( 0.7, -0.2, -0.1) | Acc: ( 0.4, -1.1, 1.5) | Rwd: 0.47
493 [Astar_planner] Segment 13 reached in 20 steps, dist=2.11, axis_error
=[1.7527709 0.22220534 1.1596642 ], reason=running
494 [Astar_planner] Segment 14 starts from actual pos [ 5.253676 0.
7763491 -20.056515 ] (planned waypoint was [ 7.    1.    -21.21561
])
495 [Astar_planner] Segment 14: [ 5.253676 0.7763491 -20.056515
] -> [ 19.    2.    -23.683458]
496 [Step 30 | Total    0] Target: ( 19.0, 2.0, -23.7) | Pos: ( 17.1, 1.6, -
23.0) | Vel: ( 0.6, -0.3, 0.2) | Acc: ( 0.6, -1.6, 0.3) | Rwd: 2.16
497 [Astar_planner] Segment 14 reached in 30 steps, dist=2.10, axis_error
=[1.9380665 0.42093933 0.6815567 ], reason=running
498 [Astar_planner] Segment 15 starts from actual pos [ 17.067312 1.
5764823 -22.999619 ] (planned waypoint was [ 19.    2.    -23.
683458])
499 [Astar_planner] Segment 15: [ 17.067312 1.5764823 -22.999619
] -> [ 29.    7.    -21.99473]
500 [Step 15 | Total    0] Target: ( 29.0, 7.0, -22.0) | Pos: ( 27.3, 5.9, -
21.2) | Vel: ( 1.0, -0.8, 0.0) | Acc: ( -0.2, -1.1, 0.7) | Rwd: 3.75
501 [Astar_planner] Segment 15 reached in 15 steps, dist=2.16, axis_error
=[1.7019672 1.100296 0.75183105], reason=running
502 [Astar_planner] Segment 16 starts from actual pos [ 27.303986 5.
895023 -21.24276 ] (planned waypoint was [ 29.    7.    -21.99473])
503 [Astar_planner] Segment 16: [ 27.303986 5.895023 -21.24276 ] -> [
38.    11.    -23.045797]
504 [Step 12 | Total    0] Target: ( 38.0, 11.0, -23.0) | Pos: ( 36.5, 11.2
, -21.9) | Vel: ( 1.2, 0.1, -1.0) | Acc: ( 0.1, -0.2, -0.6) | Rwd: 5.92
505 [Astar_planner] Segment 16 reached in 12 steps, dist=1.90, axis_error
=[1.5433731 0.19005013 1.0960045 ], reason=running
506 [Astar_planner] Segment 17 starts from actual pos [ 36.467724 11.
1912985 -21.958906 ] (planned waypoint was [ 38.    11.    -23.
045797])
507 [Astar_planner] Segment 17: [ 36.467724 11.1912985 -21.958906

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507 ] -> [ 48.    15.    -24.134064]
508 [Step 15 | Total    0] Target: ( 48.0, 15.0, -24.1) | Pos: ( 47.0, 13.8
, -23.0) | Vel: ( 1.1, -0.1, -0.7) | Acc: ( -0.3, -1.0, 0.9) | Rwd: 4.25
509 [Astar_planner] Segment 17 reached in 15 steps, dist=1.98, axis_error
=[0.9976921 1.2414083 1.1727085], reason=running
510 [Astar_planner] Segment 18 starts from actual pos [ 47.011852 13.
75731 -22.967747] (planned waypoint was [ 48.    15.    -24.
134064])
511 [Astar_planner] Segment 18: [ 47.011852 13.75731 -22.967747] -> [
58. 20. -11.3]
512 [Step 20 | Total    0] Target: ( 58.0, 20.0, -11.3) | Pos: ( 58.2, 18.7
, -10.7) | Vel: ( 0.2, 0.8, 0.1) | Acc: ( 0.2, 0.6, -1.0) | Rwd: 601.83
-> Episode End: ARRIVED!
513
514 [Astar_planner] Segment 18 reached in 20 steps, dist=1.45, axis_error
=[0.21427917 1.3082428 0.59921074], reason=arrived
515 Task 8 result: success (Success: Reached destination (19/19
segments). Shield interventions=60, emergencies=0, recoveries=10,
recovery_steps=60.)
516
517 --- Task 9/38 ---
518 From [ 23.5 140.08 -4. ] to [ 17.8 -86.92 -8.9 ]
519 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
520 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
521 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
522 Path planned via occupancy (Points: 30, Dist: 370.4m). Local target z/
alt: 0:-4.01(4.01m), 1:-20.72(20.72m), 2:-24.42(24.42m), 3:-22.16(22.
16m), 4:-22.22(22.22m), 5:-23.51(23.51m), 6:-21.65(21.65m), 7:-22.42
(22.42m), 8:-21.60(21.60m), 9:-24.76(24.76m), 10:-24.59(24.59m), 11
:-20.45(20.45m), 12:-20.65(20.65m), 13:-24.96(24.96m), 14:-24.72(24.
72m), 15:-21.68(21.68m), 16:-23.40(23.40m), 17:-23.55(23.55m), 18:-
21.31(21.31m), 19:-23.87(23.87m), 20:-24.71(24.71m), 21:-23.23(23.
23m), 22:-23.03(23.03m), 23:-25.00(25.00m), 24:-24.02(24.02m), 25:-
20.27(20.27m), 26:-23.23(23.23m), 27:-20.82(20.82m), 28:-20.01(20.
01m), 29:-8.90(8.90m)
523 [Astar_planner] Segment 0: [ 24.    140.    -4.0109267] -> [ 16
.    149.    -20.718767]
524 [Astar_planner] Placing first segment start at [ 24.    140.    -4.
0109267]
525 [Astar_planner] Initial climb to 20.0m.
526 [Step 28 | Total    0] Target: ( 16.0, 149.0, -20.7) | Pos: ( 15.1, 150.8
, -21.4) | Vel: ( -0.3, -0.2, -0.0) | Acc: ( -0.1, -0.1, 0.3) | Rwd: 0.97
527 [Astar_planner] Segment 0 reached in 28 steps, dist=2.10, axis_error
=[0.94877625 1.754242 0.65860176], reason=running
528 [Astar_planner] Segment 1 starts from actual pos [ 15.048755 150.

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528 75247 -21.377449] (planned waypoint was [ 16.    149.    -20.
    718767])
529 [Astar_planner] Segment 1: [ 15.048755 150.75247 -21.377449] -> [
    5.    151.    -24.421728]
530 [Step 29 | Total    0] Target: ( 5.0, 151.0, -24.4) | Pos: ( 5.7, 152.8
    , -22.8) | Vel: ( -0.9,  0.0, -1.1) | Acc: ( 1.2, -0.1,  1.1) | Rwd: 4.59
531 [Astar_planner] Segment 1 reached in 29 steps, dist=2.50, axis_error
    =[0.68275166 1.7814026 1.6115894 ], reason=running
532 [Astar_planner] Segment 2 starts from actual pos [ 5.671642 152.
    78165 -22.822996] (planned waypoint was [ 5.    151.    -24.
    421728])
533 [Astar_planner] Segment 2: [ 5.671642 152.78165 -22.822996] -> [ -7
    .    153.    -22.162275]
534 [Step 23 | Total    0] Target: ( -7.0, 153.0, -22.2) | Pos: ( -5.1, 151.0
    , -20.6) | Vel: ( -0.4,  0.3, -1.6) | Acc: ( 0.6,  0.9,  0.2) | Rwd: 1.97
535 [Astar_planner] Segment 2 reached in 23 steps, dist=3.15, axis_error
    =[1.8740702 1.983551 1.5794334], reason=running
536 [Astar_planner] Segment 3 starts from actual pos [ -5.1281853 151.
    0181 -20.59206 ] (planned waypoint was [ -7.    153.    -22.
    162275])
537 [Astar_planner] Segment 3: [ -5.1281853 151.0181 -20.59206 ] -> [-
    20.    153.    -22.223454]
538 [Step 31 | Total    0] Target: (-20.0, 153.0, -22.2) | Pos: (-20.0, 151.
    1, -21.3) | Vel: ( -0.6,  0.5,  0.1) | Acc: ( 0.1, -1.1,  0.1) | Rwd: -0.02
539 [Astar_planner] Segment 3 reached in 31 steps, dist=2.12, axis_error
    =[0.0265274 1.9029541 0.9238167], reason=running
540 [Astar_planner] Segment 4 starts from actual pos [-19.978714 151.
    1014 -21.298845] (planned waypoint was [-20.    153.    -22.
    223454])
541 [Astar_planner] Segment 4: [-19.978714 151.1014 -21.298845] -> [-
    25.    143.    -23.508478]
542 [Step 13 | Total    0] Target: (-25.0, 143.0, -23.5) | Pos: (-23.1, 141.
    3, -22.3) | Vel: ( -0.2, -0.4, -0.0) | Acc: ( 0.3, -2.0,  0.9) | Rwd: -1.41
543 [Astar_planner] Segment 4 reached in 13 steps, dist=2.82, axis_error
    =[1.935049 1.6724854 1.183403 ], reason=running
544 [Astar_planner] Segment 5 starts from actual pos [-23.06636 141.
    32532 -22.325197] (planned waypoint was [-25.    143.    -23.
    508478])
545 [Astar_planner] Segment 5: [-23.06636 141.32532 -22.325197] -> [-
    25.    131.    -21.647543]
546 [Step 12 | Total    0] Target: (-25.0, 131.0, -21.6) | Pos: (-25.0, 131.
    9, -20.3) | Vel: ( 0.0, -0.9,  1.0) | Acc: ( -0.0, -2.0,  0.2) | Rwd: 608.
    47 -> Episode End: ARRIVED!
547
548 [Astar_planner] Segment 5 reached in 12 steps, dist=1.64, axis_error
    =[0.01448441 0.9278717 1.3512096 ], reason=arrived
549 [Astar_planner] Segment 6 starts from actual pos [-25.01429 131.
    91968 -20.287476] (planned waypoint was [-25.    131.    -21.

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549 647543])
550 [Astar_planner] Segment 6: [-25.01429 131.91968 -20.287476] -> [-
25.    118.    -22.416246]
551 [Step 23 | Total    0] Target: (-25.0, 118.0, -22.4) | Pos: (-25.3, 117.
4, -20.6) | Vel: ( 0.3, -0.2, 0.4) | Acc: ( 0.6, -1.8, -0.0) | Rwd: 600.
72 -> Episode End: ARRIVED!
552
553 [Astar_planner] Segment 6 reached in 23 steps, dist=1.95, axis_error
=[0.34909058 0.57151794 1.8302708 ], reason=arrived
554 [Astar_planner] Segment 7 starts from actual pos [-25.346182 117.
4269 -20.581911] (planned waypoint was [-25.    118.    -22.
416246])
555 [Astar_planner] Segment 7: [-25.346182 117.4269 -20.581911] -> [-
26.    106.    -21.596603]
556 [Step 13 | Total    0] Target: (-26.0, 106.0, -21.6) | Pos: (-24.4, 106.
9, -20.2) | Vel: ( -0.3, -0.6, 1.1) | Acc: ( -0.1, -1.0, 0.2) | Rwd: 5.23
557 [Astar_planner] Segment 7 reached in 13 steps, dist=2.33, axis_error
=[1.6239185 0.8797302 1.416254 ], reason=running
558 [Astar_planner] Segment 8 starts from actual pos [-24.378729 106.
87435 -20.170359] (planned waypoint was [-26.    106.    -21.
596603])
559 [Astar_planner] Segment 8: [-24.378729 106.87435 -20.170359] -> [-
28.    94.    -24.756617]
560 [Step 173 | Total    0] Target: (-28.0, 94.0, -24.8) | Pos: (-21.5, 91.0
, -21.6) | Vel: ( -0.0, 0.0, 0.0) | Acc: ( -0.1, 0.4, -0.1) | Rwd: -0.55
561 [Astar_planner] Segment 8 reached in 173 steps, dist=7.85, axis_error
=[6.5381985 2.9557114 3.1778374], reason=stuck_near_intermediate
562 [Astar_planner] Segment 9 starts from actual pos [-21.461893 91.
04463 -21.578718] (planned waypoint was [-28.    94.    -24.
756617])
563 [Astar_planner] Segment 9: [-21.461893 91.04463 -21.578718] -> [-
29.    82.    -24.592972]
564 [Step 68 | Total    0] Target: (-29.0, 82.0, -24.6) | Pos: (-27.7, 80.0
, -23.7) | Vel: ( -0.1, -0.0, -0.0) | Acc: ( 0.2, -1.9, 0.4) | Rwd: -0.85
565 [Astar_planner] Segment 9 reached in 68 steps, dist=2.51, axis_error
=[1.3065109 1.9558411 0.8779068], reason=running
566 [Astar_planner] Segment 10 starts from actual pos [-27.69396 80.
04377 -23.715118] (planned waypoint was [-29.    82.    -24.
592972])
567 [Astar_planner] Segment 10: [-27.69396 80.04377 -23.715118] -> [-
38.    73.    -20.451252]
568 [Step 14 | Total    0] Target: (-38.0, 73.0, -20.5) | Pos: (-36.2, 72.4
, -19.7) | Vel: ( -0.2, -0.4, -0.0) | Acc: ( 0.3, -1.8, 0.0) | Rwd: 0.31
569 [Astar_planner] Segment 10 reached in 14 steps, dist=2.07, axis_error
=[1.8237801 0.59228516 0.7674961 ], reason=running
570 [Astar_planner] Segment 11 starts from actual pos [-36.178257 72.
40421 -19.68379 ] (planned waypoint was [-38.    73.    -20.
451252])

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571 [Astar_planner] Segment 11: [-36.178257 72.40421 -19.68379] -> [-
    48.    66.    -20.647228]
572 [Step 20 | Total    0] Target: (-48.0, 66.0, -20.6) | Pos: (-46.6, 66.6
    , -20.1) | Vel: ( -1.2, -0.5, 0.3) | Acc: ( -2.0, 0.2, 0.3) | Rwd: 4.32
573 [Astar_planner] Segment 11 reached in 20 steps, dist=1.62, axis_error
    =[1.4052544 0.5687561 0.581686 ], reason=running
574 [Astar_planner] Segment 12 starts from actual pos [-46.605488 66.
    56433 -20.063017] (planned waypoint was [-48.    66.    -20.
    647228])
575 [Astar_planner] Segment 12: [-46.605488 66.56433 -20.063017] -> [-
    57.    57.    -24.955118]
576 [Step 17 | Total    0] Target: (-57.0, 57.0, -25.0) | Pos: (-56.4, 57.4
    , -23.0) | Vel: ( -0.9, -0.3, -0.2) | Acc: ( -1.2, 0.1, 0.8) | Rwd: 603.07
    -> Episode End: ARRIVED!
577
578 [Astar_planner] Segment 12 reached in 17 steps, dist=2.08, axis_error
    =[0.6279297 0.35498047 1.9553986 ], reason=arrived
579 [Astar_planner] Segment 13 starts from actual pos [-56.379948 57.
    352146 -23.001047] (planned waypoint was [-57.    57.    -24.
    955118])
580 [Astar_planner] Segment 13: [-56.379948 57.352146 -23.001047] -> [-
    59.    47.    -24.718668]
581 [Step 12 | Total    0] Target: (-59.0, 47.0, -24.7) | Pos: (-57.3, 47.0
    , -23.9) | Vel: ( -0.4, -0.6, 0.4) | Acc: ( -0.3, -1.9, -1.6) | Rwd: 3.94
582 [Astar_planner] Segment 13 reached in 12 steps, dist=1.89, axis_error
    =[1.689476 0.04687119 0.8401108 ], reason=running
583 [Astar_planner] Segment 14 starts from actual pos [-57.31427 47.
    04138 -23.874804] (planned waypoint was [-59.    47.    -24.
    718668])
584 [Astar_planner] Segment 14: [-57.31427 47.04138 -23.874804] -> [-
    50.    38.    -21.679861]
585 [Step 16 | Total    0] Target: (-50.0, 38.0, -21.7) | Pos: (-51.4, 37.9
    , -21.6) | Vel: ( 0.3, -0.2, -0.0) | Acc: ( 0.6, -2.0, 0.2) | Rwd: 601.69
    -> Episode End: ARRIVED!
586
587 [Astar_planner] Segment 14 reached in 16 steps, dist=1.38, axis_error
    =[1.3752518 0.05789948 0.12169075], reason=arrived
588 [Astar_planner] Segment 15 starts from actual pos [-51.37231 37.
    94066 -21.558271] (planned waypoint was [-50.    38.    -21.
    679861])
589 [Astar_planner] Segment 15: [-51.37231 37.94066 -21.558271] -> [-
    48.    26.    -23.400963]
590 [Step 22 | Total    0] Target: (-48.0, 26.0, -23.4) | Pos: (-48.9, 24.0
    , -22.3) | Vel: ( 0.5, 0.2, 0.3) | Acc: ( 0.8, 0.3, 0.6) | Rwd: 0.78
591 [Astar_planner] Segment 15 reached in 22 steps, dist=2.45, axis_error
    =[0.8919792 1.9846954 1.133522 ], reason=running
592 [Astar_planner] Segment 16 starts from actual pos [-48.88786 24.
    016632 -22.264648] (planned waypoint was [-48.    26.    -23.

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592 400963])
593 [Astar_planner] Segment 16: [-48.88786 24.016632 -22.264648] -> [-
39. 17. -23.550135]
594 [Step 30 | Total 0] Target: (-39.0, 17.0, -23.6) | Pos: (-38.6, 15.7
, -22.3) | Vel: ( 0.2, 1.2, -0.7) | Acc: ( 0.6, 0.8, 2.0) | Rwd: 607.50
-> Episode End: ARRIVED!
595
596 [Astar_planner] Segment 16 reached in 30 steps, dist=1.82, axis_error
=[0.4180107 1.2670145 1.238554 ], reason=arrived
597 [Astar_planner] Segment 17 starts from actual pos [-38.579765 15.
743546 -22.317276] (planned waypoint was [-39. 17. -23.
550135])
598 [Astar_planner] Segment 17: [-38.579765 15.743546 -22.317276] -> [-
31. 8. -21.30798]
599 [Step 15 | Total 0] Target: (-31.0, 8.0, -21.3) | Pos: (-32.4, 7.8, -
20.8) | Vel: ( 0.7, -0.3, 0.2) | Acc: ( 1.0, -1.2, 0.1) | Rwd: 603.48
-> Episode End: ARRIVED!
600
601 [Astar_planner] Segment 17 reached in 15 steps, dist=1.50, axis_error
=[1.3819885 0.15283823 0.5573654 ], reason=arrived
602 [Astar_planner] Segment 18 starts from actual pos [-32.37569 7.
844625 -20.748898] (planned waypoint was [-31. 8. -21.30798
])
603 [Astar_planner] Segment 18: [-32.37569 7.844625 -20.748898] -> [-
21. 1. -23.870647]
604 [Step 89 | Total 0] Target: (-21.0, 1.0, -23.9) | Pos: (-20.5, 2.3, -
22.6) | Vel: ( 0.3, -0.6, 0.6) | Acc: (-0.4, 0.0, 1.2) | Rwd: 604.85
-> Episode End: ARRIVED!
605
606 [Astar_planner] Segment 18 reached in 89 steps, dist=1.90, axis_error
=[0.49341583 1.3130288 1.2799187 ], reason=arrived
607 [Astar_planner] Segment 19 starts from actual pos [-20.50366 2.
3074677 -22.585554 ] (planned waypoint was [-21. 1. -23.
870647])
608 [Astar_planner] Segment 19: [-20.50366 2.3074677 -22.585554
] -> [-10. -4. -24.714823]
609 [Step 26 | Total 0] Target: (-10.0, -4.0, -24.7) | Pos: (-11.8, -5.3, -
23.4) | Vel: ( 0.3, -0.1, -0.0) | Acc: ( 0.1, -1.0, 0.8) | Rwd: 4.17
610 [Astar_planner] Segment 19 reached in 26 steps, dist=2.59, axis_error
=[1.8065462 1.3147187 1.3085575], reason=running
611 [Astar_planner] Segment 20 starts from actual pos [-11.803956 -5.
31569 -23.406408] (planned waypoint was [-10. -4. -24.
714823])
612 [Astar_planner] Segment 20: [-11.803956 -5.31569 -23.406408] -> [-
9. -16. -23.225033]
613 [Step 13 | Total 0] Target: ( -9.0, -16.0, -23.2) | Pos: ( -9.9, -14.2, -
21.6) | Vel: ( 0.4, -0.7, 0.1) | Acc: ( 0.6, -0.7, -0.6) | Rwd: 6.16
614 [Astar_planner] Segment 20 reached in 13 steps, dist=2.59, axis_error

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614 =[0.91938496 1.7755213 1.6510925 ], reason=running
615 [Astar_planner] Segment 21 starts from actual pos [ -9.915628 -14.
230491 -21.572464] (planned waypoint was [ -9.    -16.    -23.
225033])
616 [Astar_planner] Segment 21: [ -9.915628 -14.230491 -21.572464] -> [ -
9.    -28.    -23.025625]
617 [Step 19 | Total    0] Target: ( -9.0, -28.0, -23.0) | Pos: ( -8.7, -28.2, -
21.2) | Vel: ( 0.6, -0.9, 0.5) | Acc: ( 0.7, -0.0, -0.7) | Rwd: 601.58
-> Episode End: ARRIVED!
618
619 [Astar_planner] Segment 21 reached in 19 steps, dist=1.86, axis_error
=[0.2556076 0.20168304 1.8284912 ], reason=arrived
620 [Astar_planner] Segment 22 starts from actual pos [ -8.739268 -28.
2099 -21.19234 ] (planned waypoint was [ -9.    -28.    -23.025625
])
621 [Astar_planner] Segment 22: [ -8.739268 -28.2099 -21.19234 ] -> [ -9
.    -41.    -24.996428]
622 [Step 14 | Total    0] Target: ( -9.0, -41.0, -25.0) | Pos: ( -9.9, -40.2, -
23.1) | Vel: ( 0.2, -0.8, -0.3) | Acc: ( 0.6, -1.6, -0.1) | Rwd: 605.77
-> Episode End: ARRIVED!
623
624 [Astar_planner] Segment 22 reached in 14 steps, dist=2.29, axis_error
=[0.90141106 0.84534836 1.9305668 ], reason=arrived
625 [Astar_planner] Segment 23 starts from actual pos [ -9.899289 -40.
1621 -23.068058] (planned waypoint was [ -9.    -41.    -24.
996428])
626 [Astar_planner] Segment 23: [ -9.899289 -40.1621 -23.068058] -> [ -9
.    -53.    -24.017012]
627 [Step 12 | Total    0] Target: ( -9.0, -53.0, -24.0) | Pos: ( -9.8, -51.9, -
22.8) | Vel: ( 0.2, -1.1, -0.3) | Acc: ( 0.0, -0.6, 0.4) | Rwd: 607.51
-> Episode End: ARRIVED!
628
629 [Astar_planner] Segment 23 reached in 12 steps, dist=1.83, axis_error
=[0.82279015 1.0696716 1.2321033 ], reason=arrived
630 [Astar_planner] Segment 24 starts from actual pos [ -9.820414 -51.
943153 -22.787575] (planned waypoint was [ -9.    -53.    -24.
017012])
631 [Astar_planner] Segment 24: [ -9.820414 -51.943153 -22.787575] -> [ -
9.    -66.    -20.2745]
632 [Step 16 | Total    0] Target: ( -9.0, -66.0, -20.3) | Pos: (-10.3, -66.0
, -19.7) | Vel: ( 0.7, -0.8, -0.2) | Acc: ( 0.4, -1.0, 1.6) | Rwd: 601.27
-> Episode End: ARRIVED!
633
634 [Astar_planner] Segment 24 reached in 16 steps, dist=1.45, axis_error
=[1.3167486 0.0057373 0.61657333], reason=arrived
635 [Astar_planner] Segment 25 starts from actual pos [-10.307977 -66.
01551 -19.659895] (planned waypoint was [ -9.    -66.    -20.2745])
636 [Astar_planner] Segment 25: [-10.307977 -66.01551 -19.659895] -> [-

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636 10.    -78.    -23.232107]
637 [Step 11 | Total    0] Target: (-10.0, -78.0, -23.2) | Pos: ( -9.5, -77.6
    , -22.0) | Vel: ( -0.2, -1.1,  0.1) | Acc: ( 0.2,  1.0,  1.0) | Rwd: 606.02
    -> Episode End: ARRIVED!
638
639 [Astar_planner] Segment 25 reached in 11 steps, dist=1.41, axis_error
    =[0.4853611 0.43289185 1.2517529 ], reason=arrived
640 [Astar_planner] Segment 26 starts from actual pos [ -9.5168495 -77.
    58004 -21.979206 ] (planned waypoint was [-10.    -78.    -23.
    232107])
641 [Astar_planner] Segment 26: [ -9.5168495 -77.58004 -21.979206
    ] -> [ -7.    -88.    -20.822796]
642 [Step 19 | Total    0] Target: ( -7.0, -88.0, -20.8) | Pos: ( -5.5, -86.3, -
    19.8) | Vel: ( -0.0, -0.8, -0.0) | Acc: ( 0.9, -1.4,  0.7) | Rwd: 0.60
643 [Astar_planner] Segment 26 reached in 19 steps, dist=2.48, axis_error
    =[1.4512138 1.7472153 1.0019588], reason=running
644 [Astar_planner] Segment 27 starts from actual pos [ -5.5490737 -86.
    260155 -19.821198 ] (planned waypoint was [ -7.    -88.    -20.
    822796])
645 [Astar_planner] Segment 27: [ -5.5490737 -86.260155 -19.821198
    ] -> [ 6.    -88.    -20.012146]
646 [Step 35 | Total    0] Target: ( 6.0, -88.0, -20.0) | Pos: ( 4.5, -89.6, -
    18.0) | Vel: ( 0.5, -0.2,  0.2) | Acc: ( 1.9, -1.9,  0.1) | Rwd: -0.23
647 [Astar_planner] Segment 27 reached in 35 steps, dist=2.98, axis_error
    =[1.5428634 1.6096878 1.9702492], reason=running
648 [Astar_planner] Segment 28 starts from actual pos [ 4.4619 -89.
    61154 -18.0397 ] (planned waypoint was [ 6.    -88.    -20.012146])
649 [Astar_planner] Segment 28: [ 4.4619 -89.61154 -18.0397 ] -> [ 18. -
    87. -8.9]
650 [Step 15 | Total    0] Target: ( 18.0, -87.0, -8.9) | Pos: ( 16.9, -86.0
    , -7.7) | Vel: ( 1.2,  0.0, -0.2) | Acc: ( 2.0,  0.1,  1.7) | Rwd: 599.89
    -> Episode End: ARRIVED!
651
652 [Astar_planner] Segment 28 reached in 15 steps, dist=1.93, axis_error
    =[1.1029816 0.9978256 1.2260094], reason=arrived
653 Task 9 result: success (Success: Reached destination (29/29
    segments). Shield interventions=12, emergencies=0, recoveries=2,
    recovery_steps=12.)
654
655 --- Task 10/38 ---
656 From [ 17.8 -86.92 -8.9 ] to [-10.5 163.38 -8.94]
657 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
658 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
659 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
660 Path planned via occupancy (Points: 26, Dist: 332.0m). Local target z/

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660 alt: 0:-8.90(8.90m), 1:-22.34(22.34m), 2:-23.25(23.25m), 3:-20.12(20.
    12m), 4:-23.20(23.20m), 5:-22.90(22.90m), 6:-21.36(21.36m), 7:-23.45
    (23.45m), 8:-21.64(21.64m), 9:-20.93(20.93m), 10:-23.48(23.48m), 11
    :-21.84(21.84m), 12:-20.10(20.10m), 13:-24.86(24.86m), 14:-22.65(22.
    65m), 15:-21.44(21.44m), 16:-23.31(23.31m), 17:-21.22(21.22m), 18:-
    21.96(21.96m), 19:-20.85(20.85m), 20:-20.35(20.35m), 21:-23.03(23.
    03m), 22:-22.04(22.04m), 23:-23.31(23.31m), 24:-24.12(24.12m), 25:-
    8.94(8.94m)
661 [Astar_planner] Segment 0: [ 18.    -87.    -8.900034] -> [ 5.    -88
    .    -22.33961]
662 [Astar_planner] Placing first segment start at [ 18.    -87.    -8.
    900034]
663 [Astar_planner] Initial climb to 20.0m.
664 [Step 18 | Total    0] Target: ( 5.0, -88.0, -22.3) | Pos: ( 6.4, -88.7, -
    21.4) | Vel: ( -1.2, -0.2, -0.1) | Acc: ( -1.5, -0.4, 0.0) | Rwd: 5.01
665 [Astar_planner] Segment 0 reached in 18 steps, dist=1.84, axis_error
    =[1.3795476 0.7299042 0.9839134], reason=running
666 [Astar_planner] Segment 1 starts from actual pos [ 6.368715 -88.
    73169 -21.35666 ] (planned waypoint was [ 5.    -88.    -22.33961])
667 [Astar_planner] Segment 1: [ 6.368715 -88.73169 -21.35666 ] -> [ -8
    .    -88.    -23.253216]
668 [Step 25 | Total    0] Target: ( -8.0, -88.0, -23.3) | Pos: ( -6.2, -87.2, -
    22.0) | Vel: ( -0.3, -0.0, -0.0) | Acc: ( -0.9, 0.1, -0.7) | Rwd: -0.45
669 [Astar_planner] Segment 1 reached in 25 steps, dist=2.32, axis_error
    =[1.8169045 0.7783737 1.2161064], reason=running
670 [Astar_planner] Segment 2 starts from actual pos [ -6.185676 -87.
    22171 -22.037275] (planned waypoint was [ -8.    -88.    -23.
    253216])
671 [Astar_planner] Segment 2: [ -6.185676 -87.22171 -22.037275] -> [-11
    .    -76.    -20.124262]
672 [Step 18 | Total    0] Target: (-11.0, -76.0, -20.1) | Pos: (-12.3, -76.0
    , -18.4) | Vel: ( 0.6, 0.7, 0.7) | Acc: ( 0.3, 1.9, -1.5) | Rwd: 605.20
    -> Episode End: ARRIVED!
673
674 [Astar_planner] Segment 2 reached in 18 steps, dist=2.14, axis_error
    =[1.2569399 0.03881073 1.7342758 ], reason=arrived
675 [Astar_planner] Segment 3 starts from actual pos [-12.25137 -76.
    03273 -18.383818] (planned waypoint was [-11.    -76.    -20.
    124262])
676 [Astar_planner] Segment 3: [-12.25137 -76.03273 -18.383818] -> [-10
    .    -63.    -23.200201]
677 [Step 180 | Total    0] Target: (-10.0, -63.0, -23.2) | Pos: ( -7.8, -66.8
    , -19.6) | Vel: ( 0.1, -0.3, 0.0) | Acc: ( 1.0, -2.0, -0.2) | Rwd: -2.15
678 [Astar_planner] Segment 3 timeout, dist=5.68
679 [Astar_planner] Stop after failed segment 3.
680 Task 10 result: failed (Failed at segment 3: segment_timeout (3/25
    reached). Shield interventions=48, emergencies=0, recoveries=8,
    recovery_steps=48.)

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681
682 --- Task 11/38 ---
683 From [-123.9 -90.92 -4. ] to [-117.3  1.48 -9. ]
684 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
685 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
686 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
687 Path planned via occupancy (Points: 9, Dist: 113.3m). Local target z/alt
: 0:-4.01(4.01m), 1:-22.76(22.76m), 2:-20.24(20.24m), 3:-20.90(20.
90m), 4:-23.52(23.52m), 5:-22.34(22.34m), 6:-24.72(24.72m), 7:-20.54
(20.54m), 8:-8.98(8.98m)
688 [Astar_planner] Segment 0: [-124.    -91.    -4.006912] -> [-122
.    -79.    -22.761257]
689 [Astar_planner] Placing first segment start at [-124.    -91.    -4.
006912]
690 [Astar_planner] Initial climb to 20.0m.
691 [Step  44 | Total    0] Target: (-122.0, -79.0, -22.8) | Pos: (-121.7, -82
.6, -19.6) | Vel: ( 0.0, -0.1,  0.1) | Acc: ( 0.0, -0.1, -0.5) | Rwd: -1.44

692 [Astar_planner] Segment 0 reached in 44 steps, dist=4.83, axis_error
=[0.2665329 3.6412888 3.1592922], reason=stuck_near_intermediate
693 [Astar_planner] Segment 1 starts from actual pos [-121.733406 -82.
64206 -19.601358] (planned waypoint was [-122.    -79.    -22.
761257])
694 [Astar_planner] Segment 1: [-121.733406 -82.64206 -19.601358
] -> [-119.    -68.    -20.235476]
695 [Step  28 | Total    0] Target: (-119.0, -68.0, -20.2) | Pos: (-117.6, -69
.9, -19.3) | Vel: ( -0.0,  0.5,  0.0) | Acc: ( 0.4,  1.9,  0.3) | Rwd: 0.47
696 [Astar_planner] Segment 1 reached in 28 steps, dist=2.51, axis_error
=[1.4250107 1.8564835 0.9190922], reason=running
697 [Astar_planner] Segment 2 starts from actual pos [-117.57531 -69.
851585 -19.316353] (planned waypoint was [-119.    -68.    -20.
235476])
698 [Astar_planner] Segment 2: [-117.57531 -69.851585 -19.316353
] -> [-111.    -59.    -20.899202]
699 [Step  53 | Total    0] Target: (-111.0, -59.0, -20.9) | Pos: (-112.4, -59
.5, -19.4) | Vel: ( 0.5, -0.5,  0.5) | Acc: ( 0.6, -1.5, -1.0) | Rwd: 602.
31 -> Episode End: ARRIVED!
700
701 [Astar_planner] Segment 2 reached in 53 steps, dist=2.13, axis_error
=[1.3849258 0.5418587 1.5250835], reason=arrived
702 [Astar_planner] Segment 3 starts from actual pos [-112.37839 -59.
547924 -19.367859] (planned waypoint was [-111.    -59.    -20.
899202])
703 [Astar_planner] Segment 3: [-112.37839 -59.547924 -19.367859
] -> [-110.    -46.    -23.521152]

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704 [Step 42 | Total 0] Target: (-110.0, -46.0, -23.5) | Pos: (-109.2, -46
.8, -21.6) | Vel: ( -0.8, 0.6, -1.8) | Acc: ( 0.8, -1.0, 1.2) | Rwd: 607.
42 -> Episode End: ARRIVED!
705
706 [Astar_planner] Segment 3 reached in 42 steps, dist=2.27, axis_error
=[0.84477234 0.77568436 1.9632511 ], reason=arrived
707 [Astar_planner] Segment 4 starts from actual pos [-109.162476 -46.
770447 -21.573545] (planned waypoint was [-110. -46. -23.
521152])
708 [Astar_planner] Segment 4: [-109.162476 -46.770447 -21.573545
] -> [-110. -34. -22.335045]
709 [Step 12 | Total 0] Target: (-110.0, -34.0, -22.3) | Pos: (-110.0, -35
.7, -21.0) | Vel: ( -0.4, 0.9, -1.1) | Acc: ( 1.1, -1.1, 0.6) | Rwd: 3.65

710 [Astar_planner] Segment 4 reached in 12 steps, dist=2.19, axis_error
=[0.02424622 1.7021179 1.3821259 ], reason=running
711 [Astar_planner] Segment 5 starts from actual pos [-110.02743 -35.
6944 -20.962677] (planned waypoint was [-110. -34. -22.
335045])
712 [Astar_planner] Segment 5: [-110.02743 -35.6944 -20.962677] -> [-
110. -21. -24.715239]
713 [Step 16 | Total 0] Target: (-110.0, -21.0, -24.7) | Pos: (-108.4, -20
.7, -23.6) | Vel: ( -1.1, 0.7, -0.5) | Acc: ( -0.6, 1.2, 2.0) | Rwd: 3.87

714 [Astar_planner] Segment 5 reached in 16 steps, dist=1.98, axis_error
=[1.6113052 0.3248005 1.0969238], reason=running
715 [Astar_planner] Segment 6 starts from actual pos [-108.39522 -20.
67112 -23.62106] (planned waypoint was [-110. -21. -24.
715239])
716 [Astar_planner] Segment 6: [-108.39522 -20.67112 -23.62106] -> [-
112. -10. -20.541553]
717 [Step 10 | Total 0] Target: (-112.0, -10.0, -20.5) | Pos: (-111.9, -11
.1, -20.3) | Vel: ( -0.2, 1.8, -0.1) | Acc: ( 0.7, 0.8, 0.6) | Rwd: 603.
70 -> Episode End: ARRIVED!
718
719 [Astar_planner] Segment 6 reached in 10 steps, dist=1.16, axis_error
=[0.09158325 1.139986 0.19340706], reason=arrived
720 [Astar_planner] Segment 7 starts from actual pos [-111.91055 -11.
123807 -20.349215] (planned waypoint was [-112. -10. -20.
541553])
721 [Astar_planner] Segment 7: [-111.91055 -11.123807 -20.349215
] -> [-117. 1. -8.975344]
722 [Step 18 | Total 0] Target: (-117.0, 1.0, -9.0) | Pos: (-115.6, 0.3
, -9.1) | Vel: ( -0.9, 0.1, 0.1) | Acc: ( 0.2, 0.3, 0.2) | Rwd: 1.05
723 [Astar_planner] Segment 7 reached in 18 steps, dist=1.54, axis_error
=[1.3759537 0.6746711 0.16795254], reason=running
724 Task 11 result: success (Success: Reached destination (8/8 segments
). Shield interventions=18, emergencies=0, recoveries=3,

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724 recovery_steps=18.)
725
726 --- Task 12/38 ---
727 From [-141.4 -191.82 -9. ] to [-125.1 185.28 -4. ]
728 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
729 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
730 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
731 Path planned via occupancy (Points: 33, Dist: 447.4m). Local target z/
alt: 0:-8.99(8.99m), 1:-22.69(22.69m), 2:-22.78(22.78m), 3:-21.83(21.
83m), 4:-22.49(22.49m), 5:-24.54(24.54m), 6:-20.21(20.21m), 7:-24.14
(24.14m), 8:-23.35(23.35m), 9:-21.04(21.04m), 10:-20.50(20.50m), 11
:-24.24(24.24m), 12:-23.72(23.72m), 13:-20.07(20.07m), 14:-23.39(23.
39m), 15:-21.09(21.09m), 16:-22.74(22.74m), 17:-23.94(23.94m), 18:-
23.45(23.45m), 19:-21.29(21.29m), 20:-23.76(23.76m), 21:-23.13(23.
13m), 22:-23.62(23.62m), 23:-21.13(21.13m), 24:-23.44(23.44m), 25:-
22.75(22.75m), 26:-22.27(22.27m), 27:-23.52(23.52m), 28:-24.15(24.
15m), 29:-24.05(24.05m), 30:-24.65(24.65m), 31:-24.29(24.29m), 32:-
4.00(4.00m)
732 [Astar_planner] Segment 0: [-141.    -192.    -8.994189] -> [-152
.    -186.    -22.69117]
733 [Astar_planner] Placing first segment start at [-141.    -192.    -8.
994189]
734 [Astar_planner] Initial climb to 20.0m.
735 [Step 180 | Total    0] Target: (-152.0, -186.0, -22.7) | Pos: (-148.4, -
190.0, -19.5) | Vel: ( 0.2, -0.1, 0.1) | Acc: ( 0.3, -0.3, -0.9) | Rwd: -
0.53
736 [Astar_planner] Segment 0 timeout, dist=6.20
737 [Astar_planner] Stop after failed segment 0.
738 Task 12 result: failed (Failed at segment 0: segment_timeout (0/32
reached). Shield interventions=0, emergencies=0, recoveries=0,
recovery_steps=0.)
739
740 --- Task 13/38 ---
741 From [103.4 79.58 -9. ] to [-111.5 118.98 -4. ]
742 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
743 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
744 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
745 Path planned via occupancy (Points: 23, Dist: 316.3m). Local target z/
alt: 0:-8.99(8.99m), 1:-23.51(23.51m), 2:-24.75(24.75m), 3:-20.24(20.
24m), 4:-24.89(24.89m), 5:-21.20(21.20m), 6:-20.41(20.41m), 7:-23.86
(23.86m), 8:-20.11(20.11m), 9:-23.90(23.90m), 10:-24.86(24.86m), 11
:-22.12(22.12m), 12:-22.91(22.91m), 13:-21.65(21.65m), 14:-22.85(22.

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745 85m), 15:-24.15(24.15m), 16:-24.96(24.96m), 17:-23.46(23.46m), 18:-
    24.89(24.89m), 19:-21.69(21.69m), 20:-24.39(24.39m), 21:-20.82(20.
    82m), 22:-4.00(4.00m)
746 [Astar_planner] Segment 0: [103.    80.    -8.986727] -> [105.
    93.    -23.51339]
747 [Astar_planner] Placing first segment start at [103.    80.    -8.
    986727]
748 [Astar_planner] Initial climb to 20.0m.
749 [Step 32 | Total    0] Target: (105.0, 93.0, -23.5) | Pos: (103.7, 94.6
    , -21.8) | Vel: ( -0.7, -0.9, 0.1) | Acc: ( -1.4, 1.0, -0.7) | Rwd: 3.93
750 [Astar_planner] Segment 0 reached in 32 steps, dist=2.63, axis_error
    =[1.2504272 1.5934601 1.6832771], reason=running
751 [Astar_planner] Segment 1 starts from actual pos [103.74322  94.
    585785 -21.829128] (planned waypoint was [105.    93.    -23.51339
    ])
752 [Astar_planner] Segment 1: [103.74322  94.585785 -21.829128] -> [
    105.    107.    -24.745348]
753 [Step 36 | Total    0] Target: (105.0, 107.0, -24.7) | Pos: (108.5, 92.
    5, -21.9) | Vel: ( 0.1, -0.3, -0.8) | Acc: ( -0.2, 0.5, -0.7) | Rwd: -910.
    77 -> Episode End: CRASHED!
754
755 [Astar_planner] Segment 1 ended: collision
756 [Astar_planner] Stop after failed segment 1.
757 Task 13 result: failed (Failed at segment 1: collision (1/22 reached).
    Shield interventions=18, emergencies=1, recoveries=4,
    recovery_steps=18.)
758
759 --- Task 14/38 ---
760 From [-111.5 118.98 -4. ] to [63.1 4.08 -7.7 ]
761 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
762 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
763 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
764 Path planned via occupancy (Points: 24, Dist: 295.5m). Local target z/
    alt: 0:-4.01(4.01m), 1:-24.25(24.25m), 2:-20.85(20.85m), 3:-21.74(21.
    74m), 4:-23.20(23.20m), 5:-20.32(20.32m), 6:-20.50(20.50m), 7:-23.28
    (23.28m), 8:-23.11(23.11m), 9:-21.49(21.49m), 10:-22.29(22.29m), 11
    :-23.21(23.21m), 12:-24.73(24.73m), 13:-21.33(21.33m), 14:-24.71(24.
    71m), 15:-22.32(22.32m), 16:-22.71(22.71m), 17:-22.84(22.84m), 18:-
    22.65(22.65m), 19:-20.08(20.08m), 20:-24.29(24.29m), 21:-22.92(22.
    92m), 22:-21.97(21.97m), 23:-7.70(7.70m)
765 [Astar_planner] Segment 0: [-112.    119.    -4.008858] -> [-103
    .    110.    -24.248726]
766 [Astar_planner] Placing first segment start at [-112.    119.    -4.
    008858]
767 [Astar_planner] Initial climb to 20.0m.

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768 [Step 30 | Total 0] Target: (-103.0, 110.0, -24.2) | Pos: (-104.3,
    108.8, -23.5) | Vel: ( 0.4, -0.1, 0.3) | Acc: ( 0.8, -1.2, 0.2) | Rwd: 2
    .89
769 [Astar_planner] Segment 0 reached in 30 steps, dist=1.90, axis_error
    =[1.2931747 1.1534348 0.7828598], reason=running
770 [Astar_planner] Segment 1 starts from actual pos [-104.289345 108.
    84577 -23.463428] (planned waypoint was [-103. 110. -24.
    248726])
771 [Astar_planner] Segment 1: [-104.289345 108.84577 -23.463428
    ] -> [-100. 99. -20.854763]
772 [Step 14 | Total 0] Target: (-100.0, 99.0, -20.9) | Pos: (-101.5,
    100.8, -20.2) | Vel: ( 0.3, 0.1, -0.0) | Acc: ( 1.8, 1.1, 0.7) | Rwd: 0
    .90
773 [Astar_planner] Segment 1 reached in 14 steps, dist=2.40, axis_error
    =[1.4826889 1.7781525 0.6185055], reason=running
774 [Astar_planner] Segment 2 starts from actual pos [-101.480034 100.
    77877 -20.236368] (planned waypoint was [-100. 99. -20.
    854763])
775 [Astar_planner] Segment 2: [-101.480034 100.77877 -20.236368
    ] -> [-93. 90. -21.743645]
776 [Step 35 | Total 0] Target: (-93.0, 90.0, -21.7) | Pos: (-93.6, 90.1
    , -19.8) | Vel: ( 0.2, 0.0, 0.5) | Acc: ( 0.5, -0.8, -0.8) | Rwd: 600.59
    -> Episode End: ARRIVED!
777
778 [Astar_planner] Segment 2 reached in 35 steps, dist=2.00, axis_error
    =[0.6293106 0.10232544 1.8999157 ], reason=arrived
779 [Astar_planner] Segment 3 starts from actual pos [-93.62716 90.
    102715 -19.839142] (planned waypoint was [-93. 90. -21.
    743645])
780 [Astar_planner] Segment 3: [-93.62716 90.102715 -19.839142] -> [-
    81. 89. -23.199396]
781 [Step 44 | Total 0] Target: (-81.0, 89.0, -23.2) | Pos: (-82.5, 89.8
    , -21.3) | Vel: ( 1.0, 0.1, -1.6) | Acc: ( 0.8, -0.5, 0.8) | Rwd: 7.46
782 [Astar_planner] Segment 3 reached in 44 steps, dist=2.57, axis_error
    =[1.5383606 0.79013824 1.900034 ], reason=running
783 [Astar_planner] Segment 4 starts from actual pos [-82.529434 89.
    79105 -21.31341 ] (planned waypoint was [-81. 89. -23.
    199396])
784 [Astar_planner] Segment 4: [-82.529434 89.79105 -21.31341 ] -> [-68
    . 89. -20.315985]
785 [Step 18 | Total 0] Target: (-68.0, 89.0, -20.3) | Pos: (-66.4, 89.4
    , -19.1) | Vel: ( -0.5, 0.1, -0.0) | Acc: ( -0.4, 1.2, 0.5) | Rwd: -0.24
786 [Astar_planner] Segment 4 reached in 18 steps, dist=2.06, axis_error
    =[1.6175079 0.3835144 1.22089 ], reason=running
787 [Astar_planner] Segment 5 starts from actual pos [-66.386696 89.
    383995 -19.095495] (planned waypoint was [-68. 89. -20.
    315985])
788 [Astar_planner] Segment 5: [-66.386696 89.383995 -19.095495] -> [-

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788 56.      89.      -20.496033]
789 [Step 18 | Total    0] Target: (-56.0, 89.0, -20.5) | Pos: (-54.6, 90.8
, -19.8) | Vel: ( -0.8, -0.4, 0.3) | Acc: ( -1.2, -1.2, -1.9) | Rwd: 3.37
790 [Astar_planner] Segment 5 reached in 18 steps, dist=2.34, axis_error
=[1.3733864 1.7825851 0.6510372], reason=running
791 [Astar_planner] Segment 6 starts from actual pos [-54.63347 90.
77903 -19.842392] (planned waypoint was [-56.      89.      -20.
496033])
792 [Astar_planner] Segment 6: [-54.63347 90.77903 -19.842392] -> [-46
.      82.      -23.279573]
793 [Step 13 | Total    0] Target: (-46.0, 82.0, -23.3) | Pos: (-47.5, 82.8
, -22.4) | Vel: ( 0.6, -0.6, 0.6) | Acc: ( 1.1, -1.6, 0.2) | Rwd: 4.37
794 [Astar_planner] Segment 6 reached in 13 steps, dist=1.91, axis_error
=[1.5090332 0.7956772 0.85162354], reason=running
795 [Astar_planner] Segment 7 starts from actual pos [-47.503323 82.
790344 -22.422592] (planned waypoint was [-46.      82.      -23.
279573])
796 [Astar_planner] Segment 7: [-47.503323 82.790344 -22.422592] -> [-
41.      72.      -23.107847]
797 [Step 25 | Total    0] Target: (-41.0, 72.0, -23.1) | Pos: (-39.1, 73.8
, -22.5) | Vel: ( 0.3, -0.6, -0.0) | Acc: ( 0.4, -2.0, 0.8) | Rwd: 1.33
798 [Astar_planner] Segment 7 reached in 25 steps, dist=2.68, axis_error
=[1.9017105 1.7662048 0.65476036], reason=running
799 [Astar_planner] Segment 8 starts from actual pos [-39.09593 73.
760414 -22.453362] (planned waypoint was [-41.      72.      -23.
107847])
800 [Astar_planner] Segment 8: [-39.09593 73.760414 -22.453362] -> [-
49.      65.      -21.48696]
801 [Step 17 | Total    0] Target: (-49.0, 65.0, -21.5) | Pos: (-48.6, 65.9
, -20.5) | Vel: ( -1.1, -0.5, 0.1) | Acc: ( -2.0, -0.6, 0.4) | Rwd: 607.08
-> Episode End: ARRIVED!
802
803 [Astar_planner] Segment 8 reached in 17 steps, dist=1.40, axis_error
=[0.37794876 0.9315109 0.9796066 ], reason=arrived
804 [Astar_planner] Segment 9 starts from actual pos [-48.63172 65.
927376 -20.506254] (planned waypoint was [-49.      65.      -21.48696
])
805 [Astar_planner] Segment 9: [-48.63172 65.927376 -20.506254] -> [-
58.      56.      -22.290522]
806 [Step 59 | Total    0] Target: (-58.0, 56.0, -22.3) | Pos: (-56.7, 57.8
, -21.7) | Vel: ( -0.3, -0.3, -0.0) | Acc: ( 0.1, -0.1, 0.7) | Rwd: 0.92
807 [Astar_planner] Segment 9 reached in 59 steps, dist=2.30, axis_error
=[1.259758 1.8383026 0.5678654], reason=running
808 [Astar_planner] Segment 10 starts from actual pos [-56.74298 57.
835625 -21.722788] (planned waypoint was [-58.      56.      -22.
290522])
809 [Astar_planner] Segment 10: [-56.74298 57.835625 -21.722788] -> [-
57.      45.      -23.208641]

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810 [Step 16 | Total 0] Target: (-57.0, 45.0, -23.2) | Pos: (-58.9, 46.3
, -22.9) | Vel: ( 0.3, -0.2, 0.1) | Acc: ( 0.3, -1.1, -1.3) | Rwd: 2.89
811 [Astar_planner] Segment 10 reached in 16 steps, dist=2.29, axis_error
=[1.9040947 1.251812 0.27051926], reason=running
812 [Astar_planner] Segment 11 starts from actual pos [-58.90181 46.
250454 -22.937578] (planned waypoint was [-57. 45. -23.
208641])
813 [Astar_planner] Segment 11: [-58.90181 46.250454 -22.937578] -> [-
50. 36. -24.731552]
814 [Step 20 | Total 0] Target: (-50.0, 36.0, -24.7) | Pos: (-51.2, 35.5
, -24.0) | Vel: ( 0.5, -0.3, 0.5) | Acc: ( 0.6, -2.0, 0.1) | Rwd: 603.25
-> Episode End: ARRIVED!
815
816 [Astar_planner] Segment 11 reached in 20 steps, dist=1.48, axis_error
=[1.2140121 0.46232605 0.7192116 ], reason=arrived
817 [Astar_planner] Segment 12 starts from actual pos [-51.20943 35.
534756 -24.008087] (planned waypoint was [-50. 36. -24.
731552])
818 [Astar_planner] Segment 12: [-51.20943 35.534756 -24.008087] -> [-
47. 25. -21.327856]
819 [Step 13 | Total 0] Target: (-47.0, 25.0, -21.3) | Pos: (-48.8, 25.9
, -21.1) | Vel: ( 0.4, -0.3, -0.0) | Acc: ( 0.6, -2.0, 0.6) | Rwd: 2.77
820 [Astar_planner] Segment 12 reached in 13 steps, dist=2.00, axis_error
=[1.7828178 0.8822937 0.21094513], reason=running
821 [Astar_planner] Segment 13 starts from actual pos [-48.77936 25.
87925 -21.11706] (planned waypoint was [-47. 25. -21.327856
])
822 [Astar_planner] Segment 13: [-48.77936 25.87925 -21.11706] -> [-38
. 16. -24.70684]
823 [Step 27 | Total 0] Target: (-38.0, 16.0, -24.7) | Pos: (-38.2, 14.1
, -23.5) | Vel: ( 0.2, 0.9, -0.7) | Acc: ( -0.4, -0.4, 2.0) | Rwd: 6.70
824 [Astar_planner] Segment 13 reached in 27 steps, dist=2.25, axis_error
=[0.2319336 1.9039545 1.1854115], reason=running
825 [Astar_planner] Segment 14 starts from actual pos [-38.22996 14.
103852 -23.527225] (planned waypoint was [-38. 16. -24.70684
])
826 [Astar_planner] Segment 14: [-38.22996 14.103852 -23.527225] -> [-
30. 7. -22.320124]
827 [Step 15 | Total 0] Target: (-30.0, 7.0, -22.3) | Pos: (-31.7, 7.0, -
21.5) | Vel: ( 0.6, -0.1, 0.4) | Acc: ( 0.6, 0.4, 1.1) | Rwd: 3.16
828 [Astar_planner] Segment 14 reached in 15 steps, dist=1.94, axis_error
=[1.7485466 0.03255367 0.839056 ], reason=running
829 [Astar_planner] Segment 15 starts from actual pos [-31.743448 6.
966974 -21.477335] (planned waypoint was [-30. 7. -22.
320124])
830 [Astar_planner] Segment 15: [-31.743448 6.966974 -21.477335] -> [-
20. 0. -22.712255]
831 [Step 16 | Total 0] Target: (-20.0, 0.0, -22.7) | Pos: (-21.9, 0.4, -

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831 21.0) | Vel: ( 0.4, -0.1, 0.0) | Acc: ( -1.5, 0.1, 0.7) | Rwd: 5.07
832 [Astar_planner] Segment 15 reached in 16 steps, dist=2.57, axis_error
    =[1.9121208 0.42996517 1.6651897 ], reason=running
833 [Astar_planner] Segment 16 starts from actual pos [-21.908833 0.
    42879277 -21.046453 ] (planned waypoint was [-20. 0. -22.
    712255])
834 [Astar_planner] Segment 16: [-21.908833 0.42879277 -21.046453
    ] -> [-9. -2. -22.843994]
835 [Step 120 | Total 0] Target: ( -9.0, -2.0, -22.8) | Pos: (-10.0, -3.4, -
    21.3) | Vel: ( 0.3, -0.3, -0.6) | Acc: ( 0.2, -1.6, -0.1) | Rwd: 4.83
836 [Astar_planner] Segment 16 reached in 120 steps, dist=2.32,
    axis_error=[1.0141659 1.3500581 1.5837593], reason=running
837 [Astar_planner] Segment 17 starts from actual pos [-10.011388 -3.
    352638 -21.265882] (planned waypoint was [-9. -2. -22.
    843994])
838 [Astar_planner] Segment 17: [-10.011388 -3.352638 -21.265882] -> [
    2. 1. -22.65034]
839 [Step 20 | Total 0] Target: ( 2.0, 1.0, -22.7) | Pos: ( 0.4, -0.0, -
    21.5) | Vel: ( 0.4, 0.3, -0.2) | Acc: ( -0.8, 0.5, 1.0) | Rwd: 3.54
840 [Astar_planner] Segment 17 reached in 20 steps, dist=2.23, axis_error
    =[1.6457752 1.0216268 1.1140919], reason=running
841 [Astar_planner] Segment 18 starts from actual pos [ 3.5772943e-01 -1.
    9321069e-02 -2.1537466e+01] (planned waypoint was [ 2. 1. -
    22.65034])
842 [Astar_planner] Segment 18: [ 3.5772943e-01 -1.9321069e-02 -2.
    1537466e+01] -> [ 15. 1. -20.079212]
843 [Step 18 | Total 0] Target: ( 15.0, 1.0, -20.1) | Pos: ( 13.7, 0.2, -
    19.5) | Vel: ( 0.8, 0.4, 0.1) | Acc: ( -0.0, 0.6, -1.3) | Rwd: 3.32
844 [Astar_planner] Segment 18 reached in 18 steps, dist=1.63, axis_error
    =[1.2915726 0.7897602 0.61628914], reason=running
845 [Astar_planner] Segment 19 starts from actual pos [ 13.713463 0.
    21253072 -19.462202 ] (planned waypoint was [ 15. 1. -20.
    079212])
846 [Astar_planner] Segment 19: [ 13.713463 0.21253072 -19.462202
    ] -> [ 27. 1. -24.294935]
847 [Step 36 | Total 0] Target: ( 27.0, 1.0, -24.3) | Pos: ( 25.0, 0.3, -
    23.4) | Vel: ( 0.6, -0.0, 0.2) | Acc: ( 0.4, -1.3, 1.4) | Rwd: 2.40
848 [Astar_planner] Segment 19 reached in 36 steps, dist=2.28, axis_error
    =[1.9828835 0.7159943 0.86961555], reason=running
849 [Astar_planner] Segment 20 starts from actual pos [ 25.022854 0.
    28385264 -23.42376 ] (planned waypoint was [ 27. 1. -24.
    294935])
850 [Astar_planner] Segment 20: [ 25.022854 0.28385264 -23.42376
    ] -> [ 39. 1. -22.91596]
851 [Step 14 | Total 0] Target: ( 39.0, 1.0, -22.9) | Pos: ( 37.1, 0.8, -
    22.2) | Vel: ( 1.1, 0.4, -0.3) | Acc: ( -0.2, -0.6, 0.2) | Rwd: 3.61
852 [Astar_planner] Segment 20 reached in 14 steps, dist=2.09, axis_error
    =[1.939228 0.23888427 0.7380276 ], reason=running

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853 [Astar_planner] Segment 21 starts from actual pos [ 37.070236  0.
      7650141 -22.18088 ] (planned waypoint was [ 39.      1.      -22.91596
      ])
854 [Astar_planner] Segment 21: [ 37.070236  0.7650141 -22.18088
      ] -> [ 51.      2.      -21.97241]
855 [Step 12 | Total  0] Target: ( 51.0,  2.0, -22.0) | Pos: ( 49.2,  1.8, -
      21.9) | Vel: ( 1.7,  0.0,  0.5) | Acc: ( 0.5, -2.0,  1.8) | Rwd: 4.65
856 [Astar_planner] Segment 21 reached in 12 steps, dist=1.81, axis_error
      =[1.8039589 0.15167677 0.07382965], reason=running
857 [Astar_planner] Segment 22 starts from actual pos [ 49.206142  1.
      8484527 -21.895864 ] (planned waypoint was [ 51.      2.      -21.
      97241])
858 [Astar_planner] Segment 22: [ 49.206142  1.8484527 -21.895864
      ] -> [63.      4.      -7.6993003]
859 [Step 24 | Total  0] Target: ( 63.0,  4.0, -7.7) | Pos: ( 61.7,  4.2, -
      8.2) | Vel: ( 1.3, -0.6, -0.1) | Acc: ( -1.5, -0.3, -1.6) | Rwd: 601.82 ->
      Episode End: ARRIVED!
860
861 [Astar_planner] Segment 22 reached in 24 steps, dist=1.41, axis_error
      =[1.3036423 0.23572636 0.47767496], reason=arrived
862 Task 14 result: success (Success: Reached destination (23/23
      segments). Shield interventions=48, emergencies=0, recoveries=8,
      recovery_steps=48.)
863
864 --- Task 15/38 ---
865 From [ 78.3 -148.32 -9.4 ] to [101.7 130.78 -5.3 ]
866 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
      200.0, 189.0], occupied=72869 cells (53.4%)
867 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
      (53.4%)
868 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
869 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
870 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
871 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.0
872 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
873 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=0.6
874 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
875 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=0.3
876 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
      clearance penalties.

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877 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
878 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4
    %, safety_margin=1.5
879 [Astar_planner] WARNING: using legacy occupancy fallback; local
    targets may be closer to obstacles.
880 Path planned via occupancy_legacy (Points: 39, Dist: 429.4m). Local
    target z/alt: 0:-8.94(8.94m), 1:-24.29(24.29m), 2:-22.23(22.23m), 3:-24.
    74(24.74m), 4:-24.60(24.60m), 5:-22.81(22.81m), 6:-24.32(24.32m), 7
    :-20.52(20.52m), 8:-20.05(20.05m), 9:-24.19(24.19m), 10:-20.41(20.
    41m), 11:-22.72(22.72m), 12:-22.32(22.32m), 13:-24.84(24.84m), 14:-
    23.81(23.81m), 15:-22.67(22.67m), 16:-20.54(20.54m), 17:-21.34(21.
    34m), 18:-20.45(20.45m), 19:-20.31(20.31m), 20:-20.06(20.06m), 21:-
    20.79(20.79m), 22:-23.49(23.49m), 23:-22.28(22.28m), 24:-23.53(23.
    53m), 25:-23.55(23.55m), 26:-22.15(22.15m), 27:-22.37(22.37m), 28:-
    23.27(23.27m), 29:-21.43(21.43m), 30:-20.20(20.20m), 31:-20.42(20.
    42m), 32:-24.43(24.43m), 33:-20.48(20.48m), 34:-23.90(23.90m), 35:-
    21.10(21.10m), 36:-24.04(24.04m), 37:-20.62(20.62m), 38:-5.30(5.
    30m)
881 [Astar_planner] Segment 0: [ 88.    -178.    -8.942908] -> [ 90
    .    -180.    -24.28616]
882 [Astar_planner] Placing first segment start at [ 88.    -178.    -8.
    942908]
883 [Astar_planner] Initial climb to 20.0m.
884 [Step   4 | Total   0] Target: ( 90.0, -180.0, -24.3) | Pos: ( 88.9, -178.
    1, -23.0) | Vel: ( 1.3, -0.1, -1.1) | Acc: ( 2.0, -1.8, 0.0) | Rwd: 2.30
885 [Astar_planner] Segment 0 reached in 4 steps, dist=2.52, axis_error=[
    1.0506058 1.8915405 1.2986641], reason=running
886 [Astar_planner] Segment 1 starts from actual pos [ 88.956985 -178.
    10898 -22.994123] (planned waypoint was [ 90.    -180.    -24.
    28616])
887 [Astar_planner] Segment 1: [ 88.956985 -178.10898 -22.994123
    ] -> [ 95.    -183.    -22.226027]
888 [Step  20 | Total   0] Target: ( 95.0, -183.0, -22.2) | Pos: ( 94.9, -181
    .6, -20.5) | Vel: ( 2.6, -0.2, -1.9) | Acc: ( -1.0, 1.7, 0.9) | Rwd: 602.
    60 -> Episode End: ARRIVED!
889
890 [Astar_planner] Segment 1 reached in 20 steps, dist=2.21, axis_error
    =[0.06103516 1.3838654 1.7170429], reason=arrived
891 [Astar_planner] Segment 2 starts from actual pos [ 94.96192 -181.
    61766 -20.526354] (planned waypoint was [ 95.    -183.    -22.
    226027])
892 [Astar_planner] Segment 2: [ 94.96192 -181.61766 -20.526354] -> [
    102.    -186.    -24.737263]
893 [Step   7 | Total   0] Target: (102.0, -186.0, -24.7) | Pos: (102.1, -
    184.1, -23.8) | Vel: ( 1.8, -1.2, -0.9) | Acc: ( -0.3, 0.6, 0.5) | Rwd:
    2.39
894 [Astar_planner] Segment 2 reached in 7 steps, dist=2.16, axis_error=[

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894 0.08853149 1.9413605 0.94273376], reason=running
895 [Astar_planner] Segment 3 starts from actual pos [ 102.09906 -184.
06569 -23.799837] (planned waypoint was [ 102.    -186.    -24.
737263])
896 [Astar_planner] Segment 3: [ 102.09906 -184.06569 -23.799837
] -> [ 103.    -186.    -24.59625]
897 [Step  1 | Total    0] Target: (103.0, -186.0, -24.6) | Pos: (103.7, -
185.3, -23.6) | Vel: ( 0.0, -0.6,  0.5) | Acc: ( 0.4, -2.0,  0.1) | Rwd:
598.36 -> Episode End: ARRIVED!
898
899 [Astar_planner] Segment 3 reached in 1 steps, dist=1.43, axis_error=[
0.65249634 0.73916626 1.0289402 ], reason=arrived
900 [Astar_planner] Segment 4 starts from actual pos [ 103.65264 -185.
26462 -23.564291] (planned waypoint was [ 103.    -186.    -24.
59625])
901 [Astar_planner] Segment 4: [ 103.65264 -185.26462 -23.564291
] -> [ 107.    -180.    -22.806826]
902 [Step 29 | Total    0] Target: (107.0, -180.0, -22.8) | Pos: (107.4, -
180.1, -20.9) | Vel: ( 0.0,  0.2,  0.4) | Acc: ( 1.7,  0.1,  0.3) | Rwd:
600.22 -> Episode End: ARRIVED!
903
904 [Astar_planner] Segment 4 reached in 29 steps, dist=1.96, axis_error
=[0.40052795 0.14891052 1.914156 ], reason=arrived
905 [Astar_planner] Segment 5 starts from actual pos [ 107.40069 -180.
14708 -20.888716] (planned waypoint was [ 107.    -180.    -22.
806826])
906 [Astar_planner] Segment 5: [ 107.40069 -180.14708 -20.888716
] -> [ 111.    -173.    -24.321058]
907 [Step 27 | Total    0] Target: (111.0, -173.0, -24.3) | Pos: (112.4, -
171.4, -23.3) | Vel: ( -0.0, -1.2, -1.5) | Acc: ( 0.9, -0.2,  1.5) | Rwd:
1.23
908 [Astar_planner] Segment 5 reached in 27 steps, dist=2.35, axis_error
=[1.3973083 1.5973969 1.012188 ], reason=running
909 [Astar_planner] Segment 6 starts from actual pos [ 112.397064 -171.
41002 -23.317902] (planned waypoint was [ 111.    -173.    -24.
321058])
910 [Astar_planner] Segment 6: [ 112.397064 -171.41002 -23.317902
] -> [ 115.    -166.    -20.519064]
911 [Step 12 | Total    0] Target: (115.0, -166.0, -20.5) | Pos: (114.0, -
167.0, -18.9) | Vel: ( 0.8,  1.0, -0.8) | Acc: ( -1.8,  0.8,  1.7) | Rwd:
605.09 -> Episode End: ARRIVED!
912
913 [Astar_planner] Segment 6 reached in 12 steps, dist=2.19, axis_error
=[1.049324 1.030304 1.6273823], reason=arrived
914 [Astar_planner] Segment 7 starts from actual pos [ 113.95759 -167.
02156 -18.899084] (planned waypoint was [ 115.    -166.    -20.
519064])
915 [Astar_planner] Segment 7: [ 113.95759 -167.02156 -18.899084

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915 ] -> [ 119.    -159.    -20.050772]
916 [Step 12 | Total    0] Target: (119.0, -159.0, -20.1) | Pos: (117.4, -
    158.1, -19.0) | Vel: ( 0.7,  1.0, -0.4) | Acc: (-0.1, -0.4,  2.0) | Rwd:
    1.94
917 [Astar_planner] Segment 7 reached in 12 steps, dist=2.15, axis_error
    =[1.6025772 0.9349365 1.0847454], reason=running
918 [Astar_planner] Segment 8 starts from actual pos [ 117.40179 -158.
    0592 -18.968435] (planned waypoint was [ 119.    -159.    -20.
    050772])
919 [Astar_planner] Segment 8: [ 117.40179 -158.0592 -18.968435] -> [
    122.    -154.    -24.193047]
920 [Step 11 | Total    0] Target: (122.0, -154.0, -24.2) | Pos: (120.6, -
    152.2, -23.4) | Vel: ( 1.2,  0.1, -0.7) | Acc: ( 0.7, -1.5,  0.5) | Rwd: 1
    .26
921 [Astar_planner] Segment 8 reached in 11 steps, dist=2.41, axis_error
    =[1.4416733 1.7690277 0.78972816], reason=running
922 [Astar_planner] Segment 9 starts from actual pos [ 120.565346 -152.
    2303 -23.407154] (planned waypoint was [ 122.    -154.    -24.
    193047])
923 [Astar_planner] Segment 9: [ 120.565346 -152.2303 -23.407154
    ] -> [ 122.    -144.    -20.410046]
924 [Step 30 | Total    0] Target: (122.0, -144.0, -20.4) | Pos: (123.0, -
    146.0, -19.9) | Vel: (-0.1,  0.3, -0.0) | Acc: (-1.1,  0.3,  2.0) | Rwd:
    0.78
925 [Astar_planner] Segment 9 reached in 30 steps, dist=2.29, axis_error
    =[1.0307541 1.9898071 0.48828125], reason=running
926 [Astar_planner] Segment 10 starts from actual pos [ 123.0299 -145.
    98674 -19.921993] (planned waypoint was [ 122.    -144.    -20.
    410046])
927 [Astar_planner] Segment 10: [ 123.0299 -145.98674 -19.921993
    ] -> [ 122.    -137.    -22.72262]
928 [Step 46 | Total    0] Target: (122.0, -137.0, -22.7) | Pos: (124.0, -
    138.5, -21.6) | Vel: (-0.2, -0.0, -0.0) | Acc: (-0.2, -1.8, -0.2) | Rwd: -
    0.73
929 [Astar_planner] Segment 10 reached in 46 steps, dist=2.69, axis_error
    =[1.9524994 1.488678 1.1044312], reason=running
930 [Astar_planner] Segment 11 starts from actual pos [ 123.9511 -138.
    48886 -21.618292] (planned waypoint was [ 122.    -137.    -22.
    72262])
931 [Astar_planner] Segment 11: [ 123.9511 -138.48886 -21.618292
    ] -> [ 133.    -126.    -22.318758]
932 [Step 86 | Total    0] Target: (133.0, -126.0, -22.3) | Pos: (133.3, -
    127.7, -21.6) | Vel: ( 0.0,  0.3,  0.0) | Acc: (-0.9, -1.8,  1.3) | Rwd: 0
    .79
933 [Astar_planner] Segment 11 reached in 86 steps, dist=1.85, axis_error
    =[0.29129028 1.6737671 0.7226391 ], reason=running
934 [Astar_planner] Segment 12 starts from actual pos [ 133.29143 -127.
    67069 -21.595915] (planned waypoint was [ 133.    -126.    -22.

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934 318758])
935 [Astar_planner] Segment 12: [ 133.29143 -127.67069 -21.595915
] -> [ 144. -115. -24.841885]
936 [Step 25 | Total 0] Target: (144.0, -115.0, -24.8) | Pos: (143.3, -
116.8, -24.0) | Vel: ( 1.0, 0.5, -0.4) | Acc: ( -1.0, 0.3, 2.0) | Rwd: 2
.32
937 [Astar_planner] Segment 12 reached in 25 steps, dist=2.12, axis_error
=[0.6660614 1.8245316 0.8521862], reason=running
938 [Astar_planner] Segment 13 starts from actual pos [ 143.33974 -116.
82169 -23.99192] (planned waypoint was [ 144. -115. -24.
841885])
939 [Astar_planner] Segment 13: [ 143.33974 -116.82169 -23.99192] -> [
155. -104. -23.812609]
940 [Step 23 | Total 0] Target: (155.0, -104.0, -23.8) | Pos: (156.0, -
105.9, -23.3) | Vel: ( 1.2, 0.4, -0.3) | Acc: ( -0.9, -0.4, 1.9) | Rwd:
0.60
941 [Astar_planner] Segment 13 reached in 23 steps, dist=2.23, axis_error
=[0.9574127 1.9412994 0.53640366], reason=running
942 [Astar_planner] Segment 14 starts from actual pos [ 155.9609 -105.
93997 -23.277105] (planned waypoint was [ 155. -104. -23.
812609])
943 [Astar_planner] Segment 14: [ 155.9609 -105.93997 -23.277105
] -> [160. -99. -22.671406]
944 [Step 180 | Total 0] Target: (160.0, -99.0, -22.7) | Pos: (160.1, -
119.3, -1.9) | Vel: ( 0.1, 0.3, 0.0) | Acc: ( 0.9, 1.4, 0.0) | Rwd: -0.
07
945 [Astar_planner] Segment 14 timeout, dist=29.06
946 [Astar_planner] Stop after failed segment 14.
947 Task 15 result: failed (Failed at segment 14: segment_timeout (14/38
reached). Shield interventions=70, emergencies=2, recoveries=12,
recovery_steps=70.)
948
949 --- Task 16/38 ---
950 From [150.3 37.28 -10.4 ] to [ 72.7 -198.02 -13.3 ]
951 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
952 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
953 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(20, 237), goal=(97, 2), search_radius=36.
954 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
955 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(20, 237), goal=(97, 2), search_radius=36.
956 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.0
957 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(20, 237), goal=(97, 2), search_radius=36.

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958 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=0.6
959 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(20, 237), goal=(97, 2), search_radius=36.
960 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=0.3
961 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
    clearance penalties.
962 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4
    %, safety_margin=1.5
963 [Astar_planner] WARNING: using legacy occupancy fallback; local
    targets may be closer to obstacles.
964 Path planned via occupancy_legacy (Points: 30, Dist: 286.7m). Local
    target z/alt: 0:-10.40(10.40m), 1:-20.40(20.40m), 2:-21.27(21.27m), 3
    :-20.86(20.86m), 4:-24.48(24.48m), 5:-21.26(21.26m), 6:-22.92(22.
    92m), 7:-22.83(22.83m), 8:-24.02(24.02m), 9:-24.51(24.51m), 10:-23.
    73(23.73m), 11:-23.82(23.82m), 12:-24.57(24.57m), 13:-21.73(21.
    73m), 14:-20.24(20.24m), 15:-21.46(21.46m), 16:-24.01(24.01m), 17
    :-20.92(20.92m), 18:-22.76(22.76m), 19:-20.68(20.68m), 20:-22.56(
    22.56m), 21:-24.53(24.53m), 22:-21.03(21.03m), 23:-22.66(22.66m),
    24:-24.59(24.59m), 25:-22.66(22.66m), 26:-24.43(24.43m), 27:-25.00
    (25.00m), 28:-23.95(23.95m), 29:-13.30(13.30m)
965 [Astar_planner] Segment 0: [150.    37.    -10.404803] -> [150
    .    22.    -20.398901]
966 [Astar_planner] Placing first segment start at [150.    37.    -10.
    404803]
967 [Astar_planner] Initial climb to 20.0m.
968 [Step 24 | Total    0] Target: (150.0, 22.0, -20.4) | Pos: (151.7, 23.
    9, -20.6) | Vel: ( 0.0, -0.7, 0.0) | Acc: ( -0.0, 0.5, 0.2) | Rwd: 1.18

969 [Astar_planner] Segment 0 reached in 24 steps, dist=2.54, axis_error
    =[1.7012329 1.8692379 0.22240067], reason=running
970 [Astar_planner] Segment 1 starts from actual pos [151.7015 23.
    86298 -20.620903] (planned waypoint was [150.    22.    -20.
    398901])
971 [Astar_planner] Segment 1: [151.7015 23.86298 -20.620903] -> [
    150.    7.    -21.274797]
972 [Step 17 | Total    0] Target: (150.0, 7.0, -21.3) | Pos: (150.2, 8.5
    , -21.0) | Vel: ( 0.4, -1.3, -0.7) | Acc: ( -0.6, 0.5, -0.6) | Rwd: 2.95
973 [Astar_planner] Segment 1 reached in 17 steps, dist=1.57, axis_error
    =[0.2026825 1.5380125 0.24909592], reason=running
974 [Astar_planner] Segment 2 starts from actual pos [150.205 8.
    529959 -21.029957] (planned waypoint was [150.    7.    -21.
    274797])
975 [Astar_planner] Segment 2: [150.205 8.529959 -21.029957] -> [
    150.    -8.    -20.855953]
976 [Step 25 | Total    0] Target: (150.0, -8.0, -20.9) | Pos: (152.0, -7.
    9, -20.1) | Vel: ( -0.7, -1.0, -0.1) | Acc: ( 0.0, -1.9, 0.3) | Rwd: 2.

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976 35
977 [Astar_planner] Segment 2 reached in 25 steps, dist=2.10, axis_error
    =[1.9548187 0.11682177 0.76706123], reason=running
978 [Astar_planner] Segment 3 starts from actual pos [151.9483 -7.
    892259 -20.089828] (planned waypoint was [150. -8. -20.
    855953])
979 [Astar_planner] Segment 3: [151.9483 -7.892259 -20.089828] -> [
    150. -23. -24.484362]
980 [Step 24 | Total 0] Target: (150.0, -23.0, -24.5) | Pos: (151.8, -24
    .7, -23.6) | Vel: ( -0.7, -0.9, -0.1) | Acc: ( -0.3, -0.4, 0.9) | Rwd: -0.
    75
981 [Astar_planner] Segment 3 reached in 24 steps, dist=2.60, axis_error
    =[1.7831573 1.6856937 0.8713894], reason=running
982 [Astar_planner] Segment 4 starts from actual pos [151.77686 -24.
    693499 -23.614122] (planned waypoint was [150. -23. -24.
    484362])
983 [Astar_planner] Segment 4: [151.77686 -24.693499 -23.614122] -> [
    149. -30. -21.262007]
984 [Step 15 | Total 0] Target: (149.0, -30.0, -21.3) | Pos: (149.3, -31
    .6, -20.8) | Vel: ( 0.1, 1.0, -0.3) | Acc: ( -1.2, 0.3, 1.4) | Rwd: 1.
    42
985 [Astar_planner] Segment 4 reached in 15 steps, dist=1.65, axis_error
    =[0.2922058 1.5694256 0.42071915], reason=running
986 [Astar_planner] Segment 5 starts from actual pos [149.29271 -31.
    56362 -20.843262] (planned waypoint was [149. -30. -21.
    262007])
987 [Astar_planner] Segment 5: [149.29271 -31.56362 -20.843262] -> [
    146. -37. -22.915005]
988 [Step 11 | Total 0] Target: (146.0, -37.0, -22.9) | Pos: (147.8, -37
    .5, -22.3) | Vel: ( -0.7, -1.3, -0.0) | Acc: ( 0.1, -1.2, 0.8) | Rwd: 2.
    94
989 [Astar_planner] Segment 5 reached in 11 steps, dist=1.96, axis_error
    =[1.7945404 0.48453522 0.6165161 ], reason=running
990 [Astar_planner] Segment 6 starts from actual pos [147.78831 -37.
    49607 -22.298454] (planned waypoint was [146. -37. -22.
    915005])
991 [Astar_planner] Segment 6: [147.78831 -37.49607 -22.298454] -> [
    143. -44. -22.826925]
992 [Step 8 | Total 0] Target: (143.0, -44.0, -22.8) | Pos: (144.8, -44.
    6, -21.8) | Vel: ( -0.7, -0.9, -0.5) | Acc: ( 0.2, -0.4, 1.9) | Rwd: 3.
    74
993 [Astar_planner] Segment 6 reached in 8 steps, dist=2.17, axis_error
    =[1.842865 0.55327606 1.0133266 ], reason=running
994 [Astar_planner] Segment 7 starts from actual pos [144.83842 -44.
    558876 -21.816656] (planned waypoint was [143. -44. -22.
    826925])
995 [Astar_planner] Segment 7: [144.83842 -44.558876 -21.816656] -> [
    141. -59. -24.023531]

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996 [Step 15 | Total 0] Target: (141.0, -59.0, -24.0) | Pos: (142.0, -57
    .1, -22.5) | Vel: ( -0.7, -0.0, 1.0) | Acc: ( -1.7, 2.0, 2.0) | Rwd: 4.
    49
997 [Astar_planner] Segment 7 reached in 15 steps, dist=2.58, axis_error
    =[1.0143738 1.8581848 1.4780121], reason=running
998 [Astar_planner] Segment 8 starts from actual pos [142.0082 -57.
    14192 -22.53684] (planned waypoint was [141. -59. -24.
    023531])
999 [Astar_planner] Segment 8: [142.0082 -57.14192 -22.53684] -> [141
    . -74. -24.509577]
1000 [Step 24 | Total 0] Target: (141.0, -74.0, -24.5) | Pos: (142.8, -73
    .5, -23.7) | Vel: ( -0.6, -1.9, -1.0) | Acc: ( 0.1, -1.4, -0.5) | Rwd: 3.
    49
1001 [Astar_planner] Segment 8 reached in 24 steps, dist=2.01, axis_error
    =[1.778122 0.4519577 0.81952477], reason=running
1002 [Astar_planner] Segment 9 starts from actual pos [142.77255 -73.
    56496 -23.699223] (planned waypoint was [141. -74. -24.
    509577])
1003 [Astar_planner] Segment 9: [142.77255 -73.56496 -23.699223] -> [
    139. -89. -23.728832]
1004 [Step 20 | Total 0] Target: (139.0, -89.0, -23.7) | Pos: (140.9, -88
    .1, -22.6) | Vel: ( -0.4, -1.3, -1.5) | Acc: ( 0.7, 0.5, -0.1) | Rwd: 3.
    42
1005 [Astar_planner] Segment 9 reached in 20 steps, dist=2.40, axis_error
    =[1.9179688 0.92362213 1.1016102 ], reason=running
1006 [Astar_planner] Segment 10 starts from actual pos [140.91481 -88.
    08783 -22.640678] (planned waypoint was [139. -89. -23.
    728832])
1007 [Astar_planner] Segment 10: [140.91481 -88.08783 -22.640678
    ] -> [ 139. -104. -23.820179]
1008 [Step 13 | Total 0] Target: (139.0, -104.0, -23.8) | Pos: (140.2, -
    102.1, -23.8) | Vel: ( 0.1, -1.4, -0.0) | Acc: ( 0.7, 1.4, 0.6) | Rwd:
    2.09
1009 [Astar_planner] Segment 10 reached in 13 steps, dist=2.21,
    axis_error=[1.1879272 1.8610001 0.02466011], reason=running
1010 [Astar_planner] Segment 11 starts from actual pos [ 140.18881 -102.
    147484 -23.844845] (planned waypoint was [ 139. -104. -23.
    820179])
1011 [Astar_planner] Segment 11: [ 140.18881 -102.147484 -23.844845
    ] -> [ 139. -115. -24.572018]
1012 [Step 16 | Total 0] Target: (139.0, -115.0, -24.6) | Pos: (140.9, -
    114.6, -23.9) | Vel: ( -0.5, -1.3, -1.1) | Acc: ( 1.3, 0.4, 0.4) | Rwd
    : 2.30
1013 [Astar_planner] Segment 11 reached in 16 steps, dist=2.07,
    axis_error=[1.9091339 0.4477234 0.6741848], reason=running
1014 [Astar_planner] Segment 12 starts from actual pos [ 140.90451 -114.
    564316 -23.90711 ] (planned waypoint was [ 139. -115. -24.
    572018])

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1015 [Astar_planner] Segment 12: [ 140.90451 -114.564316 -23.90711
] -> [ 135.    -121.    -21.726013]
1016 [Step 12 | Total    0] Target: (135.0, -121.0, -21.7) | Pos: (136.7, -
121.6, -20.9) | Vel: ( -0.6, -1.0, -0.0) | Acc: ( 1.1, 1.5, 0.8) | Rwd
: 2.08
1017 [Astar_planner] Segment 12 reached in 12 steps, dist=2.02,
axis_error=[1.7455444 0.5543213 0.8459091], reason=running
1018 [Astar_planner] Segment 13 starts from actual pos [ 136.73999 -121.
56286 -20.880146] (planned waypoint was [ 135.    -121.    -21.
726013])
1019 [Astar_planner] Segment 13: [ 136.73999 -121.56286 -20.880146
] -> [ 129.    -127.    -20.23577]
1020 [Step 22 | Total    0] Target: (129.0, -127.0, -20.2) | Pos: (127.7, -
125.3, -19.5) | Vel: ( -0.2, -0.9, 0.3) | Acc: ( 1.8, 0.5, 1.9) | Rwd
: -0.14
1021 [Astar_planner] Segment 13 reached in 22 steps, dist=2.30,
axis_error=[1.323288 1.7178955 0.7782154], reason=running
1022 [Astar_planner] Segment 14 starts from actual pos [ 127.67529 -125.
28726 -19.455872] (planned waypoint was [ 129.    -127.    -20.
23577])
1023 [Astar_planner] Segment 14: [ 127.67529 -125.28726 -19.455872
] -> [ 122.    -134.    -21.457224]
1024 [Step 15 | Total    0] Target: (122.0, -134.0, -21.5) | Pos: (120.3, -
133.0, -20.0) | Vel: ( 0.5, -0.9, -1.6) | Acc: ( 1.2, 1.0, 1.7) | Rwd:
3.89
1025 [Astar_planner] Segment 14 reached in 15 steps, dist=2.48,
axis_error=[1.7378998 1.0329742 1.4346714], reason=running
1026 [Astar_planner] Segment 15 starts from actual pos [ 120.26497 -132.
97218 -20.032078] (planned waypoint was [ 122.    -134.    -21.
457224])
1027 [Astar_planner] Segment 15: [ 120.26497 -132.97218 -20.032078
] -> [ 122.    -149.    -24.007235]
1028 [Step 25 | Total    0] Target: (122.0, -149.0, -24.0) | Pos: (123.8, -
150.6, -22.8) | Vel: ( -0.6, -0.9, 0.4) | Acc: ( 1.1, -0.6, 0.6) | Rwd
: -1.62
1029 [Astar_planner] Segment 15 reached in 25 steps, dist=2.63,
axis_error=[1.7509995 1.5672455 1.1780396], reason=running
1030 [Astar_planner] Segment 16 starts from actual pos [ 123.745964 -150
.57571 -22.826212] (planned waypoint was [ 122.    -149.    -24.
007235])
1031 [Astar_planner] Segment 16: [ 123.745964 -150.57571 -22.826212
] -> [ 122.    -154.    -20.923706]
1032 [Step 2 | Total    0] Target: (122.0, -154.0, -20.9) | Pos: (123.1, -
152.0, -21.6) | Vel: ( -0.2, -1.0, 0.5) | Acc: ( 0.9, -0.4, 1.1) | Rwd
: 1.89
1033 [Astar_planner] Segment 16 reached in 2 steps, dist=2.37, axis_error
=[1.1148224 1.9751129 0.7032795], reason=running
1034 [Astar_planner] Segment 17 starts from actual pos [ 123.113914 -152

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1034 .03085 -21.62422 ] (planned waypoint was [ 122.    -154.    -20.
923706])
1035 [Astar_planner] Segment 17: [ 123.113914 -152.03085 -21.62422
]-> [ 121.    -156.    -22.763605]
1036 [Step  4 | Total    0] Target: (121.0, -156.0, -22.8) | Pos: (122.8, -
155.2, -20.8) | Vel: ( -0.1, -2.0, -2.1) | Acc: ( 0.5, -1.9, 0.0) | Rwd
: 0.33
1037 [Astar_planner] Segment 17 reached in 4 steps, dist=2.75, axis_error
=[1.7664948 0.778183 1.9537678], reason=running
1038 [Astar_planner] Segment 18 starts from actual pos [ 122.765686 -155
.23378 -20.822197] (planned waypoint was [ 121.    -156.    -22.
763605])
1039 [Astar_planner] Segment 18: [ 122.765686 -155.23378 -20.822197
]-> [ 117.    -162.    -20.68389]
1040 [Step 10 | Total    0] Target: (117.0, -162.0, -20.7) | Pos: (117.7, -
160.4, -19.9) | Vel: ( -0.7, -0.9, 0.4) | Acc: ( 0.7, 0.9, 1.2) | Rwd:
0.54
1041 [Astar_planner] Segment 18 reached in 10 steps, dist=1.93,
axis_error=[0.7474747 1.6171417 0.74430084], reason=running
1042 [Astar_planner] Segment 19 starts from actual pos [ 117.74307 -160.
38802 -19.936996] (planned waypoint was [ 117.    -162.    -20.
68389])
1043 [Astar_planner] Segment 19: [ 117.74307 -160.38802 -19.936996
]-> [ 114.    -168.    -22.555008]
1044 [Step 12 | Total    0] Target: (114.0, -168.0, -22.6) | Pos: (114.6, -
167.4, -21.0) | Vel: ( -0.2, -1.0, -0.8) | Acc: ( -0.2, 1.0, 2.0) | Rwd:
604.92 -> Episode End: ARRIVED!
1045
1046 [Astar_planner] Segment 19 reached in 12 steps, dist=1.82,
axis_error=[0.57056427 0.63809204 1.6036777 ], reason=arrived
1047 [Astar_planner] Segment 20 starts from actual pos [ 114.569435 -167
.3678 -20.956095] (planned waypoint was [ 114.    -168.    -22.
555008])
1048 [Astar_planner] Segment 20: [ 114.569435 -167.3678 -20.956095
]-> [ 113.    -169.    -24.528181]
1049 [Step  3 | Total    0] Target: (113.0, -169.0, -24.5) | Pos: (113.2, -
168.0, -23.3) | Vel: ( -0.4, 0.1, 0.2) | Acc: ( 1.8, 0.1, 1.8) | Rwd:
604.22 -> Episode End: ARRIVED!
1050
1051 [Astar_planner] Segment 20 reached in 3 steps, dist=1.62, axis_error
=[0.22531891 1.0189972 1.2415543 ], reason=arrived
1052 [Astar_planner] Segment 21 starts from actual pos [ 113.22193 -167.
98004 -23.28428] (planned waypoint was [ 113.    -169.    -24.
528181])
1053 [Astar_planner] Segment 21: [ 113.22193 -167.98004 -23.28428
]-> [ 110.    -175.    -21.030071]
1054 [Step 29 | Total    0] Target: (110.0, -175.0, -21.0) | Pos: (111.8, -
173.3, -19.4) | Vel: ( -0.4, -1.2, 1.3) | Acc: ( 0.7, -1.3, 0.6) | Rwd

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1054 : -2.00
1055 [Astar_planner] Segment 21 reached in 29 steps, dist=2.96,
axis_error=[1.7959366 1.7219238 1.6068115], reason=running
1056 [Astar_planner] Segment 22 starts from actual pos [ 111.793236 -173
.28511 -19.415726] (planned waypoint was [ 110. -175. -21.
030071])
1057 [Astar_planner] Segment 22: [ 111.793236 -173.28511 -19.415726
]-> [ 109. -176. -22.661]
1058 [Step 7 | Total 0] Target: (109.0, -176.0, -22.7) | Pos: (110.9, -
175.6, -21.8) | Vel: ( -0.3, -0.9, -1.4) | Acc: ( 0.2, -0.2, 1.1) | Rwd
: 0.80
1059 [Astar_planner] Segment 22 reached in 7 steps, dist=2.13, axis_error
=[1.9106903 0.36038208 0.8724766 ], reason=running
1060 [Astar_planner] Segment 23 starts from actual pos [ 110.90908 -175.
64523 -21.796553] (planned waypoint was [ 109. -176. -22.661
])
1061 [Astar_planner] Segment 23: [ 110.90908 -175.64523 -21.796553
]-> [ 106. -182. -24.585974]
1062 [Step 16 | Total 0] Target: (106.0, -182.0, -24.6) | Pos: (107.9, -
182.9, -23.7) | Vel: ( -0.9, -0.2, -0.0) | Acc: ( -0.6, 0.2, 1.9) | Rwd
: 1.03
1063 [Astar_planner] Segment 23 reached in 16 steps, dist=2.31,
axis_error=[1.9171219 0.9237671 0.897274 ], reason=running
1064 [Astar_planner] Segment 24 starts from actual pos [ 107.90934 -182.
92566 -23.688923] (planned waypoint was [ 106. -182. -24.
585974])
1065 [Astar_planner] Segment 24: [ 107.90934 -182.92566 -23.688923
]-> [ 104. -184. -22.659548]
1066 [Step 6 | Total 0] Target: (104.0, -184.0, -22.7) | Pos: (106.0, -
185.3, -21.1) | Vel: ( -0.8, -0.0, -0.7) | Acc: ( -0.0, -0.4, -0.2) | Rwd
: 3.74
1067 [Astar_planner] Segment 24 reached in 6 steps, dist=2.84, axis_error
=[1.9704895 1.3091431 1.564621 ], reason=running
1068 [Astar_planner] Segment 25 starts from actual pos [ 105.96566 -185.
30936 -21.09919] (planned waypoint was [ 104. -184. -22.
659548])
1069 [Astar_planner] Segment 25: [ 105.96566 -185.30936 -21.09919
]-> [ 103. -186. -24.430378]
1070 [Step 6 | Total 0] Target: (103.0, -186.0, -24.4) | Pos: (104.9, -
187.1, -23.7) | Vel: ( -0.8, -0.5, -0.6) | Acc: ( -0.5, -1.7, 1.6) | Rwd
: 1.37
1071 [Astar_planner] Segment 25 reached in 6 steps, dist=2.36, axis_error
=[1.9199524 1.1464233 0.74041176], reason=running
1072 [Astar_planner] Segment 26 starts from actual pos [ 104.91539 -187.
1496 -23.693438] (planned waypoint was [ 103. -186. -24.
430378])
1073 [Astar_planner] Segment 26: [ 104.91539 -187.1496 -23.693438
]-> [ 97. -187. -24.998014]

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1074 [Step 32 | Total 0] Target: ( 97.0, -187.0, -25.0) | Pos: (100.3, -
191.1, -21.8) | Vel: ( 0.0, -0.1, 0.1) | Acc: ( 0.0, -0.1, -0.5) | Rwd
: -0.61
1075 [Astar_planner] Segment 26 reached in 32 steps, dist=6.13,
axis_error=[3.270111 4.058502 3.2210655], reason=
stuck_near_intermediate
1076 [Astar_planner] Segment 27 starts from actual pos [ 100.27016 -191.
05896 -21.776411] (planned waypoint was [ 97. -187. -24.
998014])
1077 [Astar_planner] Segment 27: [ 100.27016 -191.05896 -21.776411
]-> [ 86. -198. -23.946743]
1078 [Step 15 | Total 0] Target: ( 86.0, -198.0, -23.9) | Pos: ( 87.9, -
197.2, -22.1) | Vel: ( -0.7, -1.1, -2.4) | Acc: ( 1.3, 0.0, 0.0) | Rwd
: 3.21
1079 [Astar_planner] Segment 27 reached in 15 steps, dist=2.76,
axis_error=[1.8777695 0.79626465 1.8609123 ], reason=running
1080 [Astar_planner] Segment 28 starts from actual pos [ 87.873375 -197.
21011 -22.100235] (planned waypoint was [ 86. -198. -23.
946743])
1081 [Astar_planner] Segment 28: [ 87.873375 -197.21011 -22.100235
]-> [ 73. -198. -13.29868]
1082 [Step 29 | Total 0] Target: ( 73.0, -198.0, -13.3) | Pos: ( 71.9, -
196.9, -12.3) | Vel: ( -0.7, -0.7, 0.1) | Acc: ( -1.9, 0.9, 0.2) | Rwd
: -0.71
1083 [Astar_planner] Segment 28 reached in 29 steps, dist=1.85,
axis_error=[1.1045761 1.08992 1.0051718], reason=running
1084 Task 16 result: success (Success: Reached destination (29/29
segments). Shield interventions=0, emergencies=0, recoveries=0,
recovery_steps=0.)
1085
1086 --- Task 17/38 ---
1087 From [ 84.8 -145.42 -4. ] to [ -64.4 -146.72 -4. ]
1088 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
1089 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
1090 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
1091 Path planned via occupancy (Points: 14, Dist: 181.6m). Local target z
/alt: 0:-4.00(4.00m), 1:-23.54(23.54m), 2:-21.80(21.80m), 3:-22.74(22
.74m), 4:-24.17(24.17m), 5:-23.98(23.98m), 6:-22.93(22.93m), 7:-21.
23(21.23m), 8:-21.91(21.91m), 9:-20.35(20.35m), 10:-21.95(21.95m
), 11:-24.12(24.12m), 12:-20.01(20.01m), 13:-4.00(4.00m)
1092 [Astar_planner] Segment 0: [ 85. -145. -4.] -> [ 77. -154. -
23.539171]
1093 [Astar_planner] Placing first segment start at [ 85. -145. -4.]
1094 [Astar_planner] Initial climb to 20.0m.
1095 [Step 14 | Total 0] Target: ( 77.0, -154.0, -23.5) | Pos: ( 78.8, -

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1095 153.1, -23.5) | Vel: ( -0.8, -0.9, 0.2) | Acc: ( 0.7, -0.6, 0.2) | Rwd
: 2.01
1096 [Astar_planner] Segment 0 reached in 14 steps, dist=2.03, axis_error
=[1.7933121 0.9440613 0.02338791], reason=running
1097 [Astar_planner] Segment 1 starts from actual pos [ 78.788704 -153.
06134 -23.514725] (planned waypoint was [ 77. -154. -23.
539171])
1098 [Astar_planner] Segment 1: [ 78.788704 -153.06134 -23.514725
]-> [ 65. -156. -21.800009]
1099 [Step 16 | Total 0] Target: ( 65.0, -156.0, -21.8) | Pos: ( 63.6, -
157.7, -20.7) | Vel: ( -0.1, 0.5, 0.5) | Acc: ( -0.4, 1.8, 0.0) | Rwd:
2.86
1100 [Astar_planner] Segment 1 reached in 16 steps, dist=2.44, axis_error
=[1.3837051 1.6921387 1.0907497], reason=running
1101 [Astar_planner] Segment 2 starts from actual pos [ 63.615654 -157.
68886 -20.706001] (planned waypoint was [ 65. -156. -21.
800009])
1102 [Astar_planner] Segment 2: [ 63.615654 -157.68886 -20.706001
]-> [ 52. -156. -22.739079]
1103 [Step 28 | Total 0] Target: ( 52.0, -156.0, -22.7) | Pos: ( 51.3, -
156.6, -21.1) | Vel: ( 0.3, 0.6, 0.1) | Acc: ( -0.5, 2.0, 0.2) | Rwd:
604.94 -> Episode End: ARRIVED!
1104
1105 [Astar_planner] Segment 2 reached in 28 steps, dist=1.85, axis_error
=[0.67300797 0.6405182 1.5952625 ], reason=arrived
1106 [Astar_planner] Segment 3 starts from actual pos [ 51.329243 -156.
63489 -21.142632] (planned waypoint was [ 52. -156. -22.
739079])
1107 [Astar_planner] Segment 3: [ 51.329243 -156.63489 -21.142632
]-> [ 39. -156. -24.174868]
1108 [Step 30 | Total 0] Target: ( 39.0, -156.0, -24.2) | Pos: ( 37.1, -
156.3, -22.3) | Vel: ( 0.4, 0.1, -0.8) | Acc: ( -0.1, -0.6, 0.6) | Rwd
: 1.74
1109 [Astar_planner] Segment 3 reached in 30 steps, dist=2.69, axis_error
=[1.9100609 0.28190613 1.8764515 ], reason=running
1110 [Astar_planner] Segment 4 starts from actual pos [ 37.094208 -156.
28105 -22.307096] (planned waypoint was [ 39. -156. -24.
174868])
1111 [Astar_planner] Segment 4: [ 37.094208 -156.28105 -22.307096
]-> [ 26. -156. -23.97833]
1112 [Step 24 | Total 0] Target: ( 26.0, -156.0, -24.0) | Pos: ( 27.6, -
155.7, -23.2) | Vel: ( -0.5, 0.2, 0.6) | Acc: ( -1.6, 2.0, 1.1) | Rwd:
3.18
1113 [Astar_planner] Segment 4 reached in 24 steps, dist=1.83, axis_error
=[1.612709 0.31962585 0.79813385], reason=running
1114 [Astar_planner] Segment 5 starts from actual pos [ 27.60858 -155.
67851 -23.17453] (planned waypoint was [ 26. -156. -23.
97833])

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1115 [Astar_planner] Segment 5: [ 27.60858 -155.67851 -23.17453] -> [
14.    -157.    -22.929487]
1116 [Step 60 | Total    0] Target: ( 14.0, -157.0, -22.9) | Pos: ( 14.9, -
158.7, -22.6) | Vel: ( -0.2,  0.2, -0.0) | Acc: ( -0.8,  0.7,  0.6) | Rwd
: 1.56
1117 [Astar_planner] Segment 5 reached in 60 steps, dist=1.99, axis_error
=[0.90296936 1.7313232 0.3630371 ], reason=running
1118 [Astar_planner] Segment 6 starts from actual pos [ 14.9010725 -158.
72998  -22.566498 ] (planned waypoint was [ 14.    -157.    -22.
929487])
1119 [Astar_planner] Segment 6: [ 14.9010725 -158.72998  -22.566498
] -> [ 3.    -161.    -21.23055]
1120 [Step 68 | Total    0] Target: ( 3.0, -161.0, -21.2) | Pos: ( 4.6, -162
.2, -21.3) | Vel: ( -1.1,  0.2,  0.0) | Acc: ( -0.3, -0.1, -0.6) | Rwd: 3.
87
1121 [Astar_planner] Segment 6 reached in 68 steps, dist=1.99, axis_error
=[1.6260719 1.1519775 0.0307827], reason=running
1122 [Astar_planner] Segment 7 starts from actual pos [ 4.6163654 -162.
14987  -21.261332 ] (planned waypoint was [ 3.    -161.    -21.
23055])
1123 [Astar_planner] Segment 7: [ 4.6163654 -162.14987  -21.261332
] -> [ -10.    -161.    -21.909758]
1124 [Step 99 | Total    0] Target: (-10.0, -161.0, -21.9) | Pos: ( 2.4, -
165.7, -26.7) | Vel: ( -0.1,  0.1,  0.0) | Acc: ( 0.2, -0.2, -0.7) | Rwd: -
900.63 -> Episode End: CRASHED!
1125
1126 [Astar_planner] Segment 7 ended: collision
1127 [Astar_planner] Stop after failed segment 7.
1128 Task 17 result: failed (Failed at segment 7: collision (7/13 reached).
Shield interventions=188, emergencies=15, recoveries=32,
recovery_steps=188.)
1129
1130 --- Task 18/38 ---
1131 From [-94.8 92.08 -4. ] to [118.1 76.88 -4. ]
1132 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
1133 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
1134 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
1135 Path planned via occupancy (Points: 25, Dist: 340.8m). Local target z
/alt: 0:-4.00(4.00m), 1:-24.80(24.80m), 2:-23.37(23.37m), 3:-21.50(21
.50m), 4:-23.55(23.55m), 5:-20.04(20.04m), 6:-23.48(23.48m), 7:-22.
42(22.42m), 8:-21.82(21.82m), 9:-22.26(22.26m), 10:-21.09(21.09m
), 11:-23.85(23.85m), 12:-24.16(24.16m), 13:-22.12(22.12m), 14:-21.
63(21.63m), 15:-22.56(22.56m), 16:-24.41(24.41m), 17:-20.40(20.
40m), 18:-23.60(23.60m), 19:-20.49(20.49m), 20:-20.86(20.86m), 21
:-22.71(22.71m), 22:-23.19(23.19m), 23:-24.66(24.66m), 24:-4.00(4.

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1135 00m)
1136 [Astar_planner] Segment 0: [-95. 92. -4.] -> [-83. 89. -24.
80464]
1137 [Astar_planner] Placing first segment start at [-95. 92. -4.]
1138 [Astar_planner] Initial climb to 20.0m.
1139 [Step 114 | Total 0] Target: (-83.0, 89.0, -24.8) | Pos: (-84.4, 89.
0, -23.0) | Vel: ( 0.2, 0.4, -0.5) | Acc: (-1.1, 1.0, 0.9) | Rwd: 602.
63 -> Episode End: ARRIVED!
1140
1141 [Astar_planner] Segment 0 reached in 114 steps, dist=2.28,
axis_error=[1.3612518 0.03990936 1.8300781 ], reason=arrived
1142 [Astar_planner] Segment 1 starts from actual pos [-84.359695 89.
04339 -22.97905 ] (planned waypoint was [-83. 89. -24.80464
])
1143 [Astar_planner] Segment 1: [-84.359695 89.04339 -22.97905 ] -> [-
69. 89. -23.3709]
1144 [Step 22 | Total 0] Target: (-69.0, 89.0, -23.4) | Pos: (-67.5, 90.
2, -22.3) | Vel: (-0.8, 0.1, -0.6) | Acc: ( 1.7, -0.5, 1.4) | Rwd: 3.
08
1145 [Astar_planner] Segment 1 reached in 22 steps, dist=2.21, axis_error
=[1.5178909 1.2058411 1.0692635], reason=running
1146 [Astar_planner] Segment 2 starts from actual pos [-67.48696 90.
20672 -22.305048] (planned waypoint was [-69. 89. -23.3709])
1147 [Astar_planner] Segment 2: [-67.48696 90.20672 -22.305048] -> [-
56. 89. -21.504381]
1148 [Step 19 | Total 0] Target: (-56.0, 89.0, -21.5) | Pos: (-54.6, 90.
5, -20.1) | Vel: (-0.8, -0.3, -0.1) | Acc: (-1.6, 0.1, -1.2) | Rwd: 4.
06
1149 [Astar_planner] Segment 2 reached in 19 steps, dist=2.45, axis_error
=[1.3602486 1.5258408 1.3570595], reason=running
1150 [Astar_planner] Segment 3 starts from actual pos [-54.6472 90.
52337 -20.147635] (planned waypoint was [-56. 89. -21.
504381])
1151 [Astar_planner] Segment 3: [-54.6472 90.52337 -20.147635] -> [-
42. 89. -23.551582]
1152 [Step 17 | Total 0] Target: (-42.0, 89.0, -23.6) | Pos: (-44.0, 88.
7, -22.2) | Vel: ( 1.2, 0.2, -1.4) | Acc: (-0.1, 0.1, 0.5) | Rwd: 5.81
1153 [Astar_planner] Segment 3 reached in 17 steps, dist=2.40, axis_error
=[1.9925194 0.3186264 1.3057194], reason=running
1154 [Astar_planner] Segment 4 starts from actual pos [-43.981773 88.
68332 -22.258526] (planned waypoint was [-42. 89. -23.
551582])
1155 [Astar_planner] Segment 4: [-43.981773 88.68332 -22.258526] -> [-
30. 94. -20.040468]
1156 [Step 19 | Total 0] Target: (-30.0, 94.0, -20.0) | Pos: (-28.4, 95.
7, -19.5) | Vel: ( 0.9, -0.5, -0.1) | Acc: (-0.0, 0.9, 0.8) | Rwd: 0.
11

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1157 [Astar_planner] Segment 4 reached in 19 steps, dist=2.38, axis_error
      =[1.6272469 1.6545639 0.5378685], reason=running
1158 [Astar_planner] Segment 5 starts from actual pos [-28.36491 95.
      649734 -19.503262] (planned waypoint was [-30. 94. -20.
      040468])
1159 [Astar_planner] Segment 5: [-28.36491 95.649734 -19.503262] -> [-
      25. 105. -23.483677]
1160 [Step 21 | Total 0] Target: (-25.0, 105.0, -23.5) | Pos: (-24.2, 103
      .6, -21.5) | Vel: ( -0.5, 0.6, -1.3) | Acc: ( -2.0, -0.3, 2.0) | Rwd: 5.
      44
1161 [Astar_planner] Segment 5 reached in 21 steps, dist=2.52, axis_error
      =[0.7780647 1.3628464 1.9694843], reason=running
1162 [Astar_planner] Segment 6 starts from actual pos [-24.226206 103.
      64238 -21.5253 ] (planned waypoint was [-25. 105. -23.
      483677])
1163 [Astar_planner] Segment 6: [-24.226206 103.64238 -21.5253 ] -> [-
      25. 118. -22.42066]
1164 [Step 18 | Total 0] Target: (-25.0, 118.0, -22.4) | Pos: (-23.6, 118
      .1, -21.0) | Vel: ( -0.9, 1.1, -0.9) | Acc: ( 0.4, -0.4, 1.9) | Rwd: 603.
      37 -> Episode End: ARRIVED!
1165
1166 [Astar_planner] Segment 6 reached in 18 steps, dist=1.97, axis_error
      =[1.4063034 0.0887146 1.375885 ], reason=arrived
1167 [Astar_planner] Segment 7 starts from actual pos [-23.598906 118.
      095406 -21.049805] (planned waypoint was [-25. 118. -22.
      42066])
1168 [Astar_planner] Segment 7: [-23.598906 118.095406 -21.049805
      ] -> [-25. 132. -21.818222]
1169 [Step 20 | Total 0] Target: (-25.0, 132.0, -21.8) | Pos: (-25.7, 133
      .7, -20.5) | Vel: ( -0.2, -0.3, -0.0) | Acc: ( -0.5, 0.3, 0.3) | Rwd: -0.
      90
1170 [Astar_planner] Segment 7 reached in 20 steps, dist=2.30, axis_error
      =[0.7401829 1.7051239 1.3511524], reason=running
1171 [Astar_planner] Segment 8 starts from actual pos [-25.741993 133.
      70273 -20.46738 ] (planned waypoint was [-25. 132. -21.
      818222])
1172 [Astar_planner] Segment 8: [-25.741993 133.70273 -20.46738 ] -> [-
      25. 145. -22.26466]
1173 [Step 26 | Total 0] Target: (-25.0, 145.0, -22.3) | Pos: (-25.7, 143
      .0, -21.0) | Vel: ( -0.5, 0.4, 0.4) | Acc: ( 0.5, 0.3, 0.6) | Rwd: -1.
      63
1174 [Astar_planner] Segment 8 reached in 26 steps, dist=2.44, axis_error
      =[0.69649506 1.9527893 1.2831154 ], reason=running
1175 [Astar_planner] Segment 9 starts from actual pos [-25.699516 143.
      04958 -20.979391] (planned waypoint was [-25. 145. -22.
      26466])
1176 [Astar_planner] Segment 9: [-25.699516 143.04958 -20.979391] -> [-
      17. 153. -21.091852]

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1177 [Step 17 | Total    0] Target: (-17.0, 153.0, -21.1) | Pos: (-18.8, 152
    .3, -20.2) | Vel: ( 0.5, 0.8, -0.2) | Acc: (-1.4, -0.1, 2.0) | Rwd: 2.
    41
1178 [Astar_planner] Segment 9 reached in 17 steps, dist=2.09, axis_error
    =[1.7579536 0.68862915 0.9060364 ], reason=running
1179 [Astar_planner] Segment 10 starts from actual pos [-18.756601 152.
    31375 -20.186323] (planned waypoint was [-17.    153.    -21.
    091852])
1180 [Astar_planner] Segment 10: [-18.756601 152.31375 -20.186323
    ] -> [-3.    153.    -23.84668]
1181 [Step 22 | Total    0] Target: (-3.0, 153.0, -23.8) | Pos: (-4.8, 152.
    0, -22.3) | Vel: ( 1.1, -0.4, -2.1) | Acc: (-0.6, 0.3, -0.4) | Rwd: 4.
    76
1182 [Astar_planner] Segment 10 reached in 22 steps, dist=2.59,
    axis_error=[1.7954702 0.9747467 1.5847301], reason=running
1183 [Astar_planner] Segment 11 starts from actual pos [-4.78544 152.
    02182 -22.280596] (planned waypoint was [-3.    153.    -23.
    84668])
1184 [Astar_planner] Segment 11: [-4.78544 152.02182 -22.280596] -> [
    10.    151.    -24.16371]
1185 [Step 13 | Total    0] Target: ( 10.0, 151.0, -24.2) | Pos: ( 10.4, 152.
    1, -22.4) | Vel: ( 2.0, -0.9, -2.1) | Acc: (-2.0, 1.2, 0.8) | Rwd: 605.
    94 -> Episode End: ARRIVED!
1186
1187 [Astar_planner] Segment 11 reached in 13 steps, dist=2.12,
    axis_error=[0.4162178 1.0875854 1.7748642], reason=arrived
1188 [Astar_planner] Segment 12 starts from actual pos [ 10.434411 152.
    07944 -22.407139] (planned waypoint was [ 10.    151.    -24.
    16371])
1189 [Astar_planner] Segment 12: [ 10.434411 152.07944 -22.407139
    ] -> [ 21.    145.    -22.117489]
1190 [Step 9 | Total    0] Target: ( 21.0, 145.0, -22.1) | Pos: ( 20.6, 146.
    5, -20.5) | Vel: ( 2.4, -0.7, -1.4) | Acc: ( 0.2, 1.8, -0.4) | Rwd: 3.
    14
1191 [Astar_planner] Segment 12 reached in 9 steps, dist=2.25, axis_error
    =[0.40189743 1.5372162 1.5885963 ], reason=running
1192 [Astar_planner] Segment 13 starts from actual pos [ 20.61947 146.
    53053 -20.541073] (planned waypoint was [ 21.    145.    -22.
    117489])
1193 [Astar_planner] Segment 13: [ 20.61947 146.53053 -20.541073
    ] -> [ 31.    137.    -21.63198]
1194 [Step 26 | Total    0] Target: ( 31.0, 137.0, -21.6) | Pos: ( 33.0, 136.
    2, -20.4) | Vel: (-0.3, 0.0, 0.0) | Acc: (-0.0, -0.6, 1.1) | Rwd: -1.
    26
1195 [Astar_planner] Segment 13 reached in 26 steps, dist=2.44,
    axis_error=[1.9897537 0.7500305 1.1938992], reason=running
1196 [Astar_planner] Segment 14 starts from actual pos [ 32.9875 136.
    25006 -20.437801] (planned waypoint was [ 31.    137.    -21.

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1196 63198])
1197 [Astar_planner] Segment 14: [ 32.9875  136.25006 -20.437801] -> [
44.    137.    -22.556517]
1198 [Step 12 | Total    0] Target: ( 44.0, 137.0, -22.6) | Pos: ( 42.9, 137.
7, -20.8) | Vel: ( 2.2, -0.8, -1.7) | Acc: ( -1.8, -2.0, 0.3) | Rwd: 603.
87 -> Episode End: ARRIVED!
1199
1200 [Astar_planner] Segment 14 reached in 12 steps, dist=2.21,
axis_error=[1.0970802 0.71287537 1.7770367 ], reason=arrived
1201 [Astar_planner] Segment 15 starts from actual pos [ 42.922466 137.
7061 -20.794441] (planned waypoint was [ 44.    137.    -22.
556517])
1202 [Astar_planner] Segment 15: [ 42.922466 137.7061 -20.794441
] -> [ 58.    137.    -24.414816]
1203 [Step 12 | Total    0] Target: ( 58.0, 137.0, -24.4) | Pos: ( 59.6, 135.
5, -22.8) | Vel: ( 1.6, 0.7, -1.4) | Acc: ( -2.0, 1.8, 2.0) | Rwd: 2.17

1204 [Astar_planner] Segment 15 reached in 12 steps, dist=2.72,
axis_error=[1.569828 1.5433197 1.593914 ], reason=running
1205 [Astar_planner] Segment 16 starts from actual pos [ 59.57914 135.
46098 -22.82929] (planned waypoint was [ 58.    137.    -24.
414816])
1206 [Astar_planner] Segment 16: [ 59.57914 135.46098 -22.82929] -> [
71.    135.    -20.403234]
1207 [Step 11 | Total    0] Target: ( 71.0, 135.0, -20.4) | Pos: ( 69.2, 133.
8, -20.4) | Vel: ( 2.2, 1.2, -0.5) | Acc: ( 2.0, 1.7, 0.2) | Rwd: 4.38

1208 [Astar_planner] Segment 16 reached in 11 steps, dist=2.18,
axis_error=[1.8009262e+00 1.2221527e+00 1.0433197e-03], reason
=running
1209 [Astar_planner] Segment 17 starts from actual pos [ 69.212135 133.
7853 -20.407393] (planned waypoint was [ 71.    135.    -20.
403234])
1210 [Astar_planner] Segment 17: [ 69.212135 133.7853 -20.407393
] -> [ 84.    135.    -23.601656]
1211 [Step 46 | Total    0] Target: ( 84.0, 135.0, -23.6) | Pos: ( 82.2, 134.
3, -22.9) | Vel: ( 0.6, 0.2, -0.1) | Acc: ( -1.9, 0.7, 2.0) | Rwd: 1.83

1212 [Astar_planner] Segment 17 reached in 46 steps, dist=2.00,
axis_error=[1.7561569 0.6646881 0.6926193], reason=running
1213 [Astar_planner] Segment 18 starts from actual pos [ 82.2472 134.
33679 -22.909838] (planned waypoint was [ 84.    135.    -23.
601656])
1214 [Astar_planner] Segment 18: [ 82.2472 134.33679 -22.909838] -> [
98.    135.    -20.490204]
1215 [Step 25 | Total    0] Target: ( 98.0, 135.0, -20.5) | Pos: ( 96.7, 133.
6, -18.6) | Vel: ( 0.4, 0.9, -1.5) | Acc: ( -2.0, 1.7, 2.0) | Rwd: 4.14

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1216 [Astar_planner] Segment 18 reached in 25 steps, dist=2.72,
axis_error=[1.3287811 1.4022064 1.9128075], reason=running
1217 [Astar_planner] Segment 19 starts from actual pos [ 96.673645 133.
60341 -18.586582] (planned waypoint was [ 98.    135.    -20.
490204])
1218 [Astar_planner] Segment 19: [ 96.673645 133.60341 -18.586582
] -> [104.    125.    -20.855328]
1219 [Step 18 | Total    0] Target: (104.0, 125.0, -20.9) | Pos: (102.5,
125.4, -19.7) | Vel: ( 0.7, 0.6, -0.1) | Acc: ( 1.5, 2.0, -0.5) | Rwd:
0.86
1220 [Astar_planner] Segment 19 reached in 18 steps, dist=1.93,
axis_error=[1.5056992 0.4353943 1.1287727], reason=running
1221 [Astar_planner] Segment 20 starts from actual pos [102.500824 125.
440506 -19.727192] (planned waypoint was [104.    125.    -20.
855328])
1222 [Astar_planner] Segment 20: [102.500824 125.440506 -19.727192
] -> [104.    111.    -22.71486]
1223 [Step 27 | Total    0] Target: (104.0, 111.0, -22.7) | Pos: (104.0,
109.2, -24.0) | Vel: ( -0.6, 1.3, -0.4) | Acc: ( -1.3, 0.4, 0.1) | Rwd
: -0.10
1224 [Astar_planner] Segment 20 reached in 27 steps, dist=2.23,
axis_error=[0.02059174 1.8368301 1.2671452 ], reason=running
1225 [Astar_planner] Segment 21 starts from actual pos [103.97559 109.
171196 -23.984095] (planned waypoint was [104.    111.    -22.
71486])
1226 [Astar_planner] Segment 21: [103.97559 109.171196 -23.984095
] -> [104.    98.    -23.192644]
1227 [Step 13 | Total    0] Target: (104.0, 98.0, -23.2) | Pos: (104.0, 99.
6, -22.0) | Vel: ( 0.2, -1.5, -0.2) | Acc: ( 0.1, -1.1, 0.7) | Rwd: 3.
00
1228 [Astar_planner] Segment 21 reached in 13 steps, dist=2.04,
axis_error=[0.00711823 1.6398087 1.2070503 ], reason=running
1229 [Astar_planner] Segment 22 starts from actual pos [103.99478 99.
62595 -21.987047] (planned waypoint was [104.    98.    -23.
192644])
1230 [Astar_planner] Segment 22: [103.99478 99.62595 -21.987047] -> [
108.    86.    -24.663282]
1231 [Step 19 | Total    0] Target: (108.0, 86.0, -24.7) | Pos: (106.2, 84.
2, -24.0) | Vel: ( 1.2, -0.4, 0.4) | Acc: ( -0.2, -1.2, 0.9) | Rwd: 0.
07
1232 [Astar_planner] Segment 22 reached in 19 steps, dist=2.65,
axis_error=[1.8029099 1.8011017 0.7127266], reason=running
1233 [Astar_planner] Segment 23 starts from actual pos [106.20445 84.
19673 -23.94809] (planned waypoint was [108.    86.    -24.
663282])
1234 [Astar_planner] Segment 23: [106.20445 84.19673 -23.94809] -> [
118. 77. -4.]
1235 [Step 28 | Total    0] Target: (118.0, 77.0, -4.0) | Pos: (117.7, 78.

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1235 1, -4.9) | Vel: ( 2.2, -0.9, 1.0) | Acc: ( 1.6, -1.8, 0.4) | Rwd: 602.
      71 -> Episode End: ARRIVED!
1236
1237 [Astar_planner] Segment 23 reached in 28 steps, dist=1.47,
      axis_error=[0.2925949 1.1307373 0.88626003], reason=arrived
1238 Task 18 result: success (Success: Reached destination (24/24
      segments). Shield interventions=17, emergencies=1, recoveries=3,
      recovery_steps=17.)
1239
1240 --- Task 19/38 ---
1241 From [107. -82.52 -4. ] to [ -63.8 -136.72 -4. ]
1242 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
1243 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
1244 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
1245 Path planned via occupancy (Points: 20, Dist: 259.6m). Local target z
      /alt: 0:-4.00(4.00m), 1:-21.48(21.48m), 2:-20.64(20.64m), 3:-23.37(23
      .37m), 4:-24.88(24.88m), 5:-20.32(20.32m), 6:-24.54(24.54m), 7:-24.
      28(24.28m), 8:-22.14(22.14m), 9:-23.53(23.53m), 10:-24.12(24.12m
      ), 11:-23.58(23.58m), 12:-20.82(20.82m), 13:-24.41(24.41m), 14:-23.
      13(23.13m), 15:-21.18(21.18m), 16:-23.53(23.53m), 17:-20.92(20.
      92m), 18:-21.00(21.00m), 19:-4.00(4.00m)
1246 [Astar_planner] Segment 0: [107. -83. -4.] -> [100. -93. -21.
      484074]
1247 [Astar_planner] Placing first segment start at [107. -83. -4.]
1248 [Astar_planner] Initial climb to 20.0m.
1249 [Step 15 | Total 0] Target: (100.0, -93.0, -21.5) | Pos: (101.5, -91
      .6, -21.3) | Vel: ( -0.6, -1.0, 0.1) | Acc: ( 0.0, -0.0, 0.5) | Rwd: 2.
      00
1250 [Astar_planner] Segment 0 reached in 15 steps, dist=2.03, axis_error
      =[1.4773178 1.369873 0.20993614], reason=running
1251 [Astar_planner] Segment 1 starts from actual pos [101.471695 -91.
      639534 -21.273716] (planned waypoint was [100. -93. -21.
      484074])
1252 [Astar_planner] Segment 1: [101.471695 -91.639534 -21.273716
      ] -> [ 100. -106. -20.635683]
1253 [Step 24 | Total 0] Target: (100.0, -106.0, -20.6) | Pos: (101.3, -
      104.2, -19.9) | Vel: ( -0.1, -1.0, 0.0) | Acc: ( -0.2, 0.6, -0.2) | Rwd
      : 1.24
1254 [Astar_planner] Segment 1 reached in 24 steps, dist=2.32, axis_error
      =[1.2744522 1.7844696 0.7628746], reason=running
1255 [Astar_planner] Segment 2 starts from actual pos [ 101.273766 -104.
      224365 -19.872433] (planned waypoint was [ 100. -106. -20.
      635683])
1256 [Astar_planner] Segment 2: [ 101.273766 -104.224365 -19.872433
      ] -> [ 94. -117. -23.374067]

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1257 [Step 16 | Total 0] Target: ( 94.0, -117.0, -23.4) | Pos: ( 93.9, -
115.2, -22.7) | Vel: ( -0.6, -1.1, 0.1) | Acc: ( -0.8, 0.3, 0.0) | Rwd
: 1.54
1258 [Astar_planner] Segment 2 reached in 16 steps, dist=1.95, axis_error
=[0.06421661 1.8316727 0.67907715], reason=running
1259 [Astar_planner] Segment 3 starts from actual pos [ 93.93082 -115.
17833 -22.694458] (planned waypoint was [ 94. -117. -23.
374067])
1260 [Astar_planner] Segment 3: [ 93.93082 -115.17833 -22.694458
] -> [ 85. -126. -24.880207]
1261 [Step 12 | Total 0] Target: ( 85.0, -126.0, -24.9) | Pos: ( 85.9, -
124.7, -24.2) | Vel: ( -0.5, -1.3, 0.3) | Acc: ( 0.6, -1.3, 1.1) | Rwd
: 0.69
1262 [Astar_planner] Segment 3 reached in 12 steps, dist=1.76, axis_error
=[0.91239166 1.3467178 0.67203903], reason=running
1263 [Astar_planner] Segment 4 starts from actual pos [ 85.909256 -124.
66102 -24.206154] (planned waypoint was [ 85. -126. -24.
880207])
1264 [Astar_planner] Segment 4: [ 85.909256 -124.66102 -24.206154
] -> [ 82. -137. -20.322447]
1265 [Step 13 | Total 0] Target: ( 82.0, -137.0, -20.3) | Pos: ( 83.2, -
135.4, -19.5) | Vel: ( 0.8, -1.0, -0.6) | Acc: ( -0.8, 0.7, -1.8) | Rwd
: 0.59
1266 [Astar_planner] Segment 4 reached in 13 steps, dist=2.13, axis_error
=[1.2387619 1.5522308 0.78006554], reason=running
1267 [Astar_planner] Segment 5 starts from actual pos [ 83.243324 -135.
45378 -19.545958] (planned waypoint was [ 82. -137. -20.
322447])
1268 [Astar_planner] Segment 5: [ 83.243324 -135.45378 -19.545958
] -> [ 80. -150. -24.542889]
1269 [Step 29 | Total 0] Target: ( 80.0, -150.0, -24.5) | Pos: ( 81.6, -
148.1, -24.0) | Vel: ( -0.2, -1.0, -0.1) | Acc: ( 1.0, 0.8, -0.1) | Rwd
: 1.57
1270 [Astar_planner] Segment 5 reached in 29 steps, dist=2.51, axis_error
=[1.5744019 1.8623962 0.58898926], reason=running
1271 [Astar_planner] Segment 6 starts from actual pos [ 81.572975 -148.
1434 -23.95437 ] (planned waypoint was [ 80. -150. -24.
542889])
1272 [Astar_planner] Segment 6: [ 81.572975 -148.1434 -23.95437
] -> [ 71. -156. -24.27695]
1273 [Step 13 | Total 0] Target: ( 71.0, -156.0, -24.3) | Pos: ( 73.0, -
155.9, -23.4) | Vel: ( -1.4, -0.9, -0.9) | Acc: ( 0.3, -1.0, 0.7) | Rwd
: 3.54
1274 [Astar_planner] Segment 6 reached in 13 steps, dist=2.14, axis_error
=[1.9500275 0.06060791 0.86722946], reason=running
1275 [Astar_planner] Segment 7 starts from actual pos [ 72.937744 -155.
94759 -23.417908] (planned waypoint was [ 71. -156. -24.
27695])

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1276 [Astar_planner] Segment 7: [ 72.937744 -155.94759 -23.417908
] -> [ 58.    -156.    -22.138924]
1277 [Step 31 | Total    0] Target: ( 58.0, -156.0, -22.1) | Pos: ( 56.2, -
156.4, -20.9) | Vel: ( 0.7, -0.5, -0.6) | Acc: ( -1.7, 0.6, 1.5) | Rwd
: 3.22
1278 [Astar_planner] Segment 7 reached in 31 steps, dist=2.21, axis_error
=[1.777977 0.44488525 1.2300167 ], reason=running
1279 [Astar_planner] Segment 8 starts from actual pos [ 56.226517 -156.
44806 -20.9123 ] (planned waypoint was [ 58.    -156.    -22.
138924])
1280 [Astar_planner] Segment 8: [ 56.226517 -156.44806 -20.9123
] -> [ 45.    -156.    -23.529596]
1281 [Step 31 | Total    0] Target: ( 45.0, -156.0, -23.5) | Pos: ( 43.5, -
157.2, -22.7) | Vel: ( 1.2, -0.3, -0.0) | Acc: ( 1.3, -0.5, 2.0) | Rwd
: 2.25
1282 [Astar_planner] Segment 8 reached in 31 steps, dist=2.07, axis_error
=[1.452179 1.1779785 0.8769169], reason=running
1283 [Astar_planner] Segment 9 starts from actual pos [ 43.554752 -157.
17949 -22.652748] (planned waypoint was [ 45.    -156.    -23.
529596])
1284 [Astar_planner] Segment 9: [ 43.554752 -157.17949 -22.652748
] -> [ 33.    -160.    -24.123043]
1285 [Step 14 | Total    0] Target: ( 33.0, -160.0, -24.1) | Pos: ( 34.9, -
160.5, -22.7) | Vel: ( -1.4, -0.8, -1.4) | Acc: ( 0.6, -1.0, 0.8) | Rwd
: 3.97
1286 [Astar_planner] Segment 9 reached in 14 steps, dist=2.38, axis_error
=[1.874939 0.5063019 1.378519 ], reason=running
1287 [Astar_planner] Segment 10 starts from actual pos [ 34.86275 -160.
51349 -22.756723] (planned waypoint was [ 33.    -160.    -24.
123043])
1288 [Astar_planner] Segment 10: [ 34.86275 -160.51349 -22.756723
] -> [ 20.    -160.    -23.584671]
1289 [Step 19 | Total    0] Target: ( 20.0, -160.0, -23.6) | Pos: ( 21.8, -
159.8, -23.1) | Vel: ( -1.1, 1.1, -0.1) | Acc: ( -0.6, 1.2, -0.0) | Rwd
: 2.75
1290 [Astar_planner] Segment 10 reached in 19 steps, dist=1.91,
axis_error=[1.8264751 0.17384338 0.52433777], reason=running
1291 [Astar_planner] Segment 11 starts from actual pos [ 21.816238 -159.
8159 -23.061523] (planned waypoint was [ 20.    -160.    -23.
584671])
1292 [Astar_planner] Segment 11: [ 21.816238 -159.8159 -23.061523
] -> [ 8.    -161.    -20.818]
1293 [Step 19 | Total    0] Target: ( 8.0, -161.0, -20.8) | Pos: ( 9.9, -162
.8, -20.4) | Vel: ( -0.1, -0.1, 0.1) | Acc: ( 1.2, -1.3, -0.7) | Rwd: 0.
41
1294 [Astar_planner] Segment 11 reached in 19 steps, dist=2.63,
axis_error=[1.8720751 1.8031921 0.4271164], reason=running
1295 [Astar_planner] Segment 12 starts from actual pos [ 9.870843 -162.

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1295 80397 -20.389822] (planned waypoint was [ 8. -161. -20.818])
1296 [Astar_planner] Segment 12: [ 9.870843 -162.80397 -20.389822
]-> [ -5. -161. -24.41321]
1297 [Step 102 | Total 0] Target: ( -5.0, -161.0, -24.4) | Pos: ( 1.4, -
163.9, -25.9) | Vel: ( 0.5, -0.7, 0.1) | Acc: ( 1.4, -0.7, -0.7) | Rwd: -
904.80 -> Episode End: CRASHED!
1298
1299 [Astar_planner] Segment 12 ended: collision
1300 [Astar_planner] Stop after failed segment 12.
1301 Task 19 result: failed (Failed at segment 12: collision (12/19 reached
). Shield interventions=111, emergencies=6, recoveries=19,
recovery_steps=111.)
1302
1303 --- Task 20/38 ---
1304 From [-75.1 -8.22 -16. ] to [101.4 29.68 -6.6 ]
1305 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
1306 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
1307 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(245, 192), goal=(69, 230), search_radius=36.
1308 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
1309 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(245, 192), goal=(69, 230), search_radius=36.
1310 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.0
1311 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(245, 192), goal=(69, 230), search_radius=36.
1312 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=0.6
1313 [Astar_planner] Endpoint nearest-free cells are disconnected; using
reachable cells start=(245, 192), goal=(69, 230), search_radius=36.
1314 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=0.3
1315 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
clearance penalties.
1316 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4
%, safety_margin=1.5
1317 [Astar_planner] WARNING: using legacy occupancy fallback; local
targets may be closer to obstacles.
1318 Path planned via occupancy_legacy (Points: 18, Dist: 205.3m). Local
target z/alt: 0:-15.99(15.99m), 1:-21.20(21.20m), 2:-23.31(23.31m), 3
:-21.52(21.52m), 4:-22.45(22.45m), 5:-22.98(22.98m), 6:-24.24(24.
24m), 7:-24.64(24.64m), 8:-21.98(21.98m), 9:-21.31(21.31m), 10:-20.
09(20.09m), 11:-22.03(22.03m), 12:-20.06(20.06m), 13:-22.95(22.
95m), 14:-21.74(21.74m), 15:-22.90(22.90m), 16:-20.01(20.01m), 17
:-6.62(6.62m)

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1319 [Astar_planner] Segment 0: [-75.    -8.    -15.987416] -> [-70.    -3.    -21.2013]
1320 [Astar_planner] Placing first segment start at [-75.    -8.    -15.987416]
1321 [Astar_planner] Initial climb to 20.0m.
1322 [Step 49 | Total    0] Target: (-70.0, -3.0, -21.2) | Pos: (-72.0, -3.4, -20.5) | Vel: ( 0.0, 0.1, 0.1) | Acc: ( -1.3, 0.4, -1.5) | Rwd: 0.28
1323 [Astar_planner] Segment 0 reached in 49 steps, dist=2.15, axis_error=[1.9830017 0.44028044 0.69142723], reason=running
1324 [Astar_planner] Segment 1 starts from actual pos [-71.982635 -3.438973 -20.50934 ] (planned waypoint was [-70.    -3.    -21.2013])
1325 [Astar_planner] Segment 1: [-71.982635 -3.438973 -20.50934 ] -> [-59.    -2.    -23.31111]
1326 [Step 43 | Total    0] Target: (-59.0, -2.0, -23.3) | Pos: (-60.7, -2.0, -22.5) | Vel: ( 0.2, -0.1, -0.0) | Acc: ( 0.3, -1.3, -0.0) | Rwd: 0.24
1327 [Astar_planner] Segment 1 reached in 43 steps, dist=1.89, axis_error=[1.7041931 0.02260971 0.80591774], reason=running
1328 [Astar_planner] Segment 2 starts from actual pos [-60.703144 -2.0231187 -22.505215 ] (planned waypoint was [-59.    -2.    -23.31111])
1329 [Astar_planner] Segment 2: [-60.703144 -2.0231187 -22.505215 ] -> [-55.    -2.    -21.516668]
1330 [Step 29 | Total    0] Target: (-55.0, -2.0, -21.5) | Pos: (-56.8, -0.7, -21.4) | Vel: ( 0.2, 0.1, -0.0) | Acc: ( 0.5, 1.0, -0.0) | Rwd: 0.13
1331 [Astar_planner] Segment 2 reached in 29 steps, dist=2.23, axis_error=[1.8337669 1.2722008 0.10177231], reason=running
1332 [Astar_planner] Segment 3 starts from actual pos [-56.832188 -0.7271725 -21.414932 ] (planned waypoint was [-55.    -2.    -21.516668])
1333 [Astar_planner] Segment 3: [-56.832188 -0.7271725 -21.414932 ] -> [-40.    -2.    -22.447466]
1334 [Step 35 | Total    0] Target: (-40.0, -2.0, -22.4) | Pos: (-42.0, -1.9, -22.7) | Vel: ( 0.3, 0.3, -0.0) | Acc: ( 0.8, 1.3, -0.0) | Rwd: 1.24
1335 [Astar_planner] Segment 3 reached in 35 steps, dist=1.98, axis_error=[1.9559479 0.11216044 0.26111984], reason=running
1336 [Astar_planner] Segment 4 starts from actual pos [-41.95286 -1.8848962 -22.708797 ] (planned waypoint was [-40.    -2.    -22.447466])
1337 [Astar_planner] Segment 4: [-41.95286 -1.8848962 -22.708797 ] -> [-25.    -2.    -22.983568]
1338 [Step 89 | Total    0] Target: (-25.0, -2.0, -23.0) | Pos: (-24.6, -0.1, -21.3) | Vel: ( 0.0, -0.1, -0.4) | Acc: ( -0.6, 2.0, 2.0) | Rwd: -3.11
1339 [Astar_planner] Segment 4 reached in 89 steps, dist=2.60, axis_error=[0.38829613 1.9417824 1.6840954 ], reason=running
1340 [Astar_planner] Segment 5 starts from actual pos [-24.611357 -0.05876147 -21.302837 ] (planned waypoint was [-25.    -2.    -22.983568])
1341 [Astar_planner] Segment 5: [-24.611357 -0.05876147 -21.302837 ]

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1341 ] -> [-10.    -2.    -24.243355]
1342 [Step 33 | Total    0] Target: (-10.0, -2.0, -24.2) | Pos: (-10.7, -3.8
, -22.7) | Vel: ( 0.6, 0.5, 0.6) | Acc: ( 0.4, 2.0, 1.0) | Rwd: 2.21
1343 [Astar_planner] Segment 5 reached in 33 steps, dist=2.47, axis_error
=[0.6976452 1.8178682 1.5142918], reason=running
1344 [Astar_planner] Segment 6 starts from actual pos [-10.694339 -3.
8148217 -22.72554 ] (planned waypoint was [-10.    -2.    -24.
243355])
1345 [Astar_planner] Segment 6: [-10.694339 -3.8148217 -22.72554
] -> [ 5.    -2.    -24.636974]
1346 [Step 175 | Total    0] Target: ( 5.0, -2.0, -24.6) | Pos: ( 3.3, -1.2
, -23.7) | Vel: ( 0.4, -0.1, 0.5) | Acc: ( 0.0, -1.7, 0.8) | Rwd: 3.02
1347 [Astar_planner] Segment 6 reached in 175 steps, dist=2.13,
axis_error=[1.7393396 0.8120811 0.9287987], reason=running
1348 [Astar_planner] Segment 7 starts from actual pos [ 3.264029 -1.
1885794 -23.70354 ] (planned waypoint was [ 5.    -2.    -24.
636974])
1349 [Astar_planner] Segment 7: [ 3.264029 -1.1885794 -23.70354
] -> [ 20.    -2.    -21.982218]
1350 [Step 17 | Total    0] Target: ( 20.0, -2.0, -22.0) | Pos: ( 18.3, -2.6
, -21.2) | Vel: ( 1.3, 0.3, -0.1) | Acc: ( 0.3, 0.2, -1.7) | Rwd: 5.15
1351 [Astar_planner] Segment 7 reached in 17 steps, dist=1.96, axis_error
=[1.6910591 0.6107924 0.7742729], reason=running
1352 [Astar_planner] Segment 8 starts from actual pos [ 18.320478 -2.
6082807 -21.208897 ] (planned waypoint was [ 20.    -2.    -21.
982218])
1353 [Astar_planner] Segment 8: [ 18.320478 -2.6082807 -21.208897
] -> [ 31.    -2.    -21.314333]
1354 [Step 11 | Total    0] Target: ( 31.0, -2.0, -21.3) | Pos: ( 29.4, -2.0
, -21.4) | Vel: ( 1.2, 0.4, 0.1) | Acc: ( -0.2, 0.9, -1.1) | Rwd: 3.73
1355 [Astar_planner] Segment 8 reached in 11 steps, dist=1.59, axis_error
=[1.5936756 0.0164547 0.05287743], reason=running
1356 [Astar_planner] Segment 9 starts from actual pos [ 29.417467 -1.
9796523 -21.36665 ] (planned waypoint was [ 31.    -2.    -21.
314333])
1357 [Astar_planner] Segment 9: [ 29.417467 -1.9796523 -21.36665
] -> [ 38.    1.    -20.087997]
1358 [Step 9 | Total    0] Target: ( 38.0, 1.0, -20.1) | Pos: ( 36.7, 0.6, -
20.2) | Vel: ( 1.0, 0.8, -0.0) | Acc: ( -0.7, -0.2, -0.4) | Rwd: 602.86
-> Episode End: ARRIVED!
1359
1360 [Astar_planner] Segment 9 reached in 9 steps, dist=1.33, axis_error
=[1.2738838 0.36133653 0.09161758], reason=arrived
1361 [Astar_planner] Segment 10 starts from actual pos [ 36.737926 0.
6480834 -20.17998 ] (planned waypoint was [ 38.    1.    -20.
087997])
1362 [Astar_planner] Segment 10: [ 36.737926 0.6480834 -20.17998
] -> [ 45.    5.    -22.03353]

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1363 [Step 19 | Total 0] Target: ( 45.0, 5.0, -22.0) | Pos: ( 43.1, 3.9
, -20.9) | Vel: ( 0.4, 0.1, -1.1) | Acc: ( -1.6, 0.8, -0.0) | Rwd: 2.89
1364 [Astar_planner] Segment 10 reached in 19 steps, dist=2.47,
axis_error=[1.8945656 1.1314983 1.108469 ], reason=running
1365 [Astar_planner] Segment 11 starts from actual pos [ 43.10926 3.
8696024 -20.934954 ] (planned waypoint was [ 45. 5. -22.
03353])
1366 [Astar_planner] Segment 11: [ 43.10926 3.8696024 -20.934954
] -> [ 60. 7. -20.058126]
1367 [Step 17 | Total 0] Target: ( 60.0, 7.0, -20.1) | Pos: ( 58.5, 7.1
, -19.5) | Vel: ( 1.1, -0.8, 0.1) | Acc: ( -0.6, -1.5, 0.7) | Rwd: 601.
46 -> Episode End: ARRIVED!
1368
1369 [Astar_planner] Segment 11 reached in 17 steps, dist=1.59,
axis_error=[1.4666595 0.12342358 0.596859 ], reason=arrived
1370 [Astar_planner] Segment 12 starts from actual pos [ 58.543457 7.
116434 -19.460896] (planned waypoint was [ 60. 7. -20.
058126])
1371 [Astar_planner] Segment 12: [ 58.543457 7.116434 -19.460896
] -> [ 75. 7. -22.953327]
1372 [Step 24 | Total 0] Target: ( 75.0, 7.0, -23.0) | Pos: ( 75.6, 8.3
, -22.1) | Vel: ( 0.1, -1.2, -0.1) | Acc: ( -1.6, -1.9, 0.6) | Rwd: 601.
53 -> Episode End: ARRIVED!
1373
1374 [Astar_planner] Segment 12 reached in 24 steps, dist=1.66,
axis_error=[0.62109375 1.3184042 0.80509377], reason=arrived
1375 [Astar_planner] Segment 13 starts from actual pos [ 75.62173 8.
307864 -22.14884 ] (planned waypoint was [ 75. 7. -22.
953327])
1376 [Astar_planner] Segment 13: [ 75.62173 8.307864 -22.14884 ] -> [
90. 7. -21.741228]
1377 [Step 43 | Total 0] Target: ( 90.0, 7.0, -21.7) | Pos: ( 88.1, 8.2
, -20.8) | Vel: ( 0.5, 0.0, 0.1) | Acc: ( -0.8, 1.3, 1.4) | Rwd: 0.10
1378 [Astar_planner] Segment 13 reached in 43 steps, dist=2.38,
axis_error=[1.8577881 1.1747084 0.91160583], reason=running
1379 [Astar_planner] Segment 14 starts from actual pos [ 88.14514 8.
174945 -20.828873] (planned waypoint was [ 90. 7. -21.
741228])
1380 [Astar_planner] Segment 14: [ 88.14514 8.174945 -20.828873] -> [
94. 14. -22.903858]
1381 [Step 13 | Total 0] Target: ( 94.0, 14.0, -22.9) | Pos: ( 92.5, 13.3
, -23.1) | Vel: ( 1.1, -0.3, 0.4) | Acc: ( -0.4, -1.5, 1.1) | Rwd: 2.67
1382 [Astar_planner] Segment 14 reached in 13 steps, dist=1.68,
axis_error=[1.4971848 0.7281437 0.18803596], reason=running
1383 [Astar_planner] Segment 15 starts from actual pos [ 92.50948 13.
269942 -23.089636] (planned waypoint was [ 94. 14. -22.
903858])
1384 [Astar_planner] Segment 15: [ 92.50948 13.269942 -23.089636

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1384 ] -> [ 97.    21.    -20.007917]
1385 [Step 21 | Total    0] Target: ( 97.0, 21.0, -20.0) | Pos: ( 96.6, 19.1
, -19.4) | Vel: ( 0.1, 0.3, -0.1) | Acc: ( -1.3, -1.1, -0.4) | Rwd: 0.87
1386 [Astar_planner] Segment 15 reached in 21 steps, dist=2.04,
axis_error=[0.35279083 1.9273415 0.58421326], reason=running
1387 [Astar_planner] Segment 16 starts from actual pos [ 96.64864 19.
075668 -19.424992] (planned waypoint was [ 97.    21.    -20.
007917])
1388 [Astar_planner] Segment 16: [ 96.64864 19.075668 -19.424992] -> [
101.    30.    -6.6168437]
1389 [Step 25 | Total    0] Target: (101.0, 30.0, -6.6) | Pos: (100.9, 30.
8, -6.3) | Vel: ( 0.9, -1.6, 0.1) | Acc: ( 1.0, -1.7, 1.2) | Rwd: 602.
68 -> Episode End: ARRIVED!
1390
1391 [Astar_planner] Segment 16 reached in 25 steps, dist=0.88,
axis_error=[0.13980103 0.81526756 0.3042984 ], reason=arrived
1392 Task 20 result: success (Success: Reached destination (17/17
segments). Shield interventions=72, emergencies=2, recoveries=13,
recovery_steps=72.)
1393
1394 --- Task 21/38 ---
1395 From [-10.97211182 -13.19131592 -14.84776611] to [ -61.1 -101.02
-3.9 ]
1396 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
1397 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
1398 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
1399 Path planned via occupancy (Points: 11, Dist: 145.9m). Local target z
/alt: 0:-14.83(14.83m), 1:-24.03(24.03m), 2:-21.84(21.84m), 3:-22.05(
22.05m), 4:-22.35(22.35m), 5:-20.11(20.11m), 6:-22.01(22.01m), 7:-
24.65(24.65m), 8:-21.08(21.08m), 9:-23.52(23.52m), 10:-3.91(3.91m)
1400 [Astar_planner] Segment 0: [-11.    -13.    -14.826836] -> [ -9
.    -25.    -24.027908]
1401 [Astar_planner] Placing first segment start at [-11.    -13.    -14.
826836]
1402 [Astar_planner] Initial climb to 20.0m.
1403 [Step 66 | Total    0] Target: ( -9.0, -25.0, -24.0) | Pos: (-10.4, -26.5
, -23.1) | Vel: ( 0.7, 0.1, 0.5) | Acc: ( 0.6, 1.8, 1.1) | Rwd: 3.28
1404 [Astar_planner] Segment 0 reached in 66 steps, dist=2.26, axis_error
=[1.3884478 1.508667 0.94664955], reason=running
1405 [Astar_planner] Segment 1 starts from actual pos [-10.384461 -26.
508116 -23.07828 ] (planned waypoint was [ -9.    -25.    -24.
027908])
1406 [Astar_planner] Segment 1: [-10.384461 -26.508116 -23.07828 ] -> [ -
9.    -39.    -21.843033]
1407 [Step 13 | Total    0] Target: ( -9.0, -39.0, -21.8) | Pos: ( -8.5, -37.0

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1407 , -21.2) | Vel: ( 0.7, -1.0, 0.3) | Acc: ( 1.4, -0.8, -0.3) | Rwd: 4.02
1408 [Astar_planner] Segment 1 reached in 13 steps, dist=2.14, axis_error
=[0.53369904 1.9841194 0.613821 ], reason=running
1409 [Astar_planner] Segment 2 starts from actual pos [ -8.462294 -37.
021843 -21.227577] (planned waypoint was [ -9.    -39.    -21.
843033])
1410 [Astar_planner] Segment 2: [ -8.462294 -37.021843 -21.227577] -> [ -
9.    -53.    -22.04624]
1411 [Step 15 | Total    0] Target: ( -9.0, -53.0, -22.0) | Pos: (-10.1, -51.9
, -21.1) | Vel: ( 0.3, -1.0, -0.1) | Acc: ( 0.4, -1.5, 0.5) | Rwd: 2.42
1412 [Astar_planner] Segment 2 reached in 15 steps, dist=1.85, axis_error
=[1.1316881 1.0794754 0.9860821], reason=running
1413 [Astar_planner] Segment 3 starts from actual pos [-10.129087 -51.
929325 -21.060953] (planned waypoint was [ -9.    -53.    -22.
04624])
1414 [Astar_planner] Segment 3: [-10.129087 -51.929325 -21.060953
] -> [ -9.    -66.    -22.346502]
1415 [Step 12 | Total    0] Target: ( -9.0, -66.0, -22.3) | Pos: ( -8.9, -64.6
, -21.4) | Vel: ( 0.4, -1.6, 0.1) | Acc: ( 0.2, -2.0, -0.4) | Rwd: 607.
42 -> Episode End: ARRIVED!
1416
1417 [Astar_planner] Segment 3 reached in 12 steps, dist=1.68, axis_error
=[0.06923866 1.3723831 0.9670563 ], reason=arrived
1418 [Astar_planner] Segment 4 starts from actual pos [ -8.925638 -64.
64686 -21.378365] (planned waypoint was [ -9.    -66.    -22.
346502])
1419 [Astar_planner] Segment 4: [ -8.925638 -64.64686 -21.378365] -> [-
11.    -80.    -20.114908]
1420 [Step 13 | Total    0] Target: (-11.0, -80.0, -20.1) | Pos: ( -9.5, -78.5
, -19.5) | Vel: ( -0.1, -1.2, -0.1) | Acc: ( 0.5, -0.8, -0.2) | Rwd: 3.08
1421 [Astar_planner] Segment 4 reached in 13 steps, dist=2.25, axis_error
=[1.5197687 1.5447083 0.60326004], reason=running
1422 [Astar_planner] Segment 5 starts from actual pos [ -9.481032 -78.
47017 -19.512375] (planned waypoint was [-11.    -80.    -20.
114908])
1423 [Astar_planner] Segment 5: [ -9.481032 -78.47017 -19.512375] -> [-
18.    -88.    -22.011099]
1424 [Step 11 | Total    0] Target: (-18.0, -88.0, -22.0) | Pos: (-16.5, -87.
2, -20.2) | Vel: ( -1.3, -0.7, -0.8) | Acc: ( -1.4, -0.7, 0.1) | Rwd: 6.
42
1425 [Astar_planner] Segment 5 reached in 11 steps, dist=2.46, axis_error
=[1.5070076 0.75662994 1.7855339 ], reason=running
1426 [Astar_planner] Segment 6 starts from actual pos [-16.500645 -87.
24745 -20.230204] (planned waypoint was [-18.    -88.    -22.
011099])
1427 [Astar_planner] Segment 6: [-16.500645 -87.24745 -20.230204] -> [-
32.    -88.    -24.653782]
1428 [Step 16 | Total    0] Target: (-32.0, -88.0, -24.7) | Pos: (-30.6, -88.

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1428 6, -23.6) | Vel: ( -0.7, 0.3, 0.2) | Acc: ( -0.7, 0.2, -0.3) | Rwd: 605.
      58 -> Episode End: ARRIVED!
1429
1430 [Astar_planner] Segment 6 reached in 16 steps, dist=1.81, axis_error
      =[1.374054 0.56544495 1.0367374 ], reason=arrived
1431 [Astar_planner] Segment 7 starts from actual pos [-30.630102 -88.
      56378 -23.615513] (planned waypoint was [-32. -88. -24.
      653782])
1432 [Astar_planner] Segment 7: [-30.630102 -88.56378 -23.615513] -> [-
      46. -88. -21.078897]
1433 [Step 16 | Total 0] Target: (-46.0, -88.0, -21.1) | Pos: (-44.5, -87.
      9, -20.9) | Vel: ( -0.9, 0.3, -0.1) | Acc: ( -1.1, 0.8, 0.0) | Rwd: 602.
      92 -> Episode End: ARRIVED!
1434
1435 [Astar_planner] Segment 7 reached in 16 steps, dist=1.49, axis_error
      =[1.4685402 0.11493683 0.20474625], reason=arrived
1436 [Astar_planner] Segment 8 starts from actual pos [-44.54176 -87.
      881966 -20.874937] (planned waypoint was [-46. -88. -21.
      078897])
1437 [Astar_planner] Segment 8: [-44.54176 -87.881966 -20.874937] -> [-
      59. -91. -23.524925]
1438 [Step 15 | Total 0] Target: (-59.0, -91.0, -23.5) | Pos: (-57.5, -92.
      3, -22.0) | Vel: ( -0.6, -0.1, -1.0) | Acc: ( 1.5, -0.9, -0.0) | Rwd: 4.
      74
1439 [Astar_planner] Segment 8 reached in 15 steps, dist=2.46, axis_error
      =[1.4644012 1.2863235 1.5069885], reason=running
1440 [Astar_planner] Segment 9 starts from actual pos [-57.54083 -92.
      28756 -22.027113] (planned waypoint was [-59. -91. -23.
      524925])
1441 [Astar_planner] Segment 9: [-57.54083 -92.28756 -22.027113] -> [-
      61. -101. -3.9072433]
1442 [Step 106 | Total 0] Target: (-61.0, -101.0, -3.9) | Pos: (-60.5, -99
      .8, -4.8) | Vel: ( 0.6, -0.4, 0.0) | Acc: ( 1.6, 0.4, -0.7) | Rwd: 600.
      12 -> Episode End: ARRIVED!
1443
1444 [Astar_planner] Segment 9 reached in 106 steps, dist=1.58,
      axis_error=[0.5087242 1.1842651 0.922348 ], reason=arrived
1445 Task 21 result: success (Success: Reached destination (10/10
      segments). Shield interventions=22, emergencies=0, recoveries=4,
      recovery_steps=22.)
1446
1447 --- Task 22/38 ---
1448 From [-106.2 125.18 -4.9 ] to [119.7 78.28 -7.4 ]
1449 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
1450 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
1451 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,

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1451 safety_margin=1.5
1452 Path planned via occupancy (Points: 27, Dist: 318.1m). Local target z
/alt: 0:-4.90(4.90m), 1:-21.31(21.31m), 2:-20.29(20.29m), 3:-22.75(22
.75m), 4:-23.87(23.87m), 5:-22.90(22.90m), 6:-24.92(24.92m), 7:-20.
60(20.60m), 8:-21.32(21.32m), 9:-21.32(21.32m), 10:-24.59(24.59m
), 11:-24.29(24.29m), 12:-24.06(24.06m), 13:-20.62(20.62m), 14:-20.
87(20.87m), 15:-23.41(23.41m), 16:-23.37(23.37m), 17:-21.83(21.
83m), 18:-21.05(21.05m), 19:-21.49(21.49m), 20:-23.92(23.92m), 21
:-24.11(24.11m), 22:-24.01(24.01m), 23:-20.67(20.67m), 24:-20.17(
20.17m), 25:-22.90(22.90m), 26:-7.40(7.40m)
1453 [Astar_planner] Segment 0: [-106.      125.      -4.9029155] -> [-
100.      134.      -21.307493]
1454 [Astar_planner] Placing first segment start at [-106.      125.      -4
.9029155]
1455 [Astar_planner] Initial climb to 20.0m.
1456 [Step 15 | Total 0] Target: (-100.0, 134.0, -21.3) | Pos: (-100.9,
132.2, -19.6) | Vel: ( 0.6, 0.8, -1.9) | Acc: ( 1.4, -1.1, -0.4) | Rwd
: 3.29
1457 [Astar_planner] Segment 0 reached in 15 steps, dist=2.61, axis_error
=[0.87314606 1.776123 1.6991272 ], reason=running
1458 [Astar_planner] Segment 1 starts from actual pos [-100.86814 132.
23105 -19.625723] (planned waypoint was [-100.      134.      -21.
307493])
1459 [Astar_planner] Segment 1: [-100.86814 132.23105 -19.625723
] -> [-100.      146.      -20.285923]
1460 [Step 20 | Total 0] Target: (-100.0, 146.0, -20.3) | Pos: (-99.8,
147.9, -18.7) | Vel: ( -0.6, -1.0, -0.8) | Acc: ( 1.9, -1.5, 0.1) | Rwd
: 2.90
1461 [Astar_planner] Segment 1 reached in 20 steps, dist=2.43, axis_error
=[0.22000122 1.8577576 1.5547218 ], reason=running
1462 [Astar_planner] Segment 2 starts from actual pos [-99.785355 147.
84879 -18.738588] (planned waypoint was [-100.      146.      -20.
285923])
1463 [Astar_planner] Segment 2: [-99.785355 147.84879 -18.738588] -> [-
95.      156.      -22.754583]
1464 [Step 22 | Total 0] Target: (-95.0, 156.0, -22.8) | Pos: (-95.6, 154
.2, -21.5) | Vel: ( 0.0, 0.3, 0.6) | Acc: ( -0.6, -0.9, 1.0) | Rwd: -1.
25
1465 [Astar_planner] Segment 2 reached in 22 steps, dist=2.29, axis_error
=[0.5886917 1.8362122 1.2344933], reason=running
1466 [Astar_planner] Segment 3 starts from actual pos [-95.58866 154.
16574 -21.516798] (planned waypoint was [-95.      156.      -22.
754583])
1467 [Astar_planner] Segment 3: [-95.58866 154.16574 -21.516798] -> [-
84.      157.      -23.872208]
1468 [Step 10 | Total 0] Target: (-84.0, 157.0, -23.9) | Pos: (-84.0, 158
.3, -21.9) | Vel: ( 1.8, 0.1, -1.3) | Acc: ( -1.3, 1.8, 1.7) | Rwd: 602.
92 -> Episode End: ARRIVED!

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1469
1470 [Astar_planner] Segment 3 reached in 10 steps, dist=2.34, axis_error
    =[0.04883575 1.3036957 1.9397602 ], reason=arrived
1471 [Astar_planner] Segment 4 starts from actual pos [-83.92963 158.
    30534 -21.948072] (planned waypoint was [-84.    157.    -23.
    872208])
1472 [Astar_planner] Segment 4: [-83.92963 158.30534 -21.948072] -> [-
    72.    156.    -22.900497]
1473 [Step 59 | Total    0] Target: (-72.0, 156.0, -22.9) | Pos: (-70.5, 158
    .0, -21.5) | Vel: ( 1.0, -0.1, 0.8) | Acc: ( 0.3, 1.9, 1.6) | Rwd: -4.
    40
1474 [Astar_planner] Segment 4 reached in 59 steps, dist=2.83, axis_error
    =[1.5109406 1.961853 1.364708 ], reason=running
1475 [Astar_planner] Segment 5 starts from actual pos [-70.4804 157.
    96101 -21.52889] (planned waypoint was [-72.    156.    -22.
    900497])
1476 [Astar_planner] Segment 5: [-70.4804 157.96101 -21.52889] -> [-60
    .    156.    -24.92383]
1477 [Step 24 | Total    0] Target: (-60.0, 156.0, -24.9) | Pos: (-59.1, 157
    .1, -23.7) | Vel: ( -1.1, -0.8, -1.1) | Acc: ( -0.9, -0.4, -0.7) | Rwd: 605.
    75 -> Episode End: ARRIVED!
1478
1479 [Astar_planner] Segment 5 reached in 24 steps, dist=1.92, axis_error
    =[0.9146347 1.1079865 1.273241 ], reason=arrived
1480 [Astar_planner] Segment 6 starts from actual pos [-59.09814 157.
    09828 -23.663813] (planned waypoint was [-60.    156.    -24.
    92383])
1481 [Astar_planner] Segment 6: [-59.09814 157.09828 -23.663813] -> [-
    48.    156.    -20.601685]
1482 [Step 14 | Total    0] Target: (-48.0, 156.0, -20.6) | Pos: (-48.4, 157
    .1, -19.9) | Vel: ( 1.9, -0.7, -0.2) | Acc: ( -2.0, -2.0, 0.3) | Rwd: 604.
    44 -> Episode End: ARRIVED!
1483
1484 [Astar_planner] Segment 6 reached in 14 steps, dist=1.34, axis_error
    =[0.3707428 1.083435 0.6880131], reason=arrived
1485 [Astar_planner] Segment 7 starts from actual pos [-48.353638 157.
    0771 -19.915842] (planned waypoint was [-48.    156.    -20.
    601685])
1486 [Astar_planner] Segment 7: [-48.353638 157.0771 -19.915842] -> [-
    37.    157.    -21.319437]
1487 [Step 10 | Total    0] Target: (-37.0, 157.0, -21.3) | Pos: (-37.7, 157
    .7, -20.7) | Vel: ( 2.0, -0.8, -0.2) | Acc: ( -2.0, -2.0, -0.1) | Rwd: 606.
    38 -> Episode End: ARRIVED!
1488
1489 [Astar_planner] Segment 7 reached in 10 steps, dist=1.21, axis_error
    =[0.73472977 0.7083435 0.656044 ], reason=arrived
1490 [Astar_planner] Segment 8 starts from actual pos [-37.71698 157.
    70157 -20.664862] (planned waypoint was [-37.    157.    -21.

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1490 319437))
1491 [Astar_planner] Segment 8: [-37.71698 157.70157 -20.664862] -> [-
27.    153.    -21.322868]
1492 [Step 12 | Total    0] Target: (-27.0, 153.0, -21.3) | Pos: (-25.5, 154
.9, -20.4) | Vel: ( 0.9, -0.3, -0.0) | Acc: (-0.8, 0.1, 1.0) | Rwd: -2.
00
1493 [Astar_planner] Segment 8 reached in 12 steps, dist=2.60, axis_error
=[1.4912071 1.9195404 0.9184761], reason=running
1494 [Astar_planner] Segment 9 starts from actual pos [-25.500307 154.
91698 -20.404848] (planned waypoint was [-27.    153.    -21.
322868])
1495 [Astar_planner] Segment 9: [-25.500307 154.91698 -20.404848] -> [-
15.    153.    -24.593775]
1496 [Step 11 | Total    0] Target: (-15.0, 153.0, -24.6) | Pos: (-16.8, 153
.0, -23.4) | Vel: ( 1.3, -0.1, -1.4) | Acc: (-0.9, -0.1, -0.1) | Rwd: 4.
72
1497 [Astar_planner] Segment 9 reached in 11 steps, dist=2.16, axis_error
=[1.7915096 0.01867676 1.2097473 ], reason=running
1498 [Astar_planner] Segment 10 starts from actual pos [-16.775375 153.
0174 -23.400583] (planned waypoint was [-15.    153.    -24.
593775])
1499 [Astar_planner] Segment 10: [-16.775375 153.0174 -23.400583
] -> [-3.    153.    -24.288404]
1500 [Step 17 | Total    0] Target: ( -3.0, 153.0, -24.3) | Pos: ( -4.6, 152.
5, -22.7) | Vel: ( 1.1, -0.6, -1.4) | Acc: ( 0.1, 0.3, 1.1) | Rwd: 3.57

1501 [Astar_planner] Segment 10 reached in 17 steps, dist=2.35,
axis_error=[1.6224055 0.4764557 1.637392 ], reason=running
1502 [Astar_planner] Segment 11 starts from actual pos [ -4.6129284 152.
51797 -22.663315 ] (planned waypoint was [ -3.    153.    -24.
288404])
1503 [Astar_planner] Segment 11: [ -4.6129284 152.51797 -22.663315
] -> [ 8.    151.    -24.056438]
1504 [Step 14 | Total    0] Target: ( 8.0, 151.0, -24.1) | Pos: ( 9.5, 150.7
, -22.1) | Vel: ( 1.1, -0.2, -1.8) | Acc: (-2.0, 0.7, 2.0) | Rwd: 604.
09 -> Episode End: ARRIVED!
1505
1506 [Astar_planner] Segment 11 reached in 14 steps, dist=2.45,
axis_error=[1.4624891 0.29634094 1.9476166 ], reason=arrived
1507 [Astar_planner] Segment 12 starts from actual pos [ 9.469148 150.
70244 -22.119467] (planned waypoint was [ 8.    151.    -24.
056438])
1508 [Astar_planner] Segment 12: [ 9.469148 150.70244 -22.119467
] -> [ 18.    148.    -20.619722]
1509 [Step 7 | Total    0] Target: ( 18.0, 148.0, -20.6) | Pos: ( 16.4, 149.
1, -18.9) | Vel: ( 2.1, -0.8, 0.4) | Acc: (-1.3, -2.0, 0.3) | Rwd: 1.
57
1510 [Astar_planner] Segment 12 reached in 7 steps, dist=2.62, axis_error

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1510 =[1.6285362 1.121582 1.7132607], reason=running
1511 [Astar_planner] Segment 13 starts from actual pos [ 16.39065 149.
11453 -18.903225] (planned waypoint was [ 18. 148. -20.
619722])
1512 [Astar_planner] Segment 13: [ 16.39065 149.11453 -18.903225
]-> [ 27. 139. -20.872046]
1513 [Step 23 | Total 0] Target: ( 27.0, 139.0, -20.9) | Pos: ( 27.1, 137.
1, -20.4) | Vel: ( 0.4, 1.0, -0.5) | Acc: ( -2.0, 1.2, 2.0) | Rwd: 1.60

1514 [Astar_planner] Segment 13 reached in 23 steps, dist=1.98,
axis_error=[0.10170174 1.9145508 0.48194695], reason=running
1515 [Astar_planner] Segment 14 starts from actual pos [ 27.103891 137.
09119 -20.393219] (planned waypoint was [ 27. 139. -20.
872046])
1516 [Astar_planner] Segment 14: [ 27.103891 137.09119 -20.393219
]-> [ 37. 136. -23.405563]
1517 [Step 10 | Total 0] Target: ( 37.0, 136.0, -23.4) | Pos: ( 35.3, 134.
8, -21.5) | Vel: ( 1.7, -0.6, -0.0) | Acc: ( -1.3, -0.9, -0.5) | Rwd: 1.
48
1518 [Astar_planner] Segment 14 reached in 10 steps, dist=2.85,
axis_error=[1.7367134 1.1710815 1.9264469], reason=running
1519 [Astar_planner] Segment 15 starts from actual pos [ 35.278725 134.
82358 -21.479397] (planned waypoint was [ 37. 136. -23.
405563])
1520 [Astar_planner] Segment 15: [ 35.278725 134.82358 -21.479397
]-> [ 49. 136. -23.367464]
1521 [Step 14 | Total 0] Target: ( 49.0, 136.0, -23.4) | Pos: ( 47.4, 134.
7, -21.5) | Vel: ( 1.4, 0.0, -1.9) | Acc: ( -1.3, 1.4, 0.2) | Rwd: 2.72

1522 [Astar_planner] Segment 15 reached in 14 steps, dist=2.79,
axis_error=[1.5708809 1.310257 1.8945732], reason=running
1523 [Astar_planner] Segment 16 starts from actual pos [ 47.44129 134.
69016 -21.489428] (planned waypoint was [ 49. 136. -23.
367464])
1524 [Astar_planner] Segment 16: [ 47.44129 134.69016 -21.489428
]-> [ 61. 136. -21.828697]
1525 [Step 14 | Total 0] Target: ( 61.0, 136.0, -21.8) | Pos: ( 60.4, 135.
1, -20.0) | Vel: ( 2.2, 0.8, -1.5) | Acc: ( 2.0, -1.2, -0.5) | Rwd: 602.
31 -> Episode End: ARRIVED!

1526
1527 [Astar_planner] Segment 16 reached in 14 steps, dist=2.13,
axis_error=[0.6321182 0.86431885 1.838068 ], reason=arrived
1528 [Astar_planner] Segment 17 starts from actual pos [ 60.381245 135.
14026 -19.999632] (planned waypoint was [ 61. 136. -21.
828697])
1529 [Astar_planner] Segment 17: [ 60.381245 135.14026 -19.999632
]-> [ 72. 135. -21.051672]
1530 [Step 8 | Total 0] Target: ( 72.0, 135.0, -21.1) | Pos: ( 71.2, 135.

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1530 9, -20.0) | Vel: ( 2.9, 0.4, -1.5) | Acc: ( -1.9, 1.6, -0.2) | Rwd: 604.
      14 -> Episode End: ARRIVED!
1531
1532 [Astar_planner] Segment 17 reached in 8 steps, dist=1.64, axis_error
      =[0.84869385 0.9173889 1.0620365 ], reason=arrived
1533 [Astar_planner] Segment 18 starts from actual pos [ 71.17718 135.
      92123 -20.003212] (planned waypoint was [ 72.    135.    -21.
      051672])
1534 [Astar_planner] Segment 18: [ 71.17718 135.92123 -20.003212
      ] -> [ 84.    135.    -21.492773]
1535 [Step 45 | Total    0] Target: ( 84.0, 135.0, -21.5) | Pos: ( 82.8, 133.
      5, -20.2) | Vel: ( 0.8, 0.6, -1.0) | Acc: ( -2.0, 1.8, 2.0) | Rwd: 3.59

1536 [Astar_planner] Segment 18 reached in 45 steps, dist=2.32,
      axis_error=[1.2456589 1.4964905 1.2667332], reason=running
1537 [Astar_planner] Segment 19 starts from actual pos [ 82.75923 133.
      50691 -20.232168] (planned waypoint was [ 84.    135.    -21.
      492773])
1538 [Astar_planner] Segment 19: [ 82.75923 133.50691 -20.232168
      ] -> [ 96.    135.    -23.922901]
1539 [Step 44 | Total    0] Target: ( 96.0, 135.0, -23.9) | Pos: ( 97.7, 135.
      9, -23.1) | Vel: ( -0.2, -1.2, 0.7) | Acc: ( 0.1, -1.8, -1.4) | Rwd: 4.
      80
1540 [Astar_planner] Segment 19 reached in 44 steps, dist=2.12,
      axis_error=[1.7286224 0.87483215 0.84994507], reason=running
1541 [Astar_planner] Segment 20 starts from actual pos [ 97.72723 135.
      86444 -23.067108] (planned waypoint was [ 96.    135.    -23.
      922901])
1542 [Astar_planner] Segment 20: [ 97.72723 135.86444 -23.067108] -> [
      103.    128.    -24.108408]
1543 [Step 80 | Total    0] Target: (103.0, 128.0, -24.1) | Pos: (101.9,
      128.1, -22.6) | Vel: ( 0.5, -0.6, -1.1) | Acc: ( -0.4, -0.8, 1.8) | Rwd:
      604.09 -> Episode End: ARRIVED!
1544
1545 [Astar_planner] Segment 20 reached in 80 steps, dist=1.86,
      axis_error=[1.0542831 0.14489746 1.520277 ], reason=arrived
1546 [Astar_planner] Segment 21 starts from actual pos [101.95004 128.
      13977 -22.598019] (planned waypoint was [103.    128.    -24.
      108408])
1547 [Astar_planner] Segment 21: [101.95004 128.13977 -22.598019
      ] -> [104.    116.    -24.014645]
1548 [Step 13 | Total    0] Target: (104.0, 116.0, -24.0) | Pos: (103.3,
      117.6, -22.8) | Vel: ( 0.5, -0.7, -1.5) | Acc: ( 0.1, 1.9, 0.1) | Rwd:
      2.62
1549 [Astar_planner] Segment 21 reached in 13 steps, dist=2.13,
      axis_error=[0.73682404 1.620163 1.1711884 ], reason=running
1550 [Astar_planner] Segment 22 starts from actual pos [103.266426 117.
      61584 -22.852404] (planned waypoint was [104.    116.    -24.

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1550 014645])
1551 [Astar_planner] Segment 22: [103.266426 117.61584 -22.852404
]-> [104. 105. -20.6659]
1552 [Step 24 | Total 0] Target: (104.0, 105.0, -20.7) | Pos: (102.1,
106.0, -20.6) | Vel: ( 0.6, -1.6, 0.0) | Acc: ( 0.5, -1.2, 0.0) | Rwd:
2.41
1553 [Astar_planner] Segment 22 reached in 24 steps, dist=2.11,
axis_error=[1.87146 0.9794159 0.06935501], reason=running
1554 [Astar_planner] Segment 23 starts from actual pos [102.13394 105.
96534 -20.596384] (planned waypoint was [104. 105. -20.6659
])
1555 [Astar_planner] Segment 23: [102.13394 105.96534 -20.596384
]-> [105. 93. -20.167118]
1556 [Step 21 | Total 0] Target: (105.0, 93.0, -20.2) | Pos: (103.7, 92.
9, -20.3) | Vel: ( 1.2, -0.7, -0.6) | Acc: ( -1.7, -0.9, 0.6) | Rwd: 602.
63 -> Episode End: ARRIVED!
1557
1558 [Astar_planner] Segment 23 reached in 21 steps, dist=1.27,
axis_error=[1.2679214 0.05815887 0.0926342 ], reason=arrived
1559 [Astar_planner] Segment 24 starts from actual pos [103.73908 92.
937874 -20.26318 ] (planned waypoint was [105. 93. -20.
167118])
1560 [Astar_planner] Segment 24: [103.73908 92.937874 -20.26318 ] -> [
110. 84. -22.90379]
1561 [Step 10 | Total 0] Target: (110.0, 84.0, -22.9) | Pos: (111.4, 85.
1, -23.5) | Vel: ( 1.1, -1.5, -1.0) | Acc: ( -1.0, 0.9, -1.1) | Rwd: -0.
17
1562 [Astar_planner] Segment 24 reached in 10 steps, dist=1.87,
axis_error=[1.4364166 1.0696564 0.5548992], reason=running
1563 [Astar_planner] Segment 25 starts from actual pos [111.44622 85.
05624 -23.467308] (planned waypoint was [110. 84. -22.
90379])
1564 [Astar_planner] Segment 25: [111.44622 85.05624 -23.467308] -> [
120. 78. -7.4]
1565 [Step 18 | Total 0] Target: (120.0, 78.0, -7.4) | Pos: (119.4, 77.
0, -6.9) | Vel: ( 1.7, -0.6, 0.3) | Acc: ( -0.6, -0.8, -0.1) | Rwd: 600.
24 -> Episode End: ARRIVED!
1566
1567 [Astar_planner] Segment 25 reached in 18 steps, dist=1.26,
axis_error=[0.58519745 0.9794464 0.5251026 ], reason=arrived
1568 Task 22 result: success (Success: Reached destination (26/26
segments). Shield interventions=42, emergencies=1, recoveries=7,
recovery_steps=42.)
1569
1570 --- Task 23/38 ---
1571 From [-106.2 125.18 -4.9 ] to [16. 3.18 -7.2 ]
1572 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)

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1573 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
1574 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
1575 Path planned via occupancy (Points: 22, Dist: 251.1m). Local target z
      /alt: 0:-4.90(4.90m), 1:-20.02(20.02m), 2:-22.08(22.08m), 3:-20.95(20
      .95m), 4:-23.76(23.76m), 5:-22.71(22.71m), 6:-20.79(20.79m), 7:-22.
      76(22.76m), 8:-21.86(21.86m), 9:-22.74(22.74m), 10:-20.69(20.69m
      ), 11:-20.73(20.73m), 12:-20.71(20.71m), 13:-23.79(23.79m), 14:-20.
      38(20.38m), 15:-24.55(24.55m), 16:-25.00(25.00m), 17:-24.08(24.
      08m), 18:-21.34(21.34m), 19:-24.07(24.07m), 20:-22.99(22.99m), 21
      :-7.20(7.20m)
1576 [Astar_planner] Segment 0: [-106.      125.      -4.903584] -> [-100
      .      116.      -20.016716]
1577 [Astar_planner] Placing first segment start at [-106.      125.      -4.
      903584]
1578 [Astar_planner] Initial climb to 20.0m.
1579 [Step 17 | Total      0] Target: (-100.0, 116.0, -20.0) | Pos: (-101.7,
      115.5, -18.9) | Vel: ( 0.6, 0.3, -0.0) | Acc: ( 0.4, 1.9, 0.6) | Rwd:
      0.89
1580 [Astar_planner] Segment 0 reached in 17 steps, dist=2.04, axis_error
      =[1.6636887 0.47428894 1.0750256 ], reason=running
1581 [Astar_planner] Segment 1 starts from actual pos [-101.65845 115.
      528336 -18.941963] (planned waypoint was [-100.      116.      -20.
      016716])
1582 [Astar_planner] Segment 1: [-101.65845 115.528336 -18.941963
      ] -> [-100.      104.      -22.084091]
1583 [Step 12 | Total      0] Target: (-100.0, 104.0, -22.1) | Pos: (-101.9,
      106.0, -20.9) | Vel: ( 0.4, -0.4, 0.0) | Acc: ( 0.8, -1.8, -0.8) | Rwd
      : 3.46
1584 [Astar_planner] Segment 1 reached in 12 steps, dist=2.97, axis_error
      =[1.8898544 1.9634247 1.1932774], reason=running
1585 [Astar_planner] Segment 2 starts from actual pos [-101.88645 105.
      96031 -20.890425] (planned waypoint was [-100.      104.      -22.
      084091])
1586 [Astar_planner] Segment 2: [-101.88645 105.96031 -20.890425
      ] -> [-97.      94.      -20.948761]
1587 [Step 146 | Total      0] Target: (-97.0, 94.0, -20.9) | Pos: (-100.9,
      96.5, -17.0) | Vel: ( 0.1, 0.1, -0.0) | Acc: ( 1.6, 1.5, -0.0) | Rwd: -
      1.62
1588 [Astar_planner] Segment 2 reached in 146 steps, dist=6.07,
      axis_error=[3.8672028 2.4704437 3.9805603], reason=
      stuck_near_intermediate
1589 [Astar_planner] Segment 3 starts from actual pos [-100.866554 96.
      47147 -16.968212] (planned waypoint was [-97.      94.      -20.
      948761])
1590 [Astar_planner] Segment 3: [-100.866554 96.47147 -16.968212
      ] -> [-88.      89.      -23.764332]

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1591 [Step 180 | Total    0] Target: (-88.0, 89.0, -23.8) | Pos: (-92.3, 92.
    1, -21.3) | Vel: ( 0.6, 0.5, 0.5) | Acc: ( 2.0, 2.0, 2.0) | Rwd: -1.38

1592 [Astar_planner] Segment 3 timeout, dist=5.90
1593 [Astar_planner] Stop after failed segment 3.
1594 Task 23 result: failed (Failed at segment 3: segment_timeout (3/21
    reached). Shield interventions=6, emergencies=0, recoveries=1,
    recovery_steps=6.)
1595
1596 --- Task 24/38 ---
1597 From [-10.1 -76.12 -9.8 ] to [-106.2 125.18 -4.9 ]
1598 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
    =[-200.0, 189.0], occupied=72869 cells (53.4%)
1599 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
    cells (53.4%)
1600 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
1601 Path planned via occupancy (Points: 23, Dist: 295.5m). Local target z
    /alt: 0:-9.80(9.80m), 1:-20.65(20.65m), 2:-24.62(24.62m), 3:-24.70(24
    .70m), 4:-21.85(21.85m), 5:-20.49(20.49m), 6:-21.91(21.91m), 7:-20.
    26(20.26m), 8:-23.63(23.63m), 9:-21.71(21.71m), 10:-20.27(20.27m
    ), 11:-21.40(21.40m), 12:-22.27(22.27m), 13:-21.22(21.22m), 14:-22.
    65(22.65m), 15:-22.48(22.48m), 16:-22.40(22.40m), 17:-21.21(21.
    21m), 18:-23.18(23.18m), 19:-22.48(22.48m), 20:-21.29(21.29m), 21
    :-24.67(24.67m), 22:-4.91(4.91m)
1602 [Astar_planner] Segment 0: [-10.    -76.    -9.796569] -> [-10
    .    -63.    -20.649675]
1603 [Astar_planner] Placing first segment start at [-10.    -76.    -9.
    796569]
1604 [Astar_planner] Initial climb to 20.0m.
1605 [Step 27 | Total    0] Target: (-10.0, -63.0, -20.6) | Pos: ( -9.5, -64.8
    , -19.8) | Vel: ( -0.2, 0.3, 0.5) | Acc: ( -1.0, 0.7, -1.1) | Rwd: 1.75
1606 [Astar_planner] Segment 0 reached in 27 steps, dist=2.06, axis_error
    =[0.47368526 1.815361 0.8618622 ], reason=running
1607 [Astar_planner] Segment 1 starts from actual pos [ -9.527966 -64.
    81244 -19.783642] (planned waypoint was [-10.    -63.    -20.
    649675])
1608 [Astar_planner] Segment 1: [ -9.527966 -64.81244 -19.783642] -> [ -
    9.    -50.    -24.617744]
1609 [Step 57 | Total    0] Target: ( -9.0, -50.0, -24.6) | Pos: ( -9.7, -48.5
    , -23.0) | Vel: ( 0.7, 0.4, 0.6) | Acc: ( -0.7, 2.0, 1.9) | Rwd: 1.36
1610 [Astar_planner] Segment 1 reached in 57 steps, dist=2.36, axis_error
    =[0.7004547 1.5326309 1.652216 ], reason=running
1611 [Astar_planner] Segment 2 starts from actual pos [ -9.694274 -48.
    46369 -22.960453] (planned waypoint was [ -9.    -50.    -24.
    617744])
1612 [Astar_planner] Segment 2: [ -9.694274 -48.46369 -22.960453] -> [ -
    9.    -37.    -24.702074]

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1613 [Step 68 | Total 0] Target: (-9.0, -37.0, -24.7) | Pos: (-11.0, -36.3
, -23.9) | Vel: ( 0.2, 0.7, 0.3) | Acc: (-0.4, 2.0, 1.9) | Rwd: 1.65
1614 [Astar_planner] Segment 2 reached in 68 steps, dist=2.23, axis_error
=[1.9785843 0.6500015 0.79206085], reason=running
1615 [Astar_planner] Segment 3 starts from actual pos [-10.976679 -36.
34403 -23.906942] (planned waypoint was [ -9.    -37.    -24.
702074])
1616 [Astar_planner] Segment 3: [-10.976679 -36.34403 -23.906942] -> [ -
9.    -24.    -21.847387]
1617 [Step 19 | Total 0] Target: (-9.0, -24.0, -21.8) | Pos: (-10.8, -23.3
, -20.6) | Vel: ( 0.7, 0.2, -0.2) | Acc: ( 1.8, 0.6, 1.4) | Rwd: -0.74
1618 [Astar_planner] Segment 3 reached in 19 steps, dist=2.28, axis_error
=[1.754425 0.70253754 1.2810936 ], reason=running
1619 [Astar_planner] Segment 4 starts from actual pos [-10.748082 -23.
2953 -20.568478] (planned waypoint was [ -9.    -24.    -21.
847387])
1620 [Astar_planner] Segment 4: [-10.748082 -23.2953 -20.568478] -> [ -
9.    -11.    -20.489824]
1621 [Step 26 | Total 0] Target: (-9.0, -11.0, -20.5) | Pos: (-10.8, -10.9
, -19.5) | Vel: ( 0.4, -0.0, -0.0) | Acc: ( 0.7, -1.0, -0.2) | Rwd: 0.83
1622 [Astar_planner] Segment 4 reached in 26 steps, dist=2.09, axis_error
=[1.8131094 0.14291 1.0319729], reason=running
1623 [Astar_planner] Segment 5 starts from actual pos [-10.809168 -10.
857422 -19.457909] (planned waypoint was [ -9.    -11.    -20.
489824])
1624 [Astar_planner] Segment 5: [-10.809168 -10.857422 -19.457909] -> [-
14.    -1.    -21.910715]
1625 [Step 15 | Total 0] Target: (-14.0, -1.0, -21.9) | Pos: (-14.1, -0.6
, -20.2) | Vel: ( 0.5, 0.1, 0.5) | Acc: ( 0.9, -0.7, 1.6) | Rwd: 600.73
-> Episode End: ARRIVED!
1626
1627 [Astar_planner] Segment 5 reached in 15 steps, dist=1.77, axis_error
=[0.05379772 0.37531167 1.72719 ], reason=arrived
1628 [Astar_planner] Segment 6 starts from actual pos [-14.0495205 -0.
62398887 -20.17854 ] (planned waypoint was [-14.    -1.    -21.
910715])
1629 [Astar_planner] Segment 6: [-14.0495205 -0.62398887 -20.17854
] -> [-26.    3.    -20.263885]
1630 [Step 19 | Total 0] Target: (-26.0, 3.0, -20.3) | Pos: (-24.6, 1.3
, -19.0) | Vel: (-0.1, 0.6, 0.6) | Acc: ( 0.9, 1.0, -0.0) | Rwd: 3.68
1631 [Astar_planner] Segment 6 reached in 19 steps, dist=2.53, axis_error
=[1.3633461 1.7343498 1.2334824], reason=running
1632 [Astar_planner] Segment 7 starts from actual pos [-24.63699 1.
2693036 -19.0269 ] (planned waypoint was [-26.    3.    -20.
263885])
1633 [Astar_planner] Segment 7: [-24.63699 1.2693036 -19.0269 ] -> [-
35.    12.    -23.630417]
1634 [Step 57 | Total 0] Target: (-35.0, 12.0, -23.6) | Pos: (-35.1, 12.

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1634 1, -21.9) | Vel: ( -0.4, 0.8, -0.5) | Acc: ( 0.1, -1.2, 1.9) | Rwd: 605.
      85 -> Episode End: ARRIVED!
1635
1636 [Astar_planner] Segment 7 reached in 57 steps, dist=1.75, axis_error
      =[0.09359741 0.05222321 1.7437801 ], reason=arrived
1637 [Astar_planner] Segment 8 starts from actual pos [-35.095894 12.
      057292 -21.889297] (planned waypoint was [-35.      12.      -23.
      630417])
1638 [Astar_planner] Segment 8: [-35.095894 12.057292 -21.889297] -> [-
      44.      21.      -21.707737]
1639 [Step 13 | Total      0] Target: (-44.0, 21.0, -21.7) | Pos: (-42.4, 21.
      0, -20.4) | Vel: ( -0.7, 0.3, 0.0) | Acc: ( -0.5, 0.4, 0.2) | Rwd: 6.49

1640 [Astar_planner] Segment 8 reached in 13 steps, dist=2.06, axis_error
      =[1.5834923 0.01708603 1.3101349 ], reason=running
1641 [Astar_planner] Segment 9 starts from actual pos [-42.42239 20.
      985836 -20.397133] (planned waypoint was [-44.      21.      -21.
      707737])
1642 [Astar_planner] Segment 9: [-42.42239 20.985836 -20.397133] -> [-
      53.      30.      -20.270073]
1643 [Step 27 | Total      0] Target: (-53.0, 30.0, -20.3) | Pos: (-51.9, 28.
      8, -18.3) | Vel: ( -0.3, -0.6, 0.9) | Acc: ( 0.2, -1.9, -2.0) | Rwd: 1.
      59
1644 [Astar_planner] Segment 9 reached in 27 steps, dist=2.51, axis_error
      =[1.0666275 1.1909904 1.9344406], reason=running
1645 [Astar_planner] Segment 10 starts from actual pos [-51.936012 28.
      803774 -18.327211] (planned waypoint was [-53.      30.      -20.
      270073])
1646 [Astar_planner] Segment 10: [-51.936012 28.803774 -18.327211
      ] -> [-60.      41.      -21.40454]
1647 [Step 38 | Total      0] Target: (-60.0, 41.0, -21.4) | Pos: (-59.2, 41.
      1, -20.3) | Vel: ( -0.5, -0.7, 0.3) | Acc: ( 0.1, -1.3, 1.1) | Rwd: 605.
      21 -> Episode End: ARRIVED!
1648
1649 [Astar_planner] Segment 10 reached in 38 steps, dist=1.42,
      axis_error=[0.84093094 0.09486771 1.1362877 ], reason=arrived
1650 [Astar_planner] Segment 11 starts from actual pos [-59.16371 41.
      08845 -20.265184] (planned waypoint was [-60.      41.      -21.
      40454])
1651 [Astar_planner] Segment 11: [-59.16371 41.08845 -20.265184] -> [-
      59.      54.      -22.274933]
1652 [Step 35 | Total      0] Target: (-59.0, 54.0, -22.3) | Pos: (-60.0, 52.
      7, -21.0) | Vel: ( -0.4, 0.4, -0.6) | Acc: ( -0.8, -1.6, 1.0) | Rwd: 4.
      48
1653 [Astar_planner] Segment 11 reached in 35 steps, dist=2.07,
      axis_error=[0.95926285 1.2667961 1.3211823 ], reason=running
1654 [Astar_planner] Segment 12 starts from actual pos [-59.962677 52.
      736866 -20.958693] (planned waypoint was [-59.      54.      -22.

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1654 274933))
1655 [Astar_planner] Segment 12: [-59.962677 52.736866 -20.958693
]-> [-51.    63.   -21.21573]
1656 [Step 16 | Total    0] Target: (-51.0, 63.0, -21.2) | Pos: (-51.8, 61.
0, -20.3) | Vel: ( 1.1, 1.3, -0.1) | Acc: ( 1.9, 1.0, -0.1) | Rwd: 3.56

1657 [Astar_planner] Segment 12 reached in 16 steps, dist=2.32,
axis_error=[0.7840576 1.9801407 0.9194374], reason=running
1658 [Astar_planner] Segment 13 starts from actual pos [-51.774006 61.
031307 -20.296896] (planned waypoint was [-51.    63.   -21.
21573])
1659 [Astar_planner] Segment 13: [-51.774006 61.031307 -20.296896
]-> [-41.    70.   -22.648464]
1660 [Step 14 | Total    0] Target: (-41.0, 70.0, -22.6) | Pos: (-41.7, 72.
0, -21.4) | Vel: ( 0.9, -0.6, -0.1) | Acc: ( 0.1, -2.0, 0.1) | Rwd: 2.
60
1661 [Astar_planner] Segment 13 reached in 14 steps, dist=2.41,
axis_error=[0.6541023 1.981369 1.212925 ], reason=running
1662 [Astar_planner] Segment 14 starts from actual pos [-41.646263 71.
97635 -21.436218] (planned waypoint was [-41.    70.   -22.
648464])
1663 [Astar_planner] Segment 14: [-41.646263 71.97635 -21.436218
]-> [-46.    82.   -22.482023]
1664 [Step 28 | Total    0] Target: (-46.0, 82.0, -22.5) | Pos: (-44.4, 80.
0, -20.6) | Vel: ( -0.5, -0.2, -1.6) | Acc: ( -1.8, -0.6, 0.3) | Rwd: 6.
29
1665 [Astar_planner] Segment 14 reached in 28 steps, dist=3.18,
axis_error=[1.6152077 1.9565964 1.9155807], reason=running
1666 [Astar_planner] Segment 15 starts from actual pos [-44.38973 80.
04137 -20.580498] (planned waypoint was [-46.    82.   -22.
482023])
1667 [Astar_planner] Segment 15: [-44.38973 80.04137 -20.580498] -> [-
55.    89.   -22.399603]
1668 [Step 16 | Total    0] Target: (-55.0, 89.0, -22.4) | Pos: (-54.1, 88.
9, -21.1) | Vel: ( -1.0, -0.0, -0.6) | Acc: ( 1.3, -0.6, 1.6) | Rwd: 604.
38 -> Episode End: ARRIVED!
1669
1670 [Astar_planner] Segment 15 reached in 16 steps, dist=1.58,
axis_error=[0.9272957 0.05943298 1.2770786 ], reason=arrived
1671 [Astar_planner] Segment 16 starts from actual pos [-54.084766 88.
94008 -21.1299 ] (planned waypoint was [-55.    89.   -22.
399603])
1672 [Astar_planner] Segment 16: [-54.084766 88.94008 -21.1299 ] -> [-
69.    89.   -21.206854]
1673 [Step 23 | Total    0] Target: (-69.0, 89.0, -21.2) | Pos: (-68.8, 90.
3, -21.0) | Vel: ( 1.3, -1.2, 0.2) | Acc: ( 1.2, 0.4, 1.8) | Rwd: 603.
67 -> Episode End: ARRIVED!
1674

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1675 [Astar_planner] Segment 16 reached in 23 steps, dist=1.33,
axis_error=[0.24958801 1.2971573 0.17936516], reason=arrived
1676 [Astar_planner] Segment 17 starts from actual pos [-68.742905 90.
28991 -21.026413] (planned waypoint was [-69. 89. -21.
206854])
1677 [Astar_planner] Segment 17: [-68.742905 90.28991 -21.026413
] -> [-82. 89. -23.178724]
1678 [Step 19 | Total 0] Target: (-82.0, 89.0, -23.2) | Pos: (-83.4, 90.
9, -21.9) | Vel: (-1.2, -1.1, -0.6) | Acc: ( 1.1, -1.9, 0.4) | Rwd: 3.
28
1679 [Astar_planner] Segment 17 reached in 19 steps, dist=2.73,
axis_error=[1.4290237 1.9344788 1.2887478], reason=running
1680 [Astar_planner] Segment 18 starts from actual pos [-83.440186 90.
924545 -21.894896] (planned waypoint was [-82. 89. -23.
178724])
1681 [Astar_planner] Segment 18: [-83.440186 90.924545 -21.894896
] -> [-94. 91. -22.481707]
1682 [Step 21 | Total 0] Target: (-94.0, 91.0, -22.5) | Pos: (-92.6, 89.
7, -21.4) | Vel: (-1.8, 0.0, -1.4) | Acc: (-1.8, -0.1, 0.7) | Rwd: 4.
42
1683 [Astar_planner] Segment 18 reached in 21 steps, dist=2.16,
axis_error=[1.3597107 1.2538071 1.1071968], reason=running
1684 [Astar_planner] Segment 19 starts from actual pos [-92.65138 89.
74621 -21.383104] (planned waypoint was [-94. 91. -22.
481707])
1685 [Astar_planner] Segment 19: [-92.65138 89.74621 -21.383104] -> [-
100. 101. -21.289549]
1686 [Step 22 | Total 0] Target: (-100.0, 101.0, -21.3) | Pos: (-98.4,
101.8, -21.2) | Vel: ( 0.1, -0.7, -0.0) | Acc: ( 1.0, -2.0, 0.1) | Rwd
: 2.14
1687 [Astar_planner] Segment 19 reached in 22 steps, dist=1.82,
axis_error=[1.6264801 0.8123169 0.0456543], reason=running
1688 [Astar_planner] Segment 20 starts from actual pos [-98.37248 101.
805885 -21.244043] (planned waypoint was [-100. 101. -21.
289549])
1689 [Astar_planner] Segment 20: [-98.37248 101.805885 -21.244043
] -> [-101. 114. -24.665932]
1690 [Step 16 | Total 0] Target: (-101.0, 114.0, -24.7) | Pos: (-100.5,
112.2, -23.6) | Vel: (-0.3, 1.3, -1.4) | Acc: (-0.9, -0.8, 0.4) | Rwd
: 3.03
1691 [Astar_planner] Segment 20 reached in 16 steps, dist=2.17,
axis_error=[0.4755783 1.7949295 1.1135826], reason=running
1692 [Astar_planner] Segment 21 starts from actual pos [-100.527275 112
.21688 -23.56489 ] (planned waypoint was [-101. 114. -24.
665932])
1693 [Astar_planner] Segment 21: [-100.527275 112.21688 -23.56489
] -> [-106. 125. -4.905461]
1694 [Step 33 | Total 0] Target: (-106.0, 125.0, -4.9) | Pos: (-107.2,

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1694 124.6, -5.5) | Vel: ( 0.9, 1.0, 1.0) | Acc: ( -0.1, 0.2, 0.5) | Rwd:
      603.18 -> Episode End: ARRIVED!
1695
1696 [Astar_planner] Segment 21 reached in 33 steps, dist=1.43,
      axis_error=[1.2412262 0.37793732 0.5968065 ], reason=arrived
1697 Task 24 result: success (Success: Reached destination (22/22
      segments). Shield interventions=42, emergencies=1, recoveries=7,
      recovery_steps=42.)
1698
1699 --- Task 25/38 ---
1700 From [-60.9 -95.62 -8.8 ] to [64.2 3.18 -7.4 ]
1701 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
1702 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
1703 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
1704 Path planned via occupancy (Points: 17, Dist: 224.3m). Local target z
      /alt: 0:-8.80(8.80m), 1:-24.29(24.29m), 2:-21.22(21.22m), 3:-20.49(20
      .49m), 4:-21.06(21.06m), 5:-24.60(24.60m), 6:-21.07(21.07m), 7:-23.
      39(23.39m), 8:-21.72(21.72m), 9:-24.86(24.86m), 10:-22.96(22.96m
      ), 11:-23.76(23.76m), 12:-23.44(23.44m), 13:-22.37(22.37m), 14:-24.
      94(24.94m), 15:-20.70(20.70m), 16:-7.40(7.40m)
1705 [Astar_planner] Segment 0: [-61.    -96.    -8.796549] -> [-54
      .    -88.    -24.294538]
1706 [Astar_planner] Placing first segment start at [-61.    -96.    -8.
      796549]
1707 [Astar_planner] Initial climb to 20.0m.
1708 [Step 180 | Total    0] Target: (-54.0, -88.0, -24.3) | Pos: (-59.2, -89.
      0, -22.1) | Vel: ( 0.1, -0.1, 0.1) | Acc: ( 0.7, -0.8, -1.0) | Rwd: 0.
      18
1709 [Astar_planner] Segment 0 timeout, dist=5.75
1710 [Astar_planner] Stop after failed segment 0.
1711 Task 25 result: failed (Failed at segment 0: segment_timeout (0/16
      reached). Shield interventions=6, emergencies=0, recoveries=1,
      recovery_steps=6.)
1712
1713 --- Task 26/38 ---
1714 From [ -61.  -139.82 -11.6 ] to [108.  -87.82 -5.4 ]
1715 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
1716 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
1717 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
1718 Path planned via occupancy (Points: 21, Dist: 245.2m). Local target z
      /alt: 0:-11.59(11.59m), 1:-24.42(24.42m), 2:-24.44(24.44m), 3:-24.82(
      24.82m), 4:-21.58(21.58m), 5:-24.21(24.21m), 6:-21.07(21.07m), 7:-

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1718 22.05(22.05m), 8:-20.56(20.56m), 9:-23.68(23.68m), 10:-21.19(21.
    19m), 11:-24.74(24.74m), 12:-22.27(22.27m), 13:-21.51(21.51m), 14
    :-23.38(23.38m), 15:-24.25(24.25m), 16:-24.48(24.48m), 17:-23.34(
    23.34m), 18:-20.80(20.80m), 19:-20.56(20.56m), 20:-5.40(5.40m)
1719 [Astar_planner] Segment 0: [ -61.    -140.    -11.593689] -> [ -59
    .    -150.    -24.424278]
1720 [Astar_planner] Placing first segment start at [ -61.    -140.    -11.
    593689]
1721 [Astar_planner] Initial climb to 20.0m.
1722 [Step 17 | Total    0] Target: (-59.0, -150.0, -24.4) | Pos: (-59.6, -
    151.8, -23.4) | Vel: ( 0.2, 0.4, 0.4) | Acc: ( -0.1, 2.0, 0.7) | Rwd:
    3.34
1723 [Astar_planner] Segment 0 reached in 17 steps, dist=2.17, axis_error
    =[0.5892792 1.8019714 1.0528679], reason=running
1724 [Astar_planner] Segment 1 starts from actual pos [ -59.58734 -151.
    79837 -23.367577] (planned waypoint was [ -59.    -150.    -24.
    424278])
1725 [Astar_planner] Segment 1: [ -59.58734 -151.79837 -23.367577
    ] -> [ -55.    -160.    -24.443222]
1726 [Step 26 | Total    0] Target: (-55.0, -160.0, -24.4) | Pos: (-56.6, -
    158.3, -22.5) | Vel: ( 0.6, 0.9, -1.4) | Acc: ( 0.8, 0.8, 0.3) | Rwd:
    6.40
1727 [Astar_planner] Segment 1 reached in 26 steps, dist=3.01, axis_error
    =[1.6026077 1.6704102 1.9165001], reason=running
1728 [Astar_planner] Segment 2 starts from actual pos [ -56.59695 -158.
    32184 -22.539097] (planned waypoint was [ -55.    -160.    -24.
    443222])
1729 [Astar_planner] Segment 2: [ -56.59695 -158.32184 -22.539097
    ] -> [ -44.    -161.    -24.815943]
1730 [Step 11 | Total    0] Target: (-44.0, -161.0, -24.8) | Pos: (-50.1, -
    163.3, -22.0) | Vel: ( -0.5, 0.0, 0.1) | Acc: ( -1.3, 1.1, -0.7) | Rwd: -
    923.18 -> Episode End: CRASHED!
1731
1732 [Astar_planner] Segment 2 ended: collision
1733 [Astar_planner] Stop after failed segment 2.
1734 Task 26 result: failed (Failed at segment 2: collision (2/20 reached).
    Shield interventions=11, emergencies=0, recoveries=2,
    recovery_steps=11.)
1735
1736 --- Task 27/38 ---
1737 From [ 57.9 19.78 -11.3 ] to [-73.2 -85.22 -5.6 ]
1738 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
    =[-200.0, 189.0], occupied=72869 cells (53.4%)
1739 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
    cells (53.4%)
1740 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
1741 Path planned via occupancy (Points: 17, Dist: 231.7m). Local target z

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1741 /alt: 0:-11.29(11.29m), 1:-22.50(22.50m), 2:-22.18(22.18m), 3:-24.05(
24.05m), 4:-24.57(24.57m), 5:-20.86(20.86m), 6:-23.11(23.11m), 7:-
20.65(20.65m), 8:-23.86(23.86m), 9:-22.47(22.47m), 10:-21.83(21.
83m), 11:-24.44(24.44m), 12:-21.34(21.34m), 13:-23.69(23.69m), 14
:-21.70(21.70m), 15:-24.92(24.92m), 16:-5.61(5.61m)
1742 [Astar_planner] Segment 0: [ 58.    20.   -11.2918] -> [ 44.    19
.   -22.496315]
1743 [Astar_planner] Placing first segment start at [ 58.    20.   -11.2918]
1744 [Astar_planner] Initial climb to 20.0m.
1745 [Step 23 | Total    0] Target: ( 44.0, 19.0, -22.5) | Pos: ( 44.4, 20.8
, -21.1) | Vel: ( -0.2, -1.0, -0.5) | Acc: ( 0.1, -1.8, -1.5) | Rwd: 7.32
1746 [Astar_planner] Segment 0 reached in 23 steps, dist=2.33, axis_error
=[0.4159584 1.8010769 1.4214153], reason=running
1747 [Astar_planner] Segment 1 starts from actual pos [ 44.414524 20.
791739 -21.079575] (planned waypoint was [ 44.    19.   -22.
496315])
1748 [Astar_planner] Segment 1: [ 44.414524 20.791739 -21.079575] -> [
31.    19.   -22.183964]
1749 [Step 20 | Total    0] Target: ( 31.0, 19.0, -22.2) | Pos: ( 32.7, 20.7
, -20.6) | Vel: ( -0.3, -0.6, 0.1) | Acc: ( -0.0, -1.8, -1.6) | Rwd: 2.84
1750 [Astar_planner] Segment 1 reached in 20 steps, dist=2.90, axis_error
=[1.7491951 1.717514 1.5431156], reason=running
1751 [Astar_planner] Segment 2 starts from actual pos [ 32.74644 20.
711983 -20.639887] (planned waypoint was [ 31.    19.   -22.
183964])
1752 [Astar_planner] Segment 2: [ 32.74644 20.711983 -20.639887] -> [
19.    12.   -24.045023]
1753 [Step 45 | Total    0] Target: ( 19.0, 12.0, -24.0) | Pos: ( 24.6, 8.9
, -21.1) | Vel: ( 0.0, 0.0, 0.0) | Acc: ( 0.1, 0.0, -0.2) | Rwd: -0.65
1754 [Astar_planner] Segment 2 reached in 45 steps, dist=7.05, axis_error
=[5.6043873 3.1215487 2.921177 ], reason=stuck_near_intermediate
1755 [Astar_planner] Segment 3 starts from actual pos [ 24.604502 8.
878517 -21.123695] (planned waypoint was [ 19.    12.   -24.
045023])
1756 [Astar_planner] Segment 3: [ 24.604502 8.878517 -21.123695] -> [
10.    3.   -24.572224]
1757 [Step 39 | Total    0] Target: ( 10.0, 3.0, -24.6) | Pos: ( 10.6, 1.9
, -22.6) | Vel: ( -0.2, -0.3, 0.5) | Acc: ( 0.4, -2.0, 0.2) | Rwd: 599.
69 -> Episode End: ARRIVED!
1758
1759 [Astar_planner] Segment 3 reached in 39 steps, dist=2.32, axis_error
=[0.6028042 1.1091214 1.9517021], reason=arrived
1760 [Astar_planner] Segment 4 starts from actual pos [ 10.601411 1.
8884492 -22.61626 ] (planned waypoint was [ 10.    3.   -24.
572224])
1761 [Astar_planner] Segment 4: [ 10.601411 1.8884492 -22.61626
] -> [ -3.    0.   -20.8572]
1762 [Step 22 | Total    0] Target: ( -3.0, 0.0, -20.9) | Pos: ( -1.4, -0.2, -

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1762 18.9) | Vel: ( -0.4, -0.2, 0.3) | Acc: ( -0.5, -1.2, -0.4) | Rwd: 0.58
1763 [Astar_planner] Segment 4 reached in 22 steps, dist=2.54, axis_error
=[1.5693763 0.23551102 1.9867802 ], reason=running
1764 [Astar_planner] Segment 5 starts from actual pos [ -1.4339528 -0.
23732105 -18.868044 ] (planned waypoint was [ -3. 0. -20.
8572])
1765 [Astar_planner] Segment 5: [ -1.4339528 -0.23732105 -18.868044
] -> [ -9. -11. -23.111141]
1766 [Step 22 | Total 0] Target: ( -9.0, -11.0, -23.1) | Pos: ( -8.0, -10.6
, -21.9) | Vel: ( -0.2, -0.5, 0.2) | Acc: ( 0.5, -1.3, -0.0) | Rwd: 605.
69 -> Episode End: ARRIVED!
1767
1768 [Astar_planner] Segment 5 reached in 22 steps, dist=1.60, axis_error
=[0.98933125 0.38948917 1.1936226 ], reason=arrived
1769 [Astar_planner] Segment 6 starts from actual pos [ -8.012086 -10.
615259 -21.91563 ] (planned waypoint was [ -9. -11. -23.
111141])
1770 [Astar_planner] Segment 6: [ -8.012086 -10.615259 -21.91563 ] -> [ -
9. -25. -20.6476]
1771 [Step 16 | Total 0] Target: ( -9.0, -25.0, -20.6) | Pos: ( -9.6, -23.3
, -19.6) | Vel: ( 0.9, -0.7, 0.1) | Acc: ( 1.9, -0.3, 1.0) | Rwd: 4.93
1772 [Astar_planner] Segment 6 reached in 16 steps, dist=2.04, axis_error
=[0.5631447 1.6638165 1.0376129], reason=running
1773 [Astar_planner] Segment 7 starts from actual pos [ -9.555392 -23.
342546 -19.608934] (planned waypoint was [ -9. -25. -20.6476])
1774 [Astar_planner] Segment 7: [ -9.555392 -23.342546 -19.608934] -> [ -
9. -39. -23.859999]
1775 [Step 117 | Total 0] Target: ( -9.0, -39.0, -23.9) | Pos: ( -7.1, -38.7
, -23.0) | Vel: ( -0.1, -0.8, -0.1) | Acc: ( 0.9, -1.6, 0.5) | Rwd: 0.97
1776 [Astar_planner] Segment 7 reached in 117 steps, dist=2.12,
axis_error=[1.9132466 0.32183456 0.8561001 ], reason=running
1777 [Astar_planner] Segment 8 starts from actual pos [ -7.0878363 -38.
6855 -23.00437 ] (planned waypoint was [ -9. -39. -23.
859999])
1778 [Astar_planner] Segment 8: [ -7.0878363 -38.6855 -23.00437
] -> [ -9. -53. -22.468786]
1779 [Step 14 | Total 0] Target: ( -9.0, -53.0, -22.5) | Pos: ( -8.3, -52.1
, -21.8) | Vel: ( 0.6, -1.1, -0.2) | Acc: ( 1.1, -1.7, 0.8) | Rwd: 602.
16 -> Episode End: ARRIVED!
1780
1781 [Astar_planner] Segment 8 reached in 14 steps, dist=1.28, axis_error
=[0.6626482 0.8692436 0.6578846], reason=arrived
1782 [Astar_planner] Segment 9 starts from actual pos [ -8.333715 -52.
137398 -21.81182 ] (planned waypoint was [ -9. -53. -22.
468786])
1783 [Astar_planner] Segment 9: [ -8.333715 -52.137398 -21.81182 ] -> [ -
9. -67. -21.83482]
1784 [Step 13 | Total 0] Target: ( -9.0, -67.0, -21.8) | Pos: ( -9.7, -65.3

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1784 , -20.8) | Vel: ( 0.2, -1.2, -0.1) | Acc: ( 0.4, -2.0, 0.7) | Rwd: 3.15
1785 [Astar_planner] Segment 9 reached in 13 steps, dist=2.10, axis_error
=[0.72587013 1.6584396 1.0683918 ], reason=running
1786 [Astar_planner] Segment 10 starts from actual pos [ -9.723294 -65.
35611 -20.768044] (planned waypoint was [ -9.    -67.    -21.83482
])
1787 [Astar_planner] Segment 10: [ -9.723294 -65.35611 -20.768044] -> [-
11.    -80.    -24.43729]
1788 [Step 13 | Total    0] Target: (-11.0, -80.0, -24.4) | Pos: (-10.2, -79.
8, -23.2) | Vel: ( -0.2, -1.3, -0.1) | Acc: ( 0.6, -0.8, 0.8) | Rwd: 606.
17 -> Episode End: ARRIVED!
1789
1790 [Astar_planner] Segment 10 reached in 13 steps, dist=1.53,
axis_error=[0.8405905 0.16000366 1.2656174 ], reason=arrived
1791 [Astar_planner] Segment 11 starts from actual pos [-10.161195 -79.
85156 -23.172062] (planned waypoint was [-11.    -80.    -24.
43729])
1792 [Astar_planner] Segment 11: [-10.161195 -79.85156 -23.172062
] -> [-19.    -88.    -21.342707]
1793 [Step 12 | Total    0] Target: (-19.0, -88.0, -21.3) | Pos: (-18.0, -88.
3, -20.6) | Vel: ( -1.3, -0.5, 0.1) | Acc: ( -1.1, -0.8, -0.1) | Rwd: 605.
48 -> Episode End: ARRIVED!
1794
1795 [Astar_planner] Segment 11 reached in 12 steps, dist=1.29,
axis_error=[1.005228 0.29981995 0.7576618 ], reason=arrived
1796 [Astar_planner] Segment 12 starts from actual pos [-18.006662 -88.
30429 -20.583519] (planned waypoint was [-19.    -88.    -21.
342707])
1797 [Astar_planner] Segment 12: [-18.006662 -88.30429 -20.583519
] -> [-33.    -88.    -23.686214]
1798 [Step 127 | Total    0] Target: (-33.0, -88.0, -23.7) | Pos: (-32.8, -89.
6, -23.8) | Vel: ( -0.3, 0.6, -0.0) | Acc: ( 0.5, 1.5, 0.6) | Rwd: 2.40

1799 [Astar_planner] Segment 12 reached in 127 steps, dist=1.60,
axis_error=[0.17361069 1.590477 0.11055374], reason=running
1800 [Astar_planner] Segment 13 starts from actual pos [-32.829548 -89.
58384 -23.797342] (planned waypoint was [-33.    -88.    -23.
686214])
1801 [Astar_planner] Segment 13: [-32.829548 -89.58384 -23.797342
] -> [-47.    -88.    -21.699425]
1802 [Step 15 | Total    0] Target: (-47.0, -88.0, -21.7) | Pos: (-45.4, -88.
0, -20.8) | Vel: ( -1.2, 0.3, 0.3) | Acc: ( -1.2, 0.3, -1.2) | Rwd: 5.
40
1803 [Astar_planner] Segment 13 reached in 15 steps, dist=1.83,
axis_error=[1.612915 0.01726532 0.85782623], reason=running
1804 [Astar_planner] Segment 14 starts from actual pos [-45.397697 -87.
98032 -20.838799] (planned waypoint was [-47.    -88.    -21.
699425])

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1805 [Astar_planner] Segment 14: [-45.397697 -87.98032 -20.838799
] -> [-61.    -88.    -24.922918]
1806 [Step 45 | Total    0] Target: (-61.0, -88.0, -24.9) | Pos: (-59.3, -89.
0, -23.8) | Vel: ( -0.4, 1.1, 0.1) | Acc: ( -0.7, 2.0, 2.0) | Rwd: 5.75

1807 [Astar_planner] Segment 14 reached in 45 steps, dist=2.30,
axis_error=[1.6884193 1.0331192 1.1623592], reason=running
1808 [Astar_planner] Segment 15 starts from actual pos [-59.31525 -89.
023 -23.759548] (planned waypoint was [-61.    -88.    -24.
922918])
1809 [Astar_planner] Segment 15: [-59.31525 -89.023 -23.759548] -> [-
73.    -85.    -5.6099644]
1810 [Step 36 | Total    0] Target: (-73.0, -85.0, -5.6) | Pos: (-72.2, -85.6
, -5.5) | Vel: ( -1.0, -0.1, -0.1) | Acc: ( -0.4, -0.3, 0.2) | Rwd: 602.07
-> Episode End: ARRIVED!
1811
1812 [Astar_planner] Segment 15 reached in 36 steps, dist=0.97,
axis_error=[0.7736969 0.5740738 0.09468222], reason=arrived
1813 Task 27 result: success (Success: Reached destination (16/16
segments). Shield interventions=126, emergencies=11, recoveries=
21, recovery_steps=126.)
1814
1815 --- Task 28/38 ---
1816 From [ 17.8 -86.92 -8.9 ] to [ 23.5 140.08 -4. ]
1817 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
1818 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
1819 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
1820 Path planned via occupancy (Points: 29, Dist: 371.0m). Local target z
/alt: 0:-8.90(8.90m), 1:-22.65(22.65m), 2:-22.36(22.36m), 3:-21.15(21
.15m), 4:-20.24(20.24m), 5:-21.60(21.60m), 6:-23.68(23.68m), 7:-24.
03(24.03m), 8:-21.52(21.52m), 9:-21.16(21.16m), 10:-20.78(20.78m
), 11:-23.01(23.01m), 12:-21.19(21.19m), 13:-21.35(21.35m), 14:-22.
45(22.45m), 15:-24.50(24.50m), 16:-24.26(24.26m), 17:-24.14(24.
14m), 18:-21.65(21.65m), 19:-21.92(21.92m), 20:-20.27(20.27m), 21
:-21.26(21.26m), 22:-20.83(20.83m), 23:-21.55(21.55m), 24:-24.72(
24.72m), 25:-23.06(23.06m), 26:-23.86(23.86m), 27:-20.98(20.98m),
28:-4.00(4.00m)
1821 [Astar_planner] Segment 0: [ 18.    -87.    -8.895351] -> [ 5.    -
88.    -22.645666]
1822 [Astar_planner] Placing first segment start at [ 18.    -87.    -8.
895351]
1823 [Astar_planner] Initial climb to 20.0m.
1824 [Step 24 | Total    0] Target: ( 5.0, -88.0, -22.6) | Pos: ( 6.7, -89.4
, -21.9) | Vel: ( -1.0, -0.1, 0.4) | Acc: ( -1.4, -0.6, -0.2) | Rwd: 3.94
1825 [Astar_planner] Segment 0 reached in 24 steps, dist=2.28, axis_error

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1825 =[1.6661296 1.3796234 0.7050228], reason=running
1826 [Astar_planner] Segment 1 starts from actual pos [ 6.659884 -89.
38005 -21.93828 ] (planned waypoint was [ 5. -88. -22.
645666])
1827 [Astar_planner] Segment 1: [ 6.659884 -89.38005 -21.93828 ] -> [ -8
-88. -22.362448]
1828 [Step 35 | Total 0] Target: ( -8.0, -88.0, -22.4) | Pos: ( -6.0, -87.2
, -22.0) | Vel: ( -0.1, 0.1, 0.0) | Acc: ( 0.6, 1.7, -0.1) | Rwd: -0.27
1829 [Astar_planner] Segment 1 reached in 35 steps, dist=2.18, axis_error
=[1.9950099 0.8174286 0.32042885], reason=running
1830 [Astar_planner] Segment 2 starts from actual pos [ -6.0054636 -87.
1816 -22.041914 ] (planned waypoint was [ -8. -88. -22.
362448])
1831 [Astar_planner] Segment 2: [ -6.0054636 -87.1816 -22.041914
] -> [-11. -77. -21.153805]
1832 [Step 47 | Total 0] Target: (-11.0, -77.0, -21.2) | Pos: (-13.0, -78.
0, -19.2) | Vel: ( 0.6, 0.3, -0.8) | Acc: ( 1.7, 2.0, -0.2) | Rwd: 7.98

1833 [Astar_planner] Segment 2 reached in 47 steps, dist=3.00, axis_error
=[1.9992723 1.0442886 1.9720154], reason=running
1834 [Astar_planner] Segment 3 starts from actual pos [-12.991847 -78.
04014 -19.191559] (planned waypoint was [-11. -77. -21.
153805])
1835 [Astar_planner] Segment 3: [-12.991847 -78.04014 -19.191559] -> [-
10. -64. -20.243284]
1836 [Step 88 | Total 0] Target: (-10.0, -64.0, -20.2) | Pos: ( -8.5, -66.0
, -19.5) | Vel: ( -0.2, 0.2, 0.0) | Acc: ( -1.1, 1.0, -0.3) | Rwd: 1.18
1837 [Astar_planner] Segment 3 reached in 88 steps, dist=2.62, axis_error
=[1.540865 1.9980545 0.7110748], reason=running
1838 [Astar_planner] Segment 4 starts from actual pos [ -8.46098 -65.
99597 -19.532063] (planned waypoint was [-10. -64. -20.
243284])
1839 [Astar_planner] Segment 4: [ -8.46098 -65.99597 -19.532063] -> [ -9
-51. -21.59852]
1840 [Step 56 | Total 0] Target: ( -9.0, -51.0, -21.6) | Pos: ( -9.3, -52.7
, -20.7) | Vel: ( -0.0, 0.4, 0.3) | Acc: ( 0.6, 0.5, 1.9) | Rwd: 2.87
1841 [Astar_planner] Segment 4 reached in 56 steps, dist=1.94, axis_error
=[0.28526402 1.6716232 0.93951225], reason=running
1842 [Astar_planner] Segment 5 starts from actual pos [ -9.285309 -52.
66774 -20.65601 ] (planned waypoint was [ -9. -51. -21.59852
])
1843 [Astar_planner] Segment 5: [ -9.285309 -52.66774 -20.65601 ] -> [ -9
-38. -23.683096]
1844 [Step 47 | Total 0] Target: ( -9.0, -38.0, -23.7) | Pos: (-10.4, -36.9
, -22.4) | Vel: ( 0.6, -0.2, -0.8) | Acc: ( -0.3, 1.1, 1.7) | Rwd: 6.26
1845 [Astar_planner] Segment 5 reached in 47 steps, dist=2.17, axis_error
=[1.3788681 1.1253014 1.242485 ], reason=running
1846 [Astar_planner] Segment 6 starts from actual pos [-10.371544 -36.

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1846 876568 -22.45034 ] (planned waypoint was [ -9.    -38.    -23.
      683096])
1847 [Astar_planner] Segment 6: [-10.371544 -36.876568 -22.45034 ] -> [ -
      9.    -25.    -24.033617]
1848 [Step 36 | Total    0] Target: ( -9.0, -25.0, -24.0) | Pos: ( -7.4, -26.0
      , -22.2) | Vel: ( -0.0,  0.8,  0.5) | Acc: (  0.8,  2.0,  0.8) | Rwd:  1.66
1849 [Astar_planner] Segment 6 reached in 36 steps, dist=2.60, axis_error
      =[1.569344  1.0172749 1.7995224], reason=running
1850 [Astar_planner] Segment 7 starts from actual pos [ -7.4309163 -26.
      01018  -22.229176 ] (planned waypoint was [ -9.    -25.    -24.
      033617])
1851 [Astar_planner] Segment 7: [ -7.4309163 -26.01018  -22.229176
      ] -> [ -9.    -12.    -21.519913]
1852 [Step 23 | Total    0] Target: ( -9.0, -12.0, -21.5) | Pos: (-10.5, -12.0
      , -20.8) | Vel: (  0.4, -0.6,  0.4) | Acc: ( -0.2, -0.8, -0.7) | Rwd:  2.50
1853 [Astar_planner] Segment 7 reached in 23 steps, dist=1.71, axis_error
      =[1.5466719  0.00489044 0.73983  ], reason=running
1854 [Astar_planner] Segment 8 starts from actual pos [-10.543018 -12.
      0000725 -20.77649 ] (planned waypoint was [ -9.    -12.    -21.
      519913])
1855 [Astar_planner] Segment 8: [-10.543018 -12.0000725 -20.77649
      ] -> [-13.    -1.    -21.160435]
1856 [Step 13 | Total    0] Target: (-13.0, -1.0, -21.2) | Pos: (-13.7, -2.7
      , -19.2) | Vel: ( -0.2,  0.7,  0.5) | Acc: ( -0.6, -0.6,  0.5) | Rwd:  4.58
1857 [Astar_planner] Segment 8 reached in 13 steps, dist=2.66, axis_error
      =[0.69434834 1.6948714 1.930378 ], reason=running
1858 [Astar_planner] Segment 9 starts from actual pos [-13.696199 -2.
      688344 -19.225576] (planned waypoint was [-13.    -1.    -21.
      160435])
1859 [Astar_planner] Segment 9: [-13.696199 -2.688344 -19.225576] -> [-
      25.    2.    -20.77974]
1860 [Step 20 | Total    0] Target: (-25.0,  2.0, -20.8) | Pos: (-24.9,  0.1
      , -19.0) | Vel: ( -0.4,  0.5,  0.0) | Acc: ( -0.8,  1.4, -0.7) | Rwd:  6.07
1861 [Astar_planner] Segment 9 reached in 20 steps, dist=2.65, axis_error
      =[0.1133728 1.9281584 1.809679 ], reason=running
1862 [Astar_planner] Segment 10 starts from actual pos [-24.89045  0.
      07640002 -18.969826 ] (planned waypoint was [-25.    2.    -20.
      77974])
1863 [Astar_planner] Segment 10: [-24.89045  0.07640002 -18.969826
      ] -> [-34.    11.    -23.005096]
1864 [Step 24 | Total    0] Target: (-34.0, 11.0, -23.0) | Pos: (-32.3,  9.4
      , -21.8) | Vel: ( -0.2,  0.0,  0.6) | Acc: ( -0.1, -0.6,  0.7) | Rwd:  2.83
1865 [Astar_planner] Segment 10 reached in 24 steps, dist=2.60,
      axis_error=[1.6724701 1.6030235 1.1885033], reason=running
1866 [Astar_planner] Segment 11 starts from actual pos [-32.329624  9.
      397356 -21.809576] (planned waypoint was [-34.    11.    -23.
      005096])
1867 [Astar_planner] Segment 11: [-32.329624  9.397356 -21.809576

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1867 ] -> [-43.    21.    -21.192364]
1868 [Step 16 | Total    0] Target: (-43.0, 21.0, -21.2) | Pos: (-41.8, 20.
    2, -19.4) | Vel: ( -0.3, 0.5, -1.1) | Acc: ( -0.2, 0.4, 1.4) | Rwd: 607.
    13 -> Episode End: ARRIVED!
1869
1870 [Astar_planner] Segment 11 reached in 16 steps, dist=2.30,
    axis_error=[1.2178459 0.8182697 1.7755756], reason=arrived
1871 [Astar_planner] Segment 12 starts from actual pos [-41.785236 20.
    186249 -19.426565] (planned waypoint was [-43.    21.    -21.
    192364])
1872 [Astar_planner] Segment 12: [-41.785236 20.186249 -19.426565
    ] -> [-50.    31.    -21.353764]
1873 [Step 22 | Total    0] Target: (-50.0, 31.0, -21.4) | Pos: (-48.8, 30.
    9, -20.1) | Vel: ( -0.3, 0.1, 0.1) | Acc: ( 0.9, -1.9, 0.8) | Rwd: 604.
    92 -> Episode End: ARRIVED!
1874
1875 [Astar_planner] Segment 12 reached in 22 steps, dist=1.72,
    axis_error=[1.1978607 0.06046486 1.23522 ], reason=arrived
1876 [Astar_planner] Segment 13 starts from actual pos [-48.804775 30.
    940765 -20.117388] (planned waypoint was [-50.    31.    -21.
    353764])
1877 [Astar_planner] Segment 13: [-48.804775 30.940765 -20.117388
    ] -> [-54.    42.    -22.452587]
1878 [Step 14 | Total    0] Target: (-54.0, 42.0, -22.5) | Pos: (-54.0, 40.
    9, -21.3) | Vel: ( -0.3, 0.1, 0.2) | Acc: ( 0.5, -1.7, 1.0) | Rwd: 603.
    71 -> Episode End: ARRIVED!
1879
1880 [Astar_planner] Segment 13 reached in 14 steps, dist=1.60,
    axis_error=[0.03382492 1.1423569 1.1199818 ], reason=arrived
1881 [Astar_planner] Segment 14 starts from actual pos [-53.969067 40.
    85858 -21.330967] (planned waypoint was [-54.    42.    -22.
    452587])
1882 [Astar_planner] Segment 14: [-53.969067 40.85858 -21.330967
    ] -> [-60.    53.    -24.502853]
1883 [Step 32 | Total    0] Target: (-60.0, 53.0, -24.5) | Pos: (-60.4, 54.
    4, -23.9) | Vel: ( 0.3, -1.0, 0.3) | Acc: ( 2.0, -1.5, 2.0) | Rwd: 603.
    84 -> Episode End: ARRIVED!
1884
1885 [Astar_planner] Segment 14 reached in 32 steps, dist=1.63,
    axis_error=[0.38207245 1.4430008 0.65195656], reason=arrived
1886 [Astar_planner] Segment 15 starts from actual pos [-60.379124 54.
    434265 -23.848253] (planned waypoint was [-60.    53.    -24.
    502853])
1887 [Astar_planner] Segment 15: [-60.379124 54.434265 -23.848253
    ] -> [-52.    62.    -24.258718]
1888 [Step 32 | Total    0] Target: (-52.0, 62.0, -24.3) | Pos: (-52.6, 60.
    5, -23.0) | Vel: ( 0.3, 0.5, -0.6) | Acc: ( 0.0, -0.4, 0.2) | Rwd: 6.28

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1889 [Astar_planner] Segment 15 reached in 32 steps, dist=2.00,
axis_error=[0.57933044 1.4614868 1.2420692 ], reason=running
1890 [Astar_planner] Segment 16 starts from actual pos [-52.577038 60.
543068 -23.021639] (planned waypoint was [-52.    62.    -24.
258718])
1891 [Astar_planner] Segment 16: [-52.577038 60.543068 -23.021639
] -> [-41.    70.    -24.142818]
1892 [Step 152 | Total    0] Target: (-41.0, 70.0, -24.1) | Pos: (-41.7, 70.
5, -22.8) | Vel: ( 1.3, -0.5, 0.2) | Acc: ( 0.7, -1.6, -0.1) | Rwd: 605.
22 -> Episode End: ARRIVED!
1893
1894 [Astar_planner] Segment 16 reached in 152 steps, dist=1.61,
axis_error=[0.7351265 0.5437546 1.324606 ], reason=arrived
1895 [Astar_planner] Segment 17 starts from actual pos [-41.72383 70.
53889 -22.816238] (planned waypoint was [-41.    70.    -24.
142818])
1896 [Astar_planner] Segment 17: [-41.72383 70.53889 -22.816238] -> [-
32.    79.    -21.651226]
1897 [Step 16 | Total    0] Target: (-32.0, 79.0, -21.7) | Pos: (-33.6, 79.
6, -20.5) | Vel: ( 1.2, 0.2, -1.2) | Acc: ( -0.0, 0.2, -0.7) | Rwd: 5.
35
1898 [Astar_planner] Segment 17 reached in 16 steps, dist=2.07,
axis_error=[1.602066 0.57865906 1.1702766 ], reason=running
1899 [Astar_planner] Segment 18 starts from actual pos [-33.591286 79.
5804 -20.491913] (planned waypoint was [-32.    79.    -21.
651226])
1900 [Astar_planner] Segment 18: [-33.591286 79.5804 -20.491913] -> [-
25.    89.    -21.919872]
1901 [Step 81 | Total    0] Target: (-25.0, 89.0, -21.9) | Pos: (-23.3, 87.
7, -20.8) | Vel: ( -0.3, -0.3, -0.0) | Acc: ( -0.0, -1.9, 0.6) | Rwd: -0.
50
1902 [Astar_planner] Segment 18 reached in 81 steps, dist=2.36,
axis_error=[1.6747208 1.262764 1.0730152], reason=running
1903 [Astar_planner] Segment 19 starts from actual pos [-23.328196 87.
73469 -20.847137] (planned waypoint was [-25.    89.    -21.
919872])
1904 [Astar_planner] Segment 19: [-23.328196 87.73469 -20.847137
] -> [-25.    102.    -20.265055]
1905 [Step 90 | Total    0] Target: (-25.0, 102.0, -20.3) | Pos: (-21.7, 99.
7, -16.9) | Vel: ( -0.0, 0.1, 0.0) | Acc: ( -0.4, 0.9, -0.3) | Rwd: -1.
27
1906 [Astar_planner] Segment 19 reached in 90 steps, dist=5.26,
axis_error=[3.3234863 2.2935867 3.3742332], reason=
stuck_near_intermediate
1907 [Astar_planner] Segment 20 starts from actual pos [-21.676716 99.
70711 -16.890678] (planned waypoint was [-25.    102.    -20.
265055])
1908 [Astar_planner] Segment 20: [-21.676716 99.70711 -16.890678

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1908 ]-> [-25.    115.    -21.263023]
1909 [Step 30 | Total    0] Target: (-25.0, 115.0, -21.3) | Pos: (-23.5, 113
    .8, -19.9) | Vel: ( -1.1,  0.6, -1.9) | Acc: ( -0.6, -1.1,  0.1) | Rwd:  4.
    08
1910 [Astar_planner] Segment 20 reached in 30 steps, dist=2.33,
    axis_error=[1.515934  1.1777573 1.324934 ], reason=running
1911 [Astar_planner] Segment 21 starts from actual pos [-23.494389 113.
    82794 -19.954895] (planned waypoint was [-25.    115.    -21.
    263023])
1912 [Astar_planner] Segment 21: [-23.494389 113.82794 -19.954895
    ]-> [-25.    128.    -20.831627]
1913 [Step 23 | Total    0] Target: (-25.0, 128.0, -20.8) | Pos: (-24.7, 129
    .9, -20.3) | Vel: ( -0.1, -0.6,  0.1) | Acc: ( -0.1, -1.7, -0.7) | Rwd:  3.
    92
1914 [Astar_planner] Segment 21 reached in 23 steps, dist=1.96,
    axis_error=[0.30504417 1.8508301 0.5762005 ], reason=running
1915 [Astar_planner] Segment 22 starts from actual pos [-24.695974 129.
    84586 -20.25439 ] (planned waypoint was [-25.    128.    -20.
    831627])
1916 [Astar_planner] Segment 22: [-24.695974 129.84586 -20.25439
    ]-> [-25.    141.    -21.54774]
1917 [Step 15 | Total    0] Target: (-25.0, 141.0, -21.5) | Pos: (-25.2, 140
    .1, -20.2) | Vel: (  0.6,  2.6, -1.3) | Acc: (  0.4, -0.3, -1.0) | Rwd: 605.
    98 -> Episode End: ARRIVED!
1918
1919 [Astar_planner] Segment 22 reached in 15 steps, dist=1.66,
    axis_error=[0.18005943 0.90719604 1.3792591 ], reason=arrived
1920 [Astar_planner] Segment 23 starts from actual pos [-25.176607 140.
    10825 -20.175999] (planned waypoint was [-25.    141.    -21.
    54774])
1921 [Astar_planner] Segment 23: [-25.176607 140.10825 -20.175999
    ]-> [-21.    153.    -24.724478]
1922 [Step 42 | Total    0] Target: (-21.0, 153.0, -24.7) | Pos: (-21.1, 151
    .2, -23.1) | Vel: ( -0.3,  0.7, -0.9) | Acc: ( -1.5, -0.0,  2.0) | Rwd:  3.
    80
1923 [Astar_planner] Segment 23 reached in 42 steps, dist=2.46,
    axis_error=[0.11174393 1.8015289 1.6648445 ], reason=running
1924 [Astar_planner] Segment 24 starts from actual pos [-21.11465 151.
    20493 -23.067852] (planned waypoint was [-21.    153.    -24.
    724478])
1925 [Astar_planner] Segment 24: [-21.11465 151.20493 -23.067852
    ]-> [-9.    153.    -23.064537]
1926 [Step 14 | Total    0] Target: ( -9.0, 153.0, -23.1) | Pos: (-10.4, 152.
    2, -22.1) | Vel: (  1.3, -0.6, -1.0) | Acc: ( -0.6, -1.2,  0.1) | Rwd:  3.
    90
1927 [Astar_planner] Segment 24 reached in 14 steps, dist=1.88,
    axis_error=[1.368043  0.84284973 0.96749306], reason=running
1928 [Astar_planner] Segment 25 starts from actual pos [-10.36  152.

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1928 15355 -22.103168] (planned waypoint was [ -9.    153.    -23.
064537])
1929 [Astar_planner] Segment 25: [-10.36    152.15355 -22.103168] -> [
4.    152.    -23.856861]
1930 [Step  20 | Total    0] Target: ( 4.0, 152.0, -23.9) | Pos: ( 3.6, 152.6
, -22.6) | Vel: ( 0.7, -0.5, 0.2) | Acc: ( -1.0, -0.7, 2.0) | Rwd: 604.
09 -> Episode End: ARRIVED!
1931
1932 [Astar_planner] Segment 25 reached in 20 steps, dist=1.50,
axis_error=[0.44682527 0.60287476 1.2950573 ], reason=arrived
1933 [Astar_planner] Segment 26 starts from actual pos [ 3.5593393 152.
59882 -22.5601 ] (planned waypoint was [ 4.    152.    -23.
856861])
1934 [Astar_planner] Segment 26: [ 3.5593393 152.59882 -22.5601
] -> [ 16.    149.    -20.975151]
1935 [Step  11 | Total    0] Target: ( 16.0, 149.0, -21.0) | Pos: ( 14.8, 147.
9, -19.8) | Vel: ( 1.3, 0.2, 0.2) | Acc: ( -1.9, 1.8, 0.8) | Rwd: 0.59

1936 [Astar_planner] Segment 26 reached in 11 steps, dist=2.02,
axis_error=[1.2294683 1.0990143 1.1747932], reason=running
1937 [Astar_planner] Segment 27 starts from actual pos [ 14.7786 147.
9022 -19.79916] (planned waypoint was [ 16.    149.    -20.
975151])
1938 [Astar_planner] Segment 27: [ 14.7786 147.9022 -19.79916] -> [ 24
.    140.    -4.001443]
1939 [Step  20 | Total    0] Target: ( 24.0, 140.0, -4.0) | Pos: ( 22.5, 139.
0, -4.4) | Vel: ( 0.0, -0.2, 0.0) | Acc: ( -1.1, -1.5, -0.5) | Rwd: -0.58

1940 [Astar_planner] Segment 27 reached in 20 steps, dist=1.81,
axis_error=[1.479414 0.97662354 0.3833127 ], reason=running
1941 Task 28 result: success (Success: Reached destination (28/28
segments). Shield interventions=102, emergencies=0, recoveries=17
, recovery_steps=102.)
1942
1943 --- Task 29/38 ---
1944 From [-10.5 163.38 -8.94] to [ 17.8 -86.92 -8.9 ]
1945 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
1946 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
1947 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
1948 Path planned via occupancy (Points: 26, Dist: 333.8m). Local target z
/alt: 0:-8.94(8.94m), 1:-24.67(24.67m), 2:-20.50(20.50m), 3:-24.61(24
.61m), 4:-21.01(21.01m), 5:-21.80(21.80m), 6:-23.80(23.80m), 7:-22.
66(22.66m), 8:-23.54(23.54m), 9:-22.76(22.76m), 10:-21.81(21.81m
), 11:-20.35(20.35m), 12:-24.24(24.24m), 13:-22.45(22.45m), 14:-23.
63(23.63m), 15:-20.15(20.15m), 16:-24.29(24.29m), 17:-23.53(23.

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1948 53m), 18:-21.04(21.04m), 19:-23.35(23.35m), 20:-22.58(22.58m), 21
:-23.69(23.69m), 22:-21.40(21.40m), 23:-24.62(24.62m), 24:-20.31(
20.31m), 25:-8.90(8.90m)
1949 [Astar_planner] Segment 0: [-10.    163.    -8.939899] -> [-19.
154.    -24.6724]
1950 [Astar_planner] Placing first segment start at [-10.    163.    -8.
939899]
1951 [Astar_planner] Initial climb to 20.0m.
1952 [Step 30 | Total    0] Target: (-19.0, 154.0, -24.7) | Pos: (-19.3, 152
.1, -23.9) | Vel: ( -0.2,  0.8, -0.5) | Acc: ( -0.2, -0.6,  0.7) | Rwd: 2.
13
1953 [Astar_planner] Segment 0 reached in 30 steps, dist=2.10, axis_error
=[0.2629242 1.9346161 0.7643051], reason=running
1954 [Astar_planner] Segment 1 starts from actual pos [-19.264378 152.
0703 -23.911285] (planned waypoint was [-19.    154.    -24.6724])
1955 [Astar_planner] Segment 1: [-19.264378 152.0703 -23.911285] -> [-
25.    143.    -20.50338]
1956 [Step 12 | Total    0] Target: (-25.0, 143.0, -20.5) | Pos: (-23.6, 144
.1, -20.6) | Vel: ( -0.1, -0.6, -0.0) | Acc: (  0.3, -2.0, -0.1) | Rwd: 2.
81
1957 [Astar_planner] Segment 1 reached in 12 steps, dist=1.80, axis_error
=[1.4202442 1.1118164 0.05953598], reason=running
1958 [Astar_planner] Segment 2 starts from actual pos [-23.581089 144.
10638 -20.56306 ] (planned waypoint was [-25.    143.    -20.
50338])
1959 [Astar_planner] Segment 2: [-23.581089 144.10638 -20.56306 ] -> [-
25.    130.    -24.607447]
1960 [Step 34 | Total    0] Target: (-25.0, 130.0, -24.6) | Pos: (-25.2, 131
.9, -23.8) | Vel: ( -0.1, -0.8, -1.2) | Acc: ( -0.4, -2.0,  1.8) | Rwd: 0.
46
1961 [Astar_planner] Segment 2 reached in 34 steps, dist=2.09, axis_error
=[0.19091415 1.9203339 0.8102646 ], reason=running
1962 [Astar_planner] Segment 3 starts from actual pos [-25.191261 131.
91563 -23.80443 ] (planned waypoint was [-25.    130.    -24.
607447])
1963 [Astar_planner] Segment 3: [-25.191261 131.91563 -23.80443 ] -> [-
25.    117.    -21.012861]
1964 [Step 18 | Total    0] Target: (-25.0, 117.0, -21.0) | Pos: (-26.6, 118
.1, -21.2) | Vel: (  0.5, -0.4,  0.0) | Acc: (  0.7, -0.7, -0.6) | Rwd: 1.
91
1965 [Astar_planner] Segment 3 reached in 18 steps, dist=1.97, axis_error
=[1.6054325 1.138977 0.1395607], reason=running
1966 [Astar_planner] Segment 4 starts from actual pos [-26.600769 118.
13513 -21.152124] (planned waypoint was [-25.    117.    -21.
012861])
1967 [Astar_planner] Segment 4: [-26.600769 118.13513 -21.152124] -> [-
25.    104.    -21.796509]
1968 [Step 15 | Total    0] Target: (-25.0, 104.0, -21.8) | Pos: (-25.3, 105

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1968 .7, -22.0) | Vel: ( 0.1, -0.5, -0.0) | Acc: ( 0.3, -1.8, -0.4) | Rwd: 3.
21
1969 [Astar_planner] Segment 4 reached in 15 steps, dist=1.69, axis_error
=[0.25667 1.6625595 0.19607925], reason=running
1970 [Astar_planner] Segment 5 starts from actual pos [-25.255411 105.
6578 -21.99261 ] (planned waypoint was [-25. 104. -21.
796509])
1971 [Astar_planner] Segment 5: [-25.255411 105.6578 -21.99261 ] -> [-
25. 91. -23.804686]
1972 [Step 23 | Total 0] Target: (-25.0, 91.0, -23.8) | Pos: (-23.2, 89.
2, -22.9) | Vel: ( -0.0, 0.3, 0.5) | Acc: ( 0.8, 2.0, -0.3) | Rwd: 1.48

1973 [Astar_planner] Segment 5 reached in 23 steps, dist=2.73, axis_error
=[1.8473263 1.8227463 0.85525703], reason=running
1974 [Astar_planner] Segment 6 starts from actual pos [-23.153015 89.
17991 -22.945314] (planned waypoint was [-25. 91. -23.
804686])
1975 [Astar_planner] Segment 6: [-23.153015 89.17991 -22.945314] -> [-
31. 80. -22.663101]
1976 [Step 43 | Total 0] Target: (-31.0, 80.0, -22.7) | Pos: (-25.2, 77.
2, -20.3) | Vel: ( -0.1, 0.1, 0.0) | Acc: ( -0.6, 0.5, 0.1) | Rwd: -0.
50
1977 [Astar_planner] Segment 6 reached in 43 steps, dist=6.86, axis_error
=[5.773575 2.8315735 2.3969955], reason=stuck_near_intermediate
1978 [Astar_planner] Segment 7 starts from actual pos [-25.226915 77.
168884 -20.26606 ] (planned waypoint was [-31. 80. -22.
663101])
1979 [Astar_planner] Segment 7: [-25.226915 77.168884 -20.26606 ] -> [-
40. 71. -23.5384]
1980 [Step 20 | Total 0] Target: (-40.0, 71.0, -23.5) | Pos: (-38.2, 69.
1, -22.7) | Vel: ( -0.3, -0.6, 0.5) | Acc: ( 0.9, -1.9, -1.2) | Rwd: -0.
28
1981 [Astar_planner] Segment 7 reached in 20 steps, dist=2.71, axis_error
=[1.7559433 1.8725586 0.8592453], reason=running
1982 [Astar_planner] Segment 8 starts from actual pos [-38.246468 69.
122246 -22.674383] (planned waypoint was [-40. 71. -23.5384])
1983 [Astar_planner] Segment 8: [-38.246468 69.122246 -22.674383] -> [-
50. 64. -22.762897]
1984 [Step 13 | Total 0] Target: (-50.0, 64.0, -22.8) | Pos: (-48.9, 64.
8, -21.0) | Vel: ( -1.4, -0.6, -0.1) | Acc: ( -1.9, -1.6, 0.2) | Rwd: 605.
23 -> Episode End: ARRIVED!
1985
1986 [Astar_planner] Segment 8 reached in 13 steps, dist=2.22, axis_error
=[1.0883675 0.7517929 1.7858486], reason=arrived
1987 [Astar_planner] Segment 9 starts from actual pos [-48.928417 64.
7444 -20.977882] (planned waypoint was [-50. 64. -22.
762897])
1988 [Astar_planner] Segment 9: [-48.928417 64.7444 -20.977882] -> [-

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1988 59.    54.    -21.806393]
1989 [Step 16 | Total    0] Target: (-59.0, 54.0, -21.8) | Pos: (-58.1, 55.
    8, -21.9) | Vel: ( -0.4, -0.4,  0.2) | Acc: ( -0.6, -0.4, -1.4) | Rwd: 2.
    34
1990 [Astar_planner] Segment 9 reached in 16 steps, dist=2.03, axis_error
    =[0.86484146 1.8318405 0.08221817], reason=running
1991 [Astar_planner] Segment 10 starts from actual pos [-58.138535 55.
    828236 -21.887043] (planned waypoint was [-59.    54.    -21.
    806393])
1992 [Astar_planner] Segment 10: [-58.138535 55.828236 -21.887043
    ] -> [-55.    43.    -20.347717]
1993 [Step 24 | Total    0] Target: (-55.0, 43.0, -20.3) | Pos: (-56.5, 42.
    8, -20.1) | Vel: ( 0.3, -0.1,  0.0) | Acc: ( 1.0, -0.9, -0.5) | Rwd: 2.
    19
1994 [Astar_planner] Segment 10 reached in 24 steps, dist=1.55,
    axis_error=[1.5139046 0.17956161 0.25352097], reason=running
1995 [Astar_planner] Segment 11 starts from actual pos [-56.511513 42.
    819878 -20.094002] (planned waypoint was [-55.    43.    -20.
    347717])
1996 [Astar_planner] Segment 11: [-56.511513 42.819878 -20.094002
    ] -> [-50.    32.    -24.241535]
1997 [Step 22 | Total    0] Target: (-50.0, 32.0, -24.2) | Pos: (-51.5, 31.
    9, -23.3) | Vel: ( 0.5, 0.9,  0.1) | Acc: ( -0.7, 1.6,  1.9) | Rwd: 3.76

1998 [Astar_planner] Segment 11 reached in 22 steps, dist=1.77,
    axis_error=[1.5057564 0.10237503 0.93190956], reason=running
1999 [Astar_planner] Segment 12 starts from actual pos [-51.49972 31.
    908558 -23.308363] (planned waypoint was [-50.    32.    -24.
    241535])
2000 [Astar_planner] Segment 12: [-51.49972 31.908558 -23.308363
    ] -> [-44.    22.    -22.446188]
2001 [Step 18 | Total    0] Target: (-44.0, 22.0, -22.4) | Pos: (-45.8, 21.
    0, -21.5) | Vel: ( 0.4, -0.2, -0.0) | Acc: ( 0.6, -2.0,  0.5) | Rwd: 0.
    67
2002 [Astar_planner] Segment 12 reached in 18 steps, dist=2.24,
    axis_error=[1.7943459 0.9690876 0.9242153], reason=running
2003 [Astar_planner] Segment 13 starts from actual pos [-45.791164 21.
    029385 -21.52209 ] (planned waypoint was [-44.    22.    -22.
    446188])
2004 [Astar_planner] Segment 13: [-45.791164 21.029385 -21.52209
    ] -> [-35.    12.    -23.625113]
2005 [Step 26 | Total    0] Target: (-35.0, 12.0, -23.6) | Pos: (-35.4, 10.
    5, -22.6) | Vel: ( 0.6, 1.2, -0.2) | Acc: ( -0.5, 2.0,  1.9) | Rwd: 4.70

2006 [Astar_planner] Segment 13 reached in 26 steps, dist=1.85,
    axis_error=[0.39819336 1.5103817 0.9984875 ], reason=running
2007 [Astar_planner] Segment 14 starts from actual pos [-35.394543 10.
    496935 -22.627987] (planned waypoint was [-35.    12.    -23.

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2007 625113])
2008 [Astar_planner] Segment 14: [-35.394543 10.496935 -22.627987
] -> [-26.    3.    -20.145798]
2009 [Step 16 | Total    0] Target: (-26.0,  3.0, -20.1) | Pos: (-27.9,  4.1
, -19.2) | Vel: ( 0.3, -0.5,  0.7) | Acc: ( -1.1, -1.1,  0.8) | Rwd:  3.33
2010 [Astar_planner] Segment 14 reached in 16 steps, dist=2.38,
axis_error=[1.8538647 1.1406178 0.9694786], reason=running
2011 [Astar_planner] Segment 15 starts from actual pos [-27.851034  4.
136472 -19.169947] (planned waypoint was [-26.    3.    -20.
145798])
2012 [Astar_planner] Segment 15: [-27.851034  4.136472 -19.169947
] -> [-14.    -1.    -24.29125]
2013 [Step 77 | Total    0] Target: (-14.0, -1.0, -24.3) | Pos: (-15.6, -2.5
, -22.5) | Vel: ( 0.6, -0.2,  0.2) | Acc: ( 0.7, -2.0, -0.1) | Rwd:  1.97
2014 [Astar_planner] Segment 15 reached in 77 steps, dist=2.84,
axis_error=[1.5749216 1.5486324 1.7882767], reason=running
2015 [Astar_planner] Segment 16 starts from actual pos [-15.5695 -2.
55062 -22.50108] (planned waypoint was [-14.    -1.    -24.29125])
2016 [Astar_planner] Segment 16: [-15.5695 -2.55062 -22.50108] -> [ -9
.   -11.   -23.533024]
2017 [Step 15 | Total    0] Target: ( -9.0, -11.0, -23.5) | Pos: (-10.2, -12.9
, -22.0) | Vel: ( 0.7,  0.3,  0.4) | Acc: ( 0.9,  1.6,  0.0) | Rwd:  2.40
2018 [Astar_planner] Segment 16 reached in 15 steps, dist=2.77,
axis_error=[1.2083626 1.9295244 1.5747757], reason=running
2019 [Astar_planner] Segment 17 starts from actual pos [-10.202309 -12.
926821 -21.954603] (planned waypoint was [ -9.   -11.   -23.
533024])
2020 [Astar_planner] Segment 17: [-10.202309 -12.926821 -21.954603
] -> [ -9.   -24.   -21.037506]
2021 [Step 13 | Total    0] Target: ( -9.0, -24.0, -21.0) | Pos: (-10.5, -22.7
, -20.1) | Vel: ( 0.7, -0.8,  0.1) | Acc: ( 1.0, -1.1, -0.3) | Rwd:  5.89
2022 [Astar_planner] Segment 17 reached in 13 steps, dist=2.20,
axis_error=[1.4624825 1.3334408 0.967474 ], reason=running
2023 [Astar_planner] Segment 18 starts from actual pos [-10.458229 -22.
671116 -20.06955 ] (planned waypoint was [ -9.   -24.   -21.
037506])
2024 [Astar_planner] Segment 18: [-10.458229 -22.671116 -20.06955
] -> [ -9.   -38.   -23.348299]
2025 [Step 19 | Total    0] Target: ( -9.0, -38.0, -23.3) | Pos: ( -9.6, -38.9
, -21.6) | Vel: ( 0.7, -0.1, -0.3) | Acc: ( 0.3, -0.4,  0.3) | Rwd: 602.
85 -> Episode End: ARRIVED!
2026
2027 [Astar_planner] Segment 18 reached in 19 steps, dist=2.00,
axis_error=[0.5792923 0.8596916 1.713747 ], reason=arrived
2028 [Astar_planner] Segment 19 starts from actual pos [ -9.570939 -38.
861057 -21.63721 ] (planned waypoint was [ -9.   -38.   -23.
348299])
2029 [Astar_planner] Segment 19: [ -9.570939 -38.861057 -21.63721

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2029 ] -> [ -9.    -51.    -22.579336]
2030 [Step 12 | Total    0] Target: ( -9.0, -51.0, -22.6) | Pos: ( -9.7, -50.5
    , -21.6) | Vel: ( 0.3, -1.5, -0.1) | Acc: ( 0.2, -2.0, -0.8) | Rwd: 608.
    14 -> Episode End: ARRIVED!
2031
2032 [Astar_planner] Segment 19 reached in 12 steps, dist=1.34,
    axis_error=[0.7094326 0.49784088 1.028038 ], reason=arrived
2033 [Astar_planner] Segment 20 starts from actual pos [ -9.70662 -50.
    51603 -21.552126] (planned waypoint was [ -9.    -51.    -22.
    579336])
2034 [Astar_planner] Segment 20: [ -9.70662 -50.51603 -21.552126] -> [ -
    9.    -64.    -23.685535]
2035 [Step 12 | Total    0] Target: ( -9.0, -64.0, -23.7) | Pos: ( -9.8, -63.3
    , -22.3) | Vel: ( 0.3, -1.7, -0.8) | Acc: ( -0.1, -2.0, 1.3) | Rwd: 608.
    75 -> Episode End: ARRIVED!
2036
2037 [Astar_planner] Segment 20 reached in 12 steps, dist=1.73,
    axis_error=[0.8243189 0.6841011 1.353056 ], reason=arrived
2038 [Astar_planner] Segment 21 starts from actual pos [ -9.820154 -63.
    33611 -22.342226] (planned waypoint was [ -9.    -64.    -23.
    685535])
2039 [Astar_planner] Segment 21: [ -9.820154 -63.33611 -22.342226] -> [-
    10.    -77.    -21.40019]
2040 [Step 11 | Total    0] Target: (-10.0, -77.0, -21.4) | Pos: (-11.0, -75.
    5, -21.6) | Vel: ( 0.2, -1.4, -0.1) | Acc: ( 0.8, -1.5, -0.7) | Rwd: 3.
    30
2041 [Astar_planner] Segment 21 reached in 11 steps, dist=1.83,
    axis_error=[0.9659519 1.5318146 0.24273682], reason=running
2042 [Astar_planner] Segment 22 starts from actual pos [-10.964074 -75.
    48048 -21.643675] (planned waypoint was [-10.    -77.    -21.
    40019])
2043 [Astar_planner] Segment 22: [-10.964074 -75.48048 -21.643675
    ] -> [ -8.    -87.    -24.615261]
2044 [Step 25 | Total    0] Target: ( -8.0, -87.0, -24.6) | Pos: ( -9.8, -85.8
    , -23.5) | Vel: ( 0.3, -0.1, -0.3) | Acc: ( -0.5, -0.8, -0.3) | Rwd: 3.71
2045 [Astar_planner] Segment 22 reached in 25 steps, dist=2.45,
    axis_error=[1.8054752 1.1890106 1.1513824], reason=running
2046 [Astar_planner] Segment 23 starts from actual pos [ -9.802752 -85.
    81209 -23.466122] (planned waypoint was [ -8.    -87.    -24.
    615261])
2047 [Astar_planner] Segment 23: [ -9.802752 -85.81209 -23.466122
    ] -> [ 5.    -88.    -20.313711]
2048 [Step 19 | Total    0] Target: ( 5.0, -88.0, -20.3) | Pos: ( 3.7, -88.4
    , -19.6) | Vel: ( 0.9, 0.0, 0.2) | Acc: ( -0.5, 0.4, 1.0) | Rwd: 603.05
    -> Episode End: ARRIVED!
2049
2050 [Astar_planner] Segment 23 reached in 19 steps, dist=1.53,
    axis_error=[1.2880967 0.4483261 0.6972904], reason=arrived

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2051 [Astar_planner] Segment 24 starts from actual pos [ 3.7222488 -88.
      44811 -19.614191 ] (planned waypoint was [ 5. -88. -20.
      313711])
2052 [Astar_planner] Segment 24: [ 3.7222488 -88.44811 -19.614191
      ] -> [ 18. -87. -8.9]
2053 [Step 16 | Total 0] Target: ( 18.0, -87.0, -8.9) | Pos: ( 17.0, -86.7
      , -8.2) | Vel: ( 1.4, 0.1, -0.2) | Acc: ( 2.0, 0.1, 1.7) | Rwd: 601.56
      -> Episode End: ARRIVED!
2054
2055 [Astar_planner] Segment 24 reached in 16 steps, dist=1.27,
      axis_error=[1.0135612 0.33255005 0.6804085 ], reason=arrived
2056 Task 29 result: success (Success: Reached destination (25/25
      segments). Shield interventions=0, emergencies=0, recoveries=0,
      recovery_steps=0.)
2057
2058 --- Task 30/38 ---
2059 From [-117.3 1.48 -9. ] to [-123.9 -90.92 -4. ]
2060 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
2061 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
2062 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
2063 Path planned via occupancy (Points: 10, Dist: 117.7m). Local target z
      /alt: 0:-8.97(8.97m), 1:-23.57(23.57m), 2:-20.59(20.59m), 3:-22.96(22
      .96m), 4:-24.78(24.78m), 5:-21.96(21.96m), 6:-21.26(21.26m), 7:-21.
      40(21.40m), 8:-23.91(23.91m), 9:-4.00(4.00m)
2064 [Astar_planner] Segment 0: [-117. 1. -8.969448] -> [-113
      . -8. -23.571178]
2065 [Astar_planner] Placing first segment start at [-117. 1. -8.
      969448]
2066 [Astar_planner] Initial climb to 20.0m.
2067 [Step 35 | Total 0] Target: (-113.0, -8.0, -23.6) | Pos: (-113.5, -7
      .1, -22.1) | Vel: ( 0.7, -0.3, 0.7) | Acc: ( 1.4, 0.3, 0.0) | Rwd: 604.
      62 -> Episode End: ARRIVED!
2068
2069 [Astar_planner] Segment 0 reached in 35 steps, dist=1.85, axis_error
      =[0.5237045 0.93648577 1.5020885 ], reason=arrived
2070 [Astar_planner] Segment 1 starts from actual pos [-113.51721 -7.
      0662537 -22.062954 ] (planned waypoint was [-113. -8. -23
      .571178])
2071 [Astar_planner] Segment 1: [-113.51721 -7.0662537 -22.062954
      ] -> [-110. -19. -20.58787]
2072 [Step 14 | Total 0] Target: (-110.0, -19.0, -20.6) | Pos: (-108.5, -
      17.5, -19.7) | Vel: ( -0.1, -0.7, -0.3) | Acc: ( 0.0, 0.6, 0.2) | Rwd:
      3.76
2073 [Astar_planner] Segment 1 reached in 14 steps, dist=2.31, axis_error
      =[1.471695 1.540863 0.899456], reason=running

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2074 [Astar_planner] Segment 2 starts from actual pos [-108.52998 -17.467098 -19.691393] (planned waypoint was [-110. -19. -20.58787])
2075 [Astar_planner] Segment 2: [-108.52998 -17.467098 -19.691393] -> [-110. -30. -22.964905]
2076 [Step 16 | Total 0] Target: (-110.0, -30.0, -23.0) | Pos: (-108.4, -30.2, -21.9) | Vel: ( -0.9, -0.5, -0.7) | Acc: ( -1.5, 1.2, 1.6) | Rwd: 3.91
2077 [Astar_planner] Segment 2 reached in 16 steps, dist=1.95, axis_error=[1.6092758 0.19449806 1.0785961 ], reason=running
2078 [Astar_planner] Segment 3 starts from actual pos [-108.39871 -30.199394 -21.892202] (planned waypoint was [-110. -30. -22.964905])
2079 [Astar_planner] Segment 3: [-108.39871 -30.199394 -21.892202] -> [-110. -41. -24.775303]
2080 [Step 12 | Total 0] Target: (-110.0, -41.0, -24.8) | Pos: (-110.7, -40.8, -23.3) | Vel: ( 0.8, -1.3, -0.4) | Acc: ( 2.0, -1.8, 1.1) | Rwd: 606.63 -> Episode End: ARRIVED!
2081
2082 [Astar_planner] Segment 3 reached in 12 steps, dist=1.65, axis_error=[0.6899414 0.23003387 1.4782124 ], reason=arrived
2083 [Astar_planner] Segment 4 starts from actual pos [-110.68285 -40.781387 -23.300478] (planned waypoint was [-110. -41. -24.775303])
2084 [Astar_planner] Segment 4: [-110.68285 -40.781387 -23.300478] -> [-110. -52. -21.963976]
2085 [Step 10 | Total 0] Target: (-110.0, -52.0, -22.0) | Pos: (-109.7, -51.1, -21.3) | Vel: ( -0.1, -1.4, 0.0) | Acc: ( -1.0, 0.6, 0.7) | Rwd: 604.32 -> Episode End: ARRIVED!
2086
2087 [Astar_planner] Segment 4 reached in 10 steps, dist=1.17, axis_error=[0.32572174 0.9255791 0.63793755], reason=arrived
2088 [Astar_planner] Segment 5 starts from actual pos [-109.67533 -51.086628 -21.325745] (planned waypoint was [-110. -52. -21.963976])
2089 [Astar_planner] Segment 5: [-109.67533 -51.086628 -21.325745] -> [-111. -63. -21.257776]
2090 [Step 9 | Total 0] Target: (-111.0, -63.0, -21.3) | Pos: (-110.1, -61.0, -21.5) | Vel: ( -0.1, -1.5, 0.0) | Acc: ( 0.3, -0.5, -0.9) | Rwd: 3.24
2091 [Astar_planner] Segment 5 reached in 9 steps, dist=2.18, axis_error=[0.90696716 1.9642982 0.26358223], reason=running
2092 [Astar_planner] Segment 6 starts from actual pos [-110.09351 -61.048996 -21.521301] (planned waypoint was [-111. -63. -21.257776])
2093 [Astar_planner] Segment 6: [-110.09351 -61.048996 -21.521301] -> [-115. -72. -21.398458]
2094 [Step 19 | Total 0] Target: (-115.0, -72.0, -21.4) | Pos: (-114.0, -

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2094 73.9, -20.6) | Vel: ( -0.5, 0.3, 0.0) | Acc: ( -0.4, -1.7, 1.6) | Rwd:
      0.86
2095 [Astar_planner] Segment 6 reached in 19 steps, dist=2.28, axis_error
      =[0.9544678 1.9149399 0.7906742], reason=running
2096 [Astar_planner] Segment 7 starts from actual pos [-114.04977 -73.
      91234 -20.607391] (planned waypoint was [-115.      -72.      -21.
      398458])
2097 [Astar_planner] Segment 7: [-114.04977 -73.91234 -20.607391
      ] -> [-120.      -82.      -23.907951]
2098 [Step 12 | Total 0] Target: (-120.0, -82.0, -23.9) | Pos: (-118.4, -
      83.1, -23.1) | Vel: ( -1.0, -1.1, -0.0) | Acc: ( -0.4, -0.5, 1.6) | Rwd:
      0.46
2099 [Astar_planner] Segment 7 reached in 12 steps, dist=2.11, axis_error
      =[1.6092529 1.1143341 0.7825699], reason=running
2100 [Astar_planner] Segment 8 starts from actual pos [-118.4001 -83.
      124504 -23.125584] (planned waypoint was [-120.      -82.      -23.
      907951])
2101 [Astar_planner] Segment 8: [-118.4001 -83.124504 -23.125584
      ] -> [-124. -91. -4.]
2102 [Step 39 | Total 0] Target: (-124.0, -91.0, -4.0) | Pos: (-122.6, -
      90.2, -3.4) | Vel: ( -0.1, -0.6, 0.0) | Acc: ( 0.7, 0.4, 0.5) | Rwd: 0.
      21
2103 [Astar_planner] Segment 8 reached in 39 steps, dist=1.72, axis_error
      =[1.4419937 0.7527008 0.5617416], reason=running
2104 Task 30 result: success (Success: Reached destination (9/9
      segments). Shield interventions=15, emergencies=0, recoveries=3,
      recovery_steps=15.)
2105
2106 --- Task 31/38 ---
2107 From [-125.1 185.28 -4. ] to [-141.4 -191.82 -9. ]
2108 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
2109 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
2110 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
2111 Path planned via occupancy (Points: 31, Dist: 447.1m). Local target z
      /alt: 0:-4.00(4.00m), 1:-23.62(23.62m), 2:-23.50(23.50m), 3:-22.34(22
      .34m), 4:-21.25(21.25m), 5:-23.08(23.08m), 6:-20.30(20.30m), 7:-24.
      70(24.70m), 8:-21.67(21.67m), 9:-24.97(24.97m), 10:-21.65(21.65m
      ), 11:-21.51(21.51m), 12:-20.03(20.03m), 13:-22.27(22.27m), 14:-21.
      24(21.24m), 15:-23.08(23.08m), 16:-24.63(24.63m), 17:-23.28(23.
      28m), 18:-24.16(24.16m), 19:-23.67(23.67m), 20:-20.80(20.80m), 21
      :-23.50(23.50m), 22:-21.39(21.39m), 23:-20.93(20.93m), 24:-21.37(
      21.37m), 25:-20.40(20.40m), 26:-24.10(24.10m), 27:-24.62(24.62m),
      28:-21.52(21.52m), 29:-24.81(24.81m), 30:-9.00(9.00m)
2112 [Astar_planner] Segment 0: [-125.      185.      -4.0039387] -> [-
      135.      175.      -23.619766]

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2113 [Astar_planner] Placing first segment start at [-125.    185.    -4
      .0039387]
2114 [Astar_planner] Initial climb to 20.0m.
2115 [Step 19 | Total    0] Target: (-135.0, 175.0, -23.6) | Pos: (-134.9,
      176.3, -22.9) | Vel: ( -0.3, -1.1,  0.3) | Acc: ( 0.6, -1.4, -0.9) | Rwd:
      605.68 -> Episode End: ARRIVED!
2116
2117 [Astar_planner] Segment 0 reached in 19 steps, dist=1.44, axis_error
      =[0.0944519 1.256897 0.70051193], reason=arrived
2118 [Astar_planner] Segment 1 starts from actual pos [-134.9083  176.
      2471  -22.916714] (planned waypoint was [-135.    175.    -23.
      619766])
2119 [Astar_planner] Segment 1: [-134.9083  176.2471  -22.916714
      ] -> [-150.    175.    -23.503052]
2120 [Step 75 | Total    0] Target: (-150.0, 175.0, -23.5) | Pos: (-149.7,
      173.5, -22.5) | Vel: ( -0.7,  0.8, -0.8) | Acc: ( -0.2, -1.6,  1.2) | Rwd
      : 2.43
2121 [Astar_planner] Segment 1 reached in 75 steps, dist=1.83, axis_error
      =[0.2918396 1.5049591 1.0041332], reason=running
2122 [Astar_planner] Segment 2 starts from actual pos [-149.7124  173.
      49997  -22.503424] (planned waypoint was [-150.    175.    -23.
      503052])
2123 [Astar_planner] Segment 2: [-149.7124  173.49997  -22.503424
      ] -> [-159.    167.    -22.33933]
2124 [Step 13 | Total    0] Target: (-159.0, 167.0, -22.3) | Pos: (-157.4,
      169.0, -21.8) | Vel: ( -1.0, -1.3, -0.5) | Acc: ( 0.4, -2.0, -0.8) | Rwd
      : 3.34
2125 [Astar_planner] Segment 2 reached in 13 steps, dist=2.62, axis_error
      =[1.6364136 1.9701843 0.5474186], reason=running
2126 [Astar_planner] Segment 3 starts from actual pos [-157.37288  168.
      95865  -21.79622] (planned waypoint was [-159.    167.    -22.
      33933])
2127 [Astar_planner] Segment 3: [-157.37288  168.95865  -21.79622] -> [-
      159.    152.    -21.25207]
2128 [Step 15 | Total    0] Target: (-159.0, 152.0, -21.3) | Pos: (-159.5,
      152.9, -20.3) | Vel: ( 0.4, -0.7, -0.0) | Acc: ( 0.6, -2.0, 0.6) | Rwd:
      602.61 -> Episode End: ARRIVED!
2129
2130 [Astar_planner] Segment 3 reached in 15 steps, dist=1.42, axis_error
      =[0.5380249 0.9212036 0.937418 ], reason=arrived
2131 [Astar_planner] Segment 4 starts from actual pos [-159.53455  152.
      91516  -20.315075] (planned waypoint was [-159.    152.    -21.
      25207])
2132 [Astar_planner] Segment 4: [-159.53455  152.91516  -20.315075
      ] -> [-165.    141.    -23.075224]
2133 [Step 22 | Total    0] Target: (-165.0, 141.0, -23.1) | Pos: (-164.3,
      139.1, -21.8) | Vel: ( -0.9,  0.3, -0.7) | Acc: ( -1.2, 0.5,  2.0) | Rwd
      : 2.47

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2134 [Astar_planner] Segment 4 reached in 22 steps, dist=2.40, axis_error
      =[0.7025604 1.9342041 1.2376461], reason=running
2135 [Astar_planner] Segment 5 starts from actual pos [-164.30307  139.
      06772 -21.841873] (planned waypoint was [-165.    141.    -23.
      075224])
2136 [Astar_planner] Segment 5: [-164.30307  139.06772 -21.841873
      ] -> [-171.    129.    -20.29513]
2137 [Step  18 | Total    0] Target: (-171.0, 129.0, -20.3) | Pos: (-169.2,
      128.4, -20.3) | Vel: ( -0.9, -0.6, -0.1) | Acc: ( -0.7,  0.6,  1.7) | Rwd
      : 0.86
2138 [Astar_planner] Segment 5 reached in 18 steps, dist=1.92, axis_error
      =[1.8304291 0.58000183 0.00479889], reason=running
2139 [Astar_planner] Segment 6 starts from actual pos [-169.17485  128.
      41658 -20.290968] (planned waypoint was [-171.    129.    -20.
      29513])
2140 [Astar_planner] Segment 6: [-169.17485  128.41658 -20.290968
      ] -> [-171.    114.    -24.699924]
2141 [Step  11 | Total    0] Target: (-171.0, 114.0, -24.7) | Pos: (-171.7,
      114.5, -23.7) | Vel: ( 0.5, -1.2,  0.8) | Acc: ( 0.8, -2.0,  0.2) | Rwd:
      607.35 -> Episode End: ARRIVED!
2142
2143 [Astar_planner] Segment 6 reached in 11 steps, dist=1.33, axis_error
      =[0.69633484 0.4619751 1.0357018 ], reason=arrived
2144 [Astar_planner] Segment 7 starts from actual pos [-171.69221  114.
      45165 -23.657412] (planned waypoint was [-171.    114.    -24.
      699924])
2145 [Astar_planner] Segment 7: [-171.69221  114.45165 -23.657412
      ] -> [-171.    100.    -21.668558]
2146 [Step  12 | Total    0] Target: (-171.0, 100.0, -21.7) | Pos: (-172.8,
      99.7, -20.4) | Vel: ( 0.7, -0.9,  0.7) | Acc: ( 0.7, -2.0,  0.3) | Rwd: 6
      .84
2147 [Astar_planner] Segment 7 reached in 12 steps, dist=2.20, axis_error
      =[1.7595673 0.29972076 1.2821293 ], reason=running
2148 [Astar_planner] Segment 8 starts from actual pos [-172.75339  99.
      69264 -20.380123] (planned waypoint was [-171.    100.    -21.
      668558])
2149 [Astar_planner] Segment 8: [-172.75339  99.69264 -20.380123
      ] -> [-172.    86.    -24.966585]
2150 [Step  180 | Total    0] Target: (-172.0, 86.0, -25.0) | Pos: (-175.1,
      75.3, -11.3) | Vel: ( -0.0, -0.0,  0.0) | Acc: ( -0.0,  0.0, -0.3) | Rwd: -
      1.14
2151 [Astar_planner] Segment 8 timeout, dist=17.63
2152 [Astar_planner] Stop after failed segment 8.
2153 Task 31 result: failed (Failed at segment 8: segment_timeout (8/30
      reached). Shield interventions=0, emergencies=0, recoveries=0,
      recovery_steps=0.)
2154
2155 --- Task 32/38 ---

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2156 From [-111.5 118.98 -4. ] to [103.4 79.58 -9. ]
2157 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
2158 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
2159 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
2160 Path planned via occupancy (Points: 26, Dist: 315.2m). Local target z
/alt: 0:-4.01(4.01m), 1:-20.40(20.40m), 2:-22.41(22.41m), 3:-20.29(20
.29m), 4:-23.04(23.04m), 5:-24.07(24.07m), 6:-24.43(24.43m), 7:-23.
06(23.06m), 8:-23.77(23.77m), 9:-20.47(20.47m), 10:-23.85(23.85m
), 11:-21.81(21.81m), 12:-21.39(21.39m), 13:-23.94(23.94m), 14:-23.
41(23.41m), 15:-20.74(20.74m), 16:-22.06(22.06m), 17:-23.27(23.
27m), 18:-24.53(24.53m), 19:-20.73(20.73m), 20:-21.43(21.43m), 21
:-22.37(22.37m), 22:-24.27(24.27m), 23:-24.87(24.87m), 24:-23.91(
23.91m), 25:-8.99(8.99m)
2161 [Astar_planner] Segment 0: [-112.      119.      -4.0114517] -> [-
103.      128.      -20.402073]
2162 [Astar_planner] Placing first segment start at [-112.      119.      -4
.0114517]
2163 [Astar_planner] Initial climb to 20.0m.
2164 [Step 16 | Total    0] Target: (-103.0, 128.0, -20.4) | Pos: (-104.9,
128.9, -18.8) | Vel: ( 1.2,  0.4, -1.4) | Acc: ( -1.9, -0.7,  0.3) | Rwd
: 3.45
2165 [Astar_planner] Segment 0 reached in 16 steps, dist=2.65, axis_error
=[1.900917 0.9430237 1.5947838], reason=running
2166 [Astar_planner] Segment 1 starts from actual pos [-104.88642 128.
9476 -18.823845] (planned waypoint was [-103.      128.      -20.
402073])
2167 [Astar_planner] Segment 1: [-104.88642 128.9476 -18.823845
]-> [-100.      139.      -22.411184]
2168 [Step 13 | Total    0] Target: (-100.0, 139.0, -22.4) | Pos: (-99.9,
138.3, -20.8) | Vel: ( 0.6,  1.6, -0.8) | Acc: ( 1.9, -1.0,  2.0) | Rwd:
605.22 -> Episode End: ARRIVED!
2169
2170 [Astar_planner] Segment 1 reached in 13 steps, dist=1.74, axis_error
=[0.07315063 0.7167816 1.5873642 ], reason=arrived
2171 [Astar_planner] Segment 2 starts from actual pos [-99.92328 138.
29301 -20.828287] (planned waypoint was [-100.      139.      -22.
411184])
2172 [Astar_planner] Segment 2: [-99.92328 138.29301 -20.828287] -> [-
99.      151.      -20.290077]
2173 [Step 11 | Total    0] Target: (-99.0, 151.0, -20.3) | Pos: (-97.9, 149
.5, -18.7) | Vel: ( 0.1,  1.2, -2.2) | Acc: ( 0.6, -1.4, -0.2) | Rwd: 2.
99
2174 [Astar_planner] Segment 2 reached in 11 steps, dist=2.44, axis_error
=[1.0514984 1.5356293 1.5734043], reason=running
2175 [Astar_planner] Segment 3 starts from actual pos [-97.94818 149.

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2175 47179 -18.729942] (planned waypoint was [-99.    151.    -20.
290077])
2176 [Astar_planner] Segment 3: [-97.94818  149.47179 -18.729942] -> [-
90.    157.    -23.040035]
2177 [Step  13 | Total    0] Target: (-90.0, 157.0, -23.0) | Pos: (-91.2, 156
.0, -21.7) | Vel: ( 0.6,  0.4, -0.8) | Acc: ( -1.2, -0.3,  2.0) | Rwd: 3.
06
2178 [Astar_planner] Segment 3 reached in 13 steps, dist=2.06, axis_error
=[1.1584473 1.0166473 1.3687611], reason=running
2179 [Astar_planner] Segment 4 starts from actual pos [-91.15316  155.
9867  -21.678055] (planned waypoint was [-90.    157.    -23.
040035])
2180 [Astar_planner] Segment 4: [-91.15316  155.9867  -21.678055] -> [-
79.    156.    -24.072628]
2181 [Step  13 | Total    0] Target: (-79.0, 156.0, -24.1) | Pos: (-78.2, 155
.9, -22.8) | Vel: ( 0.6, -0.2, -0.3) | Acc: ( -2.0,  1.0,  2.0) | Rwd: 602.
76 -> Episode End: ARRIVED!
2182
2183 [Astar_planner] Segment 4 reached in 13 steps, dist=1.52, axis_error
=[0.8428421 0.1003418 1.2569504], reason=arrived
2184 [Astar_planner] Segment 5 starts from actual pos [-78.15199  155.
89787  -22.818378] (planned waypoint was [-79.    156.    -24.
072628])
2185 [Astar_planner] Segment 5: [-78.15199  155.89787  -22.818378] -> [-
66.    156.    -24.432877]
2186 [Step  12 | Total    0] Target: (-66.0, 156.0, -24.4) | Pos: (-64.6, 156
.4, -23.0) | Vel: ( 1.2,  0.6, -0.9) | Acc: ( -2.0,  0.4,  2.0) | Rwd: 601.
52 -> Episode End: ARRIVED!
2187
2188 [Astar_planner] Segment 5 reached in 12 steps, dist=2.05, axis_error
=[1.4411545 0.36141968 1.411005 ], reason=arrived
2189 [Astar_planner] Segment 6 starts from actual pos [-64.55188  156.
36484  -23.027412] (planned waypoint was [-66.    156.    -24.
432877])
2190 [Astar_planner] Segment 6: [-64.55188  156.36484  -23.027412] -> [-
54.    156.    -23.064884]
2191 [Step  23 | Total    0] Target: (-54.0, 156.0, -23.1) | Pos: (-53.6, 156
.2, -21.6) | Vel: ( 1.3,  0.6, -0.7) | Acc: ( -1.5,  0.6,  1.5) | Rwd: 602.
85 -> Episode End: ARRIVED!
2192
2193 [Astar_planner] Segment 6 reached in 23 steps, dist=1.50, axis_error
=[0.40528488 0.24925232 1.4245872 ], reason=arrived
2194 [Astar_planner] Segment 7 starts from actual pos [-53.586674  156.
25304  -21.644262] (planned waypoint was [-54.    156.    -23.
064884])
2195 [Astar_planner] Segment 7: [-53.586674  156.25304  -21.644262] -> [-
42.    157.    -23.772278]
2196 [Step  47 | Total    0] Target: (-42.0, 157.0, -23.8) | Pos: (-43.8, 157

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2196 .9, -22.1) | Vel: ( 2.9, -0.5, 0.0) | Acc: ( 2.0, -2.0, -0.9) | Rwd: 3.
    10
2197 [Astar_planner] Segment 7 reached in 47 steps, dist=2.61, axis_error
    =[1.7804565 0.93444824 1.6668205 ], reason=running
2198 [Astar_planner] Segment 8 starts from actual pos [-43.75467 157.
    93005 -22.105104] (planned waypoint was [-42.    157.    -23.
    772278])
2199 [Astar_planner] Segment 8: [-43.75467 157.93005 -22.105104] -> [-
    31.    154.    -20.465984]
2200 [Step 16 | Total    0] Target: (-31.0, 154.0, -20.5) | Pos: (-29.0, 155
    .5, -20.6) | Vel: ( -0.2, -0.6, 0.4) | Acc: ( -0.4, -2.0, -2.0) | Rwd: 2.
    16
2201 [Astar_planner] Segment 8 reached in 16 steps, dist=2.46, axis_error
    =[1.9864311 1.4528503 0.11131096], reason=running
2202 [Astar_planner] Segment 9 starts from actual pos [-29.015142 155.
    44742 -20.573902] (planned waypoint was [-31.    154.    -20.
    465984])
2203 [Astar_planner] Segment 9: [-29.015142 155.44742 -20.573902] -> [-
    19.    153.    -23.849628]
2204 [Step 32 | Total    0] Target: (-19.0, 153.0, -23.8) | Pos: (-19.2, 151
    .3, -22.9) | Vel: ( -0.7, 0.7, -0.7) | Acc: ( -0.7, -0.5, 1.3) | Rwd: 1.
    57
2205 [Astar_planner] Segment 9 reached in 32 steps, dist=1.97, axis_error
    =[0.24875069 1.7315521 0.9091263 ], reason=running
2206 [Astar_planner] Segment 10 starts from actual pos [-19.252811 151.
    27284 -22.944836] (planned waypoint was [-19.    153.    -23.
    849628])
2207 [Astar_planner] Segment 10: [-19.252811 151.27284 -22.944836
    ] -> [ -7.    153.    -21.80783]
2208 [Step 14 | Total    0] Target: ( -7.0, 153.0, -21.8) | Pos: ( -8.3, 153.
    1, -20.9) | Vel: ( 1.4, -1.2, -0.3) | Acc: ( -0.2, -2.0, -0.4) | Rwd: 607.
    29 -> Episode End: ARRIVED!
2209
2210 [Astar_planner] Segment 10 reached in 14 steps, dist=1.62,
    axis_error=[1.3326597 0.08992004 0.90938187], reason=arrived
2211 [Astar_planner] Segment 11 starts from actual pos [ -8.32407 153.
    08289 -20.900068] (planned waypoint was [ -7.    153.    -21.
    80783])
2212 [Astar_planner] Segment 11: [ -8.32407 153.08289 -20.900068
    ] -> [ 5.    152.    -21.394756]
2213 [Step 13 | Total    0] Target: ( 5.0, 152.0, -21.4) | Pos: ( 3.3, 152.6
    , -20.3) | Vel: ( 2.0, 0.1, -1.4) | Acc: ( 0.0, -1.0, 0.7) | Rwd: 3.84
2214 [Astar_planner] Segment 11 reached in 13 steps, dist=2.14,
    axis_error=[1.740411 0.59124756 1.0999756 ], reason=running
2215 [Astar_planner] Segment 12 starts from actual pos [ 3.2713463 152.
    59192 -20.303288 ] (planned waypoint was [ 5.    152.    -21.
    394756])
2216 [Astar_planner] Segment 12: [ 3.2713463 152.59192 -20.303288

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2216 ] -> [ 16.    149.    -23.942505]
2217 [Step 12 | Total    0] Target: ( 16.0, 149.0, -23.9) | Pos: ( 15.4, 148.
    1, -22.3) | Vel: ( 1.7, -0.7, -1.1) | Acc: ( -1.9, 1.2, 2.0) | Rwd: 604.
    43 -> Episode End: ARRIVED!
2218
2219 [Astar_planner] Segment 12 reached in 12 steps, dist=2.02,
    axis_error=[0.62756634 0.9293213 1.6843014 ], reason=arrived
2220 [Astar_planner] Segment 13 starts from actual pos [ 15.382348 148.
    06659 -22.264446] (planned waypoint was [ 16.    149.    -23.
    942505])
2221 [Astar_planner] Segment 13: [ 15.382348 148.06659 -22.264446
    ] -> [ 25.    141.    -23.40706]
2222 [Step 9 | Total    0] Target: ( 25.0, 141.0, -23.4) | Pos: ( 24.3, 142.
    7, -21.8) | Vel: ( 1.9, -1.3, -0.3) | Acc: ( 0.2, 0.2, -1.3) | Rwd: 4.
    34
2223 [Astar_planner] Segment 13 reached in 9 steps, dist=2.44, axis_error
    =[0.687418 1.7210541 1.5828228], reason=running
2224 [Astar_planner] Segment 14 starts from actual pos [ 24.335281 142.
    7058 -21.82801 ] (planned waypoint was [ 25.    141.    -23.40706
    ])
2225 [Astar_planner] Segment 14: [ 24.335281 142.7058 -21.82801 ] -> [
    35.    136.    -20.743374]
2226 [Step 19 | Total    0] Target: ( 35.0, 136.0, -20.7) | Pos: ( 34.5, 134.
    0, -19.6) | Vel: ( 0.3, 0.2, -0.9) | Acc: ( 0.6, -0.7, 0.0) | Rwd: -0.
    52
2227 [Astar_planner] Segment 14 reached in 19 steps, dist=2.36,
    axis_error=[0.53871536 1.9718323 1.1860638 ], reason=running
2228 [Astar_planner] Segment 15 starts from actual pos [ 34.463345 134.
    02933 -19.562586] (planned waypoint was [ 35.    136.    -20.
    743374])
2229 [Astar_planner] Segment 15: [ 34.463345 134.02933 -19.562586
    ] -> [ 47.    136.    -22.056505]
2230 [Step 17 | Total    0] Target: ( 47.0, 136.0, -22.1) | Pos: ( 46.3, 134.
    9, -20.1) | Vel: ( 2.0, 0.3, -2.0) | Acc: ( 1.5, -1.9, -0.5) | Rwd: 602.
    99 -> Episode End: ARRIVED!
2231
2232 [Astar_planner] Segment 15 reached in 17 steps, dist=2.31,
    axis_error=[0.74274826 1.0656128 1.9084415 ], reason=arrived
2233 [Astar_planner] Segment 16 starts from actual pos [ 46.26918 134.
    93611 -20.159887] (planned waypoint was [ 47.    136.    -22.
    056505])
2234 [Astar_planner] Segment 16: [ 46.26918 134.93611 -20.159887
    ] -> [ 60.    136.    -23.27237]
2235 [Step 12 | Total    0] Target: ( 60.0, 136.0, -23.3) | Pos: ( 60.9, 135.
    3, -21.4) | Vel: ( 1.7, 1.1, -1.4) | Acc: ( -0.9, 0.2, 2.0) | Rwd: 603.
    35 -> Episode End: ARRIVED!
2236
2237 [Astar_planner] Segment 16 reached in 12 steps, dist=2.23,

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2237 axis_error=[0.9196167 0.7041168 1.9026947], reason=arrived
2238 [Astar_planner] Segment 17 starts from actual pos [ 60.934784 135.
30554 -21.381924] (planned waypoint was [ 60.    136.    -23.
27237])
2239 [Astar_planner] Segment 17: [ 60.934784 135.30554 -21.381924
]-> [ 71.    135.    -24.531542]
2240 [Step 20 | Total    0] Target: ( 71.0, 135.0, -24.5) | Pos: ( 70.4, 133.
3, -23.2) | Vel: ( 0.2, 0.7, -0.6) | Acc: ( -1.6, 0.1, 2.0) | Rwd: 3.56

2241 [Astar_planner] Segment 17 reached in 20 steps, dist=2.29,
axis_error=[0.6057892 1.7329712 1.3684502], reason=running
2242 [Astar_planner] Segment 18 starts from actual pos [ 70.39529 133.
2713 -23.16674] (planned waypoint was [ 71.    135.    -24.
531542])
2243 [Astar_planner] Segment 18: [ 70.39529 133.2713 -23.16674] -> [ 84
.    135.    -20.725513]
2244 [Step 37 | Total    0] Target: ( 84.0, 135.0, -20.7) | Pos: ( 82.1, 133.
3, -19.2) | Vel: ( 0.6, -0.0, -2.0) | Acc: ( 0.1, -1.2, 0.5) | Rwd: 1.
85
2245 [Astar_planner] Segment 18 reached in 37 steps, dist=2.98,
axis_error=[1.9207153 1.6537628 1.561491 ], reason=running
2246 [Astar_planner] Segment 19 starts from actual pos [ 82.0829 133.
346 -19.176086] (planned waypoint was [ 84.    135.    -20.
725513])
2247 [Astar_planner] Segment 19: [ 82.0829 133.346 -19.176086] -> [
96.    135.    -21.432337]
2248 [Step 25 | Total    0] Target: ( 96.0, 135.0, -21.4) | Pos: ( 94.3, 136.
5, -20.5) | Vel: ( 0.8, 1.0, -0.3) | Acc: ( -2.0, 1.9, 2.0) | Rwd: 1.20

2249 [Astar_planner] Segment 19 reached in 25 steps, dist=2.45,
axis_error=[1.7258835 1.4647217 0.9289799], reason=running
2250 [Astar_planner] Segment 20 starts from actual pos [ 94.27914 136.
4708 -20.504984] (planned waypoint was [ 96.    135.    -21.
432337])
2251 [Astar_planner] Segment 20: [ 94.27914 136.4708 -20.504984] -> [
103.    127.    -22.367544]
2252 [Step 19 | Total    0] Target: (103.0, 127.0, -22.4) | Pos: (103.4,
128.4, -21.8) | Vel: ( 1.0, -1.3, -0.5) | Acc: ( 0.3, -0.1, 0.0) | Rwd:
603.41 -> Episode End: ARRIVED!
2253
2254 [Astar_planner] Segment 20 reached in 19 steps, dist=1.52,
axis_error=[0.40934753 1.3500824 0.5761471 ], reason=arrived
2255 [Astar_planner] Segment 21 starts from actual pos [103.41818 128.
33804 -21.796001] (planned waypoint was [103.    127.    -22.
367544])
2256 [Astar_planner] Segment 21: [103.41818 128.33804 -21.796001
]-> [104.    116.    -24.273302]
2257 [Step 37 | Total    0] Target: (104.0, 116.0, -24.3) | Pos: (103.5,

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2257 117.2, -23.3) | Vel: ( 0.0, -1.6, 0.2) | Acc: ( -0.0, -2.0, -0.2) | Rwd:
607.32 -> Episode End: ARRIVED!
2258
2259 [Astar_planner] Segment 21 reached in 37 steps, dist=1.63,
axis_error=[0.46300507 1.2155304 0.97737694], reason=arrived
2260 [Astar_planner] Segment 22 starts from actual pos [103.53718 117.
20148 -23.293556] (planned waypoint was [104. 116. -24.
273302])
2261 [Astar_planner] Segment 22: [103.53718 117.20148 -23.293556
] -> [104. 103. -24.871792]
2262 [Step 15 | Total 0] Target: (104.0, 103.0, -24.9) | Pos: (105.1,
104.1, -23.4) | Vel: ( 0.4, -1.5, -1.1) | Acc: ( -0.3, 0.6, -1.6) | Rwd
: 5.71
2263 [Astar_planner] Segment 22 reached in 15 steps, dist=2.12,
axis_error=[1.0687714 1.0603867 1.497385 ], reason=running
2264 [Astar_planner] Segment 23 starts from actual pos [105.0732 104.
04275 -23.387012] (planned waypoint was [104. 103. -24.
871792])
2265 [Astar_planner] Segment 23: [105.0732 104.04275 -23.387012] -> [
105. 91. -23.911974]
2266 [Step 57 | Total 0] Target: (105.0, 91.0, -23.9) | Pos: (106.5, 89.
9, -23.1) | Vel: ( -0.8, 0.6, -0.3) | Acc: ( -0.3, -0.4, 0.7) | Rwd: 4.
33
2267 [Astar_planner] Segment 23 reached in 57 steps, dist=2.00,
axis_error=[1.4702301 1.0978699 0.796402 ], reason=running
2268 [Astar_planner] Segment 24 starts from actual pos [106.465576 89.
90598 -23.117493] (planned waypoint was [105. 91. -23.
911974])
2269 [Astar_planner] Segment 24: [106.465576 89.90598 -23.117493
] -> [103. 80. -8.989268]
2270 [Step 29 | Total 0] Target: (103.0, 80.0, -9.0) | Pos: (104.0, 80.
1, -8.8) | Vel: ( -0.3, -0.2, -0.0) | Acc: ( -0.3, -1.3, 1.3) | Rwd: 602.
37 -> Episode End: ARRIVED!
2271
2272 [Astar_planner] Segment 24 reached in 29 steps, dist=1.00,
axis_error=[0.97382355 0.09038544 0.20283318], reason=arrived
2273 Task 32 result: success (Success: Reached destination (25/25
segments). Shield interventions=75, emergencies=0, recoveries=12,
recovery_steps=72.)
2274
2275 --- Task 33/38 ---
2276 From [63.1 4.08 -7.7 ] to [-111.5 118.98 -4. ]
2277 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
2278 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
2279 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5

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2280 Path planned via occupancy (Points: 23, Dist: 295.8m). Local target z
    /alt: 0:-7.70(7.70m), 1:-21.12(21.12m), 2:-24.82(24.82m), 3:-21.06(21
    .06m), 4:-23.03(23.03m), 5:-23.71(23.71m), 6:-21.97(21.97m), 7:-22.
    79(22.79m), 8:-24.22(24.22m), 9:-20.70(20.70m), 10:-22.13(22.13m
    ), 11:-22.78(22.78m), 12:-21.70(21.70m), 13:-21.77(21.77m), 14:-24.
    57(24.57m), 15:-23.75(23.75m), 16:-23.07(23.07m), 17:-21.18(21.
    18m), 18:-21.65(21.65m), 19:-20.73(20.73m), 20:-23.53(23.53m), 21
    :-20.94(20.94m), 22:-4.00(4.00m)
2281 [Astar_planner] Segment 0: [63.    4.    -7.697733] -> [ 50.    4
    .    -21.11875]
2282 [Astar_planner] Placing first segment start at [63.    4.    -7.
    697733]
2283 [Astar_planner] Initial climb to 20.0m.
2284 [Step 28 | Total    0] Target: ( 50.0,  4.0, -21.1) | Pos: ( 51.5,  3.2
    , -20.1) | Vel: ( -0.3,  0.3,  0.0) | Acc: (  1.1, -1.4, -1.4) | Rwd:  4.55
2285 [Astar_planner] Segment 0 reached in 28 steps, dist=1.93, axis_error
    =[1.4518738 0.7816844 1.0043716], reason=running
2286 [Astar_planner] Segment 1 starts from actual pos [ 51.449547  3.
    2212527 -20.114155 ] (planned waypoint was [ 50.    4.    -21.
    11875])
2287 [Astar_planner] Segment 1: [ 51.449547  3.2212527 -20.114155
    ] -> [ 37.    4.    -24.818613]
2288 [Step 38 | Total    0] Target: ( 37.0,  4.0, -24.8) | Pos: ( 39.0,  3.3
    , -23.3) | Vel: ( -0.1, -0.2,  0.5) | Acc: (  0.2, -2.0, -1.4) | Rwd: -0.02
2289 [Astar_planner] Segment 1 reached in 38 steps, dist=2.60, axis_error
    =[1.9759636 0.71257734 1.5260601 ], reason=running
2290 [Astar_planner] Segment 2 starts from actual pos [ 38.975067  3.
    2857218 -23.288347 ] (planned waypoint was [ 37.    4.    -24.
    818613])
2291 [Astar_planner] Segment 2: [ 38.975067  3.2857218 -23.288347
    ] -> [ 24.    4.    -21.064816]
2292 [Step 30 | Total    0] Target: ( 24.0,  4.0, -21.1) | Pos: ( 25.4,  2.6
    , -19.7) | Vel: ( -0.2,  0.6, -0.7) | Acc: (  0.9,  0.1, -0.8) | Rwd:  4.68
2293 [Astar_planner] Segment 2 reached in 30 steps, dist=2.39, axis_error
    =[1.3726788 1.3611081 1.4039822], reason=running
2294 [Astar_planner] Segment 3 starts from actual pos [ 25.37075  2.
    6461825 -19.668756 ] (planned waypoint was [ 24.    4.    -21.
    064816])
2295 [Astar_planner] Segment 3: [ 25.37075  2.6461825 -19.668756
    ] -> [ 11.    4.    -23.032827]
2296 [Step 28 | Total    0] Target: ( 11.0,  4.0, -23.0) | Pos: ( 12.0,  2.3
    , -22.1) | Vel: (  0.1,  0.4,  0.4) | Acc: (  0.7,  2.0, -0.1) | Rwd:  1.12
2297 [Astar_planner] Segment 3 reached in 28 steps, dist=2.22, axis_error
    =[1.0430155 1.7449186 0.89904404], reason=running
2298 [Astar_planner] Segment 4 starts from actual pos [ 12.0441885  2.
    2589343 -22.130299 ] (planned waypoint was [ 11.    4.    -23.
    032827])
2299 [Astar_planner] Segment 4: [ 12.0441885  2.2589343 -22.130299

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2299 ] -> [-2.    4.    -23.711111]
2300 [Step 23 | Total    0] Target: ( -2.0,  4.0, -23.7) | Pos: ( -2.1,  4.7, -
    21.9) | Vel: ( -0.9,  0.3,  0.1) | Acc: ( -1.3, -1.3,  1.1) | Rwd: 602.76
    -> Episode End: ARRIVED!
2301
2302 [Astar_planner] Segment 4 reached in 23 steps, dist=1.94, axis_error
    =[0.09448624 0.65463877 1.8250828 ], reason=arrived
2303 [Astar_planner] Segment 5 starts from actual pos [-2.0996017  4.
    6564827 -21.88543 ] (planned waypoint was [-2.    4.    -23.
    711111])
2304 [Astar_planner] Segment 5: [-2.0996017  4.6564827 -21.88543
    ] -> [-12.    -2.    -21.968105]
2305 [Step 22 | Total    0] Target: (-12.0, -2.0, -22.0) | Pos: (-10.1, -2.6
    , -21.2) | Vel: ( -0.2, -0.2,  0.4) | Acc: (  0.5, -1.0,  0.9) | Rwd: 2.16
2306 [Astar_planner] Segment 5 reached in 22 steps, dist=2.18, axis_error
    =[1.9466391 0.6376586 0.75182915], reason=running
2307 [Astar_planner] Segment 6 starts from actual pos [-10.054815 -2.
    6386478 -21.213757 ] (planned waypoint was [-12.    -2.    -21.
    968105])
2308 [Astar_planner] Segment 6: [-10.054815 -2.6386478 -21.213757
    ] -> [-24.    1.    -22.794104]
2309 [Step 25 | Total    0] Target: (-24.0,  1.0, -22.8) | Pos: (-22.3,  1.3
    , -21.5) | Vel: ( -0.5,  0.3,  0.8) | Acc: ( -0.8,  0.1,  1.0) | Rwd: 2.44
2310 [Astar_planner] Segment 6 reached in 25 steps, dist=2.14, axis_error
    =[1.6875324 0.29725647 1.2807522 ], reason=running
2311 [Astar_planner] Segment 7 starts from actual pos [-22.317028  1.
    2999597 -21.506416 ] (planned waypoint was [-24.    1.    -22.
    794104])
2312 [Astar_planner] Segment 7: [-22.317028  1.2999597 -21.506416
    ] -> [-33.    10.    -24.218187]
2313 [Step 48 | Total    0] Target: (-33.0, 10.0, -24.2) | Pos: (-32.3,  9.5
    , -22.5) | Vel: ( -0.6, -1.0,  0.3) | Acc: (  0.5, -1.6, -1.3) | Rwd: 603.
    21 -> Episode End: ARRIVED!
2314
2315 [Astar_planner] Segment 7 reached in 48 steps, dist=1.89, axis_error
    =[0.6892319 0.45124626 1.7016048 ], reason=arrived
2316 [Astar_planner] Segment 8 starts from actual pos [-32.315926  9.
    540045 -22.513756] (planned waypoint was [-33.    10.    -24.
    218187])
2317 [Astar_planner] Segment 8: [-32.315926  9.540045 -22.513756] -> [-
    42.    19.    -20.698807]
2318 [Step 17 | Total    0] Target: (-42.0, 19.0, -20.7) | Pos: (-40.5, 17.
    8, -19.1) | Vel: ( -0.3,  0.4, -0.9) | Acc: ( -0.3,  0.9,  1.5) | Rwd: 7.
    92
2319 [Astar_planner] Segment 8 reached in 17 steps, dist=2.48, axis_error
    =[1.4759483 1.1847057 1.601965 ], reason=running
2320 [Astar_planner] Segment 9 starts from actual pos [-40.52732 17.
    820345 -19.107317] (planned waypoint was [-42.    19.    -20.

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2320 698807])
2321 [Astar_planner] Segment 9: [-40.52732 17.820345 -19.107317] -> [-
52. 29. -22.128696]
2322 [Step 29 | Total 0] Target: (-52.0, 29.0, -22.1) | Pos: (-50.2, 29.
1, -21.4) | Vel: ( -0.4, -0.5, 0.5) | Acc: ( 0.0, -1.8, -0.3) | Rwd: 2.
64
2323 [Astar_planner] Segment 9 reached in 29 steps, dist=1.92, axis_error
=[1.7575264 0.0764904 0.76678276], reason=running
2324 [Astar_planner] Segment 10 starts from actual pos [-50.246414 29.
07216 -21.357613] (planned waypoint was [-52. 29. -22.
128696])
2325 [Astar_planner] Segment 10: [-50.246414 29.07216 -21.357613
] -> [-60. 38. -22.780073]
2326 [Step 29 | Total 0] Target: (-60.0, 38.0, -22.8) | Pos: (-58.8, 38.
1, -21.5) | Vel: ( -0.7, 0.4, -0.6) | Acc: ( -0.4, 1.6, 0.3) | Rwd: 605.
28 -> Episode End: ARRIVED!
2327
2328 [Astar_planner] Segment 10 reached in 29 steps, dist=1.75,
axis_error=[1.1819649 0.09389496 1.2908516 ], reason=arrived
2329 [Astar_planner] Segment 11 starts from actual pos [-58.824245 38.
097057 -21.49478 ] (planned waypoint was [-60. 38. -22.
780073])
2330 [Astar_planner] Segment 11: [-58.824245 38.097057 -21.49478
] -> [-60. 51. -21.701656]
2331 [Step 16 | Total 0] Target: (-60.0, 51.0, -21.7) | Pos: (-58.3, 49.
7, -20.2) | Vel: ( 0.1, 1.0, -0.5) | Acc: ( 1.9, 2.0, 1.9) | Rwd: 3.23
2332 [Astar_planner] Segment 11 reached in 16 steps, dist=2.64,
axis_error=[1.6686783 1.33992 1.5384407], reason=running
2333 [Astar_planner] Segment 12 starts from actual pos [-58.32984 49.
671562 -20.169025] (planned waypoint was [-60. 51. -21.
701656])
2334 [Astar_planner] Segment 12: [-58.32984 49.671562 -20.169025
] -> [-53. 61. -21.766441]
2335 [Step 12 | Total 0] Target: (-53.0, 61.0, -21.8) | Pos: (-52.0, 60.
0, -20.5) | Vel: ( -0.3, 0.7, -0.4) | Acc: ( -1.5, -0.3, -0.2) | Rwd: 605.
67 -> Episode End: ARRIVED!
2336
2337 [Astar_planner] Segment 12 reached in 12 steps, dist=1.94,
axis_error=[1.0028763 1.035614 1.2932339], reason=arrived
2338 [Astar_planner] Segment 13 starts from actual pos [-52.00027 59.
972847 -20.478117] (planned waypoint was [-53. 61. -21.
766441])
2339 [Astar_planner] Segment 13: [-52.00027 59.972847 -20.478117
] -> [-43. 68. -24.566906]
2340 [Step 180 | Total 0] Target: (-43.0, 68.0, -24.6) | Pos: (-52.0, 63.
8, -18.2) | Vel: ( 0.2, -0.1, -0.0) | Acc: ( 1.7, -0.8, 0.6) | Rwd: -1.
06

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2341 [Astar_planner] Segment 13 timeout, dist=11.81
2342 [Astar_planner] Stop after failed segment 13.
2343 Task 33 result: failed (Failed at segment 13: segment_timeout (13/22
      reached). Shield interventions=36, emergencies=0, recoveries=6,
      recovery_steps=36.)
2344
2345 --- Task 34/38 ---
2346 From [101.7 130.78 -5.3 ] to [ 78.3 -148.32 -9.4 ]
2347 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
2348 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
2349 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(53, 347), goal=(82, 22), search_radius=36.
2350 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
2351 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(53, 347), goal=(82, 22), search_radius=36.
2352 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.0
2353 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(53, 347), goal=(82, 22), search_radius=36.
2354 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=0.6
2355 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(53, 347), goal=(82, 22), search_radius=36.
2356 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=0.3
2357 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
      clearance penalties.
2358 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(53, 347), goal=(82, 22), search_radius=36.
2359 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4
      %, safety_margin=1.5
2360 [Astar_planner] WARNING: using legacy occupancy fallback; local
      targets may be closer to obstacles.
2361 Path planned via occupancy_legacy (Points: 44, Dist: 421.4m). Local
      target z/alt: 0:-5.63(5.63m), 1:-22.64(22.64m), 2:-22.35(22.35m), 3:-
      21.37(21.37m), 4:-20.66(20.66m), 5:-23.66(23.66m), 6:-23.20(23.
      20m), 7:-24.22(24.22m), 8:-20.70(20.70m), 9:-23.02(23.02m), 10:-21.
      90(21.90m), 11:-20.58(20.58m), 12:-20.81(20.81m), 13:-20.19(20.
      19m), 14:-20.84(20.84m), 15:-21.23(21.23m), 16:-23.71(23.71m), 17
      :-24.55(24.55m), 18:-22.11(22.11m), 19:-23.87(23.87m), 20:-24.08(
      24.08m), 21:-22.52(22.52m), 22:-20.42(20.42m), 23:-22.35(22.35m),
      24:-20.80(20.80m), 25:-20.60(20.60m), 26:-22.70(22.70m), 27:-24.14
      (24.14m), 28:-24.02(24.02m), 29:-21.27(21.27m), 30:-20.37(20.37m
      ), 31:-22.07(22.07m), 32:-22.99(22.99m), 33:-21.16(21.16m), 34:-23.
      08(23.08m), 35:-21.69(21.69m), 36:-23.83(23.83m), 37:-24.19(24.

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2361 19m), 38:-20.81(20.81m), 39:-24.42(24.42m), 40:-21.90(21.90m), 41
:-22.02(22.02m), 42:-24.70(24.70m), 43:-9.40(9.40m)
2362 [Astar_planner] Segment 0: [117.    147.    -5.6264067] -> [129
.    147.    -22.637657]
2363 [Astar_planner] Placing first segment start at [117.    147.    -5.
6264067]
2364 [Astar_planner] Initial climb to 20.0m.
2365 [Step 17 | Total    0] Target: (129.0, 147.0, -22.6) | Pos: (127.6,
147.4, -22.5) | Vel: ( 1.3, -0.5, -0.2) | Acc: ( -1.0, 0.5, 0.3) | Rwd:
602.25 -> Episode End: ARRIVED!
2366
2367 [Astar_planner] Segment 0 reached in 17 steps, dist=1.47, axis_error
=[1.3978806 0.4245453 0.14356422], reason=arrived
2368 [Astar_planner] Segment 1 starts from actual pos [127.61019 147.
42125 -22.495508] (planned waypoint was [129.    147.    -22.
637657])
2369 [Astar_planner] Segment 1: [127.61019 147.42125 -22.495508] -> [
131.    144.    -22.351452]
2370 [Step 3 | Total    0] Target: (131.0, 144.0, -22.4) | Pos: (129.6, 145
.9, -22.1) | Vel: ( 1.0, -0.8, -0.3) | Acc: ( 0.7, 0.8, -0.3) | Rwd: 1.
96
2371 [Astar_planner] Segment 1 reached in 3 steps, dist=2.36, axis_error
=[1.4333801 1.8577881 0.23057556], reason=running
2372 [Astar_planner] Segment 2 starts from actual pos [129.57536 145.
85028 -22.123451] (planned waypoint was [131.    144.    -22.
351452])
2373 [Astar_planner] Segment 2: [129.57536 145.85028 -22.123451] -> [
131.    132.    -21.366829]
2374 [Step 14 | Total    0] Target: (131.0, 132.0, -21.4) | Pos: (129.4,
133.4, -21.2) | Vel: ( 0.5, -1.8, 0.0) | Acc: ( 0.3, -1.1, 0.4) | Rwd:
3.04
2375 [Astar_planner] Segment 2 reached in 14 steps, dist=2.14, axis_error
=[1.5742188 1.4352417 0.17351723], reason=running
2376 [Astar_planner] Segment 3 starts from actual pos [129.43005 133.
41902 -21.193247] (planned waypoint was [131.    132.    -21.
366829])
2377 [Astar_planner] Segment 3: [129.43005 133.41902 -21.193247] -> [
141.    120.    -20.661228]
2378 [Step 21 | Total    0] Target: (141.0, 120.0, -20.7) | Pos: (139.2,
120.9, -19.3) | Vel: ( 1.3, -1.8, -0.2) | Acc: ( -0.0, -0.6, -0.3) | Rwd
: 2.84
2379 [Astar_planner] Segment 3 reached in 21 steps, dist=2.43, axis_error
=[1.7936249 0.8611908 1.3938026], reason=running
2380 [Astar_planner] Segment 4 starts from actual pos [139.21394 120.
8506 -19.268536] (planned waypoint was [141.    120.    -20.
661228])
2381 [Astar_planner] Segment 4: [139.21394 120.8506 -19.268536] -> [
152.    109.    -23.65846]

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2382 [Step 49 | Total    0] Target: (152.0, 109.0, -23.7) | Pos: (150.5,
    107.8, -22.0) | Vel: ( 0.9,  0.5, -1.9) | Acc: ( -1.9, -0.5,  0.7) | Rwd
    : 3.93
2383 [Astar_planner] Segment 4 reached in 49 steps, dist=2.54, axis_error
    =[1.4834137 1.2327576 1.6493168], reason=running
2384 [Astar_planner] Segment 5 starts from actual pos [150.52194 107.
    77028 -22.020596] (planned waypoint was [152.    109.    -23.
    65846])
2385 [Astar_planner] Segment 5: [150.52194 107.77028 -22.020596] -> [
    161.    96.    -23.195728]
2386 [Step 18 | Total    0] Target: (161.0, 96.0, -23.2) | Pos: (162.9, 94.
    1, -21.6) | Vel: ( -0.1,  0.4, -0.1) | Acc: ( 0.2,  2.0,  2.0) | Rwd: -0.
    65
2387 [Astar_planner] Segment 5 reached in 18 steps, dist=3.14, axis_error
    =[1.9307251 1.8827896 1.6081371], reason=running
2388 [Astar_planner] Segment 6 starts from actual pos [162.92972 94.
    12094 -21.588709] (planned waypoint was [161.    96.    -23.
    195728])
2389 [Astar_planner] Segment 6: [162.92972 94.12094 -21.588709] -> [
    163.    93.    -24.224628]
2390 [Step 180 | Total    0] Target: (163.0, 93.0, -24.2) | Pos: (160.7, 91
    .9, -20.7) | Vel: ( -0.0,  0.2,  0.5) | Acc: ( -0.1,  1.9,  1.9) | Rwd: -0.
    17
2391 [Astar_planner] Segment 6 timeout, dist=4.35
2392 [Astar_planner] Stop after failed segment 6.
2393 Task 34 result: failed (Failed at segment 6: segment_timeout (6/43
    reached). Shield interventions=48, emergencies=0, recoveries=8,
    recovery_steps=48.)
2394
2395 --- Task 35/38 ---
2396 From [ 72.7 -198.02 -13.3 ] to [150.3  37.28 -10.4 ]
2397 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
    =[-200.0, 189.0], occupied=72869 cells (53.4%)
2398 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
    cells (53.4%)
2399 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(97, 2), goal=(20, 237), search_radius=36.
2400 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
2401 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(97, 2), goal=(20, 237), search_radius=36.
2402 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.0
2403 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(97, 2), goal=(20, 237), search_radius=36.
2404 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=0.6
2405 [Astar_planner] Endpoint nearest-free cells are disconnected; using

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2405 reachable cells start=(97, 2), goal=(20, 237), search_radius=36.
2406 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=0.3
2407 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
      clearance penalties.
2408 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4
      %, safety_margin=1.5
2409 [Astar_planner] WARNING: using legacy occupancy fallback; local
      targets may be closer to obstacles.
2410 Path planned via occupancy_legacy (Points: 23, Dist: 284.5m). Local
      target z/alt: 0:-13.30(13.30m), 1:-21.79(21.79m), 2:-23.29(23.29m), 3
      :-20.55(20.55m), 4:-23.61(23.61m), 5:-21.51(21.51m), 6:-20.25(20.
      25m), 7:-23.09(23.09m), 8:-22.02(22.02m), 9:-24.63(24.63m), 10:-24.
      05(24.05m), 11:-24.99(24.99m), 12:-24.87(24.87m), 13:-24.90(24.
      90m), 14:-20.25(20.25m), 15:-24.12(24.12m), 16:-22.30(22.30m), 17
      :-22.56(22.56m), 18:-24.60(24.60m), 19:-24.80(24.80m), 20:-22.55(
      22.55m), 21:-24.29(24.29m), 22:-10.40(10.40m)
2411 [Astar_planner] Segment 0: [ 73.    -198.    -13.296481] -> [ 86
      .    -198.    -21.792572]
2412 [Astar_planner] Placing first segment start at [ 73.    -198.    -13.
      296481]
2413 [Astar_planner] Initial climb to 20.0m.
2414 [Step 16 | Total    0] Target: ( 86.0, -198.0, -21.8) | Pos: ( 84.6, -
      199.3, -20.9) | Vel: ( 1.2,  0.0,  0.1) | Acc: ( -1.1, -1.4,  0.1) | Rwd:
      0.87
2415 [Astar_planner] Segment 0 reached in 16 steps, dist=2.07, axis_error
      =[1.3777161 1.2588348 0.90664864], reason=running
2416 [Astar_planner] Segment 1 starts from actual pos [ 84.629295 -199.
      25883 -20.885422] (planned waypoint was [ 86.    -198.    -21.
      792572])
2417 [Astar_planner] Segment 1: [ 84.629295 -199.25883 -20.885422
      ] -> [ 96.    -188.    -23.285332]
2418 [Step 21 | Total    0] Target: ( 96.0, -188.0, -23.3) | Pos: ( 94.6, -
      189.4, -21.9) | Vel: ( 0.8,  0.1, -1.0) | Acc: ( -1.9,  0.7,  2.0) | Rwd:
      3.01
2419 [Astar_planner] Segment 1 reached in 21 steps, dist=2.46, axis_error
      =[1.441597 1.4025726 1.4079933], reason=running
2420 [Astar_planner] Segment 2 starts from actual pos [ 94.565315 -189.
      40143 -21.885836] (planned waypoint was [ 96.    -188.    -23.
      285332])
2421 [Astar_planner] Segment 2: [ 94.565315 -189.40143 -21.885836
      ] -> [ 103.    -186.    -20.554028]
2422 [Step  9 | Total    0] Target: (103.0, -186.0, -20.6) | Pos: (101.8, -
      187.5, -19.7) | Vel: ( 1.5,  0.4, -0.9) | Acc: ( -2.0,  1.1,  1.0) | Rwd:
      2.93
2423 [Astar_planner] Segment 2 reached in 9 steps, dist=2.13, axis_error
      =[1.1806259 1.5212402 0.90105057], reason=running
2424 [Astar_planner] Segment 3 starts from actual pos [ 101.82842 -187.

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2424 51868 -19.658188] (planned waypoint was [ 103.    -186.    -20.
554028])
2425 [Astar_planner] Segment 3: [ 101.82842 -187.51868 -19.658188
]-> [ 107.    -180.    -23.611217]
2426 [Step 36 | Total    0] Target: (107.0, -180.0, -23.6) | Pos: (108.1, -
181.1, -21.8) | Vel: ( -0.0,  0.0,  0.9) | Acc: ( 0.3, -2.0, -0.4) | Rwd
: 1.38
2427 [Astar_planner] Segment 3 reached in 36 steps, dist=2.34, axis_error
=[1.1199188 1.0648651 1.7635803], reason=running
2428 [Astar_planner] Segment 4 starts from actual pos [ 108.119736 -181.
0645    -21.839674] (planned waypoint was [ 107.    -180.    -23.
611217])
2429 [Astar_planner] Segment 4: [ 108.119736 -181.0645    -21.839674
]-> [ 111.    -173.    -21.50803]
2430 [Step 16 | Total    0] Target: (111.0, -173.0, -21.5) | Pos: (110.0, -
171.4, -20.1) | Vel: ( 0.3, -0.1, -0.8) | Acc: ( 0.0, -1.0, -0.2) | Rwd
: 4.31
2431 [Astar_planner] Segment 4 reached in 16 steps, dist=2.38, axis_error
=[1.0157928 1.6255646 1.4103851], reason=running
2432 [Astar_planner] Segment 5 starts from actual pos [ 109.98749 -171.
37505    -20.106466] (planned waypoint was [ 111.    -173.    -21.
50803])
2433 [Astar_planner] Segment 5: [ 109.98749 -171.37505    -20.106466
]-> [ 115.    -166.    -20.247847]
2434 [Step 7 | Total    0] Target: (115.0, -166.0, -20.2) | Pos: (113.0, -
166.9, -18.7) | Vel: ( 1.6,  1.7,  0.1) | Acc: ( 0.0,  0.7,  0.4) | Rwd:
1.57
2435 [Astar_planner] Segment 5 reached in 7 steps, dist=2.66, axis_error
=[1.9641037 0.85510254 1.5749397 ], reason=running
2436 [Astar_planner] Segment 6 starts from actual pos [ 113.0454    -166.
84491    -18.672272] (planned waypoint was [ 115.    -166.    -20.
247847])
2437 [Astar_planner] Segment 6: [ 113.0454    -166.84491    -18.672272
]-> [ 119.    -159.    -23.089792]
2438 [Step 20 | Total    0] Target: (119.0, -159.0, -23.1) | Pos: (120.8, -
157.4, -22.0) | Vel: ( -0.0, -1.1,  0.6) | Acc: ( -0.5, -0.8,  1.1) | Rwd
: -1.11
2439 [Astar_planner] Segment 6 reached in 20 steps, dist=2.69, axis_error
=[1.841835 1.6025085 1.1271725], reason=running
2440 [Astar_planner] Segment 7 starts from actual pos [ 120.84164 -157.
40761    -21.957405] (planned waypoint was [ 119.    -159.    -23.
089792])
2441 [Astar_planner] Segment 7: [ 120.84164 -157.40761    -21.957405
]-> [ 122.    -154.    -22.018919]
2442 [Step 8 | Total    0] Target: (122.0, -154.0, -22.0) | Pos: (121.6, -
155.8, -20.4) | Vel: ( -0.5,  0.4, -0.5) | Acc: ( -0.8, -0.5,  1.5) | Rwd
: 7.27
2443 [Astar_planner] Segment 7 reached in 8 steps, dist=2.45, axis_error

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2443 =[0.4423828 1.7784424 1.6247673], reason=running
2444 [Astar_planner] Segment 8 starts from actual pos [ 121.55348 -155.77464 -20.398567] (planned waypoint was [ 122. -154. -22.018919])
2445 [Astar_planner] Segment 8: [ 121.55348 -155.77464 -20.398567 ] -> [ 122. -144. -24.625576]
2446 [Step 33 | Total 0] Target: (122.0, -144.0, -24.6) | Pos: (121.7, -145.9, -22.7) | Vel: ( 0.0, 0.5, -1.8) | Acc: (-1.9, 0.7, 1.8) | Rwd: 4.60
2447 [Astar_planner] Segment 8 reached in 33 steps, dist=2.77, axis_error=[0.31465912 1.9446716 1.9521275 ], reason=running
2448 [Astar_planner] Segment 9 starts from actual pos [ 121.68539 -145.94032 -22.689465] (planned waypoint was [ 122. -144. -24.625576])
2449 [Astar_planner] Segment 9: [ 121.68539 -145.94032 -22.689465 ] -> [ 122. -135. -24.05273]
2450 [Step 34 | Total 0] Target: (122.0, -135.0, -24.1) | Pos: (123.7, -136.9, -22.4) | Vel: (-0.5, 0.2, 0.5) | Acc: (-1.4, 1.5, 0.6) | Rwd: 1.06
2451 [Astar_planner] Segment 9 reached in 34 steps, dist=3.01, axis_error=[1.6524506 1.9373627 1.6102695], reason=running
2452 [Astar_planner] Segment 10 starts from actual pos [ 123.64787 -136.93535 -22.43752] (planned waypoint was [ 122. -135. -24.05273])
2453 [Astar_planner] Segment 10: [ 123.64787 -136.93535 -22.43752 ] -> [ 133. -124. -24.989351]
2454 [Step 42 | Total 0] Target: (133.0, -124.0, -25.0) | Pos: (135.0, -122.7, -23.9) | Vel: (-0.2, -1.3, -1.5) | Acc: ( 1.2, 0.2, 1.2) | Rwd : 1.51
2455 [Astar_planner] Segment 10 reached in 42 steps, dist=2.63, axis_error=[1.9908905 1.3239517 1.1063061], reason=running
2456 [Astar_planner] Segment 11 starts from actual pos [ 134.98943 -122.68763 -23.896807] (planned waypoint was [ 133. -124. -24.989351])
2457 [Astar_planner] Segment 11: [ 134.98943 -122.68763 -23.896807 ] -> [ 144. -113. -24.867386]
2458 [Step 180 | Total 0] Target: (144.0, -113.0, -24.9) | Pos: (145.4, -117.0, -19.8) | Vel: ( 0.0, -0.0, 0.1) | Acc: ( 0.7, -1.4, 0.2) | Rwd : -2.42
2459 [Astar_planner] Segment 11 timeout, dist=6.62
2460 [Astar_planner] Stop after failed segment 11.
2461 Task 35 result: failed (Failed at segment 11: segment_timeout (11/22 reached). Shield interventions=0, emergencies=0, recoveries=0, recovery_steps=0.)
2462
2463 --- Task 36/38 ---
2464 From [ -64.4 -146.72 -4. ] to [ 84.8 -145.42 -4. ]
2465 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y

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2465 =[-200.0, 189.0], occupied=72869 cells (53.4%)
2466 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
2467 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
2468 Path planned via occupancy (Points: 13, Dist: 180.0m). Local target z
      /alt: 0:-4.00(4.00m), 1:-21.18(21.18m), 2:-24.85(24.85m), 3:-21.51(21
      .51m), 4:-23.30(23.30m), 5:-22.71(22.71m), 6:-23.42(23.42m), 7:-22.
      12(22.12m), 8:-24.60(24.60m), 9:-22.37(22.37m), 10:-24.47(24.47m
      ), 11:-20.73(20.73m), 12:-4.00(4.00m)
2469 [Astar_planner] Segment 0: [ -64. -147.  -4.] -> [ -58.    -158.    -
      21.179342]
2470 [Astar_planner] Placing first segment start at [ -64. -147.  -4.]
2471 [Astar_planner] Initial climb to 20.0m.
2472 [Step 12 | Total    0] Target: (-58.0, -158.0, -21.2) | Pos: (-60.0, -
      158.0, -19.9) | Vel: ( 0.5, -0.6, -1.4) | Acc: ( -1.9,  1.8,  0.1) | Rwd
      :  5.10
2473 [Astar_planner] Segment 0 reached in 12 steps, dist=2.35, axis_error
      =[1.9541321  0.01303101 1.2970486 ], reason=running
2474 [Astar_planner] Segment 1 starts from actual pos [ -59.949223 -157.
      99251  -19.894922] (planned waypoint was [ -58.    -158.    -21.
      179342])
2475 [Astar_planner] Segment 1: [ -59.949223 -157.99251  -19.894922
      ] -> [ -45.    -161.    -24.847073]
2476 [Step 91 | Total    0] Target: (-45.0, -161.0, -24.8) | Pos: (-46.9, -
      159.8, -24.9) | Vel: ( 0.2, -0.5,  0.3) | Acc: ( 0.0, -1.8, -1.3) | Rwd
      :  1.38
2477 [Astar_planner] Segment 1 reached in 91 steps, dist=2.25, axis_error
      =[1.894104  1.220932  0.06361771], reason=running
2478 [Astar_planner] Segment 2 starts from actual pos [ -46.89196 -159.
      78398  -24.908392] (planned waypoint was [ -45.    -161.    -24.
      847073])
2479 [Astar_planner] Segment 2: [ -46.89196 -159.78398  -24.908392
      ] -> [ -31.    -161.    -21.512892]
2480 [Step 180 | Total    0] Target: (-31.0, -161.0, -21.5) | Pos: (-38.5, -
      162.3, -28.2) | Vel: ( -0.4,  0.1,  0.1) | Acc: ( 0.4, -0.1, -0.7) | Rwd
      : -1.85
2481 [Astar_planner] Segment 2 timeout, dist=10.15
2482 [Astar_planner] Stop after failed segment 2.
2483 Task 36 result: failed (Failed at segment 2: segment_timeout (2/12
      reached). Shield interventions=224, emergencies=16, recoveries=38
      , recovery_steps=224.)
2484
2485 --- Task 37/38 ---
2486 From [118.1  76.88  -4. ] to [-94.8  92.08  -4. ]
2487 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
2488 [Astar_planner] Occupancy base_grid=390x350, occupied=72869

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2488 cells (53.4%)
2489 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
2490 Path planned via occupancy (Points: 24, Dist: 336.0m). Local target z
    /alt: 0:-4.00(4.00m), 1:-24.59(24.59m), 2:-23.60(23.60m), 3:-21.94(21
    .94m), 4:-21.08(21.08m), 5:-21.98(21.98m), 6:-20.31(20.31m), 7:-21.
    44(21.44m), 8:-20.14(20.14m), 9:-21.07(21.07m), 10:-22.81(22.81m
    ), 11:-23.95(23.95m), 12:-24.26(24.26m), 13:-20.24(20.24m), 14:-23.
    77(23.77m), 15:-24.37(24.37m), 16:-22.45(22.45m), 17:-24.41(24.
    41m), 18:-20.44(20.44m), 19:-23.75(23.75m), 20:-23.48(23.48m), 21
    :-24.13(24.13m), 22:-21.06(21.06m), 23:-4.00(4.00m)
2491 [Astar_planner] Segment 0: [118. 77. -4.] -> [108.    86.   -24.
    593882]
2492 [Astar_planner] Placing first segment start at [118. 77. -4.]
2493 [Astar_planner] Initial climb to 20.0m.
2494 [Step 24 | Total    0] Target: (108.0, 86.0, -24.6) | Pos: (108.7, 85.
    7, -23.1) | Vel: ( -0.8,  0.7, -1.0) | Acc: ( 1.2,  0.9,  0.8) | Rwd: 604.
    41 -> Episode End: ARRIVED!
2495
2496 [Astar_planner] Segment 0 reached in 24 steps, dist=1.65, axis_error
    =[0.72434235 0.27853394 1.4551525 ], reason=arrived
2497 [Astar_planner] Segment 1 starts from actual pos [108.717224 85.
    72785 -23.147411] (planned waypoint was [108.    86.   -24.
    593882])
2498 [Astar_planner] Segment 1: [108.717224 85.72785 -23.147411] -> [
    105.    99.   -23.596596]
2499 [Step 13 | Total    0] Target: (105.0, 99.0, -23.6) | Pos: (105.1, 100
    .6, -21.9) | Vel: ( 0.2,  1.3, -1.2) | Acc: ( -0.1, -1.4, -0.7) | Rwd: 2.
    04
2500 [Astar_planner] Segment 1 reached in 13 steps, dist=2.34, axis_error
    =[0.13037872 1.612175  1.6960869 ], reason=running
2501 [Astar_planner] Segment 2 starts from actual pos [105.13146 100.
    62009 -21.907902] (planned waypoint was [105.    99.   -23.
    596596])
2502 [Astar_planner] Segment 2: [105.13146 100.62009 -21.907902] -> [
    105.   113.   -21.935883]
2503 [Step 57 | Total    0] Target: (105.0, 113.0, -21.9) | Pos: (105.1,
    114.8, -22.2) | Vel: ( 0.6, -0.4,  0.1) | Acc: ( -0.5,  0.2, -0.7) | Rwd
    : 0.33
2504 [Astar_planner] Segment 2 reached in 57 steps, dist=1.85, axis_error
    =[0.10626221 1.8268661 0.27405167], reason=running
2505 [Astar_planner] Segment 3 starts from actual pos [105.111755 114.
    823296 -22.209448] (planned waypoint was [105.   113.   -21.
    935883])
2506 [Astar_planner] Segment 3: [105.111755 114.823296 -22.209448
    ] -> [104.   127.   -21.082148]
2507 [Step 19 | Total    0] Target: (104.0, 127.0, -21.1) | Pos: (105.0,
    125.0, -20.3) | Vel: ( -0.2,  2.1,  2.1) | Acc: ( 0.4, -0.6, -0.4) | Rwd

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2507 : 1.42
2508 [Astar_planner] Segment 3 reached in 19 steps, dist=2.33, axis_error
=[0.95012665 1.972374 0.7946873 ], reason=running
2509 [Astar_planner] Segment 4 starts from actual pos [104.94877 125.
04032 -20.27483] (planned waypoint was [104. 127. -21.
082148])
2510 [Astar_planner] Segment 4: [104.94877 125.04032 -20.27483] -> [ 95
. 135. -21.979898]
2511 [Step 17 | Total 0] Target: ( 95.0, 135.0, -22.0) | Pos: ( 95.7, 136.
5, -20.4) | Vel: ( -0.1, -0.9, 0.6) | Acc: ( 0.5, -1.7, -1.1) | Rwd: 5.
44
2512 [Astar_planner] Segment 4 reached in 17 steps, dist=2.29, axis_error
=[0.7212677 1.5422516 1.53652 ], reason=running
2513 [Astar_planner] Segment 5 starts from actual pos [ 95.720215 136.
53413 -20.437717] (planned waypoint was [ 95. 135. -21.
979898])
2514 [Astar_planner] Segment 5: [ 95.720215 136.53413 -20.437717] -> [
81. 135. -20.310034]
2515 [Step 31 | Total 0] Target: ( 81.0, 135.0, -20.3) | Pos: ( 82.3, 133.
7, -19.2) | Vel: ( -1.4, 0.6, -1.4) | Acc: ( 0.5, 0.1, -0.8) | Rwd: 4.
53
2516 [Astar_planner] Segment 5 reached in 31 steps, dist=2.16, axis_error
=[1.3399582 1.3085785 1.0837612], reason=running
2517 [Astar_planner] Segment 6 starts from actual pos [ 82.32762 133.
69643 -19.238838] (planned waypoint was [ 81. 135. -20.
310034])
2518 [Astar_planner] Segment 6: [ 82.32762 133.69643 -19.238838] -> [
67. 136. -21.444117]
2519 [Step 18 | Total 0] Target: ( 67.0, 136.0, -21.4) | Pos: ( 67.6, 136.
7, -19.5) | Vel: ( -0.7, -0.2, 0.9) | Acc: ( 1.0, -1.1, -0.2) | Rwd: 604.
53 -> Episode End: ARRIVED!
2520
2521 [Astar_planner] Segment 6 reached in 18 steps, dist=2.20, axis_error
=[0.62047577 0.7486572 1.9717445 ], reason=arrived
2522 [Astar_planner] Segment 7 starts from actual pos [ 67.61651 136.
74744 -19.466717] (planned waypoint was [ 67. 136. -21.
444117])
2523 [Astar_planner] Segment 7: [ 67.61651 136.74744 -19.466717] -> [
53. 136. -20.139097]
2524 [Step 26 | Total 0] Target: ( 53.0, 136.0, -20.1) | Pos: ( 54.7, 135.
5, -19.1) | Vel: ( -0.4, 0.0, -0.0) | Acc: ( -1.9, 0.0, 0.4) | Rwd: 0.
88
2525 [Astar_planner] Segment 7 reached in 26 steps, dist=2.11, axis_error
=[1.7493324 0.47946167 1.0831833 ], reason=running
2526 [Astar_planner] Segment 8 starts from actual pos [ 54.74693 135.
52081 -19.056093] (planned waypoint was [ 53. 136. -20.
139097])
2527 [Astar_planner] Segment 8: [ 54.74693 135.52081 -19.056093] -> [

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2527 39.    136.    -21.072214]
2528 [Step 22 | Total    0] Target: ( 39.0, 136.0, -21.1) | Pos: ( 39.0, 135.
    8, -19.5) | Vel: ( -0.5, -0.3,  0.8) | Acc: ( -0.2, -0.5,  0.1) | Rwd: 602.
    39 -> Episode End: ARRIVED!
2529
2530 [Astar_planner] Segment 8 reached in 22 steps, dist=1.61, axis_error
    =[0.04739761 0.1852417  1.5976868 ], reason=arrived
2531 [Astar_planner] Segment 9 starts from actual pos [ 38.948086 135.
    81213 -19.467085] (planned waypoint was [ 39.    136.    -21.
    072214])
2532 [Astar_planner] Segment 9: [ 38.948086 135.81213 -19.467085] -> [
    26.    140.    -22.813864]
2533 [Step 23 | Total    0] Target: ( 26.0, 140.0, -22.8) | Pos: ( 27.7, 140.
    9, -21.8) | Vel: ( -0.5,  0.1, -0.4) | Acc: (  1.2,  0.9,  1.7) | Rwd: 2.25

2534 [Astar_planner] Segment 9 reached in 23 steps, dist=2.17, axis_error
    =[1.6725807  0.90211487 1.038435 ], reason=running
2535 [Astar_planner] Segment 10 starts from actual pos [ 27.667957 140.
    90335 -21.77893 ] (planned waypoint was [ 26.    140.    -22.
    813864])
2536 [Astar_planner] Segment 10: [ 27.667957 140.90335 -21.77893
    ] -> [ 16.    149.    -23.953745]
2537 [Step 23 | Total    0] Target: ( 16.0, 149.0, -24.0) | Pos: ( 16.9, 150.
    7, -22.2) | Vel: ( -1.8,  0.8, -1.1) | Acc: ( -1.0,  0.6,  1.9) | Rwd: 3.
    82
2538 [Astar_planner] Segment 10 reached in 23 steps, dist=2.60,
    axis_error=[0.9170227 1.7062378 1.7332401], reason=running
2539 [Astar_planner] Segment 11 starts from actual pos [ 16.90617 150.
    71127 -22.227058] (planned waypoint was [ 16.    149.    -23.
    953745])
2540 [Astar_planner] Segment 11: [ 16.90617 150.71127 -22.227058
    ] -> [  3.    152.    -24.263239]
2541 [Step 18 | Total    0] Target: (  3.0, 152.0, -24.3) | Pos: (  4.8, 152.8
    , -22.6) | Vel: ( -0.9,  0.1, -1.7) | Acc: ( -0.6, -0.1,  1.6) | Rwd:  5.00
2542 [Astar_planner] Segment 11 reached in 18 steps, dist=2.58,
    axis_error=[1.7855439 0.8428955 1.6640205], reason=running
2543 [Astar_planner] Segment 12 starts from actual pos [  4.7741737 152.
    84393 -22.619528 ] (planned waypoint was [  3.    152.    -24.
    263239])
2544 [Astar_planner] Segment 12: [  4.7741737 152.84393 -22.619528
    ] -> [-11.    153.    -20.242783]
2545 [Step 24 | Total    0] Target: (-11.0, 153.0, -20.2) | Pos: (-10.7, 151
    .3, -18.8) | Vel: ( -0.7,  0.8, -0.8) | Acc: ( -0.2, -0.4,  2.0) | Rwd:  3.
    06
2546 [Astar_planner] Segment 12 reached in 24 steps, dist=2.26,
    axis_error=[0.33560848 1.7259827 1.4187851 ], reason=running
2547 [Astar_planner] Segment 13 starts from actual pos [-10.668295 151.
    27863 -18.828869] (planned waypoint was [-11.    153.    -20.

```

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2547 242783))
2548 [Astar_planner] Segment 13: [-10.668295 151.27863 -18.828869
] -> [-24.    150.    -23.77115]
2549 [Step 15 | Total    0] Target: (-24.0, 150.0, -23.8) | Pos: (-22.2, 148
.5, -22.9) | Vel: ( -0.4, -0.0, 0.6) | Acc: ( 0.2, 0.3, 0.4) | Rwd: 1.
32
2550 [Astar_planner] Segment 13 reached in 15 steps, dist=2.52,
axis_error=[1.8033199 1.4988251 0.91073036], reason=running
2551 [Astar_planner] Segment 14 starts from actual pos [-22.201385 148.
50102 -22.853352] (planned waypoint was [-24.    150.    -23.
77115])
2552 [Astar_planner] Segment 14: [-22.201385 148.50102 -22.853352
] -> [-28.    138.    -24.372402]
2553 [Step 12 | Total    0] Target: (-28.0, 138.0, -24.4) | Pos: (-26.3, 139
.7, -22.9) | Vel: ( -0.2, -0.6, -0.0) | Acc: ( 0.1, -2.0, 0.2) | Rwd: 1.
95
2554 [Astar_planner] Segment 14 reached in 12 steps, dist=2.85,
axis_error=[1.7494392 1.6912537 1.4887981], reason=running
2555 [Astar_planner] Segment 15 starts from actual pos [-26.252058 139.
686 -22.883732] (planned waypoint was [-28.    138.    -24.
372402])
2556 [Astar_planner] Segment 15: [-26.252058 139.686 -22.883732] -> [-
28.    124.    -22.446394]
2557 [Step 23 | Total    0] Target: (-28.0, 124.0, -22.4) | Pos: (-26.2, 123
.8, -21.3) | Vel: ( -0.5, 0.5, -1.3) | Acc: ( -0.1, 2.0, 2.0) | Rwd: 2.
57
2558 [Astar_planner] Segment 15 reached in 23 steps, dist=2.13,
axis_error=[1.8022423 0.24708557 1.1167908 ], reason=running
2559 [Astar_planner] Segment 16 starts from actual pos [-26.20214 123.
75774 -21.341064] (planned waypoint was [-28.    124.    -22.
446394])
2560 [Astar_planner] Segment 16: [-26.20214 123.75774 -21.341064
] -> [-28.    110.    -24.411129]
2561 [Step 24 | Total    0] Target: (-28.0, 110.0, -24.4) | Pos: (-29.8, 110
.0, -23.8) | Vel: ( 0.7, -0.4, 0.0) | Acc: ( 0.8, -1.0, -0.1) | Rwd: 1.
80
2562 [Astar_planner] Segment 16 reached in 24 steps, dist=1.93,
axis_error=[1.8169918 0.00854492 0.6544914 ], reason=running
2563 [Astar_planner] Segment 17 starts from actual pos [-29.81059 110.
00469 -23.756266] (planned waypoint was [-28.    110.    -24.
411129])
2564 [Astar_planner] Segment 17: [-29.81059 110.00469 -23.756266
] -> [-29.    96.    -20.444525]
2565 [Step 17 | Total    0] Target: (-29.0, 96.0, -20.4) | Pos: (-29.5, 97.
3, -21.1) | Vel: ( 0.4, -0.5, -0.0) | Acc: ( 1.0, -1.4, -0.4) | Rwd: 602.
32 -> Episode End: ARRIVED!
2566
2567 [Astar_planner] Segment 17 reached in 17 steps, dist=1.54,

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2567 axis_error=[0.51399994 1.3030014 0.6505089 ], reason=arrived
2568 [Astar_planner] Segment 18 starts from actual pos [-29.510101 97.
298584 -21.095224] (planned waypoint was [-29.    96.    -20.
444525])
2569 [Astar_planner] Segment 18: [-29.510101 97.298584 -21.095224
]-> [-40.    90.    -23.746258]
2570 [Step 23 | Total    0] Target: (-40.0, 90.0, -23.7) | Pos: (-39.1, 91.
8, -22.2) | Vel: ( -0.8, -0.0, -0.5) | Acc: ( -0.5, 0.0, 2.0) | Rwd: 3.
91
2571 [Astar_planner] Segment 18 reached in 23 steps, dist=2.52,
axis_error=[0.94483566 1.7563248 1.542305 ], reason=running
2572 [Astar_planner] Segment 19 starts from actual pos [-39.062  91.
756165 -22.208273] (planned waypoint was [-40.    90.    -23.
746258])
2573 [Astar_planner] Segment 19: [-39.062  91.756165 -22.208273] -> [-
54.    90.    -23.475138]
2574 [Step 19 | Total    0] Target: (-54.0, 90.0, -23.5) | Pos: (-54.1, 91.
6, -22.0) | Vel: ( -0.7, 0.0, -0.6) | Acc: ( 1.3, -0.1, 2.0) | Rwd: 3.
17
2575 [Astar_planner] Segment 19 reached in 19 steps, dist=2.16,
axis_error=[0.07181168 1.6067352 1.4443626 ], reason=running
2576 [Astar_planner] Segment 20 starts from actual pos [-54.07812 91.
60708 -22.03624] (planned waypoint was [-54.    90.    -23.
475138])
2577 [Astar_planner] Segment 20: [-54.07812 91.60708 -22.03624] -> [-68
.    89.    -24.131464]
2578 [Step 19 | Total    0] Target: (-68.0, 89.0, -24.1) | Pos: (-66.2, 89.
3, -22.7) | Vel: ( -0.4, -0.3, -0.1) | Acc: ( 0.6, -0.7, 2.0) | Rwd: 3.
58
2579 [Astar_planner] Segment 20 reached in 19 steps, dist=2.29,
axis_error=[1.7522583 0.3412323 1.4305782], reason=running
2580 [Astar_planner] Segment 21 starts from actual pos [-66.24991 89.
33969 -22.701492] (planned waypoint was [-68.    89.    -24.
131464])
2581 [Astar_planner] Segment 21: [-66.24991 89.33969 -22.701492] -> [-
82.    89.    -21.056738]
2582 [Step 24 | Total    0] Target: (-82.0, 89.0, -21.1) | Pos: (-83.8, 89.
6, -20.1) | Vel: ( 0.7, 0.3, -0.0) | Acc: ( 1.6, 1.3, 0.6) | Rwd: -0.21
2583 [Astar_planner] Segment 21 reached in 24 steps, dist=2.15,
axis_error=[1.8044891 0.6134262 0.9921894], reason=running
2584 [Astar_planner] Segment 22 starts from actual pos [-83.79658 89.
61724 -20.064993] (planned waypoint was [-82.    89.    -21.
056738])
2585 [Astar_planner] Segment 22: [-83.79658 89.61724 -20.064993] -> [-
95.  92.  -4.]
2586 [Step 33 | Total    0] Target: (-95.0, 92.0, -4.0) | Pos: (-93.8, 91.0
, -3.2) | Vel: ( -2.0, 0.8, -1.2) | Acc: ( 0.6, 1.7, -1.7) | Rwd: 3.65

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2587 [Astar_planner] Segment 22 reached in 33 steps, dist=1.78,
axis_error=[1.220253 1.0034866 0.82001925], reason=running
2588 Task 37 result: success (Success: Reached destination (23/23
segments). Shield interventions=38, emergencies=1, recoveries=8,
recovery_steps=38.)
2589
2590 --- Task 38/38 ---
2591 From [ -63.8 -136.72 -4. ] to [107. -82.52 -4. ]
2592 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
2593 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
2594 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
2595 Path planned via occupancy (Points: 23, Dist: 260.1m). Local target z
/alt: 0:-4.00(4.00m), 1:-20.37(20.37m), 2:-23.40(23.40m), 3:-23.13(23
.13m), 4:-21.83(21.83m), 5:-21.89(21.89m), 6:-20.02(20.02m), 7:-20.
10(20.10m), 8:-20.61(20.61m), 9:-23.65(23.65m), 10:-20.58(20.58m
), 11:-23.20(23.20m), 12:-21.90(21.90m), 13:-21.65(21.65m), 14:-23.
95(23.95m), 15:-21.86(21.86m), 16:-23.89(23.89m), 17:-21.47(21.
47m), 18:-20.85(20.85m), 19:-22.19(22.19m), 20:-23.82(23.82m), 21
:-22.57(22.57m), 22:-4.00(4.00m)
2596 [Astar_planner] Segment 0: [ -64. -137. -4.] -> [ -59. -146. -
20.365273]
2597 [Astar_planner] Placing first segment start at [ -64. -137. -4.]
2598 [Astar_planner] Initial climb to 20.0m.
2599 [Step 11 | Total 0] Target: (-59.0, -146.0, -20.4) | Pos: (-60.9, -
146.7, -19.4) | Vel: ( 1.6, -1.4, -0.9) | Acc: ( 1.8, -0.4, 1.7) | Rwd
: 4.65
2600 [Astar_planner] Segment 0 reached in 11 steps, dist=2.29, axis_error
=[1.9336853 0.70680237 1.0137901 ], reason=running
2601 [Astar_planner] Segment 1 starts from actual pos [ -60.919445 -146.
71931 -19.359089] (planned waypoint was [ -59. -146. -20.
365273])
2602 [Astar_planner] Segment 1: [ -60.919445 -146.71931 -19.359089
] -> [ -59. -157. -23.402563]
2603 [Step 27 | Total 0] Target: (-59.0, -157.0, -23.4) | Pos: (-60.7, -
156.8, -22.9) | Vel: ( 0.4, -1.0, -0.2) | Acc: ( -0.3, -1.3, 0.8) | Rwd
: 2.52
2604 [Astar_planner] Segment 1 reached in 27 steps, dist=1.83, axis_error
=[1.7347641 0.24363708 0.52124023], reason=running
2605 [Astar_planner] Segment 2 starts from actual pos [ -60.730896 -156.
76538 -22.882908] (planned waypoint was [ -59. -157. -23.
402563])
2606 [Astar_planner] Segment 2: [ -60.730896 -156.76538 -22.882908
] -> [ -49. -161. -23.12712]
2607 [Step 11 | Total 0] Target: (-49.0, -161.0, -23.1) | Pos: (-49.6, -
160.3, -22.4) | Vel: ( 1.5, 0.2, 0.2) | Acc: ( 0.1, 1.9, -0.1) | Rwd:

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2607 605.12 -> Episode End: ARRIVED!
2608
2609 [Astar_planner] Segment 2 reached in 11 steps, dist=1.13, axis_error
    =[0.5596504 0.67967224 0.7080765 ], reason=arrived
2610 [Astar_planner] Segment 3 starts from actual pos [ -49.54601 -160.
    31839 -22.417309] (planned waypoint was [ -49.    -161.    -23.
    12712])
2611 [Astar_planner] Segment 3: [ -49.54601 -160.31839 -22.417309
    ] -> [ -38.    -161.    -21.825445]
2612 [Step 79 | Total    0] Target: (-38.0, -161.0, -21.8) | Pos: (-39.8, -
    160.5, -22.0) | Vel: ( 0.7, 0.4, 0.1) | Acc: ( 0.9, 1.4, -0.7) | Rwd:
    0.70
2613 [Astar_planner] Segment 3 reached in 79 steps, dist=1.86, axis_error
    =[1.7785454 0.50172424 0.1821003 ], reason=running
2614 [Astar_planner] Segment 4 starts from actual pos [ -39.774292 -160.
    4958 -22.006992] (planned waypoint was [ -38.    -161.    -21.
    825445])
2615 [Astar_planner] Segment 4: [ -39.774292 -160.4958 -22.006992
    ] -> [ -27.    -161.    -21.894249]
2616 [Step 43 | Total    0] Target: (-27.0, -161.0, -21.9) | Pos: (-28.9, -
    159.5, -21.5) | Vel: ( 0.5, -0.7, 0.0) | Acc: ( 0.9, -2.0, 0.0) | Rwd:
    1.46
2617 [Astar_planner] Segment 4 reached in 43 steps, dist=2.44, axis_error
    =[1.8688831 1.5119476 0.43980026], reason=running
2618 [Astar_planner] Segment 5 starts from actual pos [ -28.862421 -159.
    49603 -21.454071] (planned waypoint was [ -27.    -161.    -21.
    894249])
2619 [Astar_planner] Segment 5: [ -28.862421 -159.49603 -21.454071
    ] -> [ -16.    -161.    -20.019056]
2620 [Step 180 | Total    0] Target: (-16.0, -161.0, -20.0) | Pos: (-27.7, -
    164.0, -30.5) | Vel: ( -0.3, 0.1, 0.1) | Acc: ( 0.3, -0.1, -0.7) | Rwd
    : -1.71
2621 [Astar_planner] Segment 5 timeout, dist=16.01
2622 [Astar_planner] Stop after failed segment 5.
2623 Task 38 result: failed (Failed at segment 5: segment_timeout (5/22
    reached). Shield interventions=262, emergencies=8, recoveries=45,
    recovery_steps=262.)
2624
2625 =====
2626 BATCH TEST SUMMARY
2627 Total Tasks: 38
2628 Success: 20
2629 Arrival Rate: 52.63%
2630 Full results saved to: E:\Github\AirSim_RL\Src\batch_results.json
2631 =====
2632
2633 进程已结束 退出代码为0
2634

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